

वार्षिक प्रतिवेदन Annual Report 2019



ICAR-National Academy of Agricultural Research Management
Rajendranagar, Hyderabad - 500 030, India

ANNUAL REPORT

वार्षिक प्रतिवेदन

2019



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Abbreviations

AAU	Assam Agricultural University
ABF	Agri-Biotech Foundation
ABIs	Agri-Business Incubators
ADOPT	Adoption and Diffusion Outcome Prediction Tool
AI	Artificial Intelligence
AICRP	All-India Coordinated Research Project
AICTE	All India Council for Technical Education
a-IDEA	Association for Innovation Development of Entrepreneurship in Agriculture
ANGRAU	Acharya N. G. Ranga Agricultural University
AHP	Analytical Hierarchy Process
APMC	Agricultural Produce Market Committee
ARS	Agricultural Research Service
ASCI	Administrative Staff College of India
ASRB	Agricultural Scientists Recruitment Board
ASTI	Agricultural Science and Technology Indicators
ATARI	Agricultural Technology Application Research Institute
AUs	Agricultural Universities
BARC	Bangladesh Agricultural Research Council
BASU	Bihar Animal Sciences University
BIA	Bihar Industries Association
BIRAC	Biotechnology Industry Research Assistance Council
CAAST	Centre of Advanced Agricultural Science and Technology
CAI	Centre for Agri-Innovation
CART	Classification & Regression Tree algorithm
CAT	Common Admission Test
CAZRI	Central Arid Zone Research Institute
CBWE	Central Board for Workers Education
CCMB	Centre for Cellular and Molecular Biology
CSS	Centrally Sponsored Schemes
CDVL	Centre for Distance and Virtual Learning
CGIAR	Consultative Group for International Agricultural Research
CIFA	Central Institute of Freshwater Aquaculture
CIFE	Central Institute of Fisheries Education
CIFT	Central Institute of Fisheries Technology
CIIE	Centre for Innovation Incubation and Entrepreneurship
CIL	Coromandel International Limited
CIMAP	Central Institute of Medicinal & Aromatic plants

CIMMYT	International Maize and Wheat Improvement Center
CMAT	Common Management Admission Test
COLLAge	Centre for Open and Lifelong Learning in Agriculture
CPI	Consumer Price Index
CRIDA	Central Research Institute for Dry land Agriculture
CSIR	Council of Scientific & Industrial Research
CSMCRI	Central Salt and Marine Chemicals Research Institute
CTCRI	Central Tuber Crops Research Institute
CTRI	Central Tobacco Research Institute
CYP	Common Yoga Protocol
DANCE	Decoding Agri-Startup; Nurturing & Creating Entrepreneurs
DARE	Department of Agriculture Research and Education
DBT	Department of Biotechnology
DDG	Deputy Director General
DFID	Department of International Development
DPR	Detailed Project Report
DST	Department of Science and Technology
DU	Delhi University
DWRP	Developing Winning Research Proposal
EG	Enterprise Guide
EMD	Employee Master Data
EMRC	Educational Multimedia Research Centre
e-NAM	National Agriculture Market
ET	Education Technology
EVI	Enhanced Vegetation Index
FAO	Food & Agriculture Organization
FCI	Food Corporation of India
FCV	Flue Cured Virginia
FDP	Faculty Development Programme
FET	Field Experience Training
FICCI	Federation of Indian Chamber of Commerce and Industry
FLIs	Farmer Led Innovations
FMCG	Fast Moving Consumer Goods
FOCARS	Foundation Course for Agricultural Research Service
FOCAU	Foundation Course For Faculty of Agricultural Universities
FPC	Farmer Producer Companies
FPOs	Farmer Producer Organizations
FTAs	Farm Tele Advisors
FYM	Farm Yard Manure

GBM	General Body Meeting
GBPUA&T	Govind Ballabh Pant University of Agriculture and Technology
GDD	Growing Degree Days
GI	Geographical Indications
GIS	Geographic Information System
Goi	Government of India
GPS	Global Positioning System
GST	Goods and Services Tax
HAU	Haryana Agricultural University
HRD	Human Resource Development
IARI	Indian Agricultural Research Institute
IAS	Indian Administrative Service
IASRI	Indian Agricultural Statistics Research Institute
ICAR	Indian Council of Agricultural Research
IIAB	Indian Institute of Agricultural Biotechnology
IINRG	Indian Institute of Natural Resins and Gums
ICM	Institute of Cooperative Management
ICMR	Indian Council of Medical Research
ICRISAT	International Crops Research Institute for Semi-Arid Tropics
ICT	Information and Communication Technology
ICZM	Integrated Coastal Zone Management
IFPRI	International Food Policy Research Institute
IFS	Indian Forest Service
IGKV	Indira Gandhi Krishi Vishwavidyalaya
IGNFA	Indira Gandhi National Forest Academy
IGNOU	Indira Gandhi National Open University
IIIT	Indian Institute of Information Technology
IIM	Indian Institute of Management
IIMR	Indian Institute of Millets Research
IIOPR	Indian Institute of Oil Palm Research
IIPM	Indian Institute of Plantation Management
IIRR	Indian Institute of Rice Research
IMC	Institute Management Committee
IoT	Internet of Things (IoT)
IPCC	Intergovernmental Panel on Climate Change
IPE	Institute of Public Enterprise
IPTM	Intellectual Property & Technology Management
IRC	Institute Research Council
IRM	Insect Resistance Management

ISB	Indian School of Business
ISEE	Indian Society of Extension Education
ISWC	Indian Institute of Soil & Water Conservation
IT	Information Technology
IWM	Integrated Watershed Management
JAC	Joint Advisory Council
JNTU	Jawaharlal Nehru Technological University
JNU	Jawaharlal Nehru University
KCCs	Kisan Call Centers
KVKs	Krishi Vigyan Kendras
LAN	Local Area Network
LDA	Latent Dirichlet Allocation
LMS	Learning Management System
MANAGE	National Institute of Agricultural Extension Management
MCCIA	Mahratta Chamber of Commerce Industries and Agriculture
MDP	Management Development Programme
MLA	Member of Legislative Assembly
MoEFCC	Ministry of Environment, Forest and Climate Change
MOOC	Massive Open Online Courses
MORE	Marketing Officers' Resurgence & Excellence
MoU	Memorandum of Understanding
MPKV	Mahatma Phule Krishi Vidyapeeth
MPUA&T	Maharana Pratap University of Agriculture and Technology
MSP	Minimum Support Price
MWDS	Mean Weighted Discrepancy Scores
NAAC	National Assessment & Accreditation Agency
NAARM	National Academy of Agricultural Research Management
NAAS	National Academy of Agricultural Sciences
NABARD	National Bank for Agricultural and Rural Development
NAHEP	National Agricultural Higher Education Project
NAIF	National Agricultural Innovation Found
NARC	Nepal Agricultural Research Council
NARES	National Agricultural Research and Education System
NARS	National Agricultural Research System
NASF	National Agricultural Science Fund
NASI	National Academy of Sciences, India
NBA	National Board of Accreditation
NBPGR	National Bureau of Plant Genetic Resources
NBSSLUP	National Bureau of Soil Survey and Land Use Planning

NCSCM	National Centre for Sustainable Coastal Management
NDDI	Normalized Difference Drought Index
NDRI	National Dairy Research Institute
NDVI	Normalized Difference Vegetation Index
NDWI	Normalized Difference Water Index
NEAT	New Entrants Assimilation Training
NF	Natural Farming
NFDB	National Fisheries Development Board
NGOS	Non-Government Organizations
NIAP	National Institute of Agricultural Economics and Policy Research
NIDM	National Institute for Disaster management
NIF	National Innovation Foundation
NIPHM	National Institute of Plant Health Management
NIRD &PR	National Institute of Rural Development and Panchayat Raj
NITI	National Institution for Transforming India
NPAA	NAARM PG Alumni Association
NRCO	National Research Centre for Orchids
NRM	Natural Resource Management
NRRI	National Rice Research Institute
NSTEDB	National Science &Technology Entrepreneurship Development Board
OPAC	Online Public Access Catalog
OUAT	Odisha University of Agriculture and Technology
PARC	Pakistan Agricultural Research Council
PCA	Principal Component Analysis
PDP	Project Directorate on Poultry
PGDETM	Post Graduate Diploma in Education Technology Management
PGDMA	Post Graduate Diploma in Management-Agriculture
PGDTMA	Post Graduate Diploma in Technology Management in Agriculture
PJTSAU	Professor Jayashankar Telangana State Agricultural University
PLW	Project Log frame writer
PME	Priority Setting, Monitoring and Evaluation
PPV&FRA	Protection of Plant Varieties & Farmer Rights Authority
PRA	Participatory Rural Appraisal
PVNRTVU	PV Narasimha Rao Telangana Veterinary University
RAC	Research Advisory committee
RCER	Research Complex for Eastern Region
RCW	Research Concept Writer
REACH	Registration, Evaluation, Authorization and Restriction of Chemicals
RMP	Risk Management Plan

RVSKVV	Rajamata Vijayaraja Scindia Krishi Vishwa Vidyalaya
SAARC	South Asian Association for Regional Cooperation
SAS	Statistical Analysis System
SAUs	State Agricultural Universities
SDG	Sustainable Development Goals
SEED	Skill Enhancing and Employees Development
SFC	Specific Fuel Consumption
SKLTSHU	Sri Konda Laxman Telangana State Horticultural University
SLA	Students' Learning Approaches
SLACC	Sustainable Livelihoods and Adaptation to Climate Change
SPICMACAY	Society for the Promotion of Indian Classical Music and Culture Amongst Youth
SSS	Skilled Support Staff
SVVU	Sri Venkateswara Veterinary University
TANUVAS	Tamil Nadu Veterinary and Animal Sciences University
TBI	Technology Business Incubator
TCQ	Teaching Competency Questionnaire
TELAGe	Technology Enhanced Learning in Agricultural Education
TEP	Technical Evaluation Panel
TERI	The Energy and Resources Institute
TIFAC	Technology Information Forecasting and Assessment Council
TNAU	Tamil Nadu Agricultural University
ToT	Training of Trainers
TPACK	Technology Pedagogy and Content Knowledge
UAHS	University of Agricultural and Horticultural Sciences
UASB	University of Agricultural Sciences, Bangalore
UGC	University Grants Commission
UK	United Kingdom
UNDP	United Nations Development Programme
UNFCC	United Nations Framework Convention on Climate Change
UoH	University of Hyderabad
USA	United States of America
VNMKV	Vasantrao Naik Marathwada Agricultural University
VPKAS	Vivekananda Parvatiya Krishi Anusandhan Sansthan
WDP	Wheat Development Programme
WPI	Wholesale Price Index
WSP	Water Soluble Pouch
ZBNF	Zero Budget Natural Farming
ZTMU	Zonal Technology Management Unit

निदेशक की कलम से



डा. सीएच श्रीनिवास राव
निदेशक

भारत में, राष्ट्रीय कृषि अनुसंधान और शिक्षा प्रणाली (NARES) के अंतर्गत भा.कृ.अनु.पू. के कृषि वैज्ञानिकों और कृषि विश्वविद्यालयों के संकाय सदस्यों की क्षमता को अद्यतन बनाने के लिए भारतीय कृषि अनुसन्धान परिषद् के अंतर्गत भा.कृ.अनु.पू.- रा. कृ. अनु. प्र.अ.(ICAR-NAARM) एक मुख्य नोडल प्रशिक्षण और क्षमता निर्माण संस्थान के रूप में अपना योगदान दे रहा है। वर्ष 2019 के दौरान, कृषि और संबद्ध विज्ञान के क्षेत्र में क्षमता निर्माण, अनुसंधान, शिक्षा और उद्यमिता विकास के क्षेत्र में अकादमी की प्रगति और उपलब्धियों को अभिलिखित करने पर आज मुझे बेहद खुशी का अनुभव हो रहा है। अकादमी, मौजूदा प्रशिक्षण कार्यक्रमों में लगातार सुधार करके और आवश्यकता-आधारित कार्यक्रमों को जोड़कर और रणनीतिक विषयगत क्षेत्रों में विशेषज्ञों को आमंत्रित कर एवम उनके सुझाव को शामिल

करके विविध पोर्टफोलियो को विकसित करने के लिए लगातार प्रयासरत रहती है। वर्ष 2019 के दौरान, अकादमी ने विभिन्न कार्यक्रमों के माध्यम से देश भर के लगभग 3,976 प्रतिभागियों को प्रशिक्षित करने का कार्य किया है।

फाउंडेशन कोर्स फॉर एग्रीकल्चरल रिसर्च सर्विसेज (FOCARS), अकादमी का एक प्रमुख कार्यक्रम है और वर्ष 2019 के दौरान, भा.कृ.अनु.पू.-रा. कृ. अनु. प्र. अ.(ICAR-NAARM) ने 109 वें फाउंडेशन कोर्स बैच का संचालन किया है, जिसके माध्यम से कुल 32 नए भर्ती किए गए परिवीक्षाधीन वैज्ञानिकों को आधारभूत प्रशिक्षण प्रदान किया गया है। इस मुख्य कार्यक्रम के अलावा, अकादमी ने प्रबंधन विकास कार्यक्रम(एमडीपी), आवश्यकता-आधारित कार्यक्रम और ऑफ-कैंपस प्रशिक्षण कार्यक्रम भी आयोजित किए हैं। अकादमी का एमओओसी कार्यक्रम उन पेशेवरों को एक महत्वपूर्ण लिंक प्रदान करता है जो वास्तविक समय पर प्रशिक्षण कार्यक्रमों में भाग लेने में किसी कारणवश असमर्थ हैं। मुझे इस बात पर गर्व है कि ऑनलाइन पाठ्यक्रम एमओओसी में न केवल विभिन्न पाठ्यक्रमों के लिए पंजीकरण की संख्या बढ़ी है, बल्कि इन पाठ्यक्रमों के पूरा करने के प्रतिशत के मामले में भी हमारा अकादमी लगातार बेहतर प्रदर्शन कर रहा है।

अकादमी एक ऐसे शिक्षण संगठन के रूप में विकसित हो रहा है जो लगातार खुद को नवीनीकृत कर और भा.कृ.अनु.पू. एवम राष्ट्रीय कृषि अनुसंधान और शिक्षा प्रणाली के अंतर्गत अन्य संस्थानों की बदलती जरूरतों के अनुरूप अपने को जल्दी से अनुकूलित कर स्थापित कर रहा है। रा. कृ. अनु. प्र.अ. ने कृषि के क्षेत्र में और अधिक दिशात्मक भूमिका निभाने की उम्मीद से अपने आप को भा.कृ.अनु.पू. के प्रबुद्ध मंडल (Think Tank) के रूप में अपनी परिकल्पित भूमिका को स्वीकार किया है। रा. कृ. अनु. प्र.अ. को प्रबुद्ध मंडल के रूप में परिभाषित करने के उद्देश्य से कई कार्यशाला श्रृंखला का आयोजन एवम 7वीं QRT और RAC जैसे मंच पर विस्तार से चर्चा भी की गयी है। अकादमी अब कृषि के क्षेत्र में नीतिगत निर्देशों को तैयार करने के लिए महत्वपूर्ण निविष्टि जैसे - रणनीतिक, कार्यात्मक और तकनीकी विषय पर अपना योगदान भा.कृ.अनु.पू. को प्रदान कर रही है। कृषि के महत्वपूर्ण क्षेत्रों में चार नीतिगत दस्तावेज जैसे- जलवायु परिवर्तन, कृषि-उद्यमिता विकास, कृषि शिक्षा, और संकाय क्षमता विकास को सामने लाया गया है और इससे संबंधित क्षेत्रों में प्रमुख नीतिगत सिफारिशों का भी सुझाव प्रदान किया गया है।

क्षमता निर्माण कार्यक्रम पर जोर देने के अलावा, अकादमी, अनुसंधान कार्यक्रम पर भी उतना ही बल देता है, इसके तहत अनुसन्धान के परिणाम को क्षमता निर्माण कार्यक्रमों में जोड़कर और कार्यक्रमों में सुधार करके इसके प्रवाह को और अधिक तीव्र करने का लगातार प्रयास किया जा रहा है। विभिन्न सार्वजनिक और निजी एजेंसियों से अनुबंध और परामर्श परियोजनाओं के माध्यम से अकादमी, बड़ी रकम को प्राप्त करने में भी सफल हुआ है, जो न केवल अनुसंधान के लिए, बल्कि अपने हितधारकों की नज़र में संकाय सदस्यों के विश्वास को भी और अधिक बढ़ाती है। अकादमी में संविदात्मक अनुसंधान परियोजनाओं की संख्या में निरंतर बढ़ोतरी हो रही है, जो उच्च गुणवत्ता के शोध कार्यों को बढ़ा रही है जिससे इन शोध कार्यों का प्रकाशन उच्च प्रभाव वाली पत्रिकाओं में लगातार प्रकाशित हो रही है।

अकादमी का प्रमुख शिक्षा कार्यक्रम कृषि के दो वर्षीय पूर्णकालिक स्नातकोत्तर डिप्लोमा (पीजीडीएमए) है जो देश भर से बड़ी संख्या में छात्रों को आकर्षित करती है। भा.कृ.अनु.पू.- रा. कृ. अनु. प्र. अ.(ICAR-NAARM) ने इस प्रबंधन पाठ्यक्रम में उच्च गुणवत्ता शिक्षा को बनाये रखने में

भी सफलता हासिल की है जिसके परिणामस्वरूप प्रतिष्ठित कृषि-व्यवसाय कंपनियों में छात्रों का शत प्रतिशत पद स्थापन भी हुआ है। दूरस्थ शिक्षा पाठ्यक्रम, पोस्ट ग्रेजुएट डिप्लोमा इन टेक्नोलॉजी मैनेजमेंट (पीजीडीटीएमए) जो की हैदराबाद विश्वविद्यालय के सहयोग से चलाया जा रहा है, इस कार्यक्रम की भी काफी सफल प्रतिक्रियाएं प्राप्त हो रही हैं। वर्ष 2019 में एक और पोस्ट ग्रेजुएट डिप्लोमा इन एजुकेशनल टेक्नोलॉजी मैनेजमेंट (पीजीडीटीएम) कोर्स को प्रारम्भ किया गया है जिसकी भी उत्साहवर्धक प्रतिक्रियाएं प्राप्त हो रही हैं।

अकादमी में दो उत्कृष्टता केंद्र भी हैं, जिसमें कृषि-नवाचार केंद्र (CAI) और कृषि खुला व आजीवन अध्ययन केंद्र (COLLAGe) शामिल हैं। दोनों उत्कृष्ट केंद्रों ने वर्ष 2019 के दौरान महत्वपूर्ण उपलब्धियां हासिल की हैं। कृषि-नवाचार केंद्र (CAI) के तहत प्रौद्योगिकी व्यापार इनक्यूबेटर केंद्र, a-आईडिया द्वारा Agri-UDAAN 3.0 कार्यक्रम चलाया गया है जिसने बड़ी संख्या में एग्री-स्टार्टअप और कृषि उद्यमियों को अपनी ओर आकर्षित किया है। यह केंद्र, सरकारी विज्ञान व तकनीक विभागों जैसे DST, BIRAC, DBT, GoI से वित्तीय सहायता प्राप्त कर रहा है, और एग्री-बायोटेक फाउंडेशन (ABF), राजेंद्रनगर, हैदराबाद के साथ साझेदारी में कार्य कर रहा है। भारतीय कृषि अनुसन्धान परिषद् के अंतर्गत उत्कृष्टता क्षेत्र परियोजना के तहत इस वर्ष के दौरान, कृषि खुला व आजीवन अध्ययन केंद्र (COLLAGe), ने ग्रीन मैट प्रौद्योगिकी को स्थापित करके अपनी सुविधाओं में सुधार किया है और दूरस्थ व ऑनलाइन कार्यक्रमों में उपयोग किए जाने वाले डिजिटल मॉड्यूल के द्वारा चलाये जाने वाले वर्चुअल कक्षाएं, एमओओसी आदि के संचालन के लिए परिष्कृत उपकरणों के साथ काम कर रहा है। राष्ट्रीय प्रशिक्षण संस्थान, जिसमें नेशनल इंस्टीट्यूट ऑफ रूरल डेवलपमेंट (एनआईआरडी) और नेशनल इंस्टीट्यूट ऑफ प्लांट हेल्थ मैनेजमेंट (एनआईपीएचएम) शामिल हैं, ने अपने कार्यक्रमों के लिए गुणवत्ता प्रशिक्षण मॉड्यूल का निर्माण करने के लिए भा.कृ.अनु.प.- रा. कृ. अनु. प्र. अ.(ICAR-NAARM) के साथ समझौता ज्ञापन पर हस्ताक्षर भी किया है। कृषि खुला व आजीवन अध्ययन केंद्र (COLLAGe) में आधुनिक लैब सुविधा, अकादमी के लिए राजस्व को भी लगातार बढ़ा रही है।

अकादमी, डॉ त्रिलोचन महापात्र, सचिव, (कृषि अनुसन्धान एवम शिक्षा विभाग) एवम महानिदेशक, भा.कृ.अनु.प., श्री संजय कुमार सिंह, अतिरिक्त सचिव, (कृषि अनुसन्धान एवम शिक्षा विभाग) एवम सचिव, भा.कृ.अनु.प. और श्री सुशील कुमार, पूर्व अतिरिक्त सचिव, (कृषि अनुसन्धान एवम शिक्षा विभाग) एवम सचिव, भा.कृ.अनु.प. के निरंतर मार्गदर्शन और प्रोत्साहन के लिए सदा आभारी है। अकादमी, डॉ आर.सी. अग्रवाल, उप महानिदेशक (शिक्षा), भा.कृ.अनु.प., और डॉ एन.एस. राठौर, पूर्व उप महानिदेशक (शिक्षा), भा.कृ.अनु.प. को इस कार्य में उनके सहयोग और मार्गदर्शन के लिए व अकादमी की गतिविधियों को आगे बढ़ाने और अकादमी के कर्मचारियों और कर्मचारियों द्वारा की गई प्रगति को आगे बढ़ाने में सहयोग के लिए आभार व्यक्त करता है। यह सभी कार्य तभी संभव हो सकता है जब तक भा.कृ.अनु.प. के अधिकारीगण अक्सर हमारा निरंतर सहयोग करते रहें।

मैं अकादमी के संयुक्त निदेशक (प्रभारी) सभी विभागाध्यक्षों, संयुक्त निदेशक (प्रशासन) व कुलसचिव, मुख्य वित्त व लेखा अधिकारी, अनुभाग प्रभारियों और सभी कर्मचारियों के योगदान की भी सराहना करता हूँ। मैं, डॉ आई सेकर, डॉ बी.एस. सोनटक्की, डॉ रंजीत कुमार, डॉ आर.वी.एस. राव, डॉ एस.एस. विनायगम, और डॉ आलोक कुमार, के प्रयासों और सहयोग की भी प्रशंसा करता हूँ, जो इस रिपोर्ट को सामने लाने में सहायक सिद्ध हुए हैं।

मैं अकादमी परिवार के प्रमुख के तौर पर, उन सभी सहयोगियों का आभार व्यक्त करता हूँ जिनकी, कड़ी मेहनत, प्रगति और ईमानदार प्रयासों के प्रदर्शन से अकादमी ने वर्ष 2019 को सफल व लाभप्रद बनाया है।

सीएच श्रीनिवास राव

15 जून, 2020

स्थान: हैदराबाद

(सीएच श्रीनिवास राव)

निदेशक

From the Director's Desk



Dr. Ch. Srinivasa Rao
Director

ICAR-NAARM is the nodal training and capacity building institute of the ICAR and strives to continuously update and build the capacity of agricultural scientists and university faculty of National Agricultural Research Education and Extension System (NAREES) in India. I am happy to record the progress and achievements of the Academy during the year 2019 in areas of capacity building, research, education, policy development and entrepreneurship development in agriculture and allied sciences. The Academy consistently tries to improve upon itself by refining the existing training programmes and adding need-based programmes and also by inviting experts to the programmes to develop a diversified portfolio of engagements in capacity

building in strategic thematic areas of the Academy. During the year 2019, about 3,976 participants across the country have been imparted training through various programmes of the Academy.

ICAR-NAARM conducted 109th batch foundation courses for ARS Probationers during the year 2019 through which 32 newly recruited ARS Scientist Probationers have been imparted foundation training. In addition to these core programmes, the Academy also conducted MDP programmes, need-based programmes and off-campus training programmes. The MOOCs program of the academy provides a vital link to those professionals who are constrained owing to several factors from participating in real-time training programmes. I am proud to place on record that the Massive Open Online Courses (MOOCs) is continuously bettering itself not only in terms of number of registrations for various courses but also the percentage of completion of these courses.

The Academy has evolved into a learning organization that can continuously renew itself and adapt quickly to the changing needs of ICAR and other institutions of NARES. NAARM in its envisioned role as a Think Tank of ICAR, the Academy is expected to play a far more directional role in the field of agriculture. The defined role of NAARM as Think Tank of ICAR had been deliberated in series of meetings including Scoping workshop, VII QRT and RAC at NAARM. The Academy has now evolved into formulating policy directives in the field of agriculture providing significant inputs - strategic, functional and technical – to the ICAR. Four policy documents in critical areas of climate change, agri-entrepreneurship development, agricultural education, and faculty competence were brought out and key policy recommendations were suggested in respective areas.

In addition to our focus on capacity building, the academy also has research programmes, the output of which has a direct relevance on improving the capacity building programmes by adding to the flow of knowledge. The Academy is also successful in garnering large funds through contractual and consultancy projects from various public and private agencies which not only adds to research and extension but also the confidence of the faculty in the eyes of its stakeholders. Contractual research being continuous generates high-quality research output which gets published in high impact journals

The Flagship education programme of the Academy is the two-year full-time Post-Graduate Diploma in Management of Agriculture (PGDMA) which attracts a large number of students from all over the

country. ICAR-NAARM has been very successful in imparting high-quality education in this management course which resulted in 100 per cent placement of the students in reputed agri-business companies. The one-year Post-Graduate Diploma in Technology Management (PGDTMA) in collaboration with the University of Hyderabad (UOH) is receiving an overwhelming response. In 2019, we initiated another distance learning course on Post-Graduate Diploma in Educational Technology Management (PGDETM), which has also received a high level of acceptance.

The Centres of Excellence at the Academy viz. Centre for Agri-innovation (CAI) and Centre for Open and Lifelong Learning in Agriculture (COLLAGe). Both the centres have made commendable progress during the year. The Technology Business Incubator Centre viz. a-IDEA under CAI is running the Agri-UDAAN which has attracted a large number of agri-startups and Agri-preneur enthusiasts. Last year, Agri-UDAAN 3.0 was launched. This centre is receiving financial support from government S&T departments like DST, BIRAC, DBT, GoI, and runs in partnership with Agri-Biotech Foundation (ABF), Rajendranagar, Hyderabad.

Under COLLAGe, the facility developed under niche-area of excellence project of ICAR has improved its facilities by instilling Green Mat Technology during the year and has housed with sophisticated equipment for developing digital modules used in distance and online programmes and conduct of virtual classes, MOOCs, etc. National training institutions including the National Institute of Rural Development (NIRD) and National Institute of Plant Health Management (NIPHM) and National Dairy Research Institute (NDRI) have collaborated with ICAR-NAARM to produce quality training modules for their programmes. The modern lab facility at COLLAGe has been consistently generating revenue for the Academy.

The Academy is grateful to Dr Trilochan Mohapatra, Secretary-DARE and Director General, ICAR, Shri Sanjay Kumar Singh, Addl. Secretary-DARE & Secretary-ICAR and Shri Sushil Kumar, Former Addl. Secretary-DARE & Secretary-ICAR for their continuous guidance and encouragement. The Academy is thankful to Dr R. C. Agrawal, Dy. Director General (Edn.) and Dr N. S. Rathore, Former Dy. Director General (Edn.) for providing whole-hearted support in pursuing the activities of the academy and furthering the progress made by the faculty and staff of the Academy. All this could not have been possible but for the unstinted and unconditional support of ICAR.

I acknowledge the contributions of Joint Director In-charge, all Heads of Divisions, various section in charges, Joint Director (Admn.) & Registrar, Chief Finance & Accounts Officer and all staff of ICAR-NAARM in strengthening the Academy. I sincerely appreciate the efforts put in by Drs. I. Sekar, B.S. Sontakki, Ranjit Kumar, R.V.S. Rao, S.S. Vinayagam and Alok Kumar, who are instrumental in bringing out this report.

As Head of NAARM family, I place on record the hard work and sincere efforts of all in making the year 2019 successful and rewarding.

Date : 15th June 2020

Place : Hyderabad



(Ch. Srinivasa Rao)

DIRECTOR

कार्यकारी सारांश

भा.कृ.अनु.प.- राष्ट्रीय कृषि अनुसन्धान प्रबंध अकादमी, क्षमता निर्माण के लिए, भारतीय कृषि अनुसन्धान परिषद् (भा.कृ.अनु.प.) का एक मुख्य नोडल संस्थान है और भारत में संपूर्ण राष्ट्रीय कृषि अनुसंधान और शिक्षा प्रणाली (NARES) में पेशेवरों की क्षमता को बढ़ाने के कार्य में अपना योगदान दे रहा है। वर्ष 2019 के दौरान, भा.कृ.अनु.प.-राष्ट्रीय कृषि अनुसन्धान प्रबंध अकादमी (भा.कृ.अनु.प.-रा. कृ. अनु. प्र.अ.) ने कृषि क्षेत्र में नीति, कृषि अनुसंधान और शिक्षा प्रबंधन में अपना महत्वपूर्ण योगदान दिया है। भा.कृ.अनु.प.- रा. कृ. अनु. प्र.अ. के कृषि उद्यमिता विकास के क्षेत्र में किये गए महत्वपूर्ण योगदान को सभी ने सराहना की है। अकादमी ने कृषि विज्ञान केंद्र (KVK) सहित विभिन्न कृषि विश्वविद्यालयों के कृषि पेशेवरों की क्षमता को बढ़ाने में भी अपना योगदान दिया है। चूंकि अकादमी का मुख्य उद्देश्य राष्ट्रीय कृषि अनुसंधान और शिक्षा प्रणाली के कृषि अनुसंधान पेशेवरों की क्षमता निर्माण को मजबूत करना है, इसके बावजूद 2019 के दौरान, अकादमी के द्वारा, अपने प्रमुख प्रशिक्षण कार्यक्रमों के अलावा कई आवश्यकता-आधारित कार्यक्रम, ऑफ-कैंपस कार्यक्रम का भी आयोजन किया गया है।

अकादमी द्वारा आयोजित किये गए कुल 76 कार्यक्रमों में से, प्रमुख प्रशिक्षण कार्यक्रम फाउंडेशन कोर्स फॉर एग्रीकल्चर रिसर्च सर्विस (FOCARS) व अन्य के अलावा, 45 कार्यक्रम आवश्यकता-आधारित, 20 ऑफ-कैंपस कार्यक्रम और 08 कार्यशालाएं का आयोजन देश के विभिन्न हिस्सों में किया है, जिसमें देश भर के विभिन्न संस्थानों से विभिन्न प्रशिक्षण कार्यक्रमों के माध्यम से कुल 3976 प्रशिक्षु लाभान्वित हुए थे। इन प्रशिक्षण कार्यक्रमों में टीचिंग एक्सीलेंस पर 5वां विशाल ओपन ऑनलाइन कोर्स (एमओओसी) भी शामिल है, जिसमें कुल 1322 प्रतिभागी शामिल थे।

अकादमी का प्रमुख कार्यक्रम फाउंडेशन कोर्स फॉर एग्रीकल्चर रिसर्च सर्विस (FOCARS) परिवीक्षाधीन वैज्ञानिकों के लिए है। 109 वें बैच FOCARS के तहत, कुल 32 नए भर्ती किए गए एग्रीकल्चरल रिसर्च सर्विस के परिवीक्षाधीन वैज्ञानिकों को 04 जनवरी से 03 अप्रैल, 2019 के दौरान प्रशिक्षित किया गया है। इस साल अकादमी ने PJTSAU, ANGRAU, UAS, UAHS, BASU, AAU, AUJ आदि विभिन्न विश्वविद्यालयों की प्रशिक्षण आवश्यकताओं को भी पूरा किया है। अकादमी ने 2019 में विभिन्न प्रायोजित प्रशिक्षण कार्यक्रमों के माध्यम से कुल 100.78 लाख रुपये की राशि इकठ्ठा की है। फ्लैगशिप प्रोग्राम और आवश्यकता-आधारित कार्यक्रमों के अलावा, भा.कृ.अनु.प.- राष्ट्रीय कृषि अनुसन्धान प्रबंध अकादमी (भा.कृ.अनु.प.- रा. कृ. अनु. प्र.अ.) ने NARES के तहत विभिन्न स्तरों पर विभिन्न संवर्गों के प्रशिक्षण की जिम्मेदारियों को पूरा करने के लिए मैनेजमेंट डेवलपमेंट प्रोग्राम (MDP), प्री-आरएमपी (Pre-RMP) कार्यक्रम भी आयोजित किया है। इस वर्ष अकादमी ने, भा.कृ.अनु.प. के कई संस्थानों के कुल 154 तकनीकी अधिकारियों और प्रशासनिक व वित्त पेशेवरों को भी प्रशिक्षित किया है।

रा. कृ. अनु. प्र.अ., क्षमता विकास के माध्यम से व्यक्तिगत और संस्थागत दक्षताओं को बढ़ाने के उद्देश्य के अलावा, अब, कृषि में सामयिक महत्व के क्षेत्रों में अनुसंधान नीति की हिमायत करके भा.कृ.अनु.प. के प्रबुद्ध मंडल (Think Tank) के रूप में स्थापित होने का प्रयास कर रही है। भा.कृ.अनु.प. ने अकादमी को प्रबुद्ध मंडल की भूमिका के लिए रोडमैप को तैयार करने के लिए डॉ जे सी कत्याल, पूर्व उप महानिदेशक (शिक्षा) की अध्यक्षता में एक विशेषज्ञ समिति का भी गठन किया है। प्रबुद्ध मंडल विकास के उद्देश्य के लिए अकादमी ने 31 अगस्त, 2019 को एक स्कॉपिंग कार्यशाला का भी आयोजन किया गया, जिसकी अध्यक्षता डॉ तिलोचन महापात्र, सचिव, डेयर और महानिदेशक, भा.कृ.अनु.प. द्वारा की गई, जिन्होंने अकादमी को प्रबुद्ध मंडल के रूप में बदलने के लिए संस्थागत ढांचे और तंत्र को विकसित करने पर विचार-विमर्श करने का सुझाव दिया। अकादमी में बैठकों और प्रस्तुतियों की श्रृंखला के बाद वैज्ञानिकों ने अपनी बहुमूल्य अंतर्दृष्टि भी व्यक्त की। इसके अलावा 7वें QRT, RAC, IMC और वैज्ञानिक कैडर समीक्षा बैठकों सहित कई अन्य शैक्षिक मंचों में भी अकादमी की नयी भूमिका पर चर्चा की गई और संबंधित समिति से इसके लिए बहुमूल्य सुझाव भी लिए गए। अकादमी ने भा.कृ.अनु.प. के प्रबुद्ध मंडल में अपने आप को परिवर्तन करने के लिए बदलती जरूरतों के साथ कृषि के महत्वपूर्ण क्षेत्रों में महत्वपूर्ण सलाहकार, भूमिका और अनुसंधान नीति की हिमायत के लिए प्रतिबद्ध है। इस प्रक्रिया में, अकादमी ने जलवायु परिवर्तन, कृषि-उद्यमिता विकास, कृषि शिक्षा, और संकाय क्षमता विकास जैसे महत्वपूर्ण क्षेत्र के विषयों पर नीति दस्तावेज विकसित किए हैं जिससे राष्ट्रीय कृषि अनुसंधान और शिक्षा प्रणाली को और मजबूती प्रदान किया जा सके।

कृषि में स्नातकोत्तर डिप्लोमा (पीजीडीएमए) दो साल का पूरी तरह से आवासीय कार्यक्रम है जिसमें शैक्षणिक वर्ष 2019-21 के दौरान 48 छात्रों को प्रवेश दिया गया है, जो वर्तमान में अध्ययनरत हैं। वहीं कृषि में स्नातकोत्तर डिप्लोमा (पीजीडीएमए) का पिछला बैच विभिन्न कंपनियों में शत प्रतिशत पद स्थापन में सफल रहा है और प्रति वर्ष प्राप्त उच्चतम मुआवजा पैकेज करीब 11.0 लाख रुपये का है। पोस्ट ग्रेजुएट डिप्लोमा इन टेक्नोलॉजी मैनेजमेंट इन एग्रीकल्चर (PGDTMA) एक साल का कोर्स है जिसमें 2019 के दौरान 33 छात्रों को प्रवेश दिया गया है। इसके अलावा, 2019 में एक नया दूरस्थ शिक्षा कार्यक्रम, पोस्ट ग्रेजुएट डिप्लोमा इन एजुकेशनल टेक्नोलॉजी मैनेजमेंट (PGDETM) को प्रारम्भ किया गया है जिसमें अभी कुल 48 छात्रों का प्रवेश दिया गया है।

अनुसंधान के मोर्चे पर, अकादमी के पास कुल 41 अनुसंधान परियोजनाएं हैं, जिसमें 19 इन-हाउस परियोजनाएं और 22 बाहरी वित्त पोषित अनुबंध अनुसंधान और परामर्श परियोजनाएं शामिल हैं। प्रायोजित परियोजनाएं में ICAR Schemes, NASE, NSTMIS, DST, DBT World Bank,

NITI Aayog, IFPRI, RGMVP, PJTSAU, MANAGE, NIPHM, ITC Pvt. Ltd., Mahyco Monsanto Biotech (India) Pvt. Ltd., Tobacco Board, NDRI, एवम NCSCM शामिल है। अनुसंधान परियोजना, अकादमी के छह विषयगत क्षेत्रों जैसे समावेशी विकास के लिए कृषि व्यवसाय प्रबंधन, समृद्ध शिक्षा की गुणवत्ता और पर्यावरण के लिए शिक्षा प्रणाली प्रबंधन, नेतृत्व शासन के लिए क्षमता में वृद्धि, बाजार संचालित पर्यावरण में विस्तार प्रणाली प्रबंधन, नवाचार और शासन को बढ़ावा देने के लिए सूचना और संचार प्रबंधन और सतत विकास में नवाचार के लिए विज्ञान और प्रौद्योगिकी शामिल है।

2019 के दौरान, अकादमी के संकाय सदस्यों ने 164 प्रकाशनों को विभिन्न स्वरूपों में प्रकाशित किया है। अकादमी के संकाय सदस्यों के द्वारा कुल 51 पीयर रिव्यू शोध पत्र को प्रकाशित किया है, जो कुल प्रकाशनों का 31 प्रतिशत से अधिक है। अन्य प्रकाशनों में पुस्तकें, पुस्तक अध्याय, लोकप्रिय लेख, प्रशिक्षण मैनुअल, संगोष्ठी कार्यवाही, शोध रिपोर्ट आदि शामिल हैं।

भा.कृ.अनु.प. द्वारा शुरू की गई फार्मर्स फर्स्ट पहल के कंसोर्टिया भागीदारों में से भा.कृ.अनु.प.- रा. कृ. अनु. प्र.अ. एक महत्वपूर्ण इकाई है। फार्मर्स फर्स्ट कार्यक्रम के तहत, अकादमी ने क्षमता निर्माण गतिविधियों और कई कार्यशालाओं का आयोजन किया है। भा.कृ.अनु.प.- रा. कृ. अनु. प्र.अ. के संकाय सदस्यों ने देश भर में आयोजित हुए विभिन्न क्षमता निर्माण कार्यक्रमों में भाग लेकर अपने ज्ञान और कौशल को लगातार बढ़ाते रहे हैं। वर्ष 2019 में, भा.कृ.अनु.प.- रा. कृ. अनु. प्र.अ. के संकाय सदस्यों ने ऐसे 05 विभिन्न क्षमता निर्माण कार्यक्रमों में भी भाग लिया है।

कृषि-नवाचार केंद्र (CAI), जो की नार्म परिसर में स्थित है, यह प्रौद्योगिकी व्यापार इनक्यूबेटर केंद्र, a-IDEA के माध्यम से एग्री-बिजनेस इनक्यूबेटर्स को हर स्तर पर सहायता प्रदान करता है। अब तक इसके माध्यम से, 12 इनक्यूबेटर को सहायता प्रदान किया गया है और जिसमें से सात पहले से ही परिपक्व हो चुके हैं। इस वर्ष विभिन्न स्थानों पर विभिन्न कार्यक्रमों का आयोजन किया गया है, जिसमें AGRI-UDAAN 3.0, संवेदीकरण कार्यशालाएं, स्टार्टअप संवाद, स्टार्टअप 360, EDP पिच परफेक्ट और डेटा प्लेटफॉर्म पर कार्यशाला शामिल हैं।

कृषि शिक्षा में कृषि खुला व आजीवन अध्ययन केंद्र (COLLAgE) दूसरा उत्कृष्ट केंद्र है, जिसमें अत्याधुनिक कृषि शिक्षा लैब (TELLAgE) शामिल है जो डिजिटल पाठ्यक्रम विकसित करने, दूरस्थ और ऑनलाइन कार्यक्रमों की पेशकश करने, आभासी आचरण और अकादमी के कार्यक्रमों में सहयोग करने के लिए स्थापित किया गया है। यह सुविधा तकनीकी रूप से अच्छी तरह से उन्नत होने के साथ-साथ बुनियादी सुविधाओं के मामले में भी विकसित है। इस सुविधा का उपयोग अब डिजिटल सामग्री को विकसित करने और अन्य एजेंसियों को MOOCs कंसल्टेंसी सहायता देने के लिए किया जा रहा है। अन्य पड़ोसी संस्थान जैसे MANAGE, PJTSAU और NIPHM इस सुविधा का उपयोग अपने डिजिटल आउटपुट को उत्पन्न करने के लिए कर रहे हैं। इस प्रक्रिया में, 2018-19 के दौरान, इस सुविधा से कुल 21.6 लाख रुपये का राजस्व प्राप्त हुआ है। इस सुविधा का नाम बदलकर अब सेंटर फॉर लाइफ लॉन्ग लर्निंग इन एग्रीकल्चरल एजुकेशन (COLLAgE) कर दिया गया है।

वर्ष 2019 के दौरान, अकादमी का दौरा करने वाले कुछ प्रमुख गणमान्य लोगों में श्री गजेन्द्र सिंह शेखावत, केंद्रीय कृषि और किसान कल्याण राज्य मंत्री शामिल हैं, जिन्होंने 09 मार्च, 2019 को अकादमी में आयोजित संवाद कार्यक्रम में भाग लिया। संवाद कार्यक्रम के दौरान, श्री शेखावत ने सरकार की प्राथमिक इनपुट लागत को कम करके कृषि को लाभदायक बनाने और किसानों की आय बढ़ाने के लिए जोर दिया। माननीय केंद्रीय कृषि और किसान कल्याण राज्य मंत्री का विचार था कि हितधारकों को सुनने से सरकार के कामकाज और निर्णय लेने की प्रक्रिया को बढ़ाने में मदद मिलती है। श्री शेखावत ने हैदराबाद में स्थित सभी भा.कृ.अनु.प. संस्थानों के वैज्ञानिकों, अधिकारियों, किसान संगठनों, किसानों, छात्रों, स्टार्ट-अप और राज्य सरकार से संबंधित कृषि और संबद्ध क्षेत्र के विभागों और विश्वविद्यालयों, आदि के साथ भी संवाद किया। इस अवसर पर माननीय मंत्री जी द्वारा नार्म द्वारा हिंदी में लिखित पुस्तिका 'AgRIM' (अग्रिम) का विमोचन भी किया गया।

श्री नरेंद्र सिंह तोमर, माननीय कृषि, किसान कल्याण और ग्रामीण विकास मंत्री, भारत सरकार, ने अकादमी का दौरा किया, और आईजीएनएफए, देहरादून के आईएफएस प्रोबेशनर्स जो अपने प्रशिक्षण कार्यक्रम का एक हिस्सा अकादमी में पूरा कर रहे थे उनको संबोधित किया। इस अवसर पर माननीय मंत्री ने IFS प्रोबेशनर्स से पर्यावरण शासन के मूल सिद्धांतों को कायम रखते हुए आदिवासी और कृषक समुदाय को साथ लेकर संसाधनों के उपयोग करने की अपील की। श्री सूर्य प्रताप शाही, कृषि मंत्री, उत्तर प्रदेश ने अपने सरकार के वरिष्ठ अधिकारियों के साथ अकादमी का दौरा 29 अगस्त, 2019 को किया। इस अवसर पर उन्होंने कृषि के लिए सामूहिक रोड मैप विकसित करने के लिए विभिन्न राज्यों के कृषि मंत्रियों को आमंत्रित करते हुए भारत में किसानों की आय दोगुनी करने के मिशन को 2022 तक प्राप्त करने के लिए एक राष्ट्रीय कार्यशाला आयोजन करने का भी सुझाव दिया।

डॉ लिलोचन महापात्र, सचिव (डेयर) और महानिदेशक (आईसीएआर) ने 109 वें बैच के प्रोबेशनर्स कृषि वैज्ञानिकों को संबोधित करने के लिए अकादमी का दौरा 18 जनवरी, 2019 को किया। इस अवसर पर उन्होंने युवा वैज्ञानिकों से नए विचारों के साथ आगे आने और शोध कार्य में योगदान देने के लिए प्रोत्साहित किया। डॉ रेणु स्वरूप, सचिव, जैव प्रौद्योगिकी विभाग, विज्ञान और प्रौद्योगिकी मंत्रालय, भारत सरकार और अध्यक्ष, BIRAC ने अकादमी में कार्य कर रहे प्रौद्योगिकी बिजनेस इनक्यूबेटर (ए-आईडिया) का दौरा 21 फरवरी, 2019 को किया। उन्होंने प्रौद्योगिकी बिजनेस इनक्यूबेटर (ए-आईडिया) द्वारा समर्थित स्टार्ट-अप के साथ बातचीत भी की, जो कृषि-मूल्य श्रृंखला के विभिन्न क्षेत्रों में कार्य कर रहे हैं।

Executive Summary

ICAR-National Academy of Agricultural Research Management is the nodal organization of ICAR for capacity building and engaged in enhancing the capacity of professionals in the entire National Agricultural Research and Education Systems (NARES) in India. During 2019, the contribution of ICAR-NAARM, Hyderabad in agricultural education research, policy and education has been immense. ICAR-NAARM has made significant contributions in the field of entrepreneurship development in the agriculture sector, which was lauded by all. The Academy also contributed towards the capacity of agriculture professionals of different Agricultural Universities including Krishi Vigyan Kendras (KVKs). The mandate of the Academy is the capacity building of agricultural research personnel of NARES, and to serve as a 'think tank' to ICAR.

There were 76 capacity building programmes organized by the Academy in 2019 out of which, 45 programmes were need-based programmes, eight were workshops and 20 of them were off-campus programmes in addition to the flagship training programme of FOCARS benefitting 3976 trainees of different institutes across the country. The fifth Massive Open Online Course (MOOC) on "Teaching Excellence" was organized in which 1322 participants participated.

The Foundation Course of Agriculture Research Service (FOCARS) has been the flagship programme of the Academy for ARS probationers. The 109th batch FOCARS, in which a total of 32 newly recruited ARS Scientist Probationers were trained during January 04 – April 03, 2019. The Academy also cater to the training needs of various universities like PJTSAU, ANGRAU, UAS, UAHS, BASU, AAU, AU Jodhpur etc. The Academy generated approx. Rs. 100.78 lakhs in 2019 through various sponsored programmes. The Management Development Programme (MDP), pre-RMP programme were organized fulfilling NAARM's responsibilities of training different cadres at different hierarchical levels within the NARES. The Academy also trained 154 technical officers and administrative including finance professionals of various ICAR institutes by conducting need-based programmes during the year.

The Academy has now evolved into a Think Tank of ICAR by providing research policy advocacy in areas of topical significance in agriculture, besides the mandate of enhancing individual and institutional competencies through capacity development. ICAR constituted an Expert Committee with the chairmanship of Dr. J.C. Katyal, Former DDG (Education) to prepare a roadmap for the transition towards Think Tank role of NAARM. A Scoping Workshop was held at NAARM on August 31, 2019 to develop a roadmap for transitioning the Academy to a credible Think Tank of ICAR. The workshop was graced by Dr. Trilochan Mohapatra, Secretary, DARE and DG, ICAR, who suggested to deliberate on devising institutional framework and mechanism for transforming NAARM as Think Tank. The valuable insights of scientists were sought after series of meetings and presentations at NAARM. The transition role of NAARM was further discussed in several academic forums including 7th QRT, RAC, IMC, and Scientific cadre review meetings and sought the suggestions of the respective committee. The Academy has now transitioned into Think Tank of ICAR and is committed to providing a significant advisory role and research policy advocacy in critical areas of importance in agriculture with changing needs. In this process, the Academy has developed four policy documents in critical areas of importance such as climate change, agri-entrepreneurship development, agricultural education, and faculty competence and key policy prescriptions were made for further strengthening of NARES.

The Post-Graduate Diploma in Management- Agriculture (PGDMA) is a two-year fully residential programme in which 48 students were selected for admission during the academic year 2019-21. The previous batch was able to get placements in various companies with the highest compensation package of Rs. 11.0 lakhs per year. The Post-Graduate Diploma in Technology Management in Agriculture (PGDTMA), a one-year online programme admitted 33 students during 2019. The Academy also initiated a new distance education programme on Post Graduate Diploma in Educational Technology Management (PGDETM) in collaboration with University of Hyderabad (UOH) with an intake of 48 students in 2019.

On the research front, the Academy had 41 research projects, which comprised of 19 In-house projects, and 22 externally funded contract research and consultancy projects. The sponsored projects are from various organizations including ICAR Schemes, NSTMIS, DST, DBT World Bank, NITI Aayog, IFPRI, RGMVP-Rae Bareilly, PJTSAU, MANAGE, NIPHM, ITC Pvt. Ltd., Mahyco Monsanto Biotech (India) Pvt. Ltd., Tobacco Board, NASF, NDRI & NCSCM. The research projects span over six thematic areas viz. Agricultural Business Management for Inclusive growth, Education Systems Management for Enriched Learning Quality and Environment, Enhancing Capacity

for Leadership Governance, Extension Systems Management in Market-driven Environment, Information and Communication Management for Promoting Innovation and Governance and Mobilizing Science & Technology for Innovation in Sustainable Development.

The faculty members of the Academy have brought out 164 publications in 2019 in various formats. NAARM faculty published fifty-one peer review research papers, which is more than 31% of the total publications. The other publications include books, book chapters, popular articles, training manuals, seminar proceedings, research reports etc. ICAR-NAARM is one of the consortia partners of Farmers FIRST Initiative launched by ICAR. Academy was involved in the capacity building activities and organized several workshops under Farmers FIRST programme. NAARM faculty constantly upgrade their knowledge and skills through attending various capacity building programmes across the country. In 2019, NAARM faculty have attended 05 different capacity building programmes.

The Centre for Agri-Innovation (CAI) is located in the campus provides hand-holding support to the agri-business incubators through a-IDEA, the technology business incubator centre of the Academy. So far, 12 incubatees have been inducted and seven have already graduated. Different events were organized in various places which include AGRI-UDAAN 3.0, sensitization workshops, Startup Samvaad, startups 360, EDP Pitch Perfect, and workshop on data platform.

Among others, the Technology Enhanced Learning is state-of-the-art Agricultural Education Lab (TELAge), which was established in the Academy to develop digital courses, offer distance and online programmes, conduct virtual classes and support in-house programmes. This facility has advanced technologically as well as in terms of infrastructure. The facility is now being used to offer MOOCs consultancy support to other agencies in developing and hosting digital content. Other neighbouring institutions like MANAGE, PJTSAU and NIPHM have been using this facility for generating their digital output. In this process, this facility has generated revenue of Rs. 21.6 lakhs during 2018-19. This facility is now renamed as Centre for Life Long Learning in Agricultural Education (COLLAge). Number of dignitaries graced ICAR-NAARM during 2019. Some of the lead luminaries who visited the Academy include Shri Gajendra Singh Shekhawat, Union Minister of State for Agriculture & Farmers' Welfare, participated in the interactive session held at the Academy on March 09, 2019. During the session, Shri Shekhawat emphasized that the government's priority is for making the agriculture profitable by reducing the input cost and providing the remunerative prices for enhancing the farmers' income. Shri Shekhawat had an interactive meeting with the farmers, farmer associations, scientists of all the ICAR Institutes located at Hyderabad, officials, students, start-ups and State department of Agri-Allied Sector and Universities, etc. The Minister also released a NAARM Hindi publication titled "AgRIM" during the occasion.

Shri. Narendra Singh Tomar, Hon'ble Minister for Agriculture, Farmers Welfare and Rural Development, GoI, visited the Academy and addressed the IFS Probationers of IGNFA, Dehradun who underwent a part of the training programme at NAARM. The Hon'ble Minister appealed to the IFS probationers to be compassionate with the rural, tribal and farming communities while upholding the core principles of environmental governance and sustainable utilization of the resources. Hon'ble Minister highlighted that agriculture and rural development have very close linkages with the environment, and complemented NAARM for designing the module for the IFS officers, which would have great significance in their field service.

Shri Surya Pratap Shahi, Minister of Agriculture, Government of Uttar Pradesh visited ICAR-NAARM along with senior officials of Govt. of UP on 29th August, 2019. He appreciated the critical role of NAARM in country's agricultural development in terms of human resource management and research policies. He suggested for a national workshop inviting Agriculture Ministers of different states for developing a collective road map for agriculture sector in India for achieving the mission of 'Doubling the Farmers' Income in India' by 2022.

Dr. Trilochan Mohapatra, Secretary (DARE) & Director General (ICAR) on a visit to NAARM on January 18, 2019, addressed the agricultural scientists- probationers of 109th batch of Foundation Course for Agricultural Research Service (FOCARs). He urged the young scientists to come up with innovative ideas to undertake research for the benefit of stakeholders. He shared the importance of new frontier areas like artificial intelligence, and IoT in agricultural research in the areas of precision agriculture and post-harvest management. Dr. Renu Swarup, Secretary, Department of Biotechnology, Ministry of Science & Technology, GoI and Chairperson, BIRAC visited Technology Business Incubator (a-IDEA) at NAARM on Feb 21, 2019. She interacted with the start-ups who are being supported by a-IDEA in different theme areas of agri-value chain.

1 ICAR-NAARM: An Overview

The ICAR-National Academy of Agricultural Research Management (NAARM) was established by the Indian Council of Agricultural Research in 1976 at Hyderabad. The major mandate of the Academy is to build capacity in agricultural research, education and extension education systems, and provide policy advocacy for the National Agricultural Research and Education System (NARES). To fulfill these mandates, Academy organizes various capacity building programmes for researchers, academicians, extension personnel, scholars, and other stakeholders in NARES. The Academy strives to enhance individual and institutional capacity for innovation in NARES. Considering the strategic importance of agricultural research in food security and economic growth of the country, leadership, governance and innovation are emerging as prerequisite for the transformation of NARES into a more pluralistic innovation system. The Academy strives to generate a sense of fraternity and inculcate a scientific work culture amongst the agriculture scientists and faculty across the country. It plays a catalyzing role to usher planned transformation of the NARES in meeting its current challenges.

Vision

A global knowledge institution enabling National Agricultural Research and Education System (NARES) adapt to change through continuous innovation.

The Academy has leveraged its potential of excellent human resource management in the field of agricultural research into agribusiness management, technology management and education technology management to develop and nurture students. The Academy offer higher education programs viz. Post Graduate Diploma in Management- Agriculture (PGDMA) having AICTE approval and accreditation from National Board

of Accreditation (NBA), Post Graduate Diploma in Technology Management in Agriculture (PGDTMA) and Post Graduate Diploma in Education Technology Management (PGDETM) in distance mode in collaboration with University of Hyderabad (UoH). The Technology Business Incubator of the Academy, a-IDEA (Association for Innovation Development of Entrepreneurship in Agriculture) has earned laurels for its innovative approach and received national award for nurturing the Agri Startups.

Mission

To enhance leadership, governance and innovation capacities of National Agricultural Research and Education System (NARES) through capacity strengthening, education, research, consultancy and policy support.

The Academy aims to emerge as an institution par excellence to facilitate and support a culture of dynamic management in agricultural research and education and has been working with the mission of enhancing leadership, governance and innovation capacities of National Agricultural Research and Education System (NARES) through capacity strengthening, education, research and consultancy. The Academy is poised towards performing the role of a think tank for ICAR in advocating policies for strengthening National Agricultural Research and Education system and for overall development of Indian agriculture and sustainability.

Mandate

- Enhance individual and institutional competencies in managing innovation through capacity development, research and policy advocacy.
- Serve as a think tank for National Agricultural Research and Education System and facilitate strategic management of human capital.

1.1 Strategic framework

In tune with its Vision, Mission and Mandate, the Academy has developed a six-tier strategic framework during 12th Plan to steer different activities to achieve the set goals and targets. Those are:

1. Agribusiness Management for Inclusive Growth

- Enhance capacities for food and agribusiness management education and research in NARES
- Enhance capacities for entrepreneurship development in NARES and strategic management of agribusinesses

2. Education System Management for Enriched Learning Quality & Environment

- Enhance capacities for agricultural education policy, planning and evaluation in institutions of NARES
- Enhance capacities for faculty excellence and technology-enhanced learning for increasing learning effectiveness and continuous learning by deploying emerging media to create new and vibrant education environments.

3. Enhancing Capacities for Leadership and Governance

- Institutionalize a framework for leadership development at all levels of NARES – early, mid- career, senior professional and manager across all functions (research, education, extension).
- Institutionalize good governance in NARES by building capacities for effective management of institutions, performance assessment, accountability, and developing strong organizational value systems and work culture that promote innovation

4. Extension Systems Management in a Market driven Environment

- Enhance operational, adaptive and

generational capacity of front-line agricultural extension systems to address emerging challenges

- Enhance capacities for use of ICTs to provide customized knowledge, skills and solutions to farmers, farmer groups, rural communities and for social networking

5. Information and Communication Management for Promoting Innovation and Governance

- Enhance capacities for integrating ICTs and Internet of Things (IoT) in institutional management and governance
- Enhance capacities for managing data, information and knowledge in agriculture, scientific communication and institutional governance

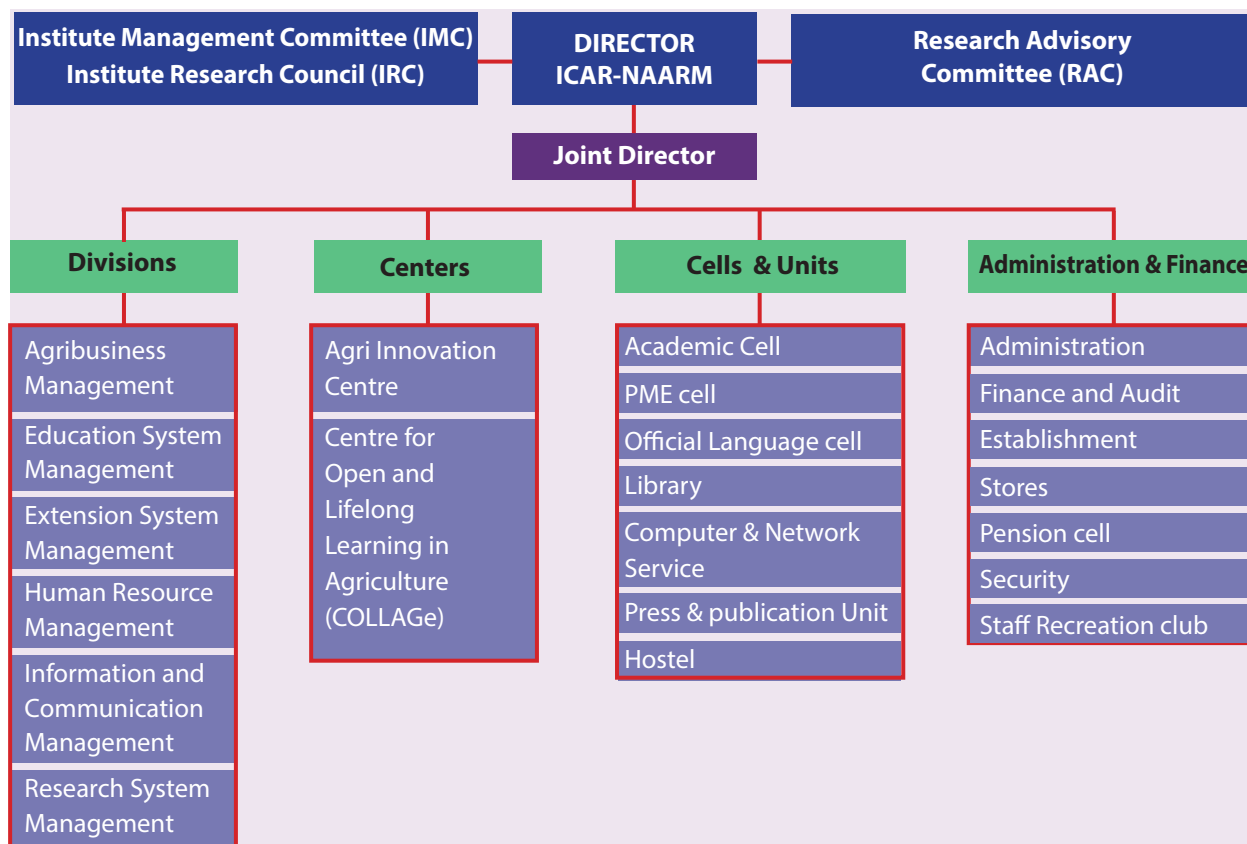
6. Mobilizing Science and Technology for Innovation and Sustainable Development

- Enhance capacities for research policy, strategy, priority setting, planning, management, monitoring and evaluation for a more pluralistic NARES, in the emerging contexts of food security, climate change and globalization.
- Enhance capacities for technology foresight and strategic management of intellectual property and commercialization of technologies

1.2 Organization and Management

The academic, research, and capacity building programmes of the Academy in tune with the above strategic framework is carried out through six divisions:

- Agribusiness Management
- Education System Management
- Extension System Management
- Human Resource Management
- Information and Communication Management
- Research System Management



Organogram of the Academy

1.2.1 Research Advisory Committee (RAC) of the Academy

The Indian Council of Agricultural Research (ICAR) constituted the Research Advisory Committee (RAC) for three years from 24.11.2016 to 23.11.2019 as given below:

Prof. S.K. Sopory Ex-Vice Chancellor, JNU, New Delhi	Chairman
Dr Ch. Srinivasa Rao, Director, ICAR-NAARM, Hyderabad	Member
Dr N.S. Rathore DDG (Agri. Education), ICAR, New Delhi	Member
Dr K. C. Bansal, Ex- Director, ICAR-NBPGR OSD (NAEP), ICAR, Agril. Education Division, New Delhi	Member
Dr P.K. Joshi, Director for South Asia International Food Policy Research Institute (IFPRI), New Delhi	Member
Dr C. Shambu Prasad Professor, Institute of Rural Management- Anand, Gujarat	Member
Dr Ishwar Murthy Professor, Decision Science & Information Systems, Bengaluru	Member
Dr R.K. Samanta Ex- Director, ICAR-NAARM, Hyderabad	Member
Shri Venkateshwarlu Vasanthapu Telangana	Member
Shri M. Dharma Rao, Ex- MLA Telangana	Member
Dr S.K. Soam Head, ICM Division, ICAR-NAARM, Hyderabad	Member Secretary

1.2.2 Institute Management Committee (IMC)

The following officials were nominated as the members of the Institute Management Committee of NAARM for a period of three years with effect from September 01, 2016.

Dr Ch. Srinivasa Rao Director, ICAR-NAARM, Hyderabad	Chairman
Additional Director of Agriculture Department of Agriculture Govt. of Telangana, Hyderabad	Member
Additional Secretary - DARE & Financial Advisor, ICAR or nominee Krishi Bhawan, New Delhi	Member
Director of Agriculture Department of Agriculture Govt. of Andhra Pradesh, Secretariat, Hyderabad	Member
Director (Research) Prof. Jayashankar Telangana State Agricultural University Rajendranagar, Hyderabad	Member
Dr A. Vishnu Vardhan Reddy, Director Indian Institute of Oilseeds Research Rajendranagar, Hyderabad	Member
Sh. Venkateshwarlu Vasanthapu Rajendranagar, Hyderabad, Telangana	Member
Sh. M. Dharma Rao, Ex-MLA, Hanuma Konda, Warangal District, Telangana State	Member
Joint Director (Admn.) & Registrar NAARM, Hyderabad	Member Secretary

1.3 Linkages/Networking/Collaborations

NAARM as an Academy has a strong network with 108 ICAR Research Institutes and 75 Agricultural Universities (AUs), and plays crucial role through capacity building of these institutions, agribusiness

industry, scientists, and academia. Its network spans across various other National as well as International institutions.

1.3.1 National Institutes/ Organizations

1. All ICAR institutes, and Agricultural Universities.
2. Various departments of the Government of India like, NITI (National Institution for Transforming India) Aayog, Department of Science and Technology (DST), Department of Biotechnology

(DBT), National Science & Technology Entrepreneurship Development Board (NSTEB).

3. University of Hyderabad, Osmania University, Jawaharlal Nehru Technological University (JNTU), Indian Institute of Information Technology (IIIT), Council of Scientific & Industrial Research (CSIR), Indian Council of Medical Research (ICMR) and various institutions located at Hyderabad.

4. MoEF & CC, NBA, PPV&FRA, NFDB and State Biodiversity Boards.
5. Management Institutions located at Hyderabad viz. the Administrative Staff College of India (ASCI), Indian School of Business (ISB), National Institute of Agricultural Extension Management (MANAGE), National Institute of Rural Development and Panchayat Raj (NIRD & PR), and Institute of Public Enterprise (IPE); and across the country such as Indian Institute of Management (IIM).
6. State Departments of Agriculture of various States (Telangana, Andhra Pradesh).
7. Private Sector Organizations (Agri Input Companies, Service Companies, etc.).
8. Non-Government Organizations (NGOs).
9. Other Organizations viz. National Innovation Foundation (NIF), Federation of Indian Chamber of Commerce and Industry (FICCI), Centre for Innovation Incubation and Entrepreneurship (CIIE) etc.
10. Krishi Vigyan Kendras (KVKs).

1.3.2 International Organizations

1. CGIAR Institutions (CIMMYT, ICRISAT, IFPRI, etc.)
2. World Bank, FAO other United Nations Organizations
3. Department of International Development (DFID, UK).
4. SAARC Agricultural Centre (SAC).
5. NARS of South Asian Countries such as Bangladesh Agricultural Research Council (BARC), Nepal Agricultural Research Council (NARC), Pakistan Agricultural Research Council (PARC), and other NARS Institutions in Afghanistan, Sri Lanka and other ASEAN countries.
6. NARS in other African and South East Asian countries such as Kenya, Mali, Tanzania, Nigeria, Malawi, Liberia, and Philippines.
7. Leading Land Grant and State Universities of USA and other Universities in Europe, Australia and other developing and developed countries.

Innovation, governance and knowledge form the fulcrum of the Academy's mission, which is driven by its Vision documents (2030 and 2050) as well as those of ICAR. Recognizing the enabling nature of role of emerging science and technologies across diverse sectors, it voices for their integration into agricultural value chain. Further, the evolving changes in agri-education sector shifting focus from 'teaching to learning outcome' signifies the requirement to develop newer institutional mechanisms and capacities of the system. The Academy also identifies the crucial need for a continuous stakeholder engagement to usher effective and efficient performance in the NARES with more transparency and accountability systems in place. Therefore, the activities of research, capacity building and education have been aligned to the six-tier strategic approach. Through a structured deployment of this strategy, the Academy envisages to play a catalytic role in the planned transformation process of the NARES. The subsequent sections present the annual progress and achievements in research, capacity building and the other activities of the Academy under the six strategic areas during the year 2019.

2.1 Agribusiness Management for Inclusive growth

With the shift in agricultural development strategies from a pure production oriented approach to a broader systems perspective, the importance of the agribusiness sector has grown significantly. Seen as an engine for growth, agribusiness and its related industries are receiving increased attention in policies and strategies that aim to promote investments in agro-enterprises and agro based value chain development. This has prompted a need for deeper understanding of agribusinesses, agro-industries and agri-

food value chains. The theme on Agribusiness management for Inclusive growth and education and consultation activities is for developing future leaders in the field of agribusiness. It broadly aims to:

- Enhance capacities for food and agribusiness management education and research in NARES
- Enhance capacities for entrepreneurship development in NARES and strategic management of agribusinesses

Rapid urbanization, rising disposable income of households, expanding and diversifying food demand and globalization of agricultural economy have given impetus to research and capacity building in the area of Agribusiness Management. In order to meet the broader objectives of the theme, the Academy focuses on the following thrust areas for research, capacity building and education:

- **Agribusiness environment:** with major focus on value chain analysis for agri-commodities and services; consumer's demands; price behavior and forecast analysis
- **Public policy and impact:** with focus on market reforms like e-NAM, GST, etc.; financial inclusion and its spillover; spread and convergence of crop insurance
- **Markets and institutional innovation:** market integration of commodities; producer companies and FPOs; inputs and service delivery systems; diversification and intensification of agriculture for high value commodities; business analytics
- **Nurturing future leaders in agribusiness:** through imparting quality education in agribusiness management

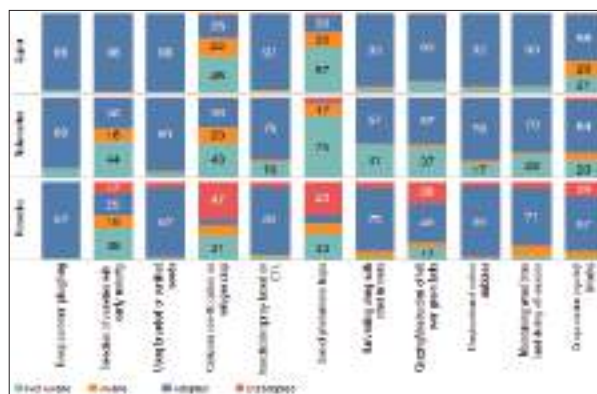
2.1.1. Agribusiness Environment

The agribusiness sector, which comprises the business activities performed from farm to fork, is a major generator of employment and income worldwide. In contrast to the dynamism evident in other sectors, investment in Indian agriculture and agribusiness-public and private, domestic and foreign- has been low and until very recently, has shown little growth. Private agribusiness investment depends on state and central regulatory policies and business environment including marketing, interstate movement, storage and taxation policies. Prospects for agricultural investment hinge largely on the environment for private investment by farmers and agribusinesses. Key factors likely to shape the climate for private investment by farmers and agribusinesses include the extent to which domestic demand for agricultural products make investment potentially profitable; a range of government regulatory, credit, tax, and other policies that affect incentives for domestic and foreign investment in Indian agriculture; and infrastructure facilities.

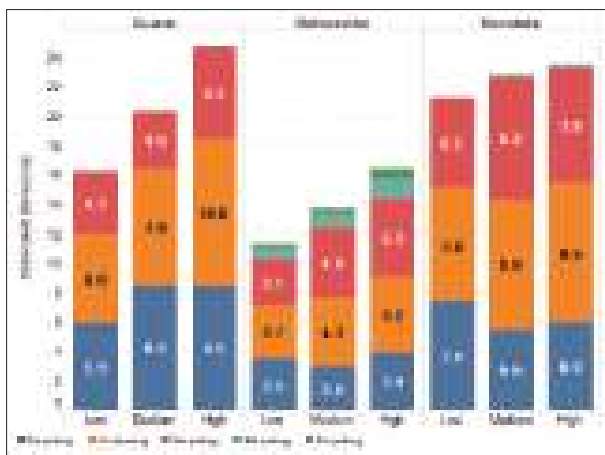
Adoption of Insect Resistance Management Practices in Bt Cotton Cultivation in India

The study was undertaken to evaluate the extent of adoption of Insect Resistance Management (IRM) measures, its effectiveness, tangible and intangible benefits derived from the IRM practice in cotton cultivation. Additionally, it is also proposed to suggest suitable mode of information dissemination for stimulating diffusion of best cultivation practices. The primary data were collected through field survey and in total, 611 sample farmer households were chosen randomly of which 304 were from Gujarat, 129 were from Maharashtra and 178 were from

Karnataka. Farmers reported infestation of the crops with several insects, primarily American Bollworm, Aphids, Jassids, Pink boll worm and spotted bollworm. Among the 11 IRM practices chosen for indicating adoption there has been serious gap in 3 indicators, namely, adoption of refuge crop, non-Bt cotton and use of pheromone trap. Apprehension of production loss with refuge/ non-Bt cotton crop and non-availability of pheromone trap in the locality coupled with lack of trust in controlling the insect menace was the major reasons for non-adoption. Further, farmers adopting most of the IRM practices reported 10-15% higher cotton yield as compared to medium level adopters. It was suggested to adopt integrated communication strategy about IRM involving state department, seed industries and research organizations; implementation of Refuge-in-Bag system; ginning mills regulation; and sanitization of field and neighboring area as short term strategies. Medium term strategies like simplification of IRM practices for the farmers; free distribution of pheromone trap; and conducting training programmes involving all the stakeholders were also suggested. As a long term strategy, it was suggested for development of suitable crop variety.



Farmers Awareness and Adoption of various IRM Practices (% of Total)



Picking wise cotton yield across different IRM adoption level

Adoption of Natural Farming and its Effect on Crop Yield and Farmers Livelihood in India

The NITI Aayog sponsored project on Natural Farming was carried out in collaboration with ICAR-CRIDA to understand the adoption of different practices under Natural Farming by the farmers and its effect on soil fertility, crop yield, and income of the household. The study encompassed survey of 295 NF (Natural Farming) and 170 non-NF farmers in Andhra Pradesh, Karnataka and Maharashtra. It was observed that with the support of the state government, it is being adopted by large number of farmers in Andhra Pradesh, while in Karnataka, adoption is less but some of these farmers are practicing NF since more than 10 years. The response in Maharashtra is mixed. Out of several practices under NF, most adopted practices are use of Jeevamritha, Beejamritha and locally made plant protection materials. Mulching and mixed cropping are also practiced sporadically by the farmers. The Natural Farming being agroecological approach, encourages use of local resources. Thereby, farmers in Andhra Pradesh also use *Ghanajeevamritha* and Azolla in paddy field. When soil samples from few fields of adopter and non-adopter farmers were analysed,

the available soil nutrients (both macro- and micro-nutrients) were observed to be declined. Though, the exact causality could not be established due to lack of information available at initial level from the same fields and smaller number of soil samples. On the basis of survey of adopter and non-adopter farmers in 3 states, it was found that the crop yield in NF is not higher as compared to conventional farming. However, when the adopter-farmers supplemented with *Ghanajeevamritha* or moderate quantity of FYM, the yield improved significantly. Moreover, there is substantial reduction in (paid-out) input cost in NF as compared to non-NF. Similarly, NF-farmers in Karnataka were able to sell the NF-product as niche product at premium price. This resulted into improved B:C ratio over conventional farms. The study recommends that systematic research may be conducted by the crop research institutes to examine the effect of natural farming on the soil nutrients' sustainability under different crop combinations and in different agro-ecological conditions. Moreover, the NF practice may offer an option for the farmers as well as consumers as organic products may be available at lower price due to lower cost of cultivation.

2.1.2 Public Policy and Impact

Financial Inclusion of Smallholders and Landless Households in Madhya Pradesh

The Government of India has undertaken several initiatives to enhance financial inclusiveness of households which are poor or socially backward. However, formal financial institutions are out of reach of many poor households even now. In this context, the study was undertaken with the objectives of assessing the extent and level of financial inclusions for smallholders and landless households in Madhya Pradesh; estimating the extent of income gains to the smallholders

and landless households with better financial inclusion; and finding out the determinants of financial inclusion in the study area. Morena and Jabalpur districts of Madhya Pradesh were purposively selected as they are categorized as the most backward district and most financially included district, respectively. The primary data from 220 households of Morena and Jabalpur has been Collected.

The result showed that resource poor and landless households were less financially included in both the districts viz. Morena and Jabalpur. The results of binary logistic model reveal that for every one additional year of education, there will be 0.37 per cent increase in financial inclusion (table 1). The study recommends to explore for non-farm income generating activities and more investment in education.

Determinants of Financial Inclusion: Binary Logistic Regression Model

Variable	B	S.E.	Wald	df	Sig.	Exp(B)	VIF
District	-0.25	0.47	0.28	1	0.60	0.78	1.94
Education	0.32	0.15	4.37	1	0.04	1.37	1.18
Age	0.01	0.01	0.80	1	0.37	1.01	1.11
Caste			1.90	2	0.39		
OBC	1.04	0.79	1.76	1	0.18	2.84	3.18
General	0.72	0.76	0.90	1	0.34	2.05	3.5
Occupation	-0.5	0.49	1.30	1	0.25	0.57	1.15
Income	0.07	0.00	11.79	1	0.00	1.01	1.25
Land holding	0.34	0.16	4.64	1	0.03	1.40	1.52
Constant	-3.76	1.10	11.54	1	0.00	0.02	
Morena in Districts, SC&ST among Caste and Non-Agriculture among occupation are taken as reference category							

2.1.3 Markets and Institutional Innovation

National Agriculture Market: Effect on Inclusiveness, Efficiency and Profitability

National Agriculture Market (NAM) is a Pan-India electronic trading portal, eNAM which networks the existing APMC *mandis* to create a unified national market for agricultural commodities. The aim of the project is to examine the extent and pattern of participation by the small holders in the new market system and to estimate the benefits or losses arising out from the participation. The study covered select markets in Telangana,

Andhra Pradesh, Madhya Pradesh, Rajasthan and Maharashtra. A survey was conducted covering 400 farmers spread in about 100 km from the *mandi*. The salient findings are:

The level of implementation of eNAM in the selected states was at primary level. It confined mainly on digitization of off-line transaction. The current assaying process of all the commodities are almost manual and cumbersome, which makes it almost impossible to enforce assaying for all the lots arriving in *mandi*, particularly during peak harvest season.

The number of farmers trading through *mandi* was *much* higher than the traditional way of selling through village merchant, though, few farmers were found to sell better quality produces directly to trader to get higher prices.

Farmers also realized better prices (for many commodities studied) when they sold through *mandi*.

The study also proposed a conceptual framework of establishment of Smart Micro Mandi at village cluster level to remove the existing frictions in implementation of eNAM. (https://naarm.org.in/wp-content/uploads/2018/09/E_Policy_Brief_SMM04082018.pdf)

eNAM Implementation Status in Selected Mandi

S. No.	Facilities Infrastructure + processed	Atamand:	Shadhinagar:	Jambhvan:	Parbhani:	Pajurthar:
1	Accounting equipment	✓	✓	✓	✓	✓
2	Warehouse	✓	✓	✓	✓	✓
3	Cold storage	✗	✗	✗	✗	✗
4	Driver (electrical driver, solar driver, etc.)	✓	✗	✗	✗	✗
5	Closed bidding system (before eNAM)	✓	✗	✗	✗	✗
6	Electronic weighing machines at yard	✓	✓	✓	✓	✓
7	Gate entry (barcode based)	✓	✓	✗	✓	✗
8	Inter-mandi trading system	✗	✗	✗	✗	✗
9	Traded price communicated to CA & farmer through SMS	✗	✗	✗	✗	✗
10	Transacting proceeds through bank	✓	✓	✓	✓	✗
11	Cash transactions permitted	✓	✓	✓	✓	✓

Farmer Producer Companies: A Study on Management Practices and Business Potential

Farmer Producer Companies (FPC), registered under the Indian Companies Act of 1956, enacted in 2003, has been projected as an institutional arrangement to solve the problems of the small producers through collective actions. In FPC, the members are primary producers like agricultural produce, forest produce, artisanal products or any other local produce. They enjoy the advantages of both of a professional entity and cooperative entity. Realizing the importance of FPCs, this study was taken up to examine ongoing business and identify the business opportunities of selected FPCs. The distribution of FPCs revealed

that, out of about 3600 FPCs, Maharashtra tops the list with over 800 FPCs constituting about 23 per cent. Uttar Pradesh (13%), Madhya Pradesh (10%), Rajasthan (8%) and Tamil Nadu (7%) are the other important states in terms of number of FPCs. In the first phase, three FPCs were studied, viz., Kazhani Farmer Producer Company Ltd. and Perundurai Farmer Producer Company Ltd in Erode and Coimbatore districts of Tamil Nadu. These FPCs were promoted by MYRADA and supported by NABARD with technical guidance from KVK Erode. These are functional and on the expansion mode through opening of outlets to sell the products including the processed foods. Skill development and advisory services are the

other services provided by them to the members. However, the area of operation is restricted to output marketing and not venturing into other areas like input purchase, consumption, finance, etc. The finance and management are managed well by the executive committee. These FPCs are struggling on various fronts like inadequacy of finance for investment, uninterested members, high cost of operations, difficulty to market with reasonable margin, lack of market information and guidance.



Visit of Project Team to Kazhani Farmer Producer Company in Tamil Nadu

Mainstreaming the Small Ruminant Farmers into Efficient Value Chains in India

Small ruminant production systems play an important role in providing livelihood for landless and resource poor farmers in our country, especially in dryland conditions as they are climate resilient animals and can thrive under harsh environment. The sector has tremendous potential in sustaining their income and standard of living. Considering this, the study was undertaken with the objectives of enumerating the socio-economic profile of the small ruminant farmers and carry out livelihood analysis of them; to study their production and marketing practices with respect to small ruminants; to identify their constraints in small ruminant value chains in the study area; and to suggest strategies and policies for small ruminant farmers to increase their income through efficient

value chains. About 120 sheep farmers (60 each in Mahbubnagar and Nalgonda districts of Telangana and 10 value chain actors in each district (village traders, butchers & consumers) were surveyed. All the sample farmers in the study area were rearing sheep in extensive system. The mean grazing time (hrs.) and distance (km) in the study area was 9.41 hrs. and 5.98 km per day respectively.

Majority (82.50%) of sheep farmers didn't follow supplementary feeding, only 17.50 per cent of the sheep farmers were providing supplementary feeding. Homegrown ewes were the source of ewes for the majority (92.50%) of the sheep farmers. The average feed cost per animal in the study area was Rs. 45, health/treatment cost was Rs. 161, labour cost was Rs. 101, cost related to sheds, equipment and land lease was Rs. 238, and cost related to miscellaneous items was Rs. 41. Thus, the average cost of production of a sheep in the study area worked out to be Rs. 586 per annum. Marketing pattern revealed that nearly one-third of the farmers sold their sheep to the local /outstation traders, followed by local traders (29.17%) and local traders + consumers (15.83%) (Fig. 8). Majority (91.67%) of the shepherds sold their sheep at their homestead followed by livestock market and haat. Majority of the farmers (75.00%) expressed that price for marketable stock was fixed on the basis of physical appearance.

Training Programme on Advances in Market Intelligence

An off-campus training programme on 'Advances in Market Intelligence for Faculty of TANUVAS' was organized at Madras Veterinary College, Chennai during Dec 13-18, 2019. The objectives of the training programme was to acquaint the participants with advancements in the field of market intelligence; to impart skills in designing

survey instruments and conducting surveys; to make the participants having practical exercises in handling and analyzing the data; and to equip the trainees with advanced statistical tools. The training had mix of lectures, practical, hands-on expertise with software using R, Tableau and discussion. There were 25 participants in total from different colleges of TANUVAS.



Participants of Training Programme on Advances in Market Intelligence at TANUVAS, Chennai

2.1.4 Nurturing Future Leaders in Agribusiness

Agri Startups in India: Decoding the Determinants and Success Factors

Start-up has been a buzz word in the business arena in the recent years. Government of India has taken many initiatives to promote start-ups like Start-up India, Skill India, Atal Innovation Mission, establishment of start-up incubation centers and many more. As many as 22 states have come up with start-up policies. It is quite evident from the fact that investment in start-up in the country has reached \$ 3.9 billion in the first half of 2019 vis-a-vis 2016 and 2017 which were \$ 4.2 billion and \$ 4.3 billion respectively. Agriculture is among the top 5 sectors where startups operate. While there are ample number of opportunities, there are equal and tough challenges in setting up and running an Agri-startup. Hence, it becomes very much imperative to take up a study to answer

questions like: Which factors play important role for an individual to plunge in Agri startup? What Strategies Agri startups have adopted for success? Is present ecosystem conducive for Agri startups? and What are the aspirations of these Agri startups?

In order to find the answer of above questions, 30 Agri-tech start-ups incubating at alDEA, ICAR-NAARM are planned to be interviewed using structured questionnaire. So far, comprehensive survey instrument for meeting the specified objectives have been prepared. Several rounds of interview with 9 Start-ups incubated at ICAR-NAARM have been conducted for collecting the desired information.

Entrepreneurship Development through Farmer Led Innovations in Plantation Sector

The study has been initiated with the objective to identify potential Farmer Led Innovations (FLIs) for entrepreneurship development in Plantation Sector. Two workshops were organized at IIPM, Bangalore and ICAR-CPCRI, Kasaragod, Kerala by inviting consortium partners and experts in plantation sector covering the innovations in Coffee, Tea, Arecanut, Coconut, Rubber, Cocoa, Turmeric, Garlic and Ginger. The study documented 148 FLIs in a structured format under the broad areas of Crop Production, Crop Management and Crop Protection. The experts representing various specializations like breeding, agronomy, engineering, statistics, economics assessed the rationality of documented FLIs by assigning the weightages for the three identified indices viz., Novelty, Usability and Viability as 30, 40, and 30 respectively. Finally, 28 FLIs were screened for further investigation and field testing.



Director, NAARM with PI and Stakeholders during Sensitization Workshop under NASF Project

Training Programme on Entrepreneurial Skill Development among Agri-Graduates of ANGRAU

ICAR-NAARM initiated first-of-its-kind training programme to orient young minds towards entrepreneurship through a 21-days intensive residential training programme for two batches of students of ANGRAU on 'Entrepreneurial Skill Development Programme for Agri-graduates' during 14th February – 06th March, 2019 and from March 20 to April 09, 2019. The programme was specifically designed for Agri-graduates who were willing to take up entrepreneurship related to agribusiness. Forty-five final year graduate-students in each batch of Agriculture and allied discipline participated in the programme. The objectives of the programme are to promote and nurture entrepreneurship abilities in agricultural students, and to equip them with management tools needed for making successful entrepreneurs. This programme has a right blend of teaching methods including lectures, hands-on practices, exposure visits to institutions and industry, experiential sharing sessions by entrepreneurs and group assignments. The contents included Agribusiness overview; Business & Entrepreneurship; Soft Skills- Analytics Skill; Soft Skills- Personality & HR and Field Exposure. These budding entrepreneurs had visited two markets- one dealing mostly with

vegetables and flowers and the other dealing with grains. The sessions included modules on entrepreneurship concepts and opportunities in agribusiness sector, marketing, data analysis, business communication and soft skills, ICT use, business models, and business plan preparation and many more. There were sessions by many emerging start-ups or entrepreneurs who shared their experiences of starting the business and the opportunities as well as impediments facing the entrepreneurs. There were visits to start-up enterprises in and around Hyderabad. Visit to ICRISAT was also arranged where students got the opportunity to visit AIP and i-Hub, the business incubators of ICRISAT, wherein Dr. Peter Carberry, Director General, ICRISAT interacted with the student trainees. The young minds were provided with the opportunities to brainstorm for identifying problems with the opportunities for entrepreneurial innovations, working with their team for validating their hypothesis, getting feedback from start-up experts, refining the business plan and finally making pitch presentation.



Agri-Graduates of ANGRAU with Faculty of the Academy

Training Programme on Entrepreneurship Development for Veterinary & Dairy Science Students

The six-day Training Programme on 'Entrepreneurial Skill Development of Graduating Students of SVVU' organized at the

ICAR-National Academy of Agricultural Research Management, Hyderabad during Oct 21-26, 2019. The objectives of the programme was to promote and nurture entrepreneurship abilities in Veterinary and allied sector students, and to equip them with management tools needed for making successful entrepreneurs. The programme was sponsored by Sri Venkateswara Veterinary University, Tirupati under the National Agricultural Higher Education Project (NAHEP). Total 34 students attended the programme, comprising 24 girls and 10 boys from five constituent colleges of the University, viz. College of Veterinary Science-Tirupati, NTR College of Veterinary Science-Gannavaram, College of Veterinary Science-Proddutur, College of Dairy Technology-Tirupati and College of Fisheries Science-Nellore. The student-trainees comprised 23 veterinaries, 5 dairy and 6 fishery students, of which 15 were UG and 19 were PG students. The training programme covered different aspects entrepreneurship such as idea generation, incubation, business plan development, financial management, business communication, etc. The pedagogy included classroom lectures, hands-on exercises, educational games and field visit to make them understand different nuances of businesses. At the end of the programme, student-groups made presentation of their business ideas in the field of agriculture and allied sectors.

Master Trainers' Programme on Soft Skills and Agri-Entrepreneurship

The Academy organized 'Training of Trainers workshop (ToT)' during Sep. 17-18, 2019 for faculty members of State Agricultural Universities (SAUs) to create a pool of master trainers who would in turn train the Agri-graduates in their respective universities on Soft skills and Entrepreneurship

among Agri-graduates. Thirty-four faculty members representing 23 SAUs belonging to 16 states namely Telangana, Andhra Pradesh, Karnataka, Maharashtra, Bihar, Jharkhand, Odisha, Himachal Pradesh, Uttar Pradesh, Uttarakhand, Madhya Pradesh, West Bengal, Haryana, Punjab, Rajasthan and Assam participated in the workshop. Consequent upon the completion of master trainers' programme, each participant will organize one-day workshop on "Development of Soft skills for Entrepreneurship among Agri Graduates" at their respective universities covering at least 100 students.



Director, NAARM with Participants of ToT Workshop at ICAR-NAARM

Workshop on Development of Soft skills for Entrepreneurship among Veterinary & Dairy Science -Graduates

ICAR-National Academy of Agricultural Research Management, Hyderabad in collaboration with Sri Venkateswara Veterinary University (SVVU), Tirupati organized a one-day awareness workshop on soft skills for entrepreneurship development for 100 final year veterinary and dairy technology students of College of Veterinary Science and College of Dairy Technology, Tirupati on Dec 30, 2019. This one-day awareness workshop, as a part of National Agricultural Higher Education

Project (NAHEP)- Component 2, was organized with an object of creating awareness among agri-graduates on soft skills, innovativeness, and entrepreneurship. This is the first workshop in a series of 26 such workshops to be organised during Jan-Feb, 2020 by NAARM. Faculty of 26 Universities were trained as master trainers by NAARM for organising these workshops targeting 3000 students. Dr. Y Hari Babu, Vice Chancellor, SVVU inaugurated the awareness workshop and he highlighted the importance of soft skills in the context of agri-preneurship and enthused students with a few successful cases of entrepreneurship.



Workshop on Development of Soft skills at SVVU, Tirupati

Stakeholders Meeting on Entrepreneurship Development on Horticulture/ Orchid Crops

The stakeholders meeting on Entrepreneurship Development on Horticulture/ Orchid Crops was held at NRCO, Sikkim on 07 May 2019 and was attended by officials from various development departments and scientists from all the research organizations in Sikkim. The meeting, inaugurated



Dr. Panjab Singh, President, NAAS with Participants of the Stakeholders Meeting on Entrepreneurship Development in Sikkim

by the Governor of Sikkim, was conducted under the Chairmanship of Dr. Panjab Singh, President, National Academy of Agricultural Sciences (NAAS), New Delhi.

Training cum Exposure Visit on Agriculture & Agri-Market Scenario for NABARD Officers

A Training Programme-cum-Exposure Visit on 'Agriculture and Agri-Market Scenario in India' was conducted during 26-28 June, 2019 at the Academy for the NABARD Officers in Grade B. The programme was attended by 26 officers at the levels of Managers and Assistant General Managers from 11 regional/zonal offices of NABARD. The programme was conducted in collaboration with National Bank Staff College, Lucknow. The programme contained classroom lectures, experiences sharing and exposure visits to related organizations. The content covered on three aspects, viz. agricultural policies and schemes, cutting edge and advanced technologies in agricultural research and agri-market and start up scenario in India. Specific topics such as crop sector policies and schemes, allied sector policies, applications of biotechnology, GIS and artificial intelligence and IoT in agriculture, agribusiness ecosystem, Agri-clinics and agribusiness scheme, FPCs and their relevance in agri-sector were covered. The trainees were taken for exposure visits to MANAGE, NIRD & PR and a-IDEA, the technology business incubator of NAARM for



Faculty and Participants of Training cum-Exposure Visit on Agriculture & Agri-Market Scenario for NABARD Officers

showing them the activities of these institutions in promoting and nurturing the agri-preneurship and start-ups in agriculture and allied sectors in the country.

Capability Building Programme for Young Agribusiness Professionals Coromandel International Ltd.

The objective of the program was to provide business management skills and to develop personality for taking higher responsibilities in sales and marketing. The topics covered include agronomy aspects of major field crops, Balanced Nutrients Landscape, Rural marketing, Agribusiness and marketing plan, competitor analysis, channel management, market research, field survey, farmers buying behavior and customer feedback and GIS applications. Participants learned target setting, delegation and creativity, inter team interactions through experiential learning and advanced analytics through hands-on exercises.

Post Graduate Diploma in Management Agriculture (PGDMA)

The 11th batch of Post Graduate Diploma in Management - Agriculture (PGDMA) commenced at the Academy with 49 students including 21 girls taking admission. For the admission, the applicants were screened on the basis of CAT/CMAT scores and short-listed candidates were called for Precis writing, Group discussion and Personal interview at the Academy. The inauguration and orientation programmes for the newly admitted students were organized on June 17-18, 2019, while the new academic session started from 19th June, 2019. The batch comprises of students from 13 states representing 25 different universities and 12 disciplines (mainly agriculture and allied sciences) at undergraduate level.

Further, the students of 10th batch completed their 8-week long Summer Internship and joined back the Academy on June 19, 2019. Drs. Sanjiv Kumar and MB Dastagiri are the Academic Coordinators of the 10th Batch, while Drs. PC Meena and N. Sivaramane are for the 11th Batch.



The 11th Batch of PGDMA Students at ICAR-NAARM

Sankalp'19: Annual Agribusiness Festival

SANKALP '19, an annual agribusiness festival of NAARM was organized during Sep 06-07, 2019 at the Academy. This is an annual business competition conducted by PGDMA students of the Academy. Over 3,250 students from 200 plus campuses including B-schools, agricultural universities, and premier engineering colleges had registered for different events. About 100 short-listed students presented their cases in six contests, viz. Samadhaan (Case Study), Adhishtatha (Best Manager), Samiksha (Contemporary Business Solution), Analytical (Data Analytics), Advitiya (B-Plan) and Brand Master (Branding). A Workshop on 'Decoding Agri-startup: Nurturing & Creating Entrepreneurs' (DANCE 2.0) was conducted in association with a-IDEA in which 3 successful Agri-startups shared their experiences. The winners of the events were selected by a panel of juries comprising of NAARM faculty and corporate professionals representing Coromandel International, Samunnati Financials,

Rasi Seeds, Rallis India, Seed Works and Green bliss Agro. Dr. Pradeep Kumar, VP (HR), Coromandel International and Mr. Nitin Chaudhary, Director, Samunnati Agro-Services graced the inaugural ceremony. The winners of different events organized were felicitated with cash prize by Mr. Ramesh Kumar Kaul, Head, International Business, Rasi Seeds and Mr. Ajay Juneja, Business

Head, Vegetable Crops, Seed works International. Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM in his valedictory address urged the students and participants to empathize with the farmers to work on their problems. The event acted as a platform for co-learning and an intellectual delight for the participants.



Events and Award Ceremony during Sankalp '19

Reunirse'19: Alumni Meet of NAARM-PGDMA

Alumni Meet of PGDMA of ICAR-NAARM, Reunirse'19 was organized with zeal on Sep 14, 2019 at the Academy. About 75 Alumni working in different agribusiness industries came from different parts of the country to attend the Meet. These Alumni shared their life journey and experiences working in different sectors. Addressing the Alumni, Dr. Ranjit Kumar, President of the NAARM PG Alumni Association (NPAA) informed the Alumni about changes and innovations introduced in the ongoing PGDMA programme at the Academy. It was also emphasized that past 10 years of PGDMA journey had been very exciting and fruitful, however I

foresight, next 10 years will be crucial for PGDMA programme to position itself as different than other B-schools in India. To increase the deeper engagement between Alumni and in-campus students, new series of 'Alumni Talk' has been proposed. The General Body Meeting (GBM) held in the evening elected new Executive Committee and several decisions were taken to strengthen the brand of PGDMA. It was decided to start six Alumni Chapters at Bangalore, Chennai, Delhi, Hyderabad and Pune. Besides, from next Alumni Meet, a special session by select Alumni, who completed 10 years post-PGDMA, will be organized.

2.2 Education System Management for Enriched Learning Quality and Environment

This theme is one of the core themes of the Academy, which essentially aims to enhance capacities for agricultural education policy, planning and evaluation in institutions of NARES and to enhance capacities for faculty excellence and technology-enhanced learning for increasing learning effectiveness and continuous learning by deploying emerging media to create new and vibrant education environments. In order to advance the objectives of this broad theme, the Academy focused on the following thrust areas for its research, academic and capacity building activities.

- 2.2.1 Lifelong Learning for Professional Development
- 2.2.2 Education beyond Higher Educational Institutions
- 2.2.3 Technology Enhanced Learning
- 2.2.4 Policy Advocacy on Education Systems Management

2.2.1 Lifelong learning for Professional Development

Professional development and competency enhancement are a continuous phenomenon for any organization for achieving par excellence in the respective domains. The major thrust of the Academy is providing opportunities for lifelong learning for National Agricultural Research and Education System. In this endeavor, multi-pronged strategies are adopted. Processes and methodologies are envisioned to enhance the quality of agricultural education in the country. A unique initiative in this direction is competency enhancement of faculty through Foundation Courses and need based programmes. The

outcomes, feedback from participants backed up by research studies formed the basis for designing the courses for lifelong learning.

Students' Learning Approaches (SLAs) for enhanced learning outcomes in Agriculture Education: A Critical Analysis

An applied research project was taken up with the objectives to analyze Students' Learning Approaches (SLAs) in agriculture and allied sciences for enhanced learning outcomes, to analyze the changes in Students' Learning Approaches over time, to investigate the factors affecting Students' Learning Approaches for effective learning outcomes, and to develop strategies for enhancing the quality of learning outcomes in agriculture education. An instrument 'ASSIST' developed by Marton and Säljö (1976) and revised by Entwistle (1997) was modified and pre testing was conducted. Further, a schedule has been developed for identifying the factors affecting learning.

The data pertaining to students learning approaches from 450 students of Tamil Nadu Agricultural University, Coimbatore, Bihar Agricultural University, Sabour, Mahatma Phule Krishi Vidyapeeth, Rahuri, University of Agricultural Sciences, Bangalore, Assam Agricultural University, Jorhat has been collected. The data collected for temporal dimensions from PGDMA students of NAARM has been correlated with academic performance. The instrument for studying the factors affecting Students' Learning Approaches for effective learning outcomes was developed. One of the objectives of the project has been integrated as an activity into National Agricultural Higher Education Project on 'Investments in Indian Council of Agricultural Research Leadership in Agricultural Higher Education'.

Assessment of Technological Interventions in Teaching Learning Process

The study was undertaken to study faculty perceptions on technology experiences and expectations through Education Technology (ET), to study the students' overall attributes towards technology adoption in learning process and to develop suitable interventions for strengthening ET cells. The study was conducted in IGKV, Raipur & PJTSAU, Hyderabad and analysed the extent of utilization of ET resources by the teachers and students, access online journals, use of multimedia software's, computer lab with all gadgets and application of statistical software using semi structured questionnaire. It was reported that the training need of students from IGKV is more than PJTSAU. Further, the analysis indicated that there is no significant difference between teachers of IGKV and PJTSAU at 5 per cent level of significance in respect of variables such as knowledge, training needs, attitude towards ET, expertise on ET and job satisfaction. There is significant positive correlation between 'extent of use of ET' with 'training needs' and 'mass media exposure'.

Technology Enhanced Learning in Agricultural Education

The study aims at developing protocols for developing digital content for various stakeholders using innovative, and implementable technology interventions. Having established state of art Technology Enhanced Education (TEL) Lab, a new programme on digital teaching techniques is offered both in off-campus mode at UAS, Bengaluru and at institute level. The protocols were used to develop four digital courses on 'Principles of Agronomy and Agricultural Meteorology', 'Pests of Crops and their

Management', 'Veterinary Histology', 'Veterinary Pharmacology and Toxicology'. The project also has provided Content development for Distance Education for Post Graduate Diploma in Educational Technology Management (PGDETM) which had the following major deliverables: Four courses for semester I (59 modules) were digitally recorded and hosted on eLearning platform and portal was effective from 1-3-2019. The courses were Educational Technology -1, Psychology of Learning, Instructional Design and Course Development and Dimensions of Teaching. Fifth Massive Open Online Course is successfully offered with a record number of 1329 participants during November 2019. The Lab was also used for capturing panel discussions of participants like IFS probationers and special effects were used to enrich content quality.

Panel discussion of IFS probationers on contemporary issues



Content- Before special effects



Content-After special effects

Digital Teaching Evaluation in Agricultural Education using TPACK Model

The study was undertaken to assess the Digital Teaching Aptitude among Agricultural University (AU) teachers using TPACK Model and identify the issues and possible solutions to address the Digital teaching for implementation in AUs. The famous Technology Pedagogy and Content Knowledge (TPACK) Model was used to investigate digital teaching practices on those who were exposed to various training programs at NAARM. The survey instrument was modified to suit the NARES to investigate on various aspects of TPACK preferences. The data collection was administered online and collected. Data from a total of 50 participants was collected during the survey. Data processing and analysis is in progress.

Ag-Academy: A Framework for Scaling up e-learning platform

Ag-academy is a cognitive education technology platform with a vision to create a framework that enables educators to create content and engage with learners. The framework will be able to enable educators to develop courses with a Mobile App and can be shared with different stake holders. Different mobile applications can be developed for educators as well as learners. Ag-academy vision is to get the best minds from the country to share their knowledge in an easy-to-understand form with the following objectives of developing an Artificial Intelligence driven framework for scaling up of electronic learning platform for Indian Agriculture, developing the robust cloud based platform for hosting the courses developed by end users, developing proper strategic, marketing and communication methodologies for popularizing the developed platform. developing Learning path construction for individual and

customized learning of each and every end user, and ensuring financially sustainable platform and assessing adaptation and Impact of the platform among stakeholders. The technologies required for meeting the objectives were identified. Developing a frame work by involving different stake holders is in progress. The possibility tying up with private company or a-Idea incubated companies or empaneled firms of the academy is being explored as a robust cloud based platform need to be developed which requires huge IT resources.

2.2.2 Education beyond Higher Education Institutions

Development of MOOCs on Commercial Dairy Farming and Milk Processing and Value Addition

The consultancy project was undertaken to enhance the capacity of the Dairy industry professionals and farmers/entrepreneurs for efficient and knowledge driven management of dairying and their products through e-content for implementation of MOOC courses by NDRI. As per the project requirement, 66 digital content modules are to be produced. In this regard, to orient about content development and the required inputs for digital content modules preparation, Dr. G.R.K. Murthy, Co-PI of the project organized one-day sensitization training at NDRI regional station, Bangalore. In continuation of the same both NDRI regional station, Bangalore and NDRI, Karnal have identified experts for the content development.

Development of MOOCs on Rodent Control and Household Pests

The main objective of the consultancy project was to enhance the capacity of the Plant Health professionals and farmers/entrepreneurs for e-content for implementation of MOOC courses

by NIPHM. As part of the project, 21 digital content modules were developed using the expertise of the professionals identified by NIPHM. The content was recorded and edited and packaged using the protocols developed at TELAgE Lab, for online access. It generated a revenue of Rs 7.43 lakhs.



Development of MOOCs on Library Information Systems

The consultancy project was taken up for sensitizing the research and education professionals, students for understanding on how to use technology for knowledge access in libraries and research repositories. This was funded by PJTSAU, Hyderabad under NAHEP. It involved content development for 15 digital content modules, which were developed involving librarians, scientists across educational organizations as identified by NIPHM. The content was recorded and edited and packaged using the protocols developed at TELAgE Lab, for online access. It generated a revenue of Rs. 5.31 lakhs.



Post Graduate Diploma in Educational Technology and Management

The Academy has started one-year Post Graduate Diploma in Educational Technology and Management (PGDETM) with the objective to develop capable and competent teachers/potential teachers/trainers of universities and academic institutions through distance education mode in collaboration with University of Hyderabad (UoH). The programme of PGDETM-2019 was launched from March 01, 2019 with the enrolment of 48 candidates representing various organizations and states. The 1st semester contact classes were organized and examinations for PGDETM-2019 batch were held during Aug 10-14, 2019 for 27 candidates. The resource persons from NAARM and external faculty were invited to deliver sessions during the contact classes on four subjects. The admission for 2nd batch of PGDETM-2020 has been initiated during the year 2019 to start the program from January 2020.



Director, NAARM with the Participants of PGDETM Contact Classes

Joint Advisory Council meeting for Post Graduate Diploma in Educational Technology and Management

The meeting of Sixth Joint Advisory Council (JAC) meeting for Post Graduate Diploma in Educational Technology and Management (PGDETM) was held on April 24, 2019 under the Chairmanship of Dr. Ch Srinivasa Rao, Director, NAARM. Dr. S Senthil Vinayagam, Course Coordinator of PGDETM

presented the details of 48 candidates enrolled for the course. Prof. Jeelani, Director, CDVL, UoH appreciated the efforts of NAARM in e-content development and digital modules developed under the course to reach to a larger group of students. It was also decided to organize Internal Assessment during the last week of July 2019 and contact classes followed by examination for the first semester of PGDETM 2019 batch between Aug 10-14, 2019.



Sixth Joint Advisory Council Meeting at ICAR-NAARM

The seventh JAC for Post Graduate Diploma in Educational Technology and Management (PGDETM) was held in the Director's Committee hall on 27 September 2019 under the Chairmanship of Dr. S.K. Soam, Joint Director, ICAR - NAARM. It was decided to start the 2nd batch of PGDETM from Jan 2020 and organise 2nd semester contact classes and examinations for 1st batch of PGDETM during Feb 2020.



Seventh Joint Advisory Council Meeting at ICAR-NAARM

2.2.3 Technology Enhanced Learning

Competency Analysis among Faculty of Agricultural Universities in Technology Enhanced Learning

The study was undertaken to study the institutional practices in support of Technology Enhanced Learning (TEL) and to understand faculty

competency about their technology experiences and expectations for faculty development. Under the project developed Questionnaire for assessing the institutional practices and competency level in Technology Enhanced Learning (TEL) and validated the questionnaire through pilot study conducted in TNAU. Accordingly, refined questionnaire and data were collected from faculty of HAU, TANUVAS, PJTSAU, IGKV and AAU. Data entry is under process. Data collection from GBPUA&T has to be done. After data collection from all six universities analysis and report submission will be completed.

ICAR Short Course on Innovations in Educational Technology

The Indian Council of Agricultural Research (ICAR) sponsored Short Course on 'Innovations in Educational Technology' was organized at ICAR-National Academy of Agricultural Research Management from November 13 to 22, 2019. The course was designed to sensitize and update on the latest methods of teaching and learning; to provide practical experience on Technology Enhanced Learning in agricultural education and to introduce innovative technology-based methodology in agricultural education. A total of twenty-two participants including thirteen men and nine women from nine states covering fifteen Disciplines were attended. An intensive training schedule was followed with blended learning of theoretical lectures and hands-on- practical exposure with the help of internal as well as external experts. The lectures on overview of educational technology, approaches to educational planning, system approaches in educational planning, models of evaluation in educational technology, open source methodologies in education, social media in agricultural education, psychology of teaching learning process, introduction to

TEL and MOOCs, introduction to LMS through MOODLE, micro teaching production and evaluation etc. were covered. In addition, visits to Educational Multimedia Research Centre (EMRC), Osmania University and MedRec lab (Multimedia), Hyderabad were also made to expose trainees to available facilities. A training manual was prepared covering all the theoretical and practical lectures covered during the course.



Participants of ICAR Short Course Innovations in Educational Technology

Training Programme on Digital Teaching Techniques

The training programme on 'Digital Teaching Techniques' was aimed at enhancing pedagogical methods with digital technology and to sensitize on digital learning management. The programme developed the competency of teachers in using the innovative technology options in their teaching methods besides producing the digital content for online learning. The teaching faculty of AUs, and DUs of NARES were the target audience. The programme covered various aspects on Technology Enhanced Learning concepts, innovative methods of Technology based teaching, Microteaching, communication skills for digital class, e-learning, Open source philosophy for education, Learning management using MOODLE.

Training Programme on ICTs for Livestock Development

This programme was developed on demand from SVVU to enhance the competency of teaching faculty of SVVU in using ICTs for education. As part of the programme, various ICT tools relevant for education were introduced besides interacting with domain specific faculty who are using innovative ICTs for learning and knowledge sharing. Exposure to e-learning in Medical education was also provided for giving insight into use of high end ICTs for education.



Training programme on ICTs for Livestock Development for SVVU Faculty

Training Programme on Digital Teaching Techniques for PJTSAU Faculty

The course was designed to apprise the teaching faculty of PJTSAU on use of technology for education and information retrieval. The programme encompassed concepts like pedagogical methods with digital technology, E-Learning, IPR for Digital Content and National Agricultural Education System, Basics of Teaching Learning and Teaching Methodologies, Digital Teaching and Creation Methods, Micro teaching, Learning Management, Digital Information Retrieval.

Training Programme on Digital Teaching Techniques at UAS, Bangalore

The course was aimed to apprise the trainees about student centered learning and communication, to enhance pedagogical methods with digital technology and to sensitize on digital learning management. The lectures on Teaching Basics, Methods & Technologies, Curriculum Design & Microteaching, E-Learning, Digital Teaching Tools, Content Capturing in Digital Teaching, Open Education Resources, Learning Management System, Course management through LMS and its implementation, Cloud based digital teaching, Cloud based Platforms for individual participant were covered.

Training programme on Advances in Educational Technology (NAARM Sponsored under Sub Plan)

This programme was organized with the objective of orienting new recruits of Assam Agricultural University to advances in educational technology. The programme contents were a carefully chosen with a mix of fundamentals of learning and teaching processes, managing their classrooms, effective communication skills to reach students, latest educational technology and information systems approaches, with interspersed orientation topics on university rules, financial norms



Dean of College of Veterinary Science, Khanapara, Guwahati presenting course completion certificate to the participant

and administrative procedures. Major areas of coverage also included management of research, extension and human resources. Specific topics were knit around these major areas which were covered in about 23 hour sessions spread over 03 working days. The teaching strategy consisted of lectures, demonstrations, discussions and extensive in-built examples.

2.2.4 Policy Advocacy on Education Systems Management

Investments in Indian Council of Agricultural Research Leadership in Agricultural Higher Education

The National Agricultural Higher Education Project entitled '*Investment in ICAR Leadership in Agricultural Higher Education*' under Component 2A, executed by ICAR-NAARM focused on five important themes – establishing partnerships with globally recognized agricultural higher education institutions; promotion of establishment of career development centre at agricultural universities; improving curricula review processes and methods to consolidate & disseminate global best practices in agricultural education and assist agricultural universities in strengthening their linkages with industry and structuring dialogue with state government to catalyse their participation in raising the quality and relevance of agricultural higher education (<https://naarm.org.in/nahep/>).

As a part of the project, a National Workshop on 'Academic Excellence and Resources Generation' and a Training workshop on 'Education Management and Academic Leadership' were organized and published the proceedings and recommendations of the workshop. Besides, five workshops were organized on 'Academia-Industry-Government on Linkages'

and proceedings and recommendations of workshops on Academia-Industry-Government on linkages were developed. Technical Report on 'Global Best Practices in Higher Education' was published. Database on scientists trained abroad was developed. A study on 'Curriculum Revision involving BSMA members' was conducted. On the whole, 541 participants benefited from 14 various programmes.

Technical Committee Meeting cum Workshops under NAHEP Project

The first technical committee meeting-cum-workshop for the project 'Investments in Indian Council of Agricultural Research Leadership in Agricultural Higher Education' under National Agricultural Higher Education Project (NAHEP) was held on March 30, 2019 under the Chairmanship of Dr. Ch. Srinivasa Rao, Director, NAARM. The primary objective of this meeting-cum-workshop was to make the stakeholders and faculty members of the Academy aware about the project and its modalities. Apart from NAARM faculty members, representatives from SAUs like Deans, Directors and Professors of PJTSAU, PVNRTVU, SKLTSHU and ANGRAU also participated along with Dr. VP Sharma, Expert Member and Former Director of MANAGE and Dr. Prabhat Kumar, National Coordinator, NAHEP Component 2A. The issues pertaining to implementation of project and other concerns were discussed.

The second technical committee meeting-cum-workshop for the project was held on September 18, 2019 under the Chairmanship of Dr. Ch. Srinivasa Rao, Director, NAARM at ICAR-NAARM. The objective of this meeting was to discuss about the progress of the project activities and the targets for the next six months. Dr V P Sharma as an Expert Member participated in the meeting and shared his valuable suggestions.

National Workshop on Academic Excellence through Building Partnerships and Resources Generation

A National Workshop on Academic Excellence through Building Partnerships and Resources Generation of NAHEP Project 'Investments in Indian Council of Agricultural Research Leadership in Agricultural Higher Education' was held from April 30 to May 1, 2019. Dr. Narendra Singh Rathore, DDG (Education), ICAR, New Delhi was the Chief Guest of the inaugural function. The objectives of the workshop were to discuss, deliberate and brainstorm on how to escalate revenue generation in AUs and to build partnerships in strengthening National Agricultural Higher Education and Research and Extension Systems, and to sensitize the participants about the activities of the NAHEP Component 2A project.

The ICAR Nodal Officers and Principal Investigators of NAHEP participated. In total, 78 officers participated actively in two days' workshop and deliberated mainly on developing resources and mechanism for supporting infrastructure, faculty and student advancement, and providing means for better governance and management of Agricultural Universities (AUs) in India. It was also focused to establish a partnership with different institutions globally in order to improve the quality of education and resource generation at AUs. In addition, the workshop was encouraged to access AUs and strengthening the linkages with industries and universities for resource generation. Scope of building interactions with AUs and key stakeholders to improve quality of agricultural higher education, was one of the discussions during the workshop. The workshop proceedings were released by Shri Kailash Choudhary, Minister of State, Ministry of Agriculture & Farmers Welfare, Govt of India during 1st Annual Review Meeting of NAHEP, held at New Delhi.



DDG (Edu.), ICAR and Director, NAARM along with Participants of NAHEP Workshop at ICAR-NAARM

Annual Review Workshop of National Agricultural Higher Education Project

The first Annual Review Workshop of the National Agricultural Higher Education Project was organized at the National Agricultural Science Centre, New Delhi during 5-6 August, 2019. The Chief Guest, Shri Kailash Choudhary, Union Minister of State for Agriculture & Farmers' Welfare stated that a comprehensive review of the work done in the past year, achievements and the ways to deal with the future challenges of the project etc. would be discussed during the two-day Workshop. Dr T. Mohapatra, Secretary (DARE) & DG (ICAR) underlined the role of ICAR's various educational plans for the betterment of educational standards along with the main aim of the National Agricultural Higher Education Project, its inception and main achievements. Mr. Edward W. Bresnayan, Team Leader of the World Bank underlined benefits of the Project. He also highlighted the role of ICAR in the Project.



Shri Kailash Choudhary, Hon'ble Union minister of State, Ministry of Agriculture and Farmer Welfare, GOI releasing the workshop proceedings on 5th August 2019 at NASC, New Delhi

NAHEP Review Meeting on Investments in ICAR Leadership in Agricultural Higher Education

The first review meeting of the World bank and ICAR sponsored project being implemented by ICAR-NAARM was organized on June 29, 2019. Dr. R. C Agrawal, National Director of NAHEP was the Chairperson and Dr. Prabhat Kumar, National Coordinator of NAHEP Component 2, Dr. Dr. S. K Soam, Joint Director & CPI, Dr. D. Thammi Raju, Dr. N. Srinivasa Rao, Dr. Alok Kumar, Dr. Surya Rathore, Dr. V. V. Sumanth Kumar, Dr. Sanjiv Kumar, all Research Associates and Senior Research Fellows working under this project participated in the meeting. The project targets, achievements, outputs, outcomes, and the indicators were discussed in length and decisions were taken for further execution of the project.



NAHE Project Investigators with Coordinators of NAHE Project during a Review Meeting at ICAR-NAARM

Training Workshop on Education Management and Academic Leadership

The major objective of the workshop was to enhance the efficiency of Indian Agricultural Education System through effective education management and academic leadership. Participants represented five disciplines, 16 SAUs, one Deemed University and one central university, belonged to 10 states. Topics related to Education Management such as Building Excellence in Higher Education Institutions through NAHEP, New vistas in Technology Enhanced

Learning, Impact of Curriculum on the Quality of Agricultural Education; Academia - Industry Linkages for Academic Excellence, Institutional Autonomy for Education Management, Managing the Branding and Marketing of Institutions, Ethics and Values in Higher Education were covered. Under Academic leadership, the topics viz. Emotional Intelligence for Enhancing the Efficiency of Agri Leaders, Conflict Management Skills for Academic Leadership, Leadership Challenges in Agri Universities, Finding One's Strengths, Collaboration for Academic Leadership, Leadership Attributes were covered.



Training Workshop under NAHEP at ICAR-NAARM

Academia-Industry-Government Linkages for Quality Agricultural Education

During 2019, five Consultative Workshops on 'Academia-Industry-Government Linkages for Quality Agricultural Education' were organized across the country.

1st Consultative Workshop on Academia-Industry-Government Linkages for Quality Agricultural Education at ICAR-NAARM, Hyderabad

The workshop was organized as a part of World Bank sponsored National Agricultural Higher Education Project Component 2A. It was organized to achieve two of the objectives of the project viz. assisting agricultural universities in strengthening their linkages with industry; and structuring

dialogue with state government to catalyze their participation in raising the quality and relevance of agricultural higher education. A total of 46 participants from academia (Agricultural universities), industry and government from the states of Telangana, Andhra Pradesh, Tamil Nadu and Puducherry were participated the workshop. Major themes discussed during the workshop were Existing Linkage mechanism of Agricultural Universities with Industries and State Government: Universities Perspectives, State Industry and State Government Perspectives. Brain storming session on Expectations from – Academia, Industry, State governments and vice versa, areas of expertise available and willing to help Academia, Industry, State governments and vice versa, Brain storming session on way forward for strengthening of linkages between Academia- Industry – State Government for quality agricultural education, Issues and Challenges in MoUs, Mechanisms and also needs and expectations from the stakeholders etc.



Consultative Workshop on Academia-Industry-Government Linkages for Quality Agricultural Education at ICAR - NAARM

2nd Consultative Workshop on Academia Industry Government Linkages for Quality Agricultural Education (Under NAHEP) at OUAT, Bhubaneshwar, Odisha

ICAR-NAARM and Odisha University of Agriculture and Technology (OUAT), Bhubaneshwar jointly organised the consultative workshop on 'Academia - Industry - Government Linkages for Quality Agricultural Education' to improve

the quality of agricultural higher education. The main objective of the workshop was to assess the existing linkages between State Universities in Agriculture, Horticulture, Veterinary and Fisheries with industries and State Governments for strengthening Agricultural Higher Education and to understand the existing support system from State Government in raising the quality and relevance of Agricultural Higher Education. The workshop was organized as a part of the World Bank sponsored National Agricultural Higher Education Project Component - 2A. There were total 52 participants in all from industry, state government and agricultural universities of West Bengal, Chhattisgarh and Bhubaneswar. Fruitful deliberations among the delegates regarding existing linkage between academia, government and industry as well as each other's expectations and expertise available which can be of use for other stakeholders.

3rd Consultative workshop on Academia-Industry- Government linkages for Quality Agricultural Education at MPKV, Rahuri

The Academy has organised the Consultative workshop on Academia-Industry- Government linkages for Quality Agricultural Education jointly with MPKV at Rahuri during September 23-24, 2019. This workshop has been organized with an attempt to get input for developing a framework to strengthen the relationship between three stakeholders i.e. Academia, Industry and Government for improving quality of Agricultural Education. The workshop was organized as a part of the World Bank sponsored National Agricultural Higher Education Project Component - 2A. Fifty-Six participants representing Agri-Universities, Industry and Government of Maharashtra and Gujarat participated in the workshop. Hon'ble Vice Chancellor of MPKV Rahuri,

Prof KP Viswanatha underscored the need to strengthening the relations between university, industry and government and highlighted the efforts undertaken at MPKV in this direction. He also urged the faculty of MPKV to have an access to more funds from industry for doing research as there is fund crunch in SAUs and this can be achieved by strengthening the relations which can be sustained only by having a formal framework in place.

4th Consultative workshop on Academia-Industry- Government linkages for Quality Agricultural Education at UAS Bangalore

The Academy has organised the 4th Consultative workshop on Academia-Industry- Government linkages for Quality Agricultural Education jointly with UAS, Bangalore at GKVK Campus, Bengaluru during November 06-07, 2019. This workshop has been organized with an attempt to get inputs for developing a framework to strengthen the relationship between three stakeholders i.e. Academia (AUs), Industry and Government for improving quality of Agricultural Education. The workshop was organized as a part of the World Bank sponsored National Agricultural Higher Education Project Component-2A. Sixty participants representing Agricultural Universities, Industry and Governments of state of Karnataka, Kerala and Jharkhand participated in the workshop. Hon'ble Vice Chancellor of UAS Bangalore, Prof S Rajendra Prasad underscored the need to strengthening the relations between university, industry and government and highlighted efforts undertaken at UASB in this direction. He also asked the faculty of UASB to access more funds from industry for doing research as there is fund crunch in SAUs and this can be achieved by strengthening the relations which can be sustained only by having a permanent framework in place.



4th Consultative workshop on Academia-Industry-Government linkages for Quality Agricultural Education at UAS Bangalore

5th Consultative workshop on Academia-Industry- Government linkages for Quality Agricultural Education at SVPUAT, Meerut

The fifth workshop was organized jointly with Sardar Vallabhai Patel University of Agriculture and Technology, Meerut, Uttar Pradesh during 27-28 November, 2019. There were 70 (Female: 8) participants representing agri-universities, industry and government from the states of Uttar Pradesh, Uttarakhand, Madhya Pradesh and Bihar. Dr. R.K. Mittal, Vice-Chancellor, SVPUA&T, Meerut applauded the ICAR-NAARM's efforts in bringing the 3 pillars responsible for quality agricultural education in the present scenario at one place. These pillars enable the agricultural universities to strengthen their relations with industry and state governments.



Academia- Industry- Governments of Uttar Pradesh, Uttarakhand, Madhya Pradesh and Bihar

Dr. N.K. Taneja, Vice-Chancellor, Chaudhary Charan Singh University, Meerut; Dr. Rameshwar Singh, Vice-Chancellor, Bihar Animal Sciences

University, Patna; Dr. S.K. Garg, Former Vice-Chancellor, U.P. Pandit Deen Dayal Upadhyaya Pashu Chikitsa Vigyan Vishwavidyalaya Evam Go-Anusandhan Sansthan, Mathura and Dr. A.P. Garg, Vice-Chancellor, Shobhit University, Meerut were also present during the occasion. The workshop was organized to get the inputs for developing a framework to strengthen the relationship between the three parties, that is, Agricultural Universities, Industry and State Government to improve the quality of Agricultural Education.

International Workshop on Strengthening International Agribusiness Trade

TERI School of Advanced Studies in partnership with Western Sydney University, ICAR - National Academy of Agricultural Research Management (NAARM) and National Agricultural Higher Education Project (NAHEP) organized a two-day international workshop on Strengthening International Agribusiness Trade in collaboration with Global Centre for Land-Based Innovation during 14-15 November, 2019 at TERI, SAS, New Delhi. The central theme of the workshop revolved on strengthening international agribusiness trade from emerging countries to developed countries, facilitating dialogue and partnerships between actors of food supply chain for achieving SDG2 and SDG 12 and to build partnerships among academia- industry- government for increasing the quality and relevance of agricultural higher education for achievement of SDG's. The NAHEP specially focused on exclusive session on fostering academia- industry- government for excellence in higher agricultural education. The salient recommendations of the session are introducing new interdisciplinary, industry-focused academic programmes and short-term courses for industry professionals, designing new

dual degree programmes in collaboration with prestigious foreign Universities, and mapping skill gaps identifies through NAHEP with capabilities of foreign institutes in terms of how new ideas are replicated in the Indian context after they return back. A few other recommendations are curriculum revision for all academic programmes in ICAR after every three years in line with UGC/NAAC requirements, increasing awareness and use of innovative technologies to farmers and industries. Developing a Centre for women leadership under NAHEP is also a part of recommendations.

2.3 Enhancing Capacities for Leadership and Governance

Human Resource Management division has made noteworthy efforts to initiate and infuse a competency-based management research system through training, performance management, and teaching competency framework for achieving organizational goals. The division is involved in the development of competencies and capacities of scientists and research leaders in the ICAR and Agricultural Universities. Leadership and governance are the two important drivers to promote organizational effectiveness. Realizing this, the division is engaged in research and capacity building in these areas and associated areas of human resource management, competency management, teamwork, job and organizational stress, and emotional intelligence. Through these endeavours the division aims to develop individual and institutional capacities to deal with contemporary issues and achieve individual and organizational excellence.

The strategic framework of the division envisages enhancing capacities for leadership and governance by institutionalizing a framework

for leadership development at all levels of NARES- early, mid-career, senior professional and manager across all functions (research, education, extension), and institutionalizing good governance in NARES by building capacities for effective management of institutions, performance assessment, accountability, and developing strong organizational value systems and work culture that promote innovation. To realize the above objectives, the following activities related to research and capacity building programs were carried out during the year 2019.

Emotional Intelligence and Organizational Effectiveness

The objectives of this project are to measure the emotional intelligence of scientific personnel and their organizational effectiveness in terms of teamwork, conflict resolution and leadership attributes, and to suggest strategies to improve organizational effectiveness in NARES. Psychometric data were collected from 212 Senior Scientists and Principal Scientists of ICAR and SAU's regarding their status of emotional intelligence, team effectiveness, Thomas-Kilmann conflict modes and 180-degree leadership effectiveness attributes. Preliminary analysis indicated that the emotional intelligence of scientific personnel in NARES was 'Average'. The team effectiveness of scientists was found to be 'Low'. The majority of the scientists of NARES resorted to 'Avoiding (withdrawal)' and 'Accommodating (soothing)' techniques of conflict resolution. People Enablement is the most dominant leadership trait of the Scientists. The project team plans to increase the data sets and analyze the same to establish the relationship between the emotional intelligence and organizational behavior of scientific manpower in NARES.

Emotional Intelligence for Personal and Work Excellence

A training program on 'Emotional Intelligence for personal and work excellence' was conducted during February 11-13, 2019 for 22 senior faculty members of Central Institute of Fisheries Education (CIFE), Mumbai with the following objectives:

- To facilitate participants to understand their own self and personality dimensions.
- To enable participants to learn and develop emotional intelligence.
- To strengthen the emotional orientation of participants for personal and work excellence.

The topics covered under this 3-days training program were: Types of Intelligence (IQ, EQ and SQ) – An Overview, Assessment and Interpretation of EQ and SQ, Effectiveness to Greatness: An Interactive Discussion, Assessment and Interpretation of Organization Effectiveness Traits, Emotional Intelligence for Work Performance, Layers of Personality and Self Awareness, Personal and Work Excellence Through Self-actualization, Relationship Management, Stress Management and Positive Thinking.

FDP for Improving Efficiency and Effectiveness in Teaching, Research and Extension

A Faculty Development Programme (FDP) for improving Efficiency and Effectiveness in Teaching, Research and Extension at the University of Agricultural and Horticultural Sciences (UAHS), Shivamogga, Karnataka was organized during February 04-08, 2019 at UAHS, Shivamogga. Specific objectives of the programme were to reflect on and renew the knowledge and skills of the faculty of the University for improving the quality of higher education and human resources

management and to impart knowledge and skill in teaching, research and extension. Areas of deliberations during the training programme included Student-Centered Education Technology, Effective Teaching-Learning, Evaluation and testing, Positive thinking, Stress management, Personality Development for leadership, Motivation, Interpersonal Relationship and Role Perception, Scientific communication, ICTs in extension, Research Project prioritization, IPR and Technology Management. A total of 31 participants were trained in this programme.

Teamwork and its effect on Organizational Performance in NARES

A study on the teamwork and its effect on organizational performance in NARES revealed that out of 2744 respondents, 63% were low in teamwork skills, followed by 35% and 2 % in medium teamwork skills and high teamwork skills. Analysis of team orientation behavior inventory (TOBI) of 416 respondents indicated that 59% are fully functional team members, 19% are educable retarded, 5% are lone rangers and 17% are trainer/consultant. Nine ICAR teamwork award winners were interviewed and their experiences are being synthesized and analyzed.

Competency Management System for Agricultural Research and Extension

Functional and behavioral domains of research and extension scientists to develop the competency framework were identified under this project. Data was collected from 253 respondents from the various institutions covering six subject matter divisions of ICAR through google based survey /questionnaires. Competency domains were then ranked at different levels of scientific cadre. Interim report indicated that the subject orientation got the first rank at the scientist level for the development of core strength in the area of specialization. Research communication and

research methodology domains emerged on top priority at senior scientist and principal scientist levels. At the pre-RMP level, project management and governance were identified as priority domains. Extension scientists gave top priority to research communication.

Development of Teaching Competency Framework for Quality Agricultural Education

The study was conducted among 292 respondents consisting of 30 educational administrators/experts, 97 faculty members and 165 students representing 12 agricultural and allied sciences (Veterinary, Horticulture, Fishery and Home Sciences) universities. A Teaching Competency Questionnaire (TCQ) with 28-items was administered separately for each sample group. The respondents were asked to rate their perceptions on relevance and knowledge levels of each of the teaching competencies on Likert's five-point scale (1 = Low Knowledge/Relevance; 5 = High Knowledge/Relevance). Data were analyzed using descriptive statistics and the ranking procedure described by Edwards & Briers (1999). Borich (1980) needs assessment model was used to identify the training needs of the faculty by calculating mean weighted discrepancy scores (MWDS). Results indicated that there are differences in perceptions of educational administrators, faculty and students towards the relevance and teachers' knowledge levels of teaching competencies. However, based on the average perception of all the stakeholders, the competency requirement of faculty by cadre-wise were identified. Based on the MWDS of each teaching competency, the training needs of faculty were prioritized. The study suggests to undertake periodic competency need assessment of faculty at all levels (early, mid-career and senior) and develop appropriate programmes for their development.

HR Interventions and their influence on Institutional Innovations

An extensive review of literature on HR Interventions and Institutional Innovations was done under this project. A questionnaire was developed to collect the data from ICAR Institutions and others and enlisted the HR drivers for institutional innovations i.e. Training & Development, Motivation, Team Spirit, Reward Systems, Participative Decision Making, Performance Appraisal, Career Planning and Development, Information Sharing, Remuneration, Management and Leadership, Employee Wellness Programs, Talent Management, Organizational Trust, Manpower Planning, Coaching and Mentoring.

Training Programme on Stress Management

A training programme on 'Stress Management' was conducted during June 26-29, 2019, which was attended by 21 participants representing senior scientific and administrative cadres of ICAR and SAUs. The objectives of this training program were to develop an in-depth understanding of personal and psychological causes of stress/anxiety and its effects – physical, social and organizational behavior, to understand the mechanism of occurrence of various stress-related disorders and to develop constructive means for coping up with distress and achieve productive and enriched life-style. This four days' program emphasized practicing yoga, medical examination, psychological and medical aspects of stress and its management, and the role of emotional and spiritual intelligence in understanding and managing stress holistically. Dr. R.C. Agrawal, National Director, NAHEP interacted with the participants and given his presidential remarks during the valedictory session.



Training Programme on Stress Management

Management Development Programme on Leadership Development

The MDP on Leadership Development was organized for Heads of the Divisions and other senior scientific officers, who are in Pre-Research Management Positions (Pre-RMPs) in the National Agricultural Research and Education System (NARES), to prepare them for leadership responsibilities. Two such programmes were organized during June 11-22, 2019 and December 02-13, 2019 and these programmes were attended by 25 and 22 participants respectively. The programme provided an opportunity to the participants, who are in the mid-phase of their career, to enable the smooth and gradual transformation to step into future leadership positions. The major objectives of the programme were to help the participants understand their personality dimensions, to explore the leadership challenges and core competencies needed in present-day context, to develop understanding and skills in research and technology management, information and knowledge management, and administration and finance management, and to adapt ideas of diverse skills for sharpening capacities to effectively execute various roles in the organization. The programme covered the

areas of leadership and personality development; Research and technology management; Information and knowledge management; and administration, finance, vigilance management, lessons from epics and yoga practice.



Faculty Interaction with Participants of MDP Programme on Leadership Development at ICAR-NAARM

MDP for HRD Nodal Officers of ICAR for Effective Implementation of Training Functions

This training programme was conducted during March 14-16, 2019 with 23 participants from ICAR with the objectives of enhancing competencies of the HRD Nodal Officers to provide support and guidance in the implementation of capacity building programmes and training functions. Most of the participants were of Principal scientist cadre of ICAR Institutes. The sessions covered were systematic and competency approach to training, training need analysis (TNA) of employees of all categories, design and development of training programme, development and implementation of cadre training plan and annual training plan for all categories of employees, monitoring and evaluation of trainings at the institute level, developing training schedule/calendar for the institute, and impact assessment of training. The above topics were covered by different pedagogical approaches consisting of presentations, video lectures, interactive

discussions, etc. Different practical ways and means of developing leadership characteristics were suggested to enhance one's own personal and organizational effectiveness. Apart from classroom sessions, regular early morning yoga sessions were conducted for the participants.

Motivation, Positive Thinking and Communication Skills for Technical Officers of ICAR

An off-campus training programme on 'Motivation, Positive Thinking and Communication Skills' for Technical Officers (T5 & above) of ICAR Institutes was held at Indian Institute of Soil and Water Conservation, Dehradun during 13-19 March 2019 which was attended by 24 participants from 17 ICAR institutes representing 13 different states. The main objective of this training programme was to improve the skills and efficiency of the participants in discharging functions and providing constructive technical support for meeting the organizational goals and targets. The programme consisted of a blend of lectures, interactive sessions and visits. The training programme included topics on the role and importance of technical officers, communication skills, motivation, time management, personality profiling, socio-psychological issues and use of tools like MS Excel and MS PowerPoint for better communication. The programme also included experiential learning visits to Sainji Watershed, Mussoorie and Indira Gandhi National Forest Academy.



Training Programme on Motivation, Positive Thinking and Communication Skills for Technical Officers of ICAR

Enhancing Workplace Efficiency for Junior Group B & A Officials of NAARM

A training programme on 'Enhancing Workplace Efficiency for Junior Group-B and Group-A Officials of ICAR-NAARM' was organized during April 25-28, 2019, as a consultancy programme at ASCI under the existing Memorandum of Understanding with the Academy. A total of 25 Group B & A Officers were trained, out of whom 4 were women participants.



Training Programme on 'Enhancing Workplace Efficiency for Junior Group B & A Officials of NAARM'

Trainers Development Programme for Various Cadres of Administrative and Finance Staff

This programme was planned jointly by ICAR-NAARM and ADG (HRM) and sponsored by ICAR to create a pool of Master Trainers to develop and organize training programmes for administrative and finance staff of ICAR in the areas of administration and finance management. The programme was conducted at the Academy from December 04-09, 2019, with the following specific objectives:

- To equip participants with knowledge and skills on design, implementation and evaluation of training
- To provide hands-on practice in training design and management; and
- To facilitate networking among the participants for training management.

Two DoPT certified trainers along with core faculty of the Academy provided academic inputs. Progressive and experiential learning strategies were followed to enable smooth learning progression among participants from known to unknown and immediately applying the learned KAS through structured exercises (individual and team). A learning log was used to track the formative evaluation. Twenty-two officers (19 males and 3 females) of administration & finance sections were successfully trained in this programme.

Workshop on Competency Framework for Agricultural Research and Extension Scientists at ICAR-IINRG Ranchi

ICAR-NAARM has organized a one-day workshop on 'Competency Framework for Agricultural Research and Extension Scientists' in collaboration with ICAR-IINRG, Ranchi involving 40 scientists from four ICAR institutes including regional stations located at Ranchi namely, ICAR-IINRG, ICAR-IIAB, ICAR-RCER, ICAR-NBPGR on March 06, 2019. The purpose of this workshop was to facilitate discussions and deliberations to identify attributes of different competencies required at scientists, senior scientists and principal scientist levels. The major competencies were Discipline orientation, Research attributes (Cognitive),



Workshop on 'Competency Framework for Agricultural Research and Extension Scientists' at ICAR-IINRG, Ranchi

Research Methodologies, Project Management & Governance skills, Research Communication skills and Research Application & Extension attributes under the category of functional competencies. The behavioural competencies included Managing Relationships, Leadership and Networking. The workshop was organized at ICAR-IINRG, Ranchi.

ICAR-NAARM and Coromandel International Limited (CIL) Collaborative Capability Building Programme

Two types of programmes were offered under this collaborative initiative. The objectives of the program on 'New Entrants Assimilation Training (NEAT)' for Fertilizer Division (4 batches) and 'Skill Enhancing and Employees Development (SEED)' for Retail Division (2 batches) were to expose the participants with current trends in agriculture, soil health and nutrition management and to provide knowledge in Coromandel product portfolio and to bring about attitudinal and behavioural changes to enhance efficiency and effectiveness. Providing a basic understanding of crop production and crop protection practices, and upgrading the knowledge and skill of the participants in the sales and marketing domain were the motives behind the training programme. The topics covered were agriculture and Indian economy, agronomy aspects of major field crops, soil health-based nutrient marketing, basics in plant health management and plant protection chemicals, irrigation and fertigation management, organic fertilizers, ICT and social media in sales, and value-added selling skills. The behavioural aspects such as personality development and emotional intelligence also covered in the training programmes. A total number of 184 employees of CIL were trained during January - March 2019.

A programme on 'Marketing Officers' Resurgence & Excellence (MORE)' was organized during 9-13 November 2019 and for Agronomists during 16-18 December 2019. There were 30 and 51 participants in the respective programmes. The second type of programmes exposed the participants with advances in agriculture technologies, soil health and nutrition management and to provide knowledge in Coromandel product portfolio and to bring attitudinal & behavioural changes to enhance Business management skills to make them ready to take higher responsibilities. The topics included Advances in agronomy practices of major crops, Climate change and balanced nutrition landscape, Farmer Producers Organizations (FPOs), Data Analytics & Forecasting. Agribusiness Management Skill development sessions on Business Model Canvas, Marketing Plan and Brand Management & Promotion strategies. The soft skills on Relationship Management, Negotiation and Conflict Management also covered during the training programme.

2.4 Extension Systems Management in Market-driven Environment

This is one of the core themes of the Academy, which essentially aims to:

- Enhance operational, adaptive and generational capacity of front-line agricultural extension systems to address emerging challenges
- Enhance their capacities for use of ICTs to provide customized knowledge, skills and solutions to farmers, farmer groups, rural communities and for social networking.

The Academy focuses on the following specific areas for its research, academic and capacity building, and policy advocacy activities.

2.4.1 Extension Policy, Planning and Management

2.4.2 Extension Information Systems

The core research areas envisaged under this theme are assessment of research role and performance of extension scientists in ICAR institutes; evaluation and enhancement of new frontline extension initiatives of ICAR; ecosystem analysis of functional effectiveness and convergence and effectiveness of Krishi Vigyan Kendras; development of quality resources for capacity building based on empirical data and case studies and provide policy inputs for enhancing the efficiency and effectiveness of frontline extension systems of ICAR and SAUs. The identified research themes are being addressed by undertaking institutional and externally funded research projects in interdisciplinary approach.

2.4.1 Extension Policy, Planning and Management

Research Role Analysis of Extension Scientists in ICAR

In the face of the dynamic agricultural landscape, emerging challenges and changing priorities, continuous assessment and improvement of the architecture of Agricultural Extension is important. A strong, vibrant and responsive extension research system can play pivotal role in enabling the country to achieve the planned objective of faster, sustainable and more inclusive agricultural growth. To address this issue, the current institutional research project has been taken up.

- Broad priority areas for extension research at national, regional and state levels have been delineated using Delphi with a sample of about 50 experts.

- An online survey was initiated to collect primary data from the extension scientists working in ICAR (sample size of 192).
- Interim analysis of responses received from 50% respondents revealed that at the scientist level, research is perceived as the most important role. However, role conflict was also reported by the respondents.

A WhatsApp group created for the purpose was used for healthy online discussion on the theme 'research role of extension scientists in ICAR'. Content analysis of the discussion threads revealed that there is need for updating the knowledge and skills of extension scientists on advanced analytical tools and methodological rigour to improve quality and relevance of extension research.

Management and Impact Assessment of Farmer FIRST Project

The ICAR has been implementing a flagship frontline extension programme 'Farmer FIRST' to promote farmer-centric and farmer-participatory research. This programme is being implemented by 51 NARES institutions representing all the agro-ecological regions. The Academy is a key partner in the Project Management Consortium, specifically to design and conduct capacity building programmes for the project personnel of implementation centres. During the reported period the following activities were carried out. Two need-based capacity building programmes viz., i) Innovative Practices in Extension Research and Evaluation, and ii) Socio- Economic Impact Assessment of Research Programmes were conducted. A book on farmers' innovations and success stories across FFP implementing centres was published. Feasible and successful technological interventions implemented by

FFP centres were identified for assessing their potential for upscaling and ADOPT (Adoption and Diffusion Outcome Prediction Tool) software was used for assessing the attributes and spillover of technological interventions for scaling up.

Innovative Practices in Extension Research and Evaluation

The programme was designed to cater to an immediate need to build capacity of extension scientists and faculty and research scholars of NARS to conduct quality research in extension. This four-days programme was conducted during Nov 04-08, 2019, with the objectives of equipping the participants with knowledge and skills needed for conducting impact-oriented and quality extension research and evaluation; providing hands-on practice in writing convincing research proposals in extension; and facilitating networking among the participants for inter-institutional research in extension. The programme was attended by 17 participants consisting of six female participants. Progressive and experiential learning strategies were followed to allow participants to progress in learning from known to unknown and immediately applying the learnt KAS through structured exercises (individual and team). Participants were organized in to four teams to work on writing concept note on an assigned theme integrating 2-3 innovative practices covered in the workshop and the guidance of mentors.

Training program on PRA and Communication Strategies including Integrated Watershed Management for Indian Forest Service (IFS) Probationers

The Academy has organized a tailor-made training program on "Participatory Rural Appraisal (PRA) and Communication Strategies including Integrated Watershed Management (IWM)" during 26-29 September 2019 at its premises for the IFS

Probationers (2018-20 batch), from Indira Gandhi National Forest Academy (IGNFA), Dehradun. The participants included 60 IFS Probationers and two officers belonging to Royal Bhutan Forest Service. The training was organized as a part of Memorandum of Understanding (MoU) signed between ICAR-NAARM and IGNFA.

The probationers were introduced to various tools of PRA and were taken to a village to use them for understanding the issues, perspectives and aspirations of the villagers towards IWM interventions. As a part of the communication module, the participants simulated a panel discussion and played different roles during an environmental conflict situation, which was recorded and appraised by communication experts. The training included a hands-on exercise on using analytical hierarchy process (AHP) analyzer, an online application developed by NAARM, for prioritization of administrative decisions. The probationers were introduced to the emerging issues in natural resource management with respect to climate change and were sensitized on current trends in IWM. They were also sensitized on the need to work with researchers and citizens as partners, so as to harness the benefits of synergy, by presenting relevant case studies.

2.4.2 Extension Information Systems

Agricultural extension is the primary knowledge and advisory delivery system to farmers. Without relevant and reliable agricultural information, improvement in agricultural production may not be achieved. Extension organizations have the important role of linking farmers to information on improved agricultural technologies. One of

the prime ingredients for rural development in developing countries is information access. Although the Information and Communication Technology (ICT) revolution in these countries has gained momentum, most of the farming communities still have no or limited access to value added information. Academy focuses on the agricultural research on extension information system to enhance their knowledge by increased stakeholder's participation in research.

Gender-based ICT led Knowledge Management in Agriculture

A prototype web based Geo-Spatial Village Information System was developed and linked 141 Soil Health Cards to the cadastral maps of S. Lingotam and Tallasingaram villages of Chautuppall Mandal. A prototype database on 'Gender disaggregated ICTs use in agriculture' was also developed including a total of 1118 respondents (514 farm women and 604 farmers) from different agro-climatic zones. Rural ICTs website was developed with 309 vernacular web pages, 286 vernacular information modules on major crops and animal husbandry for information sharing. Four vernacular video modules on homestead products for facilitating entrepreneurial development and another one on agricultural schemes for doubling farmers' income were also developed for educating farm women, youth and farmers. A mobile app 'm-Prioritize' was developed for prioritization of field/farm level problems/constraints. This app, primarily meant for extension/research professionals was pilot tested and was found to be user-friendly. This project also established that television, mobile and newspaper are the most preferred and accessed ICTs for agriculture information by both

male and female farmers. The preference of media used for agriculture, is highly correlated with the access and use of media. Therefore, exposure to new media is essential to develop preference. There is no difference regarding the preference of media use for agriculture between educated and illiterate, young and old, large, small or marginal farmers, with high or low farming experience. ICTs access and use showed significant gender differentiation, implying thereby that farm women still have limited access and use of ICTs. Similarly, the access to Wi-Fi indicated gender difference due to the differential awareness and access to the instrument. The access to the new media and training on their use are essential for promoting extensive use of ICTs by farm women, youth and farmers in agriculture.

2.5 Information and Communication Management for Promoting Innovation and Governance

Enhancing the research, capacity building and consultancy in the areas of IT based decision support systems; GIS in agricultural management; ICT applications for village knowledge centers, Development of digital multimedia and e-learning resources for agriculture; Geospatial knowledge discovery and management; ICTs in agri-supply chains; ICT policy for NARS; Emerging information technology and systems; ICTs in participatory technology development and transfer; Designing e-learning systems and Distance training are the objectives to carry out activities related to research and capacity building programmes.

In order to advance the objectives of the broad theme, the Academy focused in the following broad areas for its research, academic and capacity building activities.

2.5.1 Application of GIS, GPS and remote sensing technologies

2.5.2 Development of information systems and knowledge management

2.5.3 Statistical computing and data analytics

2.5.1 Application of GIS, GPS and Remote Sensing Technologies

Geo-Spatial Knowledge System for Monitoring Climate Change in Agro Climatic Zone-X: Southern Plateau and Hills region

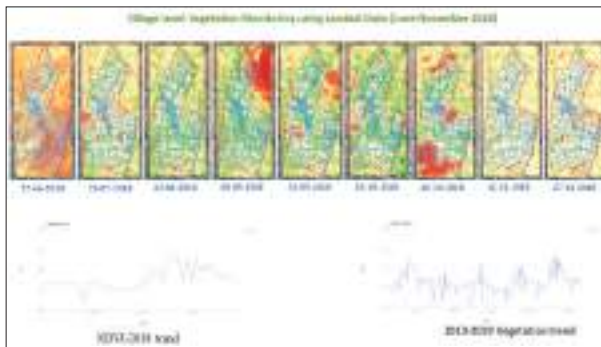
Climate variability significantly impacts the agricultural growth, stress, cropping pattern, phenophase and its vulnerability. The main objective of the study was to find the status of vegetation at village level and it plays an important role to assist policy makers as a decision support feature. Satellite derived data sets have been used to study the time series trend of NDVI (Normalized Difference Vegetation Index), NDWI (Normalized Difference Water Index) and NDDI (Normalized Difference Drought Index) for Tallasingaram village, Nalgonda district, Telangana state (NAARM adopted village). The study used Landsat 8 with 30 m resolution 8 days' interval images from 2013-2019. The results showed that, a temporal curve that summarizes the various stages that green vegetation undergoes during a complete growing season. This information helps the planners and decision makers to evaluate performance of vegetation growth at village level, in turn assess the crop insurance claims by the farmers. Enhanced Vegetation Index (EVI) was also developed as an alternative vegetation index to address some of the limitations of the NDVI.

Identifying, delineating and mapping land cover is important for village level monitoring studies,

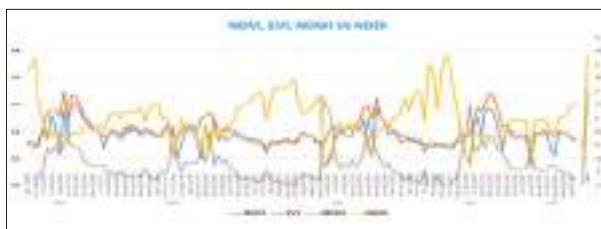
resource management and planning activities. Identification of land cover establishes the baseline from which monitoring activities can be performed. This data set provides land use and land cover classification products at 30-m resolution at village level for a particular cropping session. The data were derived from Landsat 8 and visual interpretation through google earth platform. The data were classified according to the Classification & Regression Tree (CART) algorithm.

Mapping Surface Water Area using Time Series Landsat Imagery:

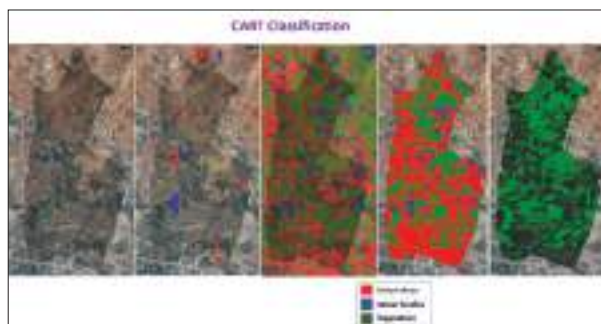
Surface water levels monitoring is a prerequisite for studying hydrological and ecological developments. Remote sensing (RS) data sets provide spatial and temporal data which aid in mapping the water bodies at the regional and global scale. In the current study, the temporal changes in the extent of surface water bodies in Telangana state were monitored through index methods, using indices like Normalized



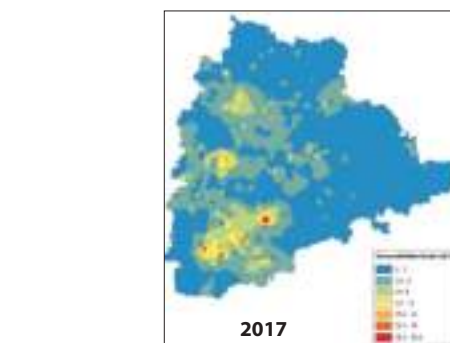
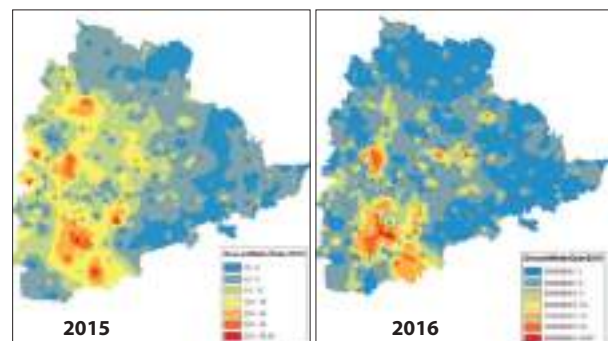
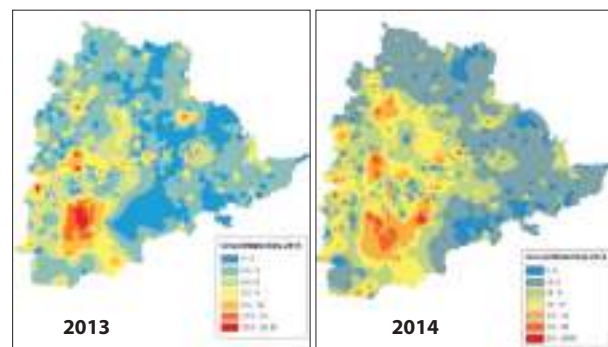
Transition in NDVI and Vegetation



Trends in NDVI, EVI, NDWI and NDDI



Results from CART classification



Ground water level variations in Telangana state during 2013-17

Difference Vegetation Index (NDVI), Normalized Difference Water Index (NDWI) and Modified NDWI (MNDWI) in Landsat 5, 7 and 8 scenes. Google Earth Engine (GEE), a free and web-based cloud computing platform was used for processing earth observation data, based on the time series Landsat images and compared with real-time groundwater levels. The results showed a significant increase ($p < 0.01$) in both surface water area and groundwater levels in Telangana state.

2.5.2 Development of Information Systems and Knowledge Management

Text Analytics of the Queries raised at Kisan Call Centres in South India

In order to harness the potential of ICT in Agriculture, Ministry of Agriculture launched the 'Kisan Call Centres (KCCs)' scheme on January 21, 2004. Under this scheme, farmers can call the Kisan Call Centres and enquire about various queries/problems related to agriculture and allied sectors. The Farm Tele Advisors (FTAs) answer the call and simultaneously record the query details which has made available a large amount of text data. This project is an attempt to study such data with the following: i) to identify the trending queries raised at Kisan Call Centres located in South India, & ii) to group the districts of South India based on the queries raised at Kisan Call Centres. As a part of the study, text analysis was carried out for the districts of Tamil Nadu state, considering the queries raised in each of the months as text corpora.

The data analysis is carried out in two stages. At first, the frequency data of the queries are analyzed using exploratory data analysis techniques. Later, topic models are fit to the data using Latent Dirichlet Allocation (LDA) technique.

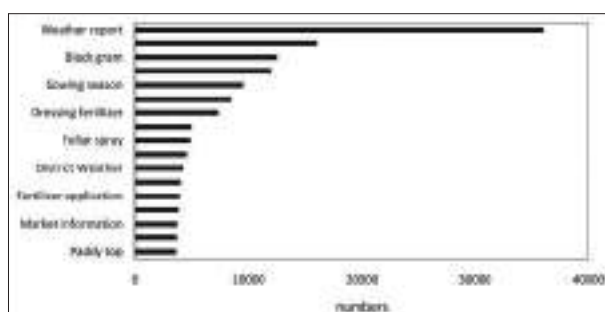
The data was cleaned for punctuation symbols and stop words, followed by the exploratory analysis for the frequency of each word appearing in the query. The most of the queries consisted the words "management", "weather", "report", "paddy" and "fertilizer", each of them appearing more than 20,000 times in the data corpus. The word cloud was plotted to see the relative appearance of these words compared to others. The bigrams were also studied to find out the pair of words which appear very often. It was found that the bigrams "weather report" appeared the maximum number of times followed by "fertilizer management", "black gram" and "market rate", each appearing more than 10,000 times in the data corpus. The bigram network plots were also drawn to study the relationships between/ co-occurrence of the words. For fitting topic models, the collection of queries for each district for each month is considered as a text document.

Since there are 30 districts in Tamilnadu and the study period has 12 months, a collection of 360 documents was obtained. The topic models were developed for this text corpus using LDA. Different values for the number of topics were tried and the optimum value was obtained by using the Harmonic Mean criteria. It was observed that there are 9 topics for the data corpus and the top 6 words that are most common with each topic are found out along with the probability of that word being generated from that topic. The probability that each document belongs in each topic is also found out. It was observed that there are many documents that belong to each topic as well as some documents not belonging to any topic indicating that documents are being well discriminated. Probability distribution within each topic was also found out which tells us each document's probability of belonging in that

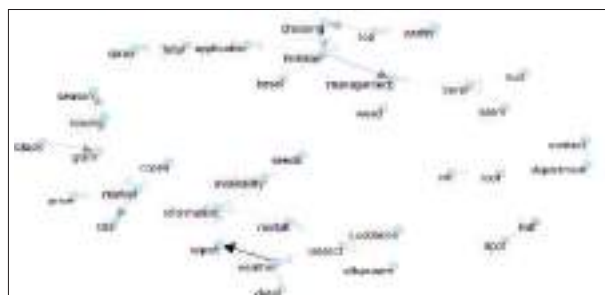
topic. The results of this study prompts for further investigations using Natural Language Processing techniques to analyze query texts to address the needs of farmers.



Text cloud of top 100 words



Top bigrams appearing frequently in the queries



Network of words appearing in the query text

Training Programme on Recent Bioinformatics Tools for Genome and Proteome Analysis

A Training programme on 'Recent Bioinformatics Tools for Genome and Proteome Analysis' was organized by ICAR-National Academy of Agricultural Research Management, Hyderabad from 17th to 21st September, 2019 at NAARM campus. This training programme was conceived with an idea to provide an insight to the

participants about the enormous potential and application of Bioinformatics and its tools management in the domain of agriculture. During the training, the participants were provided with theory and hands-on practical sessions on various software and tools on the different topics related to the emerging requirements of bioinformatics tools and software used in the agriculture and allied sectors. Besides Consultative Group on International Agricultural Research, the resource persons from the National and State Government Institutions participated in moderating the discussions. Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM in his valedictory address emphasized the importance of bioinformatics in agriculture. A total of 14 participants from various ICAR Institutes and State Agricultural Universities participated in the Training Programme were awarded certificates.



Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM interacting with the participants

National Workshop on Bioinformatics in Agriculture

In view of globally increasing importance of bioinformatics, a National workshop on 'Bioinformatics in Agriculture' was organized at ICAR-National Academy of Agricultural Research Management (NAARM) Hyderabad during 26-27 July 2019. In addition to expert deliberations, the workshop also covered theory and hands-on practical sessions on various software and tools on various topics related to emerging requirements of bioinformatics tools in agriculture and allied

sectors. The workshop included a brainstorming session on four specific issues in bioinformatics viz. constraints and opportunities in bioinformatics research; IPRs in bioinformatics; bioinformatics in teaching and education; data sharing, access and protection in bioinformatics research. Each group brought out recommendations on such issues along with supporting data constituted inputs for drafting a 'Policy Brief' on the subject. There were 22 participants from various ICAR institutes and SAUs. Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM delivered valedictory address and distributed certificates to the participants.



National Workshop on Bioinformatics in Agriculture

ICAR Research Repository for Knowledge Management

The project on 'ICAR Research Repository for Knowledge Management' is funded by ICAR to implement ICAR Data Management Policy at all ICAR institutes. The project envisages creation of repositories for different types of data (Experimental, Observational and Survey), publications, proven technologies and spatial data visualization and exploration tool in the form of Geo-Portal. The project is jointly executed by ICAR-IASRI, ICAR-NAARM, ICAR-IARI & ICAR-NBSSLUP. The contribution of NAARM is mainly on data repository and technology repository.

During the year, the experimental data repository for All-India Coordinated Research Project

(AICRP) using the AICRP Information system was implemented for AICRP on Castor, AICRP on Sunflower, AICRP on Safflower, AICRP on Small Millets and AICRP on Pearl Millets. The Information system was modified to cater to the needs of respective AICRPs. In the initial version used in AICRP on Sorghum and AICRP on Maize, the data can be entered only on those trials which were created using the system. A new facility of using the system without generation of trials through the system was implemented. Using this, the new AICRPs would start using the system from 2019 Kharif and Rabi season onwards. The data analysis module now includes analysis of block designs (Randomized Block Design, Alpha Lattice Design), Completely Randomized Design, Factorial Experiments and split plot designs. The system is in continuous use by AICRP on Sorghum and Maize and during the last year and the details of the trials are given in the table.

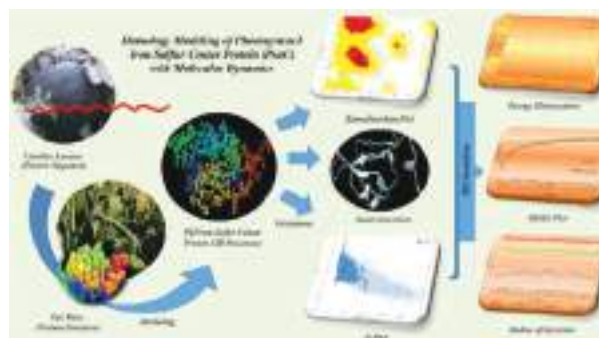
A draft policy on Quality Data acquisition was prepared which deals with steps in improving the quality of data, particularly from experiments. Another comprehensive document is under preparation to bring out ICAR-Data Management Policy, ICAR Open Access Policy, ICAR Data Licensing Policy, ICAR Intellectual Property Rights Policy to remove ambiguities and set order of precedence of these policies.

Establishment of Skill Development in Bioinformatics Infrastructure Facility (SKILL-BIF) on Agricultural Biotechnology at NAARM under BTIS net Programme

The Skill Bioinformatics infrastructure facility was developed with a Server and workstation, and 10 PCs with LAN connectivity. Under this project, one national level workshop was conducted on 'Bioinformatics in Agriculture' and a training programme on 'Recent Bioinformatics Tools for

Genome and Proteome Analysis' was conducted for the Scientists and Faculty of Bio-technology and also the scientists of FOCARS, SVUV, ANGRAU were trained. There were 40 participants from various ICAR institutes and SAUs. The centers use lecture materials, video clippings, demonstrations, tutorials and online facilities for teaching. It also periodically organizes hands-on trainings on various aspects of application of bioinformatics tools and molecular techniques in agricultural research. We have developed Molecular Modelling and Docking of Tomato Beta Galactosidase enzyme-7 (A2JGX1), In this study the ligands CME and BMA are used for in silico docking reaction with the A2JGX1(TBG-7) enzyme using Auto dock 4.2 software. Around 2 ligands were used for docking reaction only two Substrate successfully docked in the binding sites of TBG-7 of tomato. Once the binding sites predicted then Autodock4 performed to docking the TBG-7. The binding site pocket ID- 1 VAL73, PRO74, ASN99, GLY100, PRO103, SER104, TYR109, GLY112, PHE114 residues used to study the ligand interactions. Predominantly this process significantly useful for increase in tomato fruit firmness. Homology Modeling of Photosystem I Iron-Sulfur Center (PsaC) from Citrullus Lanatus using Modeller, Confusion Matrix-based Selection of Hexa-ORFs of Gene-CDS with Mutability of Central Codon etc., This study refers to the construction of 3D protein structure on the basis of comparative modeling including comprehensive validation by applying various Bioinformatics tools and technics, the corresponding model can be useful for further functional characterization of a protein regarding this species Homology modeling of capsid protein from Mung Bean Yellow Mosaic India Virus using Modeller. In this study the overall validation is being done by using UCSF chimera-

interactive visualization and analysis of molecular structure and related data, including density maps, supramolecular assemblies, sequence alignments. RAMPAGE- conformation of phi and psi angles for a possible amino acid residue in a protein, allowed and disallowed regions, RasMol-molecular graphics visualization, proSA- quality of model is indicated by QMEAN- Z score 3D model protein structure for potential errors and PDBsum (pro-check).



Homology modeling of Photosystem I

Decision Support System of Maize Inbred Germplasm

The decision support system on maize germplasm was developed. The maize germplasm database has been strengthened with the information over 420 maize inbred lines with the data on 14 traits viz. time of anthesis, time of silk emergence, density of spikelets, number of kernel rows, number of kernels per row, grain type, plant height, ear placement height, anthocyanin colouration at base of glume, anthocyanin colouration of anthers, anthocyanin colouration of silks, grain colour, kernel row arrangement and 1000-kernel weight against each inbred lines. The DSS has been updated with module on GDD for anthesis calculation. The database is being updated with additional features viz. Growing Degree Days (GDD), year and season of evaluation, heterotic group, special traits etc.

2.5.3 Statistical Computing and Data Analysis

Geospatial Analysis using QGIS & R

The main objective of this training programme was to enhance competency levels of NARES professionals in understanding GIS, GPS and RS along with QGIS and R software applying these principles in Agricultural Sciences. This programme consisted of a blend of lectures, self-exploration instruments, case analysis and hands on exercises. This training programme was meant for Scientist/Faculty/Technical Officers and above cadre, working in the area of Natural Resource Management from ICAR Institutes, State Agricultural Universities and KVK's. The training class schedule and topics covered mainly on introduction to geospatial technologies in agriculture, relational database and geodatabase concepts for GIS, Introduction to GIS, GPS and remote sensing, exploring data from open source platforms for geodata, spatial database design and implementation using quantum GIS open source software, spatial analysis like clipping, joining, buffering, interpolation, krigging etc., image processing and land use land cover using R open source software.

Analysis of Experimental Data

The objective of the training programme on 'Analysis of Experimental Data' was to orient the researchers with the basics of statistical tools and techniques mostly utilized in agricultural research. Researchers were also exposed to scientific communication of their research results. The participants were trained in performing statistical analysis of data generated from agricultural research experiments using Statistical Analysis System (SAS) and other software packages. 23 participants belonging to different states of the country attended this programme.

12 participants from ICAR, 11 from SVVU, Andhra Pradesh participated in the programme. The training consisted of both lecture cum practical demonstration followed by hands-on by the trainees themselves. A softcopy of training manual on various topics was prepared consisting of lecture materials, hand-outs, datasets and codes was provided to the participants.

Data analysis using SAS was performed to the participants using SAS foundation 9.3 and SAS Enterprise Guide (EG). The topics covered during the training programme include Principle of Experiments for Conducting Research; Exploratory Data Analysis; Introduction to SAS and Data Management, Correlation, Regression and Diagnostics; Analysis of Experimental Data using Basic as well as advanced designs, Pooled Analysis, Multivariate Data Analysis using Principal Component Analysis (PCA), Factor Analysis, Response Surface. Emphasis was given to analysis of data using appropriate statistical techniques and interpretation of the results. Data from real-life studies and simulated data were utilized to explain various statistical techniques.

Training Programme on Multivariate Analysis Using 'R'

In many experiments and research studies, observations on multiple variables are being collected. Study of interrelationship among the variables is important as it provides much more valuable information than studying individual variables separately. In this training programme, various multivariate techniques were introduced through R programming language. The multivariate techniques covered include Visualization of multivariate data, Multiple Linear Regression, Logistic Regression, Principal Component Analysis. Cluster Analysis and Factor Analysis. The techniques were introduced with

one example set and trainees were asked to do on their own to analyze another example set. This approach proved to be useful as the trainees were able to understand nuances of multivariate data analysis.



Training Programme on Multivariate Analysis Using R

Training Programme on Koha for Library staff of ICAR

ICAR – NAARM organized a training programme on 'Koha for Library staff of ICAR' during 21-26 February 2019 at this Academy with the objective to make participants understand the application of Koha in library automation, to provide hands on training on various modules of Koha related to acquisition, cataloguing, OPAC etc., and to provide the knowledge of broad aspects of Koha installation and customization as per institute requirements. This training program was designed to cover both theoretical and practical aspects on various topics. The trainees were exposed to Introduction to Linux, Installation of Ubuntu operating system, Installation and configuration of Koha and hands on training by our internal faculty. Introduction to Koha modules, Administration, Acquisition, Patron Management, Cataloguing, Circulation, Serials, OPAC and hands on training were also covered by the guest faculty Mr. Indranil Das Gupta, Member, Koha Community Release Team and Database Documentation Manager, Koha Community Project.

2.6 Mobilizing Science and Technology for Innovation and Sustainable Development

Catalyzing agricultural science and technology for innovation is important for good governance in NARES, and sustainable development in agriculture. Conducting research therefore becomes vital for use of science based interventional tools for effective research management and technology management in agriculture. Our efforts are towards infusing innovative ideas and bringing in a new milieu in the entire agricultural research system with the use of advancement in information technologies for online management system and digital framework in order to increase the productivity in research, reduce the drudgery, avoid unnecessary overlapping, and enhance the overall efficiency in research. Special focus is accorded to agricultural technology management with the application of management fundamentals for efficient management of research outputs and for reaping competitive advantage in the organization. In order to advance the objectives under the broad theme, the Academy focused on the following broad areas for its research, academic and capacity building activities:

- 2.6.1 Research Management and Policy Advocacy
- 2.6.2 Innovation and Technology Management
- 2.6.3 Research in the context of food security, environmental sustainability and globalization

2.6.1 Research Management and Policy Advocacy

Enhancing Efficiency of Agricultural Research Management in ICAR

A systematic effort has been taken up to strengthen the capacity of the young researchers

across ICAR and state agricultural universities to enhance their competence in research project management. The web-applications developed for project management viz., Research Concept Writer (RCW) and Project Logframe writer (PLW) have been institutionalized in training programs of the Academy (Module on Developing Winning Research Projects - DWRP, Foundation Courses, MDPs, etc), through which over 300 participants were provided hands-on training during 2019, under various programs. About 100 middle and senior officers involved in the research management were sensitized on the potential use of IT tools for easing their routines. Over 1750 researchers from India and 50 different countries including Singapore, USA and Germany had used these online applications for project proposal development during 2019.

As a part of collaborative research undertaken by NAARM with Department of Science and Technology, New Delhi International Food Policy Research Institute (IFPRI), Washington the data on inputs and outputs have been collected from all ICAR institutions and Agricultural Universities for the period from 2014-15 till 2017-18 including institutions under ministry of Agriculture and public sector institutions. The study also has a component to develop a Dashboard to aid in data-driven decision making in the domain of agricultural research and in avoiding duplication of efforts by multitude of agencies in assessing performance. An application to this objective has been developed jointly with one of the IT firms empaneled with NAARM. The application has innovative user interface for data collection, real-time monitoring of survey, collaborative survey, auto-generation of institute fact sheets, and a Dashboard to compare and analyze the research efficiency of various research organization(s).

A total of 113 ICAR institutions along with 11 Agricultural Technology Application Research Institute (ATARI), 67 Agricultural Universities, 24 public sector institutes and 22 institutions under Minister of Agriculture & Farmers Welfare are the sources of data collection. Data pertaining to (i) human resources deployed (ii) financial resources used (iii) research focus and (iv) research outputs were collected through a structured schedule in the application and those pertaining to the R&D performance of these institutions viz., number of publications, average citations, research areas, extent of collaboration and preferred journals were extracted from Web of Science of Clarivate Analytics.

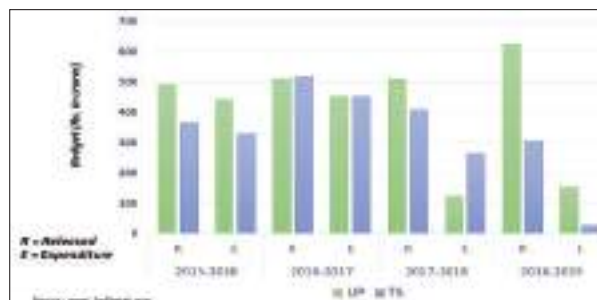
The datasets are being validated by the partner institutions independently and the same will be hosted in the Government of India's DST platform and also in the ASTI (Agricultural Science and Technology Indicators) platform of IFPRI by second quarter of 2020. Meanwhile, efforts are underway to onboard all the nodes onto the dashboard and build their capacity in using the application for institutionalizing evidence-based research management and for planning effective interventions for enhancing research efficiency of institutions at various levels.

Evidence-based Research Governance and Policy Advocacy

Natural resource management, biodiversity conservation and sustainable resource use are domains with significant national focus across various central and state ministries/ departments. Based on extensive review, interactions with experts through brainstorming sessions and systematic research and meta-analysis, the ICAR Policy on Climate Change and Indian Agriculture has been drafted and technical inputs were provided for drafting the Draft National

Mariculture Policy 2019 and National Disaster Management Plan for Agriculture. A detailed project report (DPR) on the capacity building and knowledge sharing needs of the coastal community of Karnataka to be climate resilient along with a comprehensive roadmap along with the interventions for addressing them has been prepared for implementation by the Karnataka State government under the Integrated Coastal Zone Management (ICZM) program of Government of India.

The Academy has undertaken a research study to develop a framework for mainstreaming biodiversity conservation in the development schemes of agricultural sector in India, so as to aid GoI in meeting its national and international obligations to address the challenges in biodiversity conservation, governance and NRM. The centrally sponsored schemes (CCS) under the selected Ministries viz., Ministry of Agriculture and Farmers Welfare, Ministry of Rural Development, Panchayat Raj, Ministry of Tribal Welfare, etc. have been critically reviewed, their extent of biodiversity relevance determined and the budgetary contribution of such centrally sponsored schemes in agriculture have been estimated. Telangana and Uttar Pradesh have been selected as sample states, where in the status of implementation of these identified schemes are being studied. Over 40 success stories have been documented from these states with significant contribution to biodiversity conservation. Preliminary analysis of budget allocation in biodiversity relevant components of identified public schemes of the agriculture sector in selected states as depicted in the figure below indicate that during 2017-19, the expenditure on biodiversity relevant components in the agriculture sector have been very low.



Budget utilization in biodiversity components in agricultural development schemes in UP & Telangana states

Further studies are being done to evaluate the component-wise status and to engage with the respective state functionaries to mainstream biodiversity in the development schemes of agriculture. The study envisions to recommend reforms in identified public schemes so as to ensure enhanced biodiversity outcomes and public finance for agro-biodiversity conservation and to develop a capacity building manual targeted to policy and decision makers.

Agricultural Research and Development Infrastructure and Resources Assessment

A study on assessment of agricultural research and development infrastructure was taken up in Andhra Pradesh and Telangana with the specific objectives of assessing the status and gaps of infrastructure and resources available for agriculture research in the study states, analyzing the productivity and efficiency of available resources and infrastructure and assessing the impact of agricultural research infrastructure on farmers. The two study states (Andhra Pradesh and Telangana) together have 9.37 % of the R & D institutions, slightly more than their geographical area and population share. Telangana has 39 state sector agricultural R & D institutions, while Andhra Pradesh has 63 of them. Domain and mandate analysis of some sample agricultural R &

D institutes indicates that there is some amount of commonality/ overlapping in the mandates of central institutes with either sister institutes and also with that regional / state institutions (SAUs). Among the ICAR institutes, higher vacancies (39 %) were noted in the case of the CTRI whereas in CRIDA, IIOPR, and IIMR are also having significant vacancy ranges between 20 and 31 %. Among ICAR institutes, CRIDA and IIOPR had higher publications intensity (0.55 and 0.54 papers per scientist/year). ANGRAU has released 410 crop varieties and prominent among them are the paddy varieties. Some of these varieties (Swarna, Samba Mashuri, and MTU 1010) are much popular in other states like West Bengal, Uttar Pradesh, Maharashtra, Karnataka, and Odisha. Extensive sharing of laboratory and experimental farm area infrastructure of different institutions located in the same place needed. The absence of key laboratory facilities in the state of Andhra Pradesh need to be addressed. In order to assess the impact of technologies, in Ananthapuram district, a typical rainfed area, farm ponds have been constructed as part of several schemes to ensure runoff water harvesting and use the harvested water for life saving irrigation for crops. Majority of the ponds are with a capacity of 200 cum and only 3.5 per cent of them are lined either with cement plaster or polythene sheets in the bottom. Farmers have reported a yield gain of 28 per cent due to farm ponds where groundnut is the main crop.

MDP on Priority Setting, Monitoring & Evaluation (PME) of Research Projects

The 7th Management Development Programme (MDP) on Priority setting, Monitoring and Evaluation (PME) of agricultural research projects for the In-charges of PME Cell of ICAR institutions was organized by the ICAR-National Academy of

Agricultural Research and Management (NAARM) during 18-23 July 2019.



MDP on PME with Director and Faculty Members of NAARM

The Heads of PME Cell from 32 ICAR institutions, nominated by the Director General of ICAR, participated in the program. The MDP comprised of modules on concepts and principles of prioritization, monitoring, evaluation and impact assessment of agricultural research projects. The participants were given hands-on training on the various software tools developed by ICAR-NAARM for project management. They were sensitized on the various metrics and tools used for measuring individual and institutional research performance, global and national frameworks for measuring research efficiency and ICAR guidelines pertaining to the various functions of PME Cell like managing research publications, consultancy services and contract research. The participants were provided hands-on training to use and practice various tools and applications for research review, project proposal preparation, checking plagiarism and choosing appropriate journals for research communication. The program concluded on 23 July 2019 and Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM delivered the valedictory address and distributed the certificates.

A Colloquium on Evidence-based Research Governance

A specialized sensitization program on Evidence-based Research Governance was held for the scientists and officers of National Research Centre on Orchids, NRCO, Pakyong, Sikkim during 06-

07 May 2019. The participants were provided an overview on the recent trends in measuring performance of research institutions and individuals. They were introduced to the various metrics that matter for the researchers and sensitized on the different online tools for research data processing, analysis and presentation.



Participants of the Colloquium on Evidence-based Research Governance

Training Programme on Developing Winning Research Project Proposals

NAARM organized the Training program on DWRP for the newly recruited faculty members of Professor Jayashankar Telangana State Agricultural University (PJTSAU), Hyderabad and for Bihar Animal Sciences University (BASU), Patna. The training for PJTSAU was held at NAARM in three spells and last in the series was held during 11-14 July 2019. A total of 38 faculty members representing different colleges and research stations of the university participated, of which 22 were women. For the faculty members of BASU, the program, second in the series, was held during 16-18 Aug 2019 at Bihar Veterinary College, Patna. 25 Assistant Professors belonging to different disciplines attended the program, of which, 5 were women.

The core content of the program included elements of research concept writing; preparation of project log frames. The participants were sensitized on the elements of scientific communication and various metrics used for measuring science performance. They were

provided hand-on exercises on various tools used for research review, plagiarism check, preparing illustrations, digital communication platforms, etc. The program provided a systematic overview of the elements of writing research papers, reports and poster presentation. Most exercises were through hands-on mode and participants were exposed to the research management software developed by NAARM viz., AHP Analyser, Research Concept Writer and Project Logframe Writer.

A tailor-made off-campus program was held at College of Agriculture, Indore for the faculty members of Rajmata Vijayaraje Scindia Krishi Vishwa Vidyalaya (RVSKVV), Gwalior, wherein 30 scientists and assistant professors representing different colleges and research stations of the university participated. The core content of the program included elements of research concept writing; preparation of project log frames. Most exercises were through hands-on mode and participants were exposed to the research management software developed by NAARM viz., AHP Analyser, Research Concept Writer and Project Logframe Writer.

Workshop on Developing Winning Research Proposals (DWRP) on Digital Solutions in Agriculture

NAARM organized the 40th Training-cum-Workshop on 'Developing Winning Research Proposals (DWRP) on Digital Solutions in Agriculture' under the aegis of National Agricultural Higher Education Project (NAHEP) during 19-23 Nov 2019. The focus of this module was on digital solutions in agriculture, keeping in tune with the new opportunities emerging in the sector. A total of 28 faculty members representing 11 states and 18 institutions participated, of which 03 were women. The program aimed at imparting the skills for writing research proposals that can

win funds from donors focusing on the needs of the stakeholders and sensitizing the participants on the various components of a research proposal. The core content of the program included elements of research concept writing; preparation of project logframes and making winning research proposals. Most exercises were through hands-on mode and the participants were introduced to the online applications developed by NAARM viz., AHP Analyser, Research Concept Writer and Project Logframe Writer. A Pre-Proposal Workshop on Development of Digital Solutions through application of Artificial Intelligence (AI), Block Chain Technology and IoT in Agriculture” was organized during November 19-23, 2019 at NAARM, Hyderabad under NAHEP project. This workshop was designed to enable participants to develop research proposals on potential applications of Artificial Intelligence, Machine Learning, Block Chain Technology, IoT in Agriculture.

An off-campus training workshop on ‘Developing Winning Research Proposals (DWRP) on Digital Solutions in Agriculture and Food Systems’ under



Participants of DWRP Training for the Faculty of PJTSAU



Participants of DWRP Training at Sharda University, NOIDA

NAHEP was organized during Dec 27-28, 2019 at Sharda University, NOIDA. The main objective of this training was to impart the skills for writing research proposals that can win funds from donors focusing on the needs of the stakeholders and sensitizing the participants on the various components of a research proposal. A total of 29 participants of Sharda University attended the workshop.

Foundation Course for Agricultural Research Management (FOCARS)

FOCARS is the flagship programme of the Academy and is offering the same twice a year. The Foundation course aims at building the competencies of the scientist probationers of Agricultural Research Service so as to nurture and develop the skills required to meet the professional challenges. The competency framework of FOCARS comprises the development of core discipline research skills, personal effectiveness, information management and business orientation.

The Academy organized the 109th FOCARS for 32 ARS Probationers during January 4 - April 3, 2019. The programme was organized in three phases for 73 working days. The probationers belong to sixteen ARS disciplines. The sessions were organized on core research skills, personal effectiveness, awareness and use of IT tools,



109th FOCARS Batch ARS Probationers along with DG, ICAR and Faculty, NAARM



Dr. Trilochan Mohapatra, DG, ICAR addressing 109th FOCARS Probationers

database management, awareness of research environment, orientation of organization and other skill components. Innovations brought in this FOCARS training included module on science communication and encouraged the probationers to prepare book chapters and this initiative has led to bringing out a book on 'Challenges and Emerging Opportunities in Indian Agriculture'.

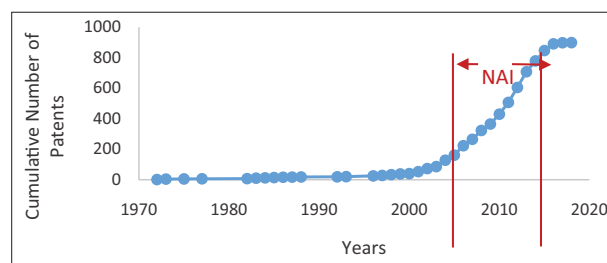
Training Programme on Science Communication and Scientific Presentation

A Training Program on 'Science Communication & Scientific Presentation' was held at ICAR-NAARM during 16-18 May 2019, exclusively for the newly joined Assistant Professors of Professor Jayashankar Telangana State Agricultural University (PJTSAU), Hyderabad. The participants were sensitized on the elements of scientific communication and various metrics used for measuring science performance. They were provided hand-on exercises on various tools used for research review, plagiarism check, preparing illustrations, digital communication platforms, etc. The program provided a systematic overview of the elements of writing research papers, reports and poster presentation. 28 participants from various colleges under PJTSAU participated in the program, of which, 05 were women. The programme consists of a blend of lectures, case analysis, discussions and hand-on exercises.

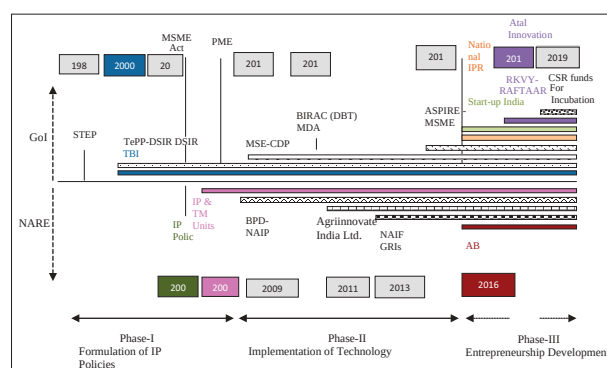
2.6.2 Innovation and Technology Management

IP Management in Agriculture

The Indian National Agricultural Research and Education System (NAARES), particularly Indian Council of Agricultural Research (ICAR), is actively involved in innovative research towards accelerated technology development and dissemination. Under Commercialization of Agricultural Technology Scheme (IPTM - ZTMC), concerted effort has been made in ICAR and these are reflected by its Intellectual Property (IP) portfolio, which seeks to provide a base to stimulate overall socio-economic growth through effective technology transfer, in agri-sector. In this study, an effort was made to understand these ongoing IP management and technology commercialization practices in ICAR and major challenges/pitfalls involved within, along with suggestive corrective measures. As recognized R&D indicators, patents



Year wise trends of patent filing in ICAR (n=899)



Advancements in IP Management & Commercialization Processes in ICAR

were used to measure innovative output. Patent data retrieved from various sources, was compiled and analyzed to decipher the trends of IP activities such as technological domain, intensity of filings, collaborations, etc. Results indicate active patent filings across all agricultural sectors, with a major spur after 2008. Technology transfer process effectively performed by Agri-Business Incubators (ABIs), created value and impact of technologies developed in the system. Information about technologies patented/non patented, commercialized by ICAR ABIs has been documented in a book 'Technology Management through IAR Agri Business Incubators: Status & Strategies'.

MDP Programme on 'Intellectual Property Valuation and Technology Management'

An MDP programme on Intellectual Property Valuation and Technology Management was conducted at the Academy to impart training on institutional innovations within NARS towards technology commercialization, entrepreneurship development and IP management. The sessions were on the S&T in emerging economy, IP valuation and monetization, technology management, transfer and commercialization. Besides, concepts on marketing and business models were discussed. To have a wider exposure of the incubation concepts, the participants visited Atal Incubation Centre-CCMB, Hyderabad and hands on training of patent analytics was provided at Patent Sciart Services. The participants gained the skills necessary for critically assessing both established and emerging approaches to business plan analysis and IP valuation and developing their own approaches to managing IP enabled businesses.



MDP Programme on 'Intellectual Property Valuation and Technology Management' at ICAR-NAARM

Post Graduate Diploma in Technology Management in Agriculture

NAARM had launched a one-year Post Graduate Diploma Programme on Technology Management in Agriculture (PGD-TMA) in 2011 on distance education mode in collaboration with the University of Hyderabad (UoH) as a stepping stone towards better management of agricultural technologies within the changing landscapes of IP regime. The course is specially designed to cater to the needs of building a strong IP portfolio along with efficient technology management, diffusion and commercialization skills among the scientists, teachers, students and professionals engaged in agriculture and related industries. The number of students admitted in 2019 is 33 with 07 female and 26 male students. These students represent 14 domain states. Most of the students are in-service candidates representing various Governmental (ICAR, CSIR & SAUs) and Non-Governmental Organizations such as Coromandel, Du Pont, Food Corporation of India (FCI), Syngenta, NABARD and State departments. Two contact sessions of 03 days' duration were conducted at the Academy during September 2019 and January 2020 covering the entire course. Resource persons from NAARM, Industry and other organizations contributed towards the lectures.



**PGDTMA Programme attending Contact Classes
at ICAR-NAARM**

2.6.3 Research in the context of food security, environmental sustainability and globalization

Economic Analysis of Food Prices and Policies for Agriculture Development

In the globalization era, government intervention through agricultural policy is important agricultural development in the country. Price policy of the government for agricultural produce seeks to ensure remunerative prices to growers for their produce and also for safeguarding the interests of consumers by making available food supplies at reasonable prices. In view of protecting the farmers, GOI emphasized on doubling farmer's income which can be partly achieved through a well-structured agricultural price policy. This understanding of the importance of agricultural prices has led to conception of the study with the following objectives: i) To analyze and estimate the price trends, growth rates and elasticities of major Indian agricultural crops, ii) To study price variability of major Indian agricultural crops and their effects on crop planning, poverty and inflation, iii) To analyze the effect of trade price transmission on major Indian agricultural crops and, iv) To evaluate the existing price policies of major crops and suggest policy options to the government.

This study analyzed MSP, WPI, CPI trends, growth rates, annual per centage change, exports and import price trends, growth rates, instability and

elasticities of major agricultural crops for the period 1990-91 to 2018-19. The study found that the WPI growth rate of pulse crops were found to be greater than the cereal and oilseed crops except sorghum. The consumer price general index has shown increasing trends where as consumer price food index has shown more or less linear trends in India. During the period of 2000-01 to 2017-18, the international prices of rice, maize, groundnut and sunflower were more than the domestic prices in India. But the domestic prices of soybean in India were more than international prices during the period of 2001-02 to 2003-04 and also 2007-08 to 2008-09. The study revealed that Government provided more MSP to pulse crops. But farmers are still interested in cultivation of staple crops like paddy and wheat. Government should support and provide incentives for farmers to balance their cropping pattern of agricultural commodities. India should design new stable foreign price policy for cereals and pulses as export import prices of all cereals and all pulses were found to be unstable.

Environmental Impact Assessment of Tobacco Curing

The study to assess the environmental impact of tobacco curing was undertaken in Andhra Pradesh and Karnataka states. Tobacco directly and indirectly provides livelihood for over 45 million people in India and more than two third of them are in the farming sector. Total area under FCV tobacco cultivation has come down by 61,000 ha during the last one decade (2007/08 to 2017/18), and the cured leaf production has reduced to the tune of 18.72 million kg. There were 87,988 FCV tobacco farmers authorized by the Tobacco Board as on March 31, 2018, of which Andhra Pradesh had 45,572 (52 %), while 42,416 (48 %) were in

Karnataka. The number of farmers has come down from 88,211 (2015-16) to 87,988 (2017-18). FCV tobacco requires a special processing i.e. curing using permanent structures called barns. In the Indian context, fuel wood is primarily used for curing in these barns. Therefore, it has been a cause of concern for several stakeholders on the source of such wood and also the impacts it might have on the environment. Eucalyptus (33.8 %), old felled trees of cashew (22.5 %) and mango (14.8 %) are the three dominant species that contributed the fuel wood used in the barns of West Godavari. Most of the fuel wood (88 %) used for tobacco curing in Prakasam district is obtained from *Prosopis juliflora*.

The tobacco barns in Mysuru district used Eucalyptus (24%) followed by Tamarind (20 %), *Prosopis cineraria* (14 %) and other material like coffee husk and coffee stems. In the case of Hassan, Pongamia and Khejri together comprised 32% of the fuel wood. Other material available in region such as coffee stems/husk is also used in the district. Eucalyptus is mainly sourced from

agroforestry plantations routed through wood traders in both the states. The second source is forest plantations duly sold through depots to the traders and farmers with transit permits. *P. juliflora* and other species originating from community lands and farm lands are supplied by the traders to the growers in most cases. The average specific fuel consumption (SFC) of traditional/ conventional barns ranged between 5.06 kg of fuelwood/ kg of cured leaf (West Godavari) and 5.27 kg of fuelwood/kg of cured leaf (Hassan). The weighted average SFC (weightage for number of barns by technology and category wise SFC) across the districts worked out to 4.27, 4.90, 5.14 and 5.22 kg. This clearly indicates the need for policy and operational thrust especially in the KLS region for energy saving. The emissions from FCV tobacco cultivation have been estimated to be in the range of 67 (NBS) - 191 (NLS) kg/ha and for the total crop area grown across the regions, it worked out to 14786 t/yr. The tangible byproduct of combustion of fuel wood in barns is the ash, which naturally gets recycled into agricultural fields.

2.7 List of Ongoing Research Projects

2.7.1 Institute Projects

Sl. No	PME Code	Title of Project	Team	Duration	
				From	To
1.	AGED/NAARM/SIL/2018/011/00150	Farmer Producer Companies in India: A Study on their Management Practices and Business Potential	N Sivaramane Ranjit Kumar PC Meena Bharat S Sontakki K Srinivas P Venkatesan	October 18	September 21
2.	AGED/NAARM/SIL/2018/009/00148	Mainstreaming the Small Ruminant Farmers into Efficient Value Chains in India	B Ganesh Kumar Sanjiv Kumar D Thammi Raju PC Meena	October 18	September 20
3.	AGED/NAARM/SIL/2017/012/00132	Team Work and its Effect on Organizational Performance in NARES.	RVS Rao Alok Kumar KH Rao	April 17	March 20

Sl. No	PME Code	Title of Project	Team	Duration	
				From	To
4.	AGED/NAARM/SIL/2017/013/00133	Competency Management System for Agricultural Research and Extension	K H Rao Alok Kumar RVS Rao P Ramesh	April 17	March 20
5.	AGED/NAARM/SIL/2018/010/00149	HR Interventions and their Influence on Institutional Innovations	Alok Kumar RVS Rao P Ramesh	October 18	September 20
6.	AGED/NAARM/SIL/2018/014/00153	Emotional Intelligence and Organizational Effectiveness	P Ramesh RVS Rao	April 18	March 20
7.	AGED/NAARM/SIL/2018/008/00147	Decision Support Database of Maize Inbred Germplasm	N Srinivasa Rao JC Shekhar N Srinivasa Rao	July 16	July 20
8.	AGED/NAARM/SIL/2018/004/00143	Text Analytics of the Queries raised at Kisan Call Centres in South India	BS Yashavanth PD Sreekanth Jabir Ali	April 18	March 20
9.	AGED/NAARM/SIL/2017/009/00129	Developing a Framework for Enhancing Agricultural Research Management Efficiency	Ch Srinivasa Rao P Krishnan I Sekar SK Soam B Ganesh Kumar K Kareemulla V V Sumanth Kumar	April 17	March 21
10.	AGED/NAARM/SIL/2018/005/00144	Assessment of Technological Interventions in Teaching-Learning Process	S Senthil Vinayagam GRK Murthy D Thammi Raju	April 18	March 20
11.	AGED/NAARM/SIL/2018/003/00142	Digital Teaching Evaluation in Agricultural Education using TRACK Model	GRK Murthy Senthil Vinayagam M Balakrishnan	April 18	March 20
12.	AGED/NAARM/SIL/2018/002/00141	Ag-Academy: A framework for Scaling up e-learning Platform	V V Sumanth Kumar M Krishnan P Krishnan SK Soam Ch Srinivasa Rao GRK Murthy NS Rao	May 18	April 22
13.	AGED/NAARM/SIL/2018/012/00151	Students' Learning Approaches (SLAs) for Enhanced Learning Outcomes in Agriculture Education: A Critical Analysis	D Thammi Raju P Ramesh Bharat S Sontakki P Krishnan	October 18	September 21
14.	AGED/NAARM/SIL/2018/012/00151	Structural Changes in Agricultural Marketing in India: Effect on Inclusiveness, Efficiency and Profitability	Ranjit Kumar B Ganesh Kumar PC Meena N Sivaramane Sanjiv Kumar	April 17	September 19

Sl. No	PME Code	Title of Project	Team	Duration	
				From	To
15.	AGED/NAARM/SIL/2017/014/00134	Geo-Spatial knowledge System for Monitoring Climate Changes in Agro-Climatic Zone X: Southern Plateau and Hill Region.	PD Sreekanth BS Yashavanth P Venkatesan	April 17	March 20
16.	AGED/NAARM/SIL/2016/007/00117	Gender-based ICT led Knowledge Management in Agriculture	N Sandhya Shenoy VKJ Rao PD Sreekanth N Srinivasa Rao BS Sontakki Surya Rathore Manju Gerard	April 16	September 19
17.	AGED/NAARM/SIL/2017/007/00127	Research Role Analysis of Extension Scientists in ICAR	VKJ Rao Barat S Sontakki Surya Rathore Alok Kumar	April 17	September 19
18.	AGED/NAARM/SIL/2019/012/00166	Agri Startups in India: Decoding the Determinants and Success Factors	Sanjiv Kumar K Srinivas Ranjit Kumar	June 19	May 21
19.	AGED/NAARM/SIL/2019/008/00162	Economic Analysis of Food Prices and Policies for the Development of Indian Agriculture	MB Dasthagiri B Ganesh Kumar A Dhandapani	June 19	July 21

2.7.2 Externally Funded Contract Research (Sponsored/Collaborative/Grant-in-Aid Projects)

Sl. No.	Title of Project	Team	Duration		Budget (in Lakhs)	Sponsored by
			From	To		
1.	Commercialization of Agricultural Technology Scheme (IPTM-ZTMC)	Manju Gerard K Srinivas	April 14	March 20	19.15	ICAR, New Delhi
2.	Agricultural Research and Development Infrastructure in select Indian States	K Kareemulla S Ravichandran	July 17	June 19	24.00	NSMIS, DST, Gol
3.	Establishment of Agri-Business Incubation (ABI) Centers under XII plan Schemes for National Agriculture Innovation Fund	K Srinivas Manju Gerard	January 16	March 20	79.15	ICAR, New Delhi
4.	Profiling the Resources Devoted to Agricultural R&D and Development of National framework for Evidence-based S&T Governance in India	P Krishnan B GaneshKumar M Balakrishnan SK Soam BS Yashavanth	October 18	September 19	29.00	DST, New Delhi

Sl. No.	Title of Project	Team	Duration		Budget (in Lakhs)	Sponsored by
			From	To		
5.	Management and Impact Assessment of Farmer FIRST Project	P Venkatesan BS Sontakki N Sivaramane	March 17	March 20	144.05	ICAR, New Delhi
6.	KRISHI: ICAR Research Data Repository for Knowledge Management	A Dhandapani SK Soam	November 15	March 20	76.00	Data Management, ICAR
7.	Economic Analysis of Eucalyptus and Subabul based Agroforestry Ecosystem in Telangana and Andhra Pradesh	SK Soam P Venkatesan N Sivaramane Ch Srinivasa Rao	September 18	August 19	11.28	ITC (P) Ltd.
8.	Establishment of Skill Development Bioinformatics Infrastructure Facility (Skill-BIF) for Agricultural Biotechnology	M Balakrishnan SK Soam	December 18	March 21	44.46	DBT, New Delhi
9.	Investments in Indian Council of Agricultural Research Leadership in Agricultural Higher Education (NAHEP)	SK Soam D Thammi Raju N Srinivasa Rao Alok Kumar VV Sumanth Kumar Sanjiv Kumar Surya Rathore	February 19	March 21	1704.17	World Bank
10.	Technology Enhanced Learning in Agriculture Education (TEAgE)	GRK Murthy S Senthil Vinayagam D Thammi Raju P Ramesh	January 14	March 20	425.00	ICAR, Niche Area of Excellence (NAE) Scheme
11.	Adoption of Insect Resistance Management Practices in Bt Cotton Cultivation in India	Ranjit Kumar N Sivaramane, A Dhandapani PC Meena Sanjiv Kumar	February 18	May 19	30.68	Mahyco Monsanto Biotech (India) Pvt. Ltd.
12.	Adoption of Natural Farming and its effect on Crop Yield and Farmers Livelihood in India	Ranjit Kumar P Ramesh PC Meena Sanjiv Kumar BS Yashavanth Indoria Sumanta Kundu Manjunath	January 19	March 20	35.00	NITI Aayog

Sl. No.	Title of Project	Team	Duration		Budget (in Lakhs)	Sponsored by
			From	To		
13.	Strengthening and Integrating Agricultural Science and Technology Indicators (ASTI) in Indian NARES	Ch. Srinivasa Rao B Ganesh Kumar P Krishnan SK Soam	October 17	June 19	30.15	IFPRI
14.	Environmental Impact Assessment of Tobacco Curing	K Kareemulla I Sekar S Ravichandran P Krishnan B Ganesh Kumar BS Yashavanth	April 17	June 19	20.87	Tobacco Board
15.	Developing Guidelines and methodologies for IP policies of PJTSAU	Manju Gerard SK Soam	April 17	May 19	5.00	PJSTAU
16.	Development of Digital Content for PGD in Agri-warehousing Management of MANAGE Hyderabad	S Senthil Vinayagam GRK Murthy	November 18	March 20	9.90	MANAGE, Hyderabad
17.	Development of Digital Content for Plant Biosecurity through MOOC Platform	GRK Murthy S Senthil Vinayagam	July 18	May 19	7.20	NIPHM, Hyderabad
18.	Entrepreneurship Development through Farmer Led Innovations- A Case Study in Plantation Sector	S Senthil Vinayagam GRK Murthy B Ganesh Kumar P Venkatesan	April 19	March 22	79.54	NASF
19.	Development of Policy Framework for Utilization of Indigenous Knowledge	P Venkatesan P Sethuraman Sivakumar	April 19	March 22	77.11	DST, CHORD (NSTMIS)
20.	Development of Digital Content for MOOC on Commercial Dairy Farming and "Milk Processing and Value Addition	S Senthil Vinayagam GRK Murthy	April 19	March 21	23.00	NDRI, Karnal
21.	Capacity building / Knowledge sharing for Institution and Community In Karnataka	Ch.Srinivasa Rao P Krishnan SK Soam BS Sontakki	September 19	Nov 19	15.00	NCSCM, Chennai

2.8 Educational Programmes

Name of the Educational Programme	Commencement Month & Year	Completion Month & Year	Coordinators	Batch Size	Participants
Post graduate Diploma in Management Agriculture (PGDMA 2017-19)	June 2017	April 2019	S Ravichandran P Krishnan	34 (19)	Students
Post graduate Diploma in Management Agriculture PGDMA 2018-20	June 2018	April 2020	D Thammi Raju (Upto March 19, 2019) MB Dastagiri Sanjiv Kumar	27 (12)	Students
Post graduate Diploma in Management Agriculture PGDMA 2019-21	June 2019	April 2021	N Sivaramane PC Meena	48 (21)	Students
Post graduate Diploma in Technology Management in Agriculture PGD-TMA	January 2019	December 2019	Manju Gerard	33 (07)	Students
Post graduate Diploma in Educational Technology and Management Agriculture PGDETM	January 2019	December 2019	Senthil Vinayagam	48 (19)	Students

Within Parenthesis-No. of Girls Students

2.8.1 Project Reports Submitted by the 9th batch of PGDMA Students

Sl. No.	Roll No.	Student Name	Title of the Project	Academy Mentor/Guide
1	1701	Adapa Teja	Development of Automatized Logistics Model	S Ravichandran
2	1702	Ankita Kala	Impact Assessment of Arize hybrid rice in Nepal and Chhattisgarh	SK Soam
3	1703	Anushree	Credit Risk Analysis for Rallis in Secunderabad and Guntur Zone	KH Rao
4	1704	Anu Choudhary	Rasi Seeds Business Opportunity, Product Mapping of Field Crops and Sales Strategy for Himachal Pradesh	D Thammi Raju
5	1705	Balambika M	Enabling Central Know Your Customer for Sammunati Financial Intermediaries and Services Private Limited	PD Sreekanth
6	1706	Bhargavi D	Assessment of Capacity Development needs of Agri Startups	I Sekar
7	1708	Chandra Shekar Cheruku	Study on Value Chain of Agroforestry Systems in Mahabubnagar & Medak Districts of Telangana	P Krishnan

Sl. No.	Roll No.	Student Name	Title of the Project	Academy Mentor/Guide
8	1709	Damini	Distribution Adequacy Analysis For Agri Inputs, Benchmarking Rallis with Key Competitors	Ranjit Kumar
9	1710	Dasari Harika	Identification of Potential Districts in Telangana for Samunnati	N Sivaramane
10	1711	Deny Devassy	Assessment of Capacity Development Needs of Agri Startups	BS Sontakki
11	1712	Dharshana V	Scope and Potential Repositories in Commodity Market In Tamilnadu	Surya Rathore
12	1713	Dhivagar B	Assessment of Direct Benefit Transfer of Fertilizer Industry in Key Districts of Tamilnadu	N Srinivasa Rao
13	1714	Dhivya B	Status, Prospects and Strategies for Business Development of Sam Agro	M Krishnan
14	1715	Goundeboyina Sheshank Yadav	Binging Operational Excellence in Trade of Samunnati	A Dhandapani
15	1716	Karthika R	Modules in Loan Origination System and Profiling of Startups	K Kareemulla
16	1717	Mallikarjun G	Identification and Categorization of Organic Farmers in Dharwad catchment, Bidar and Gulbarga, Karnataka	Alok Kumar
17	1718	Manjushree S N	A Study on Market Potential and Output Marketing Strategies of FPOs in Fruits and Vegetables in Karnataka	VKJ Rao
18	1719	Mankala Santhosh	Study on Rabi Hybrid Paddy Markets of Telangana and Odisha States	M Balakrishnan
19	1720	Marigouda	Study of Factors Influencing Adoption of eNAM	N Sivaramane
20	1721	Merin Maria Thomas	Leveraging Data Analytics to Enhance Supply Chain Efficiency	N Senthil Vinayagam
21	1722	Minati Majhi	Scope of Improvement of Farm Profitability and Productivity of Pulpwood Plantation in Malkangiri District of Odisha	P Venkatesan
22	1723	Pasula Vinaykumar	Sulphur Based Fertilisers Utilization Strategy for High Sulphur Demanding Crops in India and Its Socio-Economic Impact	Ch Srinivasa Rao
23	1724	Pooja N	Market Research Analysis of Hybrid Paddy Seeds in Bihar	B Ganesh Kumar
24	1725	Prakhar Rathi	Scope of Repository Ecosystem in Commodity Sector	S Ravichandran
25	1726	Rahul Kumar Singh	Channel Diagnostics and Channel Segmentation of Pesticide Dealers of Rallis India Ltd for Kolkata Zone.	Sandhya Shenoy
26	1727	Rajratan	Assessment of effectiveness of e-NAM in Parbhani APMC Market	B Ganesh Kumar

Sl. No.	Roll No.	Student Name	Title of the Project	Academy Mentor/Guide
27	1728	Romit Kumar Singh	Agri Extension Impact Assessment of ITC Wheat Development Programme (WDP) in Uttar Pradesh	Ch Srinivasa Rao
28	1729	Sayyeda Salima Asra	Identification of Risk Drivers and Laying a Mitigation Framework w.r.t retail sector	RVS Rao
29	1730	Shailendra Nageshwar	Understanding Institutional Lending Opportunities for FPOs and Developing a Scorecard for Creditworthiness	PC Meena
30	1731	Sharuti Dogra	Develop a Crop Acreage Prediction model and the expected Acreage of Key Crops for Rasi Seeds for the year 2019	P Ramesh
31	1732	Sruthy Antony	Assessing Credit Worthiness of Farmers and Development of Credit Score Card	Sanjiv Kumar
32	1733	Sivaranjani K	Lead value addition in the Product line operations of cut Vegetables and Packaging	S Ravichandran
33	1734	Usha B	Study of CSR Landscape and Development of CSR Collaboration Strategy for Samfins	B S Yasavanth
34	1735	Yogesh R	Analyze the Problems in Meal transportation & ways to reduce the freight charges for the Meal transportation from Latur to Tamil Nadu	M Balakrishnan

2.8.2 Summer Internship of PGDMA Students

Sl. No.	Roll No.	Student Name	Title of the Project	Academy Mentor/ Guide
1	1801	Aiswarya Sukumaran	Market Study on Strobilurin Group of Fungicides in Paddy/Chilli and Opportunities for Rallis with Ergon/KM combinations	Ch Srinivasa Rao
2	1802	Anand Singh	To evaluate the effectiveness of different campaign elements followed in promotion of Arize AZ 6633 hybrid rice	RV S Rao
3	1803	Amal Maria Jacob	Development of FPO Grading Tool	SK Soam
4	1804	Amarnath CK	Study on Mapping Trade of Commodities (Pulses & Rice) from Karnataka & Telangana to Other Markets	Ranjit Kumar
5	1805	Atul Kumar	Study on campaigning strategies for promotion of Arize Hybrid Rice	Bharat S Sontakki
6	1806	Dastageerapasha	Expansion of Osheen in Cotton And Study Of satisfaction level of User Farmers	M Krishnan
7	1807	Durgesh Chavan	Perception and acceptance of water soluble pouch (WSP) for 'Cabrio Top'	I Sekar

Sl. No.	Roll No.	Student Name	Title of the Project	Academy Mentor/ Guide
8	1808	Gayatri Gopan	Developing Financial Literacy Tool Kit for Capacity Building of Producer Group and Producer Company	B Ganesh Kumar
9	1809	G Srikanth	Identification of Multi-cropping Combinations	Manju Gerard
10	1810	Gollapalli Prasanth Mourya	Sustainable Cotton Initiative-ASSEFA creating livelihood	Alok Kumar
11	1811	Harshita Srivastava	Comprehensive Research Analysis for Identification of New Potential Markets for Hybrid Paddy in Identified Districts of Eastern Uttar Pradesh	A Dhandapani
12	1812	I C Kariappa	Preparation and implementation of Live Goat trading Business Model for FPO's in Rajasthan.	Surya Rathore
13	1814	K Sai Prajeeth Reddy	Creation of Market study template and FMCG Assortment analysis	P Krishnan
14	1815	Lingala Thrishali	Analyzing the market scenario of Plant Growth Nutrients and Water Soluble Fertilizers	S Ravichandran
15	1816	Madagoni Vamshi Krishna	Acceptance And Perceived Value Of SDHI Fungicides In Rice	Sivaramane N
16	1817	Mitali	Developing Financial Literacy Tool Kit for Capacity Building of Producer group and Producer company	KH Rao
17	1818	Naveen Yadav N	Value Chain Analysis of Turmeric	N Srinivasa Rao
18	1819	Nivedita Yogiraj Patil	To study & analyze Hybrid Marigold seeds business	P Ramesh
19	1820	Poorna Shree G	Process for on-boarding farmers into Outgrow	MB Dastagiri
20	1821	Ramya Nargund	Market size analysis of Brassinolides and CPPU in Grapes	K Kareemulla
21	1822	Rashmi Pandey	Comprehensive research analysis for identification of new potential market for Hybrid Paddy in Bihar	S Senthil Vinayagam
22	1823	Sadish Kumar	Study on Campaigning Strategies for Promotion of Arize Hybrid Rice	GR Ramakrishna Murthy
23	1824	Siddharth P	Study of Customer perception on the intended pack change for cabrio Top	PC Meena
24	1825	Shalini Singh	Complete Market Analysis of Quizalofop-Ethyl	M Balakrishnan
25	1826	Shetty Sharath Kumar	Identification of new potential market for Hybrid Maize in Karnataka	Srinivas Kondapi
26	1827	Shivani Balasaheb Doifode	Benchmarking of ADAMA India Farmer App "REACH"	PD Sreekanth
27	1828	Sumeet Singh	Assess the awareness, need and dynamics of plant nutrition products	V KJ Rao

2.8.3 Project Reports Submitted by the Students of PGD-TMA during 2019

SI No.	Roll No.	Student Name	Title of the Project	Academy Mentor/ Guide
1	33-TMA-355-15	Chinni Yerrayya	Agricultural Crop Insurance	N Sivaramane
2	33-TMA-408-17	Anil Kumar Sahu	Scopes, Opportunities and Possibilities for Geographical Indication (GI) of Naturally Occurring Rugra (mushroom) from the Forest of Jharkhand	Sudarshan Maurya
3	33-TMA-410-17	B N Bharath Raj	Isolation of Strains form Panchagavya	Ch Pavan Kumar
4	33-TMA-411-17	B Sridevi	Current Global Trends in Microbial Entophytic Research	PC Latha
5	33-TMA-448-17	Preeti Sharma	Adoption of Organic farming in Hyderabad-Karnataka region	Jaba Jagdish
6	33-TMA-451-17	Priyanka Singh	The effect of Data management on sustainable farming of LAC: Study of Jharkhand State	Ashokastana
7	33-TMA-458-17	Shankar N Karajagi	Analysis of Zero Budget Natural Farming (ZBNF) for Enhancement of Farm Production	Manju Gerard
8	33-TMA-481-18	Ishu Bansal	Organic Supply Chains for Small and Marginal Farmers: A Case Study of Mandla Organic Krishak Producer Company, Mandla, Madhya Pradesh	SRK Varshney
9	33-TMA- 488-18	Nawab Ali	A Global Market Overview, Analysis and Opportunities of Crop Protection Chemicals Business in Rice Crop	Roshan Mammen
10	33-TMA- 490-18	Praveen Narayan	Studies on Experiences of Organic Farming with Small Small Scale Farmers	Jawahar Kumar
11	33-TMA-496-18	G Selvakumar	An Analysis of the Fertilizer Control Order (1985) with Specific Reference to Biofertilizers and Organic manures	AN Ganeshamurthy
12	33-TMA-497-18	G Simi	A Case Study on the Backyard Rearing of Poultry Through School Poultry Club Scheme In Chelembra Panchayath of Kerala	CM Balusami
13	33-TMA-499-18	Syed Abuzar	Current Trends of Agrochemicals in India	Manju Gerard

2.8.4. Students (M.Sc. & Ph.D) Guided by NAARM Faculty as Major Guide/Supervisor

Sl, No	Name of the Faculty	Student	Title of the Thesis	Program MSc/PhD	University/ Institute
1	S Senthil Vinayagam	Seema Kujur	A Study on Adoption of Educational Technology in Teaching-Learning Process in Agricultural Universities.	M.Sc (Ag)	IGKV, Raipur
2	S Senthil Vinayagam	Payal Dewangan	Study on Group Dynamics of Women's Groups towards Entrepreneurial Development in Agriculture	M.Sc (Ag)	IGKV, Raipur
3	D Thammi Raju	K Vani	Effectiveness of Livestock Farmers Producer Groups (Livestock Based SHGs) in improving the Livelihood Security	MVSc	PVNRTVU, Hyderabad
4	M Balakrishnan	Md Atheeque Sultan Ghor	Development of Hybrid Neuro-Fuzzy System for Forecasting of Turmeric Yield in Telangana	Ph.D	Bharathiar University, Coimbatore
5	M Balakrishnan	M Naveen Kumar	Development of Hybrid Algorithms with an Interface for Accurate Prediction of Sugarcane Yield In Telangana	Ph.D	Bharathiar University, Coimbatore
6	A Dhandapani	Revanasiddesha	Development of R-package for Construction of Hadamard Matrices and associated balanced incomplete Block Designs	M.Sc.	IGKV, Raipur



3 NAARM as a Think Tank of ICAR

3.1 Think Tank: An Overview

Think tanks are public-policy research analysis and engagement organizations that generate policy-oriented research, analysis, and advice on domestic and international issues, thereby enabling policy makers and the public to make informed decisions about public policy. These institutions often act as a bridge between the academic and policymaking communities and between states and civil society, serving in the public interest as independent voices that translate applied and basic research into a language that is understandable, reliable, and accessible for policy makers and the public. While policy-makers may lack the tools to quickly respond to a critical policy problem, they often suffer, not from a lack of information, but from an avalanche of information that gets in the way of effective decision-making. Overcoming these obstacles often requires knowing where to turn for rigorous, reliable and accessible information and analysis. Thus, the Think Tanks act as filters and synthesizers that facilitate the identification of policy issues, the design of policy solutions, and the implementation of and feedback on policy decisions.

Think tanks today are facing increasingly intense changes in the world, both political and technological ones, which pose serious challenges and even existential crises to think tanks. In the face of rapid technological advancements, think tanks now find themselves no longer the only actor in the knowledge brokerage industry, but as only one of the competitors in the “global marketplace of ideas,” vying with other actors such as media organizations, advocacy groups, consulting and law firms. In order to catch the attention of their key audiences viz., policy makers, general public or the donors – the think tanks need to deliver the analytical information it promises in a timely and effective manner.

Think Tanks mobilize expertise and put forward evidence. The heart of a Think Tank's work is its

research and innovation. Thus, the Think Tanks need to innovate towards incorporating appropriate strategies in management, communication, financing, and technologies without undermining the quality and rigorousness of research and publication. It is only with an innovative operational model and an unwavering commitment to excellence of research could New-Age Think Tanks make a meaningful contribution in today's world.

3.2 Need of Think Tank in Agriculture & Allied Sector

Indian agriculture is pivotal in meeting the national goals of Doubling Farmers' Income by 2022, making \$5 trillion Indian economy or meeting several Sustainable Development Goals (SDGs). Moreover, agriculture sector is becoming more complex, knowledge intensive, technologically challenging and globally integrated. To address such diverse challenges, the Council desires the Academy to maintain a futuristic vision and become a 'Think Tank' in the field of agriculture.

Agriculture and its allied sectors play a critical role in ensuring food security, reducing poverty and sustaining growth in India. The sector accounts for 14% of the national GDP and provides livelihood for over 70% of its population – directly and indirectly. Agriculture, being in the concurrent list, poses governance challenges, owing to the need to synergise the roles of Central and State governments, in conceiving and implementing programs.

The dynamics of agricultural growth reflect a reduction in the share of crop sector and an increase in the share of agricultural sub-sectors. As agriculture entails risks related to production, weather, prices and policy, capitalizing the structural changes in the agriculture sector by diversifying income generating activities can mitigate the risks

and sustain growth of the economy. There is a need of shift to a new paradigm for agriculture, from being focused on production and 'sustainability of production system' to focusing on sustainability of producers' livelihood. The future agriculture should offer farmers more choices, should be resilient with an ability to adapt to climate change, environmental events, political issues, market fluctuations, changing consumer demand and the rising cost of fuel, and should be health promoting- for people, communities, the soil, ecosystems and the planet. Further with emergence of new technologies, other than yield enhancing, but more of facilitating agricultural production and marketing, like IT, IoT, AI, robotics, etc., agriculture sector is becoming more complex and knowledge intensive. This calls for setting standards of agricultural education and research, developing appropriate policies for availing the services of the best human resources, formulating frameworks for private partnerships so as to forge meaningful and beneficial collaborations, through convergence of purposes.

The Government of India has entrusted the responsibility of policy guidance pertaining to various domains of development including agriculture to NITI Aayog. However, it requires a team of policy researchers to undertake diagnostic research on various aspects of agriculture across its supply chain and create evidence-based, data-driven policy advocacy on various areas. NAARM would serve as a holistic Think Tank of the Indian agricultural system and serve as a Single-window Go-To Agency for providing policy advocacy on all aspects of agriculture - research, development, education and governance.

3.3 Transition of NAARM as a Think Tank

As an integrated institution of agricultural research management, the Academy focuses on creation, dissemination and application of knowledge through its academic, training, research, consultancy

and policy support programmes. The strength of the Academy lies in its unique blend of training, research, education and consultation activities that interactively reinforce each other. As a result, the Academy has evolved into a learning organization that can continuously renew itself and adapt quickly to the changing needs of ICAR and other institutions of NARES. ICAR-NAARM has all along contributed in a significant advisory role and in policy advocacy on various aspects of internal functioning of ICAR viz., human resource policy, transfer guidelines, career advancement, intellectual property and technology management, training needs assessment, research prioritization, etc. in addition to the core activity of capacity building. However, in its envisioned role as a Think Tank of ICAR, the Academy is expected to carry out more directional role in the field of agriculture and provide far more significant inputs - strategic, functional and technical - to the ICAR.

Key Developments:

- The National Cadre Review Committee of ICAR chaired by Hon'ble Secretary, DARE & Director General, ICAR, recommended that ICAR-NARM should maintain a futuristic vision and become a "Think Tank" in the field of agriculture.
- ICAR constituted an expert committee for cadre review of the Academy under the Chairmanship of Dr. B. Venkateshwarlu, Former VC, VNMKV, Parbhani to review the scientific cadre requirement so as to strengthen NAARM as a Think Tank of the Council.
- Dr. Mangala Rai, Former Secretary, DARE & DG, ICAR as Chairman of Seventh QRT of NAARM, envisioned that the Academy, as a Think Tank of ICAR, shall be the guiding force for the Council in identifying its future focus area, forecasting the organizational requirements, defining its priorities, strategizing to stay relevant in 2050 and beyond.

The Academy therefore, shall focus on futuristic visioning and foresight research in the field of agriculture and allied sector, knowledge management and research governance. It shall also provide evidence-based strategies to optimize the resource allocation including human resource, capital resource as well as land and water resources in such a way that agro-ecosystem remain sustainable and economically viable.

While all ICAR institutes are Think Tanks in their own spheres, the current vision of ICAR to designate NAARM as a Think Tank of ICAR would imply that it would serve as a Single-window point for ICAR for formulating all its policy directives in the field of agriculture. Thus, NAARM would function through a hub-and spokes model, by networking with all institutes under the National Agricultural Research and Education System (NARES) in India, so as to serve as the 'eyes and ears of ICAR'.

The farmers are the 'ultimate' stakeholder for ICAR while the policies and programs concerning their lives and livelihoods are framed not only from Ministry of Agriculture and Farmers' Welfare but also from various other Departments and Ministries. Thus, NAARM would strive to provide sustained, general operating support for transnational and interdisciplinary research-based support on a range of policy issues, to the respective agencies through ICAR.

Academy sought inputs from senior professionals, who have either served or are serving the NARES (ICAR & SAUs) for about 3-4 decades on the strategies for this transition. Further, a two-days brainstorming was held among all the faculty members of the Academy, wherein the perspectives of all scientists on the role of NAARM as a Think Tank were solicited. Discussions were also held with the experts of Quinquennial Review Team (QRT) and Research Advisory Committee (RAC) of NAARM. The Cadre Review Committee of NAARM had also dwelt on this domain in its report submitted to ICAR.

3.4 Scoping Workshop on 'NAARM as a Think Tank'

ICAR constituted an Expert Committee with Dr. J.C. Katyal, Former DDG (Education) as Chairman and Dr. P.K. Joshi, Dr. D. Rama Rao and Dr. N.H. Rao, Former Directors of NAARM, Dr. B. Venkateswarlu, Former Vice Chancellor, VNMKV, Parbhani as members, to prepare a roadmap for the transition. The committee held series of consultations to develop the roadmap and internally the Academy, recognizing the need for a cultural change within and outside the Academy to bring-in an organic transition from a capacity building organization to a Think Tank Body of ICAR, has taken every effort to enhance its focus and efforts towards identifying key policy issues and developing policy papers in various areas covering agricultural education, entrepreneurship, markets, etc.

A Scoping Workshop was held at NAARM on Aug 31, 2019 to develop a roadmap for transitioning the Academy to credible Think Tank of ICAR. The workshop was graced by Dr. Trilochan Mohapatra, Secretary, DARE and DG, ICAR, who complimented the Academy and suggested to deliberate on devising institutional framework and mechanism for transforming NAARM as Think Tank.

The experts invited for panel discussion included Dr. P.K. Joshi, Dr. D. Rama Rao, and Dr. N.H. Rao- former Directors of NAARM, Dr. B. Venkateswarlu, former Vice-Chancellor, VNMKV Parbhani. Besides, DDG (Engineering and NRM) Dr. Alugusundaram, and



Dr. Trilochan Mohapatra, Secretary, DARE and Director General, ICAR with Panelists

several distinguished researchers as well as former Directors and Directors of several ICAR institutes, as well as representatives from other organizations viz. MANAGE, NIRD, PJTSAU, ANGRAU, and WALMATRI participated in the deliberations.



DG, ICAR and Members of Expert Committee interacting with invited Delegates

It was recommended in the workshop that NAARM should strive to provide sustained, general operating support for transnational and interdisciplinary research-based support on a range of policy issues, to the respective agencies through ICAR. There is a need for a cultural change within and outside the Academy to bring-in an organic transition of the Academy from a Capacity Building organization to a Think Tank Body of ICAR. Such a strategic shift requires concerted efforts continuously by all the stakeholders. Accordingly, the role of NAARM as a Think Tank would be:

- The core domain for the think tank institute would be visioning i.e., envisioning the future and suggest strategies for the desired future in the wake of dynamism in agriculture and other sectors. ICAR-NAARM, as a think tank should imagine the agricultural sector in the next 20 years, identify technology gaps, prediction of supply and demand for different crops in the face of likely changes, recommendation on probable farming systems and assist ICAR to frame appropriate policies for research and development in the country.
- NAARM, as a Think Tank would be independent enough to provide non-partisan and credible inputs on various issues in research and education

governance and agriculture development. It should strive to focus on highlighting S&T inputs that can bring about positive and significant changes and promote the innovations that inspire competitiveness and effectiveness in farming.

- Providing innovative ideas and inputs for intellectual solutions to several problems and for policy making in the sector given the changes in ecosystem.
- Undertaking and promoting evidence based research for advocating data driven policies.
- The focus of the Think Tank should be more on producing scholarly work and accelerating the pace of innovation cycle in agricultural research.

3.5 Adapting to the New Paradigm

The NAARM is transitioning itself to its new role as Think Tank and engaged in research policy design and development. In the process, NAARM has undertaken research in critical areas of agriculture and brought out policy recommendations in the domain of Climate change, Agri-Entrepreneurship development, Agriculture Education and Faculty competence.

3.5.1 Climate Change and Indian Agriculture: Impacts, Coping Strategies, Programmes and Policy

The ICAR Policy document on “Climate Change Coping Strategies in Indian Agriculture” was published by the Director General, Indian Council of Agricultural Research, Department of Agricultural Research and Education, Government of India, in association with the Ministry of Agriculture and



Farmers Welfare and the Ministry of Environment, Forestry and Climate Change, Government of India. The key objectives of this document are to highlight the impacts of climate change on the economy and livelihood security as driven by the agriculture sector in Indian scenario and to make policy makers understand the possible coping strategies and its co-benefits in agriculture.

This policy paper briefs the causes for climate change and its impacts on Indian agriculture sector; climate change adaptation technologies and co-benefits of mitigation; various adaptation strategies to combat climate change; United Nations framework convention on climate change (UNFCCC) and agriculture sector-based climate change negotiation process; regional climate policies at Asian Pacific Network (APN) and South Asian Association for Regional Cooperation (SAARC) level; central and state government policy programmes for climate change adaptation and the impacts of national programmes and policies in Indian agriculture system. Some important recommendations were brought out to the government policy makers for the future decision making and technology implementations in the light of climate change risk management.

3.5.2 Digital Learning for Vocational and Informal Education in Agriculture

To promote vocational education, skill-building, and formal education in digital mode, a comprehensive framework involving various stakeholders is needed. KVKs and educational institutions under Agricultural universities, Deemed universities and Central universities need to identify their strength areas and identify the suitable Subject Matter Specialist (SMS) to develop the content. The areas



as identified will act as a primer for identifying and prioritizing the subject areas suitable to develop and knowledge partners and agencies to fund/support these activities need to be identified.

Based on the subject area, support fund organizations like the Ministry of Human Resources Development (MHRD), ICAR, Department of Science and Technology (DST), Department of Biotechnology (DBT) can be explored. Based on the subject area, the formation of action groups to develop the content needs to be done. A centralized content boosting platform which facilitates a window to offer these courses/modules has to be taken up at the national level. The digital courses offered through this platform will undergo the rigor of user testing and feedback and the same has to be considered by the concerned SMS/institutions to fine-tune, revise, upgrade and update the content from time to time.

3.5.3 Policy on 'Fostering Entrepreneurship in Agriculture through Incubation Centres in NARES'

Incubation centers play a very critical role in the success of start-ups and young entrepreneurs. A well-structured incubation programme helps start-ups right from idea refinement, imparting management skill, market identification and development, and most crucial funding support. The challenges that the incubator is facing presently in India are finding good start-ups who want to build a product/service to solve problems in the market or the value chain. Normally, the start-ups do not distinguish the value proposition that incubators can provide. Large asymmetry has been created due to the difficulty by start-ups and incubators to rightly price the "value add" at an early stage of the enterprise. Incubator normally charges for the services it provides. The charges can be in the form of fees, equity, or a combination of both. However, the majority of incubator take equity as a start-up does not have money for paying fees. If start-up fails, there is no realization for the incubator in this case.

However, the issue is more complex in terms of fees. Most of the incubates fail in paying the fees and incubators need to condone it. Government-owned incubators have the problem in this model as the receivable increases in their balance sheet.



Keeping in view of non-coherent and diverse approaches being practiced by different incubators in promoting entrepreneurship, there is an urgent need of developing systematic processes and procedures for the functioning of the incubators. The specific recommendations to promote agripreneurship through incubators: Standard Operating Procedures (SOPs): For smooth functioning and successful execution (on boarding, monitoring & evaluation, exit policy, etc.), SOPs should be framed by ICAR/SAUs in consultation with different stakeholders. Building pipeline of agripreneurs: The incubators should conduct programmes like sensitization workshops, technology fairs, ideation competitions, hackathons, accelerators, etc. for scouting and nurturing of potential entrepreneurs. Professional team for Incubators: Incubators need to engage a professional management team with relevant experience in business development, finance and strategic operations etc. for mentoring and managing incubates. Autonomy to Incubators: Incubators need to take several decisions for smooth day to day operations. Therefore, functional autonomy to the incubators should be ensured. Networking with stakeholders: Entrepreneurship requires the support of different stakeholders such as technology developers, mentors, funding agencies, legal and regulatory service provider etc. Incubators should develop and maintain a strong network with stakeholders to provide needed support to the early-stage agripreneurs.

3.5.4 Policy on 'Enhancing Competence of Agricultural Faculty: Towards Piloting Agriculture-led Development'

The research on teaching / research aptitude, academic excellence and teaching competencies lay the foundation for policy advocacy. A study conducted among the students, faculty members, administrators from various state agricultural universities to assess the training needs of faculty members based on 28 pre-identified teaching competencies showed that perceptions towards teaching competencies among the groups differed with their roles. The study indicated that the competencies related to attitude and values need to be accorded highest priority followed by that in teaching strategies / communication skills. The key recommendations are to enhance the intake of agricultural graduates to create a large pool of candidates with the minimum educational qualification, and ICAR shall enhance its fellowships to attract more graduates to higher education. The Universities shall provide for imaginative curriculum and pedagogy; flexibility in under-graduation program with scope for rigorous specialization. It is also recommended to review the PhD Course Curriculum and to offer Research Fellowships and Teaching Assistantships to Doctoral students. ICAR shall offer a certificate program in Agricultural Higher Education as Massive Open Online Course (MOOC) for all agricultural graduates, which may be considered as a desirable qualification for selecting the faculty members for the SAUs. The National Eligibility Test (NET) should be revised to test the candidates' aptitude for teaching and research, besides its current exclusive focus on testing academic and professional proficiencies. A mandatory Foundation Course for New Faculty may be offered for smooth transition of the entry-level faculty into their professional career. It is also recommended to develop guidelines for testing the communication skills of potential candidates while recruitment of new faculty and also to develop capacity building programs during mid-career of the Faculty.

4 Centres of Excellence

The Academy has two Centres of Excellence, viz Centre of Agri- Innovation (CAI) and Centre for Open and Lifelong Learning in Agriculture (COLLAge). The Centre for Agri-Innovation was established for technology management and entrepreneurship with the core activities of incubation and acceleration of startups. The second center, COLLAge was set up for promoting technology enhanced life-long learning in Agri-education.

4.1 Centre for Agri-Innovation (CAI)

The current challenges facing agriculture include not only enhancing productivity and production of agriculture produce but also building opportunities for agricultural goods in the production-consumption continuum. These call for the skill of management of research output towards technologies and up-scaling these to processes and products of commercial value. Hence, a one-stop Centre for Agri-Innovation was set up wherein technology management, entrepreneurship, and ideation, incubation, acceleration and investment with strong foresight are promoted. The initial handholding attempts made by this centre including the ICAR-funded Agri-Business Unit has led towards creating an institutional culture for creating value chains through technology transfer from ICAR institute towards entrepreneurship channels. The Centre provides handholding support to the Agri-Business incubators by inter-alia developing modules, guidelines and other forms as of learning material for capacity building for the startups; and also provides need-based support for analysis/valuation of the potentially commercial technologies; negotiation skills; monitoring of IP- rights and license agreements and marketing

instruments. The main objectives of the Centre are: to organize training & capacity building programs on technology transfer, IP management, business incubation, entrepreneurship development, and investment; to develop foresight and policy framework for governance and coordination of technology development and commercialization in the NARES and incubation development for agribusiness entrepreneurs, and to provide a platform for up-streaming and down-streaming of technologies through advisory services for IP and technology management leading to entrepreneurship development.

The Centre for Agri-Innovation of the Academy functions through the technology business incubators of ICAR-NAIF (National Agriculture Innovation Fund) and Department of Science & Technology (DST), Government of India (GoI). DST has supported technology business incubator a-IDEA (Association for Innovation Development of Entrepreneurship in Agriculture) which also supports activities of BioNest, an Agri- Bio-incubator for Agri-Biotechnology Startups. BioNest has received financial support from BIRAC, DBT, GoI, which runs in partnership with Agri-Biotech Foundation (ABF), Rajendranagar, Hyderabad.

a-IDEA fosters innovation and entrepreneurship in agriculture in India. It has been supporting startups across 14 domains of agriculture and allied sectors such as dairy, fisheries and animal husbandry since 2014. a-IDEA has helped entrepreneurs to sensitize, ideate, incubate and accelerate their innovative early-stage startups that are scalable to become competitive food and agri-business ventures through capacity building, mentoring, networking and advisory support.

These startups are in turn disrupting the value chains of agriculture across farm gate to food plate, by impacting the lives of farmers, rural and urban stakeholders. The a-IDEA is presently focusing on different sub-sectors of agriculture innovations across the value chain in agriculture as depicted in the figure. The a-IDEA has carried out activities not only for the incubation and acceleration

process but also for ecosystem development, and seed investment in agri-startups. During the year 2019, a-IDEA incubated 12 startups, 7 startups have graduated and shortlisted top 10 startups in Agri Udaan 3.0 cohort. It has also invested in one Agri Udaan startups called Delmos and few more in the pipeline

4.1.1 Incubatee Startups-2019

Sl. No	Name of Incubatee	Year of Induction	Industry Sector	Company Profile
1.	Fishy Farmer Pvt. Ltd.	April 2019	Aquaponics	Commercial Production of Exotic Greens and Vegetables Using Aquaponics
2.	AgVerse Technology Pvt. Ltd	April 2019	Animal Husbandry	Artificial Insemination Gun for Early & Easy Heat Detection & Conceiving
3.	Aadhishakti Pvt. Ltd	April 2019	Farm Mechanization	ICT based Custom Hiring Centres for Farm Machinery & Equipment to Farmers
4.	Pekhansa Pvt. Ltd	April 2019	Innovative Food Technology	Pani Puri Vending Machine
5.	Oorvi Pvt. Ltd	April 2019	Waste Management	Floral Waste Management to Value Added Products http://oorvi.org/
6.	ChemLife Innovations Pvt. Ltd	April 2019	Waste Management	Conversion of Paddy Stubble into Value Added Products http://www.chemlifeinnovations.com/
7.	Transity Digital Solutions Pvt. Ltd	January 2018	Supply Chain Technology	SaaS Aggregator Platform for Transportation of Agriculture Produce http://transity.co/
8.	Skycraft Aerospace Pvt. Ltd.	January 2018	Precision/Smart Agricultures	Addresses Commercial and Enterprise Drone Needs http://www.skycrafts.com/
9.	Robic Rufarm India Pvt. Ltd.	July 2018	ICT in Agriculture	Real-Time Pond Monitoring and Advisory Services for Aquaculture https://robicrufarm.com/
10.	Bloom Foods India	August 2018	Innovative Food Technology	Ready to Eat (RTE) Foods with a Special Focus on Traditional Non-Veg Pickles. https://bloomfoods.co.in/
11.	Urbankisaan Hydrotech Pvt. Ltd.	September 2018	Urban/Vertical Farming	Vertical Hydroponic Urban Farming
12.	JS Amey Biotech Pvt. Ltd.	September 2018	Waste Management	Food Waste Management from the Hospitality Industry



4.1.2 Graduated Startups-2019

Sl. No	Name of the Startup	Year of Induction	Industry Sector	Company Profile
1.	Saro Agri and Dairy Pvt. Ltd.	December 2015	Animal Husbandry	The supply chain of A2 Grade Cow Milk http://saronatural.in
2.	Bharat Rohan Innovations Pvt. Ltd.	April 2016	Precision/ Smart Agriculture	Airborne Hyperspectral Imagery, Analytics & Advisory http://www.bharatrohan.in/
3.	Fresco Greens Pvt. Ltd.	September 2016	Urban/Vertical Farming	Rooftop Farming Kits to Grow Pesticide-free Vegetables (Individual & Community) www.homecrop.in
4.	Stamp IT Business Solutions LLP	February 2017	ICT in Agriculture	IT Business Process Automation for Agri Companies www.stampit.biz
5.	Vertical Farming Technologies Pvt. Ltd.	November 2017	Urban/Vertical Farming	Consultancy & Training on Hydroponics for Urban Gardeners https://citygreens.in/
6.	Zoofresh Foods Pvt. Ltd.	November 2017	Fisheries	Transport & Retailing Live Fish using High-Tech Logistics http://zoofreshfoods.com/
7.	Jovaki Agro Foods India Pvt. Ltd.	June 2018	Innovative Food Technology	Value-added Processing of Seasonal Fruits from Tribal Regions https://www.jovaki.com/about/



4.1.3 Agri Udaan 3.0: Food and Agri-business Accelerator

This focused program is targeted towards startups that are trying to scale up their innovative startups. Globally out of 7500 plus incubators that exist, the majority of them focus on early-stage support to the startups through incubation programs. There are very few accelerators, especially when it comes to Accelerators in India focused on Food and Agri-Business. With this background, a-IDEA, when it started its operations in 2015, realized the importance of such programs to support Agritech startups that are facing challenges to scale up. In this endeavor, a-IDEA has launched the Agri Udaan, The Food & Agribusiness Accelerator.

Agri Udaan, Food, and Agribusiness Accelerator is the flagship accelerator program of a-IDEA, NAARM. It brings on board a range of diverse partners to create an inclusive and collaborative ecosystem focusing on catalyzing scale-up stage Food and Agribusiness startups through rigorous mentoring, industry networking and investor pitching. It is a unique platform for scale-up stage innovators, entrepreneurs and startups in Food and Agribusiness sectors to showcase their products/services and to receive valuable inputs from Mentors, Incubators, R&D Institutions, Agribusiness Industry, and Investors.

a-IDEA, NAARM launched India's first Food & Agribusiness Accelerator with CIIE, IIM-A in 2015, in this program out of ~192 startups which have applied, 20 startups were mentored and 8 startups were supported. Out of these 8 startups, 3 startups got funding to the tune of 3 Cr INR. Further in 2017, the 2nd series of Food and Agribusiness Accelerator "Agri Udaan 2.0" was launched with the support of Department of Science and Technology in association with Caspian, Yes Bank, Marico, NeML for Agri Udaan 2.0 nearly 579 startups applied, out of which 40 startups were mentored, 10 startups cohort was shortlisted and supported further, out of which 4 startups got funding to the tune of 6.25 Cr INR. In 2019, this year a-IDEA at NAARM with the support of DST is launched its Agri Udaan, Food and Agribusiness Accelerator 3.0 in association with NABARD, Caspian, Agri-Innovate, Omnivore, and other prominent partners. Agri Udaan 3.0 was launched in Hyderabad on 31st August 2019 and organized its roadshows in Bengaluru, Pune, Patna, Lucknow, Gurugram, Bhopal and Guwahati. Providing a truly Pan-India presence, Agri Udaan seeks to bring out innovations from across the country while also aiming to provide support to the startups across the country.

The following ten cohort start ups have been inducted during November 2019 through Agri-Udaan 3.0 Accelerator Programme.

Sl. No.	Name of the Startup	Industry Sector	Company Profile
1.	Hydrogreens	Animal Husbandry	Micro Climate Control System for Growing Fodder https://hydrogreenfodder.com/
2.	Poultrymon	Animal Husbandry	Smart Monitoring System for Poultry Farms and Hatcheries http://www.mlitsol.com/
3.	Amvicube	Agritech	Non-Destructive Testing Instrument for Paddy http://www.amvicube.com/
4.	SNRAS Systems	Fisheries	Recirculatory Aquaculture System https://www.snrasystems.com/
5.	Auraphyll	Agri Biotech	Bio-stimulants and Elicitors, https://www.auraphyll.com/
6.	Whrrl	Agri Fintech	Blockchain Backed Lending Ecosystem for Warehouse Finance Receipts http://www.whr.loans/
7.	Bariflo Labs	Fisheries	Hypolimnetic Aeration Device for deep Aeration, Feeding, and Screening for Shrimps https://bariflodelabs.com/
8.	Han Agrocure	Supply Chain Technology	Medicinal Mushroom Cultivation https://hanagrocure.com/
9.	Farmer Fresh Zone	Supply Chain Technology	Online Marketplace for Organic, Safe to Eat Vegetables and Fruits https://www.farmersfz.com/
10	Krishi Tantra	Soil, Water and Weather Technology	Soil, Water & Weather Technology Testing https://krishitantra.com/

4.1.4. Events conducted by a-IDEA

The a-IDEA has conducted various programs in different parts of the country so as to reach to many agri- entrepreneurs as possible. The events are listed below:

Sl. No	Name of the Event	Location	Date
1.	Sensitization Workshop	College of Agriculture, Bapatla, Andrapradesh	8 th January 2019
2.	Student Entrepreneurship Development Program	a-IDEA, NAARM	21-25 January 2019
3	STARTUP 360	a-IDEA, NAARM	23 rd January 2019
4.	Startup Entrepreneurs 2019	Hotel, Radisson Blu, Hyderabad	24 th January 2019
5.	Pitch Perfect	Hyderabad	16 th February 2019
6.	Startup Samvaad	ICAR-Vivekananda Parvatiya Krishi Anusandhan Sansthan (VPKAS), Almora, Uttarakhand, India.	25 th March 2019
7.	Startup 360	a-IDEA, NAARM	20 th May 2019
8.	Climate Launchpad Boot Camp	a-IDEA, NAARM	21-22 May 2019

Sl. No	Name of the Event	Location	Date
9.	Mentorship in Artificial Intelligence for Sustainable Organic Farming to Students	a-IDEA, NAARM	23 rd July 2019
10.	Pitch Perfect	a-IDEA, NAARM	26 th July 2019
11.	Launch of AGRI UDAAN® 3.0 Food and Agribusiness Accelerator program	HICC Novotel, HITEC City, Hyderabad.	31 st August 2019
12.	Agri Udaan's First Roadshow	91Springboard, Hosur Road, 7 th Block Koramangala, Bengaluru	6 th September 2019
13.	Agri Udaan's Second Roadshow	Mahratta Chamber of Commerce Industries and Agriculture (MCCIA), Senapati Bapat Road, Pune	13 th September 2019
14.	Agri Udaan's Third Roadshow	Bihar Industries Association(BIA), Sinha Library Road, Chajju Bagh, Patna	16 th September 2019
15.	Agri Udaan's Fourth Roadshow	Jaipuria Institute of Management, Gomti Nagar, Lucknow.	18 th September 2019
16.	Agri Udaan's Fifth Roadshow	91 Spring board, NH8-Udyog Vihar, Sector 18, Gurugram.	20 th September 2019
17.	Agri Udaan's Sixth Roadshow	B-Nest, Bhopal Incubation Centre, Govindpura, Bhopal	23 rd September 2019
18.	Agri Udaan's Seventh Roadshow	North Eastern Development Finance Corporation Ltd (NEDFi), Guwahati	27 th September 2019
19.	Global Bio India Roadshow 2019	a-IDEA, NAARM	11 th October 2019
20.	IGNITE: The Innovative Idea Competition for Students	PJTSAU Auditorium, Hyderabad	21-22 October 2019
21.	Cohort Finalization Program of Agri Udaan 3.0	a-IDEA, NAARM	31 st October & 1 st November 2019
22.	Capacity Building Workshop of Agri Udaan 3.0	a-IDEA, NAARM	11-14 November 2019
23.	BIRAC BIG'S 15th Pre-TEP Program	a-IDEA, NAARM	9 th December 2019
24.	BIRAC BIG's 15 th Technology Evaluation Panel Round	a-IDEA, NAARM	16 -18 December 2019

Glimpses of Events in a-IDEA



Launch of Agri Udaan 3.0 in 2019



Agri Udaan 3.0 website and Portal Launch



Capacity Building Workshop of Agri Udaan 3.0



IGNITE-Students Business Idea Competition



IGNITE- Student Business Idea Competition



Global Bio- India Roadshow

4.1.5 a-IDEA's Participation in Activities

S.No.	Name of the Event	Location	Date
1.	Incubators Manager Training Program	American Corner, Begumpet, Hyderabad	28-29 January 2019
2.	Bio-Economy Conclave 2019	Bangalore	31 st Jan - 1 st Feb 2019
3.	AQUAEX INDIA - 2019	Hyderabad	31 st Jan - 2 nd February 2019
4.	ISBA Managers Conclave	Mysore, Karnataka	1-3 February 2019
5.	Global R&D Summit	Hyderabad	21 st February 2019
6.	Discovery team Visit at a-IDEA office for a Documentary of Urban Kisan for Coverage of Planet Healer Program of Discovery Channel.	Hyderabad	21 st February 2019
7.	Incubators meet at Dlabs	Dlabs, Hyderabad	22 nd February 2019
8.	Showcasing Innovative Startups to Progressive farmers from Tamil Nadu	a-IDEA, Hyderabad	27 th February 2019
9.	BIRAC's BioNest Meet	India Habitat Centre, New Delhi	1 st March 2019
10.	Funding for Agribusiness Startups	Chamber of Commerce, New Delhi	7 th March 2019
11.	"Urban Kissan" being featured in Planet Healer of Discovery	ISB, Hyderabad	8 th March 2019
12.	Agri-tech panel - organized by AWS in Association with Omnivore at T-Hub to discuss on Will Agri-tech Create the next Unicorn	Hyderabad	9 th March 2019
13.	International Conference on Entrepreneurship in Agriculture and Renewable Energy Sector	Akola, Maharashtra	15 th March 2019
14.	7 th Foundation Day of BIRAC India on the theme Nurturing Innovation: Empowering India	Delhi	19-20 March 2019
15.	We-Hub Telangana office Inauguration on World Women's Day	Hyderabad	28 th March 2019
16.	Entrepreneurship Training Program for ANGRAU Students	NAARM, Hyderabad	08 th April 2019
17.	Jury of Diagnostic Panel for Grant Challenge in Agriculture	New Delhi	3 rd April 2019
18.	BioNEST Incubators Meet	University of Hyderabad, Hyderabad	24 th April 2019
19.	Agri Business Startup Awards	The Park Hotel, Hyderabad	27 th April 2019
20.	Nexus Incubator Management Training Program.	American Centre, New Delhi.	29 th April to 3 rd May 2019
21.	MEDIKO India Health Conference	HITEX Exhibition Centre Hyderabad	04 th May 2019

S.No.	Name of the Event	Location	Date
22.	India Fund Fest	Bangalore	9 -10 May 2019
23.	Monthly Incubators meet Telangana	We Hub, Hyderabad	21 st June 2019
24.	Agri-BioNEST of a-IDEA, a BIRAC-BIG Partner for its Biotechnology Ignition Grant Scheme	New Delhi	26 th June 2019
25.	Workshop on Innovations in Agribusiness for Young Entrepreneurs Sponsored by APAARI Bangkok	Taipei, Taiwan	1-5 July 2019
26.	Entrepreneurship Development Program at NRC- Meat	NRC-Meat, Hyderabad	11 th July 2019
27.	Panel discussion on Global Exposure for Home-Grown Startups at the Cafe Mandala Event	CCMB, Hyderabad	15 th July 2019
28.	13 th Annual Conference of ISBA, ISBACON2019	Kottayam, Kerala	17-19 July 2019
29.	BIRAC's 5 th BIG Conclave on Industry Connect organized by IKP	IKP Knowledge Park	25-26 July 2019
30.	Food and Agribusiness Summit	IIT, Kanpur	27 th April 2019
31.	Panel Discussion organized by Lead Business School	ICRISAT, Hyderabad	4 th September 2019
32.	FICCI Round Table	FICCI House, New Delhi	9 th September 2019
33.	Community Slate Accelerator	We Hub, Hyderabad	17 th October 2019
34.	Guest Lecture in Horticulture college on Opportunities in Horti-preneurship	SKLTSHU, Hyderabad	6 th November 2019
35.	Launch of Agri Innovation Centre and Jury for Startups at IKMC conference	IKP Hyderabad	8-9 November 2019
36.	Tie Global Summit 2019	Aero city, New Delhi	14 -15 November 2019
37.	Global Bio India 2019	New Delhi	21-23 November 2019
38.	Sankalp Forum 2019, Organized by Intelle Cap	Mumbai	26-27 November 2019
39.	Nutri Conclave IIMR	HICC, Hyderabad	29-30 November 2019
40.	Panel Discussion on Role of Incubators in IP Management for Startups Organized by STEM	Radisson, Hyderabad	7 th December 2019
41.	Farmer Success Meet Organized by Bharat Rohan Startup of a-IDEA	Barabanki, Lucknow	20 th December 2019
42.	89 th NASI Annual Conference	ICAR-NAARM, Hyderabad	21-23 December 2019

4.1.6 Achievements of Incubatees of a-IDEA

S.No	Forums
1.	Urban Kisaan Featured on The Planet Healers Program of the Discovery India Channel.
2.	Jaya Nabulla, Co-founder, Stamp IT Business Solutions LLP awarded as the best women entrepreneur by Eenadu Vasundhara.
3.	Agricx won MP Krishi Bhushan Award
4.	Chemife Innovations Shortlisted for Global Grand Finale of Climate Launchpad
5.	Oorvi Sustainable won Best Green Startup Award under Eco Ideas by Green India Awards 2019
6.	JsAmey Biotech on a National Portal for his work on Food Waste Management
7.	Bharat Rohan awarded as Best Social Enterprise in Agriculture Sector
8.	Han Agrocare won Action For India's Best Indian Social Enterprise Award 2019.
9.	Stamp IT rated as TOP 25 Tech Startup by Malaysian Global Innovation & Creativity Centre
10.	Skykraft received Best Exhibitor Award at Bengaluru Tech Summit
11.	Urban Kisaan shortlisted for Y Combinator an US-based Globally Renowned Accelerator
12.	AgNext won the TOP 5 Agri-tech companies in Asia award by Sustainable Ag Asia Challenge
13.	a-IDEA Supported Startups Agricx, Agnext, Intello Labs, Jivabhumi, Bharat Rohan featured in India's leading NASSCOM's report entitled 'Agri-tech In India-Emerging Trends in 2019'
14.	a-IDEA Supported Startups Jivabhumi, Stam IT, Flybird, Vise Organic, Swasti Agro, Bharat Rohan, Agmart, Mother AgroBiotech, Delmos, Poultrymon, Agnext, Agricx, Intello labs are featured in Compendium: Agri Startups report by FICCI
15.	Delmos Research awarded as Best Startup in the Food testing category under the FINE initiative organized by FSSAI.
16.	Inner Being Wellness featured in ET NOW's TV Program called Emerging SME – Building India
17.	Whrrl shortlisted as one of India's most Innovative Startup in DCB bank Innovation Carnival
18.	Farmer Fresh Zone Startups received Best Innovator 2018 award from Destination Kerala.
19.	Agnext represented in the Panel of National Conclave on Artificial Intelligence by CII.

Snapshots of Achievements of a-IDEA Incubatees



Best Social Enterprise in Agricultural Sector



Best Exhibitor Award at Bangalore Tech Summit



India's Most Innovative Startup



Best Green Startup Award



Best Innovator 2019 Award



National Conclave on Artificial Intelligence by CII

4.2 Centre of Lifelong Learning in Agricultural Education (COLLAge)

The centre is established as an institutionalized extension of activities of ICAR Education Division funded project on Technology enhanced Learning in Agricultural Education. The centre has wider capability of bringing a digital classroom, virtual class room, digital production centre, high end editing suite and online course management centre. The centre has evolved into a sustainable self-supporting model besides providing value and quality to the digital learning in NARES.

4.2.1 Activities of COLLAge

Digital Content Protocol Development

Developing and evolving digital learning protocols touching the aspects of production, refinement, management and consumption.

Since the technology is ever evolving, COLLAge constantly explores the various technological options available for digital content development and standardizes feasible protocols. High end PTZ camera has been added to the existing setup to enhance the quality of digital content. Besides these the editing hardware in the form of streaming server has been acquired.

Content Development in consultancy mode

Using this state of art COLLAge technology, various assignments have been taken up to develop digital content to NARES Organizations- NIPHM, MANAGE, PJTSAU, NDRI. A total of 74 modules (of 15- 20 minutes' duration) have been developed during 2019, generating revenue to the tune of Rs. 30.75 lakh. This content is being used by the respective organizations for offering online courses.



Digital Content Production on Milk Processing and Value Addition for ICAR-NDRI



Digital Content Development on Rodent and Household Pest Management for NIPHM

MOOCs Programme

Fifth MOOC on Teaching Excellence was offered during November 2019 to benefit teaching faculty to hone their teaching skills. A record high number of 1329 participants, of whom 690 participants completed the course. Provision was given to generate online certificates of completion/ Participation based on fulfilling criteria. The program could generate the revenue of approximately 6 lakh rupees through certification.

Development of Digital Courses

A total of six digital courses were developed and packaged in digital format, to benefit the UG students of Agriculture and Veterinary sciences. These courses can be useful for students in remote areas with poor connectivity.



Digital Course on Entomology



Digital Course on Veterinary Histology

Support of Distance Education Programme

Academy has started offering distance education programme- PG Diploma in Education Technology Management since March, 2019. The entire programme is offered in online mode also through COLLAGe. Entire courseware comprising 7 courses and 105 modules were digitally recorded, edited and published through eLearning portal (elearning.naarm.org.in). A total of 48 students have enrolled and accessed this portal for courseware, participation in quizzes and assignment submissions



Visit of UP Agriculture Minister Sh. Surya Pratap Shahi to COLLAGe Lab

Training and capacity building of scientific personnel in NARES is one of the mandates of the Academy. In this chapter, training programmes attended by the ICAR-NAARM employees, capacity building

programmes organized by the Faculty of NAARM and their participation in conferences, seminars, symposiums, and workshops are given below:

5.1. Training Programmes Attended by ICAR-NAARM Employees

Sl. No.	Name of the Staff	Category	Programme	Duration (in days)	Venue
1	Ch Srinivasa Rao	Scientific	Developing Internal Talent and Leadership	3 days	IIM, Ahmedabad
2	M Krishnan	Scientific	Role of Technology in Community Level Disaster Mitigation for Scientists & Technologists by Department of Science & Technology (DST), Government of India	4 days	Centre for Disaster Management, Lal Bahadur Shastri National Academy of Administration, Mussoorie
3	Manju Gerard	Scientific	Gene Editing in Agriculture: Science, Policy, Story	4 days	Hyderabad/ Cornell -Sathguru Foundation for Development
4	Manju Gerard	Scientific	MDP on Management of Creativity and Innovation	5 days	IIM, Calcutta
5	RVS Rao	Scientific	Executive Development Programme on Developing Effective Organizational Leadership for Senior Officers of ICAR	Domestic: 3 days International: 10 days	Domestic: ASCI, Hyderabad International: Netherlands, Germany, Belgium & Switzerland.
6	D Thammi Raju	Scientific	Executive Development Programme on Developing Effective Organizational Leadership for Senior officers of ICAR	Domestic: 3 days International: 10 days	Domestic: ASCI, Hyderabad International: Netherlands, Germany, Belgium & Switzerland.
7	P Ramesh	Scientific	Managerial Effectiveness	6 days	IIM, Ahmedabad
8	N Srinivas Rao	Scientific	Domestic Component of EDP on Developing Effective Organizational Leadership for Senior Officers of ICAR	Domestic: 3 days International: 10 days	ASCI, Hyderabad
9	N Srinivas Rao	Scientific	Financial Management of NAHEP Project	1 day	NASC, New Delhi
10	BS Yashavanth	Scientific	Training on NAHEP Project Monitoring and Tracking System	1 day	IARI, New Delhi
11	P Mohan Singh	Technical	Enhancing Workplace Efficiency for Technical & Administrative staff	4 days	ASCI, Hyderabad
12	Mercy Arigilla	Admin.	Enhancing Efficiency and Behavioral Skills	6 days	CPCRI, Shimla

5.2 Capacity Building Programs Organized by ICAR-NAARM

Sl. No.	Title of the Programme	Duration	Programme Directors	No. of Participants	Programme Type
1.	109 Foundation Course for Agricultural Research Service	4 Jan- 3 Apr, 2019	S Senthil Vinayagam PC Meena	32	FOCARS
2.	Off Campus Training Programme on Enhancing Efficiency and Behavioral Skills of Stenographers Grade- III, PA, PS PPS and Sr. PPS of ICAR at NBSS & LUP-Regional Station, Kolkata	7- 12 Jan, 2019	WS Bhat PG Kohad	24	Off-Campus Programme
3.	Coromandal International Limited Training Programme Skill Enhancing and Employees Development (SEED) for Retail Sales Managers (Phase - IV)	21- 23 Jan, 2019	KH Rao Ranjit Kumar	24	Need Based Programme
4.	Entrepreneurship Development Programme (Center for Agri-Innovation)	21- 25 Jan, 2019	K Srinivas Ranjit Kumar	25	Need Based Programme
5.	Training Programme on Geospatial Analysis using QGIS & R	1-6 Feb, 2019	PD Sreekanth BS Yashavanth	14	Need Based Programme
6.	Management Development Programme on Information and Communication Technologies for Farm Women	1-6 Feb, 2019	NS Shenoy VKJ Rao	11	Need Based Programme
7.	Off Campus Faculty Development Programme for Increasing Efficiency , Effectiveness in Teaching, Research & Extension at UAHS, Shivamogga	4-8 Feb, 2019	KH Rao BS Sontakki	31	Off-Campus Programme
8.	Hindi Training Programme	5 th Feb 2019	S Pradeep Singh	35	Need Based Programme
9.	Off Campus Training Programme on Emotional Intelligence for Personal and Work Excellence at CIFE, Mumbai	11-13 Feb, 2019	P Ramesh RVS Rao	22	Off-Campus Programme
10.	Students Entrepreneurship Training Programme for Students of ANGRAU	14 th Feb- 6 th Mar, 2019	Ranjit Kumar N Srivaramane	45	Need Based Programme
11.	Coromandal International Limited Training Programme for Retail Sales Managers (Phase V)	18-21 Feb, 2019	KH Rao Ranjit Kumar	53	Need Based Programme
12.	Training Programme on Analysis of Experimental Data	21- 26 Feb, 2019	A Dhandapani S Ravichandran	17	Need Based Programme
13.	Training Programme on KOHA for Library Staff of ICAR	21-26 Feb, 2019	N Srinivasa Rao SK Soam	37	Need Based Programme

Sl. No.	Title of the Programme	Duration	Programme Directors	No. of Participants	Programme Type
14.	Training Programme on Digital Teaching Techniques for the Faculty of University of Agricultural Sciences at College of Agriculture, UAS, Bangalore	26 th Feb-2 nd Mar, 2019	VV Sumath Kumar GRK Murthy	30	Off-Campus Programme
15.	Coromandal International Limited Training Programme for Young Agribusiness Professionals--Batch – I	4-8 Mar, 2019	KH Rao Ranjit Kumar	25	Need Based Programme
16.	Coromandal International Limited Training Programme Skill Enhancing and Employees Development (SEED) for Retail Sales Managers (Phase - V)	6-8 Mar, 2019	KH Rao Ranjit Kumar	28	Need Based Programme
17.	Developing Winning Research Proposals (DWRP) for Faculty of Rajmata Vijayaraje Sindjia, College of Agriculture, Indore	7-9 Mar, 2019	SK Soam P Krishnan	25	Off-Campus Programme
18.	Coromandal International Limited Training Programme for Young Agribusiness Professionals-Batch – II	11-15 Mar, 2019	KH Rao Ranjit Kumar	21	Need Based Programme
19.	Training Programme on Motivation, Positive Thinking and Communication Skills for Technical Officers of ICAR at ICAR-IISWC, Dehradun	13-19 Mar, 2019	BS Yashavanth P Vijender Reddy	24	Off-Campus Programme
20.	MDP for HRD Nodal Officers of ICAR for effective implementation of Training Functions	14-16 Mar, 2019	RVS Rao KH Rao	23	Need Based Programme
21.	Digital Teaching Techniques	14-20 Mar, 2019	VV Sumath Kumar GRK Murthy	25	Need Based Programme
22.	Coromandal International Limited Training Programme for Young Agribusiness Professionals-Batch-III	17-21 Mar, 2019	KH Rao Ranjit Kumar	33	Need Based Programme
23.	21-Days Training on Entrepreneurial Skill Development of ANGRAU Agri-Graduates	20 th Mar-9 th Apr, 2019	Ranjit Kumar K Srinivas	45	Need Based Programme
24.	Training Programme on Developing Winning Research Proposals in Agriculture/ Agricultural Engineering (For Faculty of PJTSAU)	5-7 Apr, 2019	SK Soam P Krishnan Ch Srinivasa Rao	29	Need Based Programme
25.	Training Programme on Developing Winning Research Proposals in Agriculture/ Agricultural Engineering (For Faculty of PJTSAU)	26-28 Apr, 2019	SK Soam P Krishnan Ch Srinivasa Rao	29	Need Based Programme

Sl. No.	Title of the Programme	Duration	Programme Directors	No. of Participants	Programme Type
26.	National Workshop on Academic Excellence through Building Partnerships and Resources Generation (National Agricultural Higher Education Project- NAHEP)	30 th Apr-1 st May, 2019	SK Soam D Thammi Raju N Srinivasa Rao & Alok Kumar	78	Workshop
27.	Motivation and Positive Thinking for Technical Officers of ICAR (T-5 – T-9) (Off-campus at IISWC, Dehuradun)	2-7 May, 2019	Vijender Reddy Laxman M Ahire	16	Off-Campus Programme
28.	Training Programme on Forestry Research and Education Management (For Faculty of Forest College and Research Institute, Hyderabad)	8-17 May, 2019	SK Soam Alok Kumar P Venkatesan	10	Need Based Programme
29.	Training Programme on Science Communication and Scientific Presentation (for Faculty of PJTSAU)	16-18 May, 2019	SK Soam P Krishnan Ch Srinivasa Rao	20	Need Based Programme
30.	Training Programme on ICTs for Livestock Development (for SVVU Faculty)	16-22 May, 2019	GRK Murthy S Senthil Vinayagam VV Sumanth Kumar	28	Need Based Programme
31.	Off campus Workshop on Scientific communication & Development of Winning Research Proposals at Bihar Animal Sciences University, Patna	31 st May-2 nd June, 2019	P Krishnan SK Soam	25	Off-Campus Programme
32.	Spotting Agricultural Research Innovations - Cases from NARES	3-4 June, 2019	K Kareemulla S Ravichandran	9	Need Based Programme
33.	MDP on Leadership Development (a pre-RMP programme)	11 th June, 2019	KH Rao P Ramesh	25	MDP Pre-RMP
34.	Training Programme on Administrative and Finance Management for Section Officers/AAO/AFAOs & JAOs/Assistants of ICAR Headquarters /Institutes (Off Campus at CIFT Kochin)	13- 17 June, 2019	S George WS Bhat	49	Off-Campus Programme
35.	MDP on Business Plan Development and Accelerating FPOs/FPCs	13-18 June, 2019	K Srinivas Ranjit Kumar	22	Need Based Programme
36.	Enhancing Efficiency and Behavioral Skills of Stenographers Grade-III, PA, PS, PPS and Sr. PPS of ICAR (Off-campus at CPRI, Shimla)	20-25 June, 2019	PG Kohad N Raghunath	30	Off-Campus Programme
37.	Training Programme-cum-Exposure Visit on Agriculture & Agri – Market Scenario In India (for NABARD Officers in Grade 'B')	26-28 June, 2019	B Ganesh Kumar Surya Rathore SK Soam	26	Need Based Programme
38.	Training Programme on Stress Management	26-28 June, 2019	P Ramesh A Debnath	21	Need Based Programmes

Sl. No.	Title of the Programme	Duration	Programme Directors	No. of Participants	Programme Type
39.	Training Programme on Hospitality Management (In collaboration with Institute of Hospitality Management, Catering Technology & Applied Nutrition (IHMCT & AN))	26 th June-2 nd July, 2019	Sham Bahadur	21	Need Based Programme
40.	Training Programme for Administrative and Financial Staff of Agricultural University, Jodhpur (Sponsored Programme)	26 th June-5 th July, 2019	S George JLN Das W Sreenivasa Bhat	20	Off-Campus Programme
41.	Off-Campus Training Programme on Advances in Educational Technology organized at College of Veterinary Science, Khanapara, Guwahati in collaboration with Assam Agricultural University (NAARM Sponsored)	4-6 July 2019	D Thammi Raju	34	Off-Campus Programme
42.	Training Programme on Science Communication and Scientific Presentation for Faculty of PJTSAU (Batch-II), sponsored by PJTSAU, Hyderabad	11-13 July, 2019	P Krishnan SK Soam Ch Srinivasa Rao	38	Need Based Programme
43.	Training Programme on Digital Teaching Techniques for PJTSAU Faculty	16-20 July, 2019	VV Sumanth Kumar GR K Murthy S Senthil Vinayagam	31	Need Based Programme
44.	MDP on Priority setting, Monitoring and Evaluation (PME) of Agricultural Research Projects	18- 23 July, 2019	P Krishnan Alok Kumar	34	Need Based Programme
45.	Collaborative Capability Building program for Coromandel Sales Personnel	22-26 July, 2019	KH Rao Ranjit Kumar	23	Need Based Programme
46.	National Workshop on Bioinformatics in Agriculture	26-27 July, 2019	M Balakrishnan SK Soam	21	Need Based Programme
47.	Off campus Training Programme on "Scientific Communication & Developing Winning Research Proposals" (for faculty members of Bihar Animal Sciences University) at Bihar Animal Sciences University, Patna	16-18 Aug, 2019	P Krishnan SK Soam	24	Off-Campus Programmes
48.	Training Programme on Analysis of Experimental Data	22-27 Aug, 2019	S Ravichadran A Dandapani	23	Need Based Programme
49.	Consultative Workshop on Academia Industry Government Organizations for Quality Agricultural Education Under NAHEP	23-24 Aug, 2019	Senthil Vinayagam Surya Rathore	46	Workshops
50.	Scoping Workshop on ICAR-NAARM as a Think Tank of ICAR (A Roadmap for the Transition)	31-Aug, 2019	Ranjit Kumar P Krishnan	110	Workshop

Sl. No.	Title of the Programme	Duration	Programme Directors	No. of Participants	Programme Type
51.	A Training of Trainer's (ToT) workshop on Development of Soft skills for Entrepreneurship among Agri Graduates (Under NAHEP)	17- 18 Sep, 2019	Surya Rathore Alok Kumar N Srinivasa Rao	34	Workshop
52.	Training Programme on Recent Bioinformatics Tools for Genome and Proteome Analysis	17-21 Sep, 2019	M.Balakrishnan N Srinivasa Rao	14	Need Based Programme
53.	II Consultative Workshop on Academia Industry Government Linkages for Quality Agricultural Education (Under NAHEP) at OUAT, Bhubaneswar, Odisha	20-21 Sep, 2019	Sanjiv Kumar VV Sumanth Kumar	52	Off-Campus Programme
54.	Training Workshop on Education Management and Academic Leadership	20-25 Sep, 2019	D Thammi Raju Senthil Vinayagam	27	Workshop
55.	III Consultative Workshop on Academia Industry Government Linkages for Quality Agricultural Education (Under NAHEP) at MPKV, Rahuri, Maharashtra	23-24 Sep, 2019	VV Sumanth Kumar Surya Rathore	58	Off-Campus Programme
56.	MDP on Administrative and Financial Management for Deputy Secretaries, Chief Admn. Officers, Chief Finance & Accounts Officers, SAOs, Under Secretaries & SFAOs (Off- campus at NDRI, Karnal)	24-27 Sep, 2019	C Muralidharan ZH Khilji	24	Off-Campus Programmes
57.	PRA and Communication Strategies for IFS Probationers	26-29 Sep, 2019	SK Soam BS Sontakki P Krishnan	62	Need Based Programme
58.	Training Programme on Intellectual Property Valuation and Technology Management	15-19 Oct, 2019	Manju Gerard K Srinivas	30	Need Based Programme
59.	Training Programme on Entrepreneurship Development for Students of SVVU-Batch I	21-26 Oct, 2019	Ranjit Kumar K Srinivas	34	Need Based Programme
60.	Training Workshop for Vigilance Officers of ICAR Institutes	31 st Oct- 1 st Nov, 2019	Senthil Vinayagam C Muralidharan	32	Workshop
61.	5 th Massive Open Online Course (MOOC) on Teaching Excellence	1-30 Nov, 2019	GRK Murthy Senthil Vinayagam	1322	Online Course
62.	Krishi Kala Ustav - III	4-8 Nov, 2019	M Krishnan Laxman M Ahire Namdev	25	Workshop
63.	Training Programme on Innovative practices in extension research and evaluation	5-8 Nov, 2019	BS Sontakki P Venkatesan	17	Need Based Programme

Sl. No.	Title of the Programme	Duration	Programme Directors	No. of Participants	Programme Type
64.	Consultation Workshop of Academia – Industry – Government (under NAHEP) at UAS, Bangalore	6-7 Nov, 2019	V V Sumanth Kumar	66	Off-Campus Programme
65.	Collaborative Capability Building Program for Young Agri-Business Professionals MORE (Marketing Officers' Resurgence & Excellence)	9-13 Nov, 2019	KH Rao Ranjit Kumar	30	Need Based Programme
66.	ICAR Short Course on Innovative in Education Technology	13-22 Nov, 2019	Senthil Vinayagam GRK Murthy	22	Need Based Programme
67.	International Workshop on Strengthening International Agribusiness Trade at TERI School of Advanced Studies, New Delhi	14-15 Nov, 2019	Surya Rathore SK Soam D Thammi Raju	304	Workshop
68.	40th Training-cum-Workshop on Developing Winning Research Proposals on Digital Solutions in Agriculture	19-23 Nov, 2019	SK Soam P Krishnan M Balakrishnan	28	Need Based Programme
69.	Training Programme on Multivariate Data Analysis Using R	22-27 Nov, 2019	A Dhandapani N Sivaramane	16	Need Based Programme
70.	Consultative Workshop on Academia Industry Government Linkages for Quality Agricultural Education (Under NAHEP) at SVPUAT, Meerut	27-28 Nov, 2019	Surya Rathore Sanjiv Kumar	67	Off-Campus Programme
71.	MDP on Leadership Development (a pre-RMP programme)	2-13 Dec, 2019	Alok Kumar RVS Rao	22	Need Based Programme
72.	Trainers' Development Programme for various cadres of Administrative and Finance Staff of ICAR	4-9 Dec, 2019	SK Soam BS Sontakki	21	Need Based Programme
73.	Training on e-Office File Management System (e-file)	5-7 Dec, 2019	C Muralidharan S Rajukumar	137	Need Based Programme
74.	Training Programme on Motivation Positive Thinking and Communication Skills for Technical Staff (T1 to T4) of ICAR at NIANP, Bangalore	5 - 11 Dec, 2019	MA Basith G Aneeja	19	Off-Campus Programme
75.	Advances in Market Intelligence for Faculty of Veterinary Sciences at Madras Veterinary College, TANUVAS, Chennai	13- 18 Dec, 2019	N Sivaramane Sanjiv Kumar B Ganesh Kumar	25	Off-Campus Programme
76.	Collaborative Capability Building Program for Agronomists	16- 18 Dec, 2019	KH Rao Ranjit Kumar	51	Need Based Programme
Total Number of Participants				3976	

5.3 Participation in Seminar/Symposium/ Conferences/Workshops/Meetings

	Name	Programme	Venue	Period
January 2019	Ch Srinivasa Rao	Meeting with Consultative Group on International Agricultural Research Centres (CGIAR) Organized by DARE, Ministry of Agriculture & Farmers Welfare, Govt. of India.	NASC, New Delhi	Jan 24-25, 2019
	Ch Srinivasa Rao	ICAR Directors' Conference	NASC, New Delhi	Jan 31 – Feb 01, 2019
	Surya Rathore	Meeting on hosting I Regional Agriculture Fair from Feb. 12-14, 2019 on Agriculture as a collective adventure for ensuring doubling of farmers' income theme	PJTSAU, Hyderabad	Jan 8, 2019
	Surya Rathore	State Level Workshop on Skill Development in Agriculture & Allied Sectors	Marriott, Hyderabad	Jan 29, 2019
January 2019	S Ravi Chandran	21 st Annual Conference of the Society of Statistics, Computer and Applications (SSCA)	SV College of Agriculture (ANGRAU), Tirupati	Jan 29-31, 2019.
	A Dhandapani	National Conference on Challenges and Opportunities in Statistics and Informatics for Futuristic Humanosphere in Agriculture (COSIFHA)	S.V. College of Agricultural (ANGRAU), Tirupati	Jan 29 -31, 2019
February 2019	D Thammi Raju	Workshop on Dual Vocational Education and Training Organized by Steinbeis Centre for Technology Transfer India and Telangana Academy for Skill and Knowledge	The Plaza, Hyderabad	Feb 16, 2019
	I Sekar	ICAR-Senior Officers Committee Meeting	KAB, New Delhi	Feb 18, 2019
	Surya Rathore	Workshop on Doubling the Farmers' Income by 2022	ICM, Hyderabad	Feb 15, 2019
	Surya Rathore	International Conference on Advances in Agriculture & Allied Sciences Research, Jointly organized by Samagra Vikas Welfare Society, Lucknow & Rama University, Kanpur	Rama University, Kanpur	Feb 23 – 24, 2019
	P Venkatesan	XIV Agricultural Science Congress Innovations for Agricultural Transformation Organized by ICAR-IARI & NAAS, New Delhi	ICAR-IARI New Delhi	Feb 20-23, 2019
	N Srinivasa Rao	Sixth International Conference of Asian Special Libraries on Libraries and Librarianship in Digital Plus Era	Institute of Economic Growth, New Delhi	Feb 14-16, 2019
	Ch Srinivasa Rao	Executive Council Meeting of NAAS	NAAS, New Delhi	Feb 19, 2019
	Ch Srinivasa Rao	NAAS XIV Agricultural Science Congress: Innovations for Agriculture Transformation.	NASC, New Delhi	Feb 20, 2019

	Name	Programme	Venue	Period
March 2019	Surya Rathore	National Conference on Strategic Approaches for developing World Class Agricultural Universities	NTR College of Agricultural Engineering, Bapatla, (ANGRAU),Guntur	Mar 19-20, 2019
	PD Sreekanth	Geospatial Technologies in Agriculture in Google Earth Engine user Meet	Bangalore	Mar 15-16, 2019
	PD Sreekanth	Interactive meeting cum Workshop of Scientists in IT/ Computer Applications	New Delhi	Mar 06, 2019
April 2019	S Senthil Vinayagam	Role of Science & Technology (S&T) Institutions in Promoting Entrepreneurship and Livelihoods in Rural Areas	NIRD & PR, Hyderabad	Apr 22-23, 2019
	S Senthil Vinayagam	Sensitization Workshop on Entrepreneurship Development through Farmer Led Innovations – A Case Study in Plantation Sector	ICAR-NAARM, Hyderabad	April 27, 2019
	S Senthil Vinayagam	NAHEP National Workshop on Academic Excellence	ICAR-NAARM, Hyderabad	April 30, 2019
	VV Sumanth Kumar	NAHEP National Workshop on Academic Excellence	ICAR-NAARM, Hyderabad	April 30, 2019
	Surya Rathore	National Workshop on Academic Excellence through Building Partnerships and Resource Generation	ICAR- NAARM, Hyderabad	April 30 – May 1, 2019
	PD Sreekanth	Workshop on Academic Excellence through Building Partnerships and Resources Generation under NAHEP	ICAR-NAARM, Hyderabad	April 30 – May 1, 2019
	Ch Srinivasa Rao	Experts Meet on Soil and Land Use Policy under the Chairmanship of President, NAAS	NAAS, New Delhi	April 13, 2019
	Ch Srinivasa Rao	Brainstorming Session on Agrobiodiversity Indices and Ecosystem Services Organized by Bioversity International in collaboration with ICAR	NASC, New Delhi	April 15-16, 2019
May 2019	SK Soam	ICAR – Senior Officers Committee Meeting	KAB, New Delhi ICAR	May 06, 2019
	SK Soam	Meeting of Trainer Development Programme of ICAR	KAB II, Pusa, New Delhi ICAR	May 30, 2019
	Ch Srinivasa Rao	Meeting of the Programme Committee of NAAS	NAAS, New Delhi	May 30, 2019

	Name	Programme	Venue	Period
June 2019				
	S Senthil Vinayagam	Symposium on Climate Change Mitigation in CIRDAP Member Countries	NIRDPR, Hyderabad	June 26, 2019
	D Thammi Raju	30 th Extension Education Advisory Council Meeting	TANUVAS, Chennai	June 26, 2019
	Alok Kumar	Monitoring & Evaluation Sensitization Workshop	NASC, New Delhi	June 28, 2019
	SK Soam	ICAR – Senior Officers Committee Meeting	KAB, New Delhi	June 12, 2019
	SK Soam	Meeting of Training Board of ICAR	KAB, New Delhi	June 18, 2019
	SK Soam	Meeting of ABI Screening Committee	KAB I, New Delhi	June 14, 2019
	A Dhandapani	National Statistical Day Meet	IGKV, Raipur	June 29, 2019
	Ch Srinivasa Rao	26 th General Body Meeting and Foundation Day Programme of National Academy of Agricultural Sciences (NAAS)	NAAS, New Delhi	June 04-05, 2019
	Ch Srinivasa Rao	ICAR Committee Meeting for Screening of Proposals for New Agribusiness Incubation (ABI) Centres in ICAR Institutes	KAB-I, New Delhi	June 14, 2019
July 2019	Manju Gerard	3 rd Agri Biotech India Summit	Inventicon, Hyderabad	July 25-26, 2019
	SK Soam	Consultative Group Meeting on Geographical Indications	IP Office, Kolkatta	July 11, 2019
	SK Soam	Meeting of ABI Screening Committee by ICAR	KAB I, New Delhi,	July 10, 2019
	SK Soam	International Conference on Positive Impact of AI & IoT on Job creation in India with Special focus on Agriculture and Healthcare in Rural areas	Taj Mansingh, New Delhi	July 03, 2019
	Ch Srinivasa Rao	Meeting of the Steering Committee for Establishing New ABIs	KAB I, New Delhi	July 10, 2019
	Ch Srinivasa Rao	9 th Agriculture Leadership Conclave 2019	The Taj Palace, New Delhi	July 11, 2019
	Ch Srinivasa Rao	3 rd Meeting of NASI Council (2019) for 89 th Annual Session of NASI	NASI, Prayagraj, Allahabad	July 31, 2019
August 2019	S Senthil Vinayagam	International Conference on Innovating Pedagogy 2019: Language, Literature and Vocational Educational	Bharathidasan University, Trichi, Tamil Nadu	Aug 17-18, 2019
	VV Sumanth Kumar	1 st Annual Review Meeting under National Higher Agricultural Education Project	NASC, New Delhi.	Aug 5-6, 2019
	D Thammi Raju	ICAR-NAARM as a Think Tank of ICAR: Scoping Workshop	ICAR-NAARM, Hyderabad	Aug 30, 2019
	D Thammi Raju	1 st Annual Review Meeting under National Higher Agricultural Education Project	NASC, New Delhi	Aug 5-6, 2019

	Name	Programme	Venue	Period
August 2019	Alok Kumar	Annual Review Workshop of the National Agricultural Higher Education Project	NASC, New Delhi	Aug 05-06, 2019
	Manju Gerard	Food & Agri Business Accelerator: Agri Udaan 3.0	a-IDEA/NAARM, Hyderabad	Aug 31, 2019
	BS Sontakki	Scoping Workshop on ICAR-NAARM as a Think Tank of ICAR - A Roadmap for the Transition	ICAR-NAARM, Hyderabad	Aug 31, 2019
	Ch Srinivasa Rao	Senior Executive Development Programme on Developing Effective Organizational Leadership for Senior Officers of ICAR (for Batch-I and Batch-II)	ASCI, Hyderabad	Aug 2 & 9, 2019
	Ch Srinivasa Rao	Brainstorming Session on Zero Budget Natural Farming, Myths and Reality	NAAS, New Delhi	Aug 21, 2019
	Ch Srinivasa Rao	Meeting with DG, ICAR to discuss on Job Description of RMPs post	KAB, New Delhi	Aug 23, 2019
	SK Soam	Meeting for Finalization of Work Plan under MoU Between ICAR-ICFRE for Research and Capacity Building Collaboration.	KAB II, New Delhi	Aug 20, 2019
	N Srinivasa Rao	1 st Annual Review Meeting of NAHEP	NASC, New Delhi	Aug 5-6, 2019
	N Srinivasa Rao	Workshop on Academia-Industry-Government linkages for Quality Agricultural Education	ICAR-NAARM, Hyderabad	Aug 23-24, 2019
	I Sekar	Scoping Workshop on ICAR-NAARM as a Think Tank of ICAR - A Roadmap for the Transition	ICAR-NAARM, Hyderabad	Aug 31, 2019
	M Balakrishnan	International Conference on Agriculture and Animal Husbandry	UoH, Hyderabad	Aug 28-30, 2019
	PD Sreekanth	4 th International conference on Agriculture and Animal Husbandry	UoH, Hyderabad	Aug 28-30, 2019
	PD Sreekanth	International Conference on ESRI User Meet	Hyderabad	Aug 20, 2019
	S Ravichandran	Workshop on Fisheries R&D Infrastructure in Andhra Pradesh	Kakinada, Andhra Pradesh	Aug 16, 2019
	S Ravichandran	Workshop on Horticulture R&D Infrastructure in Andhra Pradesh	YSR Horticulture University, Andhra Pradesh	Aug 17, 2019
September 2019	VV Sumanth Kumar	NAHEP - 2 nd Technical Committee Meeting	ICAR-NAARM, Hyderabad	Sep 18, 2019
	S Senthil Vinayagam	NAHEP - 2 nd Technical Committee Meeting	ICAR-NAARM, Hyderabad	Sep 18, 2019
	S Senthil Vinayagam	7 th Joint Advisory Council Meeting of PGDETM	ICAR-NAARM, Hyderabad	Sep 27, 2019
	GRK Murthy	3 rd Consultative workshop on Entrepreneurship Development through Farmer Led Innovations – A Case Study in Plantation Sector	ICAR-CPCRI, Kasargod	Sep 05-06, 2019
	Surya Rathore	NAHEP 2 nd Technical Committee Meeting	ICAR-NAARM, Hyderabad	Sep 18, 2019

	Name	Programme	Venue	Period
September 2019	Ch Srinivasa Rao	39 th Meeting of Central Joint Staff Council (CJSC)	NASC Complex, New Delhi	Sep 4, 2019
	Ch Srinivasa Rao	Meeting of Sectional Committee on Natural Resources Management	NASC, New Delhi	Sep 11-12, 2019
	Ch Srinivasa Rao	109 th Executive Council Meeting of NAAS	NASC, New Delhi	Sep 13, 2019
October 2019	B Ganesh Kumar	Institute Management Committee Meeting at ICAR-NRC on Meat	NRC Meat, Hyderabad	Oct 18, 2019
	I Sekar	Brainstorming Meet on City Knowledge and Innovation Cluster at Hyderabad	The Trident, Hyderabad	Oct 18, 2019
	D Thammi Raju	Brainstorming Session on Budget Utilization under NAHEP Component 1C – Innovation Grants	KAB, New Delhi	Oct 10, 2019
	GRK Murthy	XIII Review Meeting of Niche Area Projects	KAB, New Delhi	Oct 24, 2019
	GRK Murthy	Expert Panel Committee Meeting for Selection of RA and JRF	NIPHM, Hyderabad	Oct 21, 2019
	Manju Gerard	Regional Expert Consultation Meeting on Gene Editing in Agriculture and its Regulation	ICRISAT, Hyderabad	Oct 10-11, 19
	Manju Gerard	NAIF Screening Committee Meeting	ICAR-NAARM, Hyderabad	Oct 19, 2019
November 2019	Surya Rathore	Stakeholder Meeting on Economic Analysis of Eucalyptus and Subabul based Agroforestry Ecosystem in Telangana and Andhra Pradesh	ICAR-NAARM, Hyderabad	Oct 22, 2019
	SK Soam	Meeting for ICAR ABIs under National Agriculture Innovation Fund (NAIF)	ICAR-NAARM, Hyderabad	Oct 19, 2019
	M Balakrishnan	Hyderabad City Cluster Meeting	The Trident, Hyderabad	Oct 18, 2019
	Ch Srinivasa Rao	Workshop and Review Meeting of National Agriculture Innovation Fund (NAIF) Scheme	NASC, New Delhi	Oct 4-5, 2019
	Ch Srinivasa Rao	First meeting of Regional Advisory Group (RAG) on Farms, Farmers and Rural areas for Telangana State	NABARD Regional Office, Hyderabad	Oct 15, 2019
	Ch Srinivasa Rao	Brainstorming Meet on City Knowledge and Innovation Cluster at Hyderabad	The Trident, Hyderabad	Oct 18, 2019
	Ch Srinivasa Rao	Briefing Meeting for UNFCCC COP-25 in Chile under the Chairmanship of Joint Secretary (Climate Change), MoEFCC.	Indira Paryavaran Bhavan, New Delhi	Oct 24, 2019
	Ranjit Kumar, Sanjiv Kumar	79 th Annual Conference of Indian Society of Agricultural Economics	Indira Gandhi Krishi Vishwavidyalaya, Raipur	Nov 21- 23, 2019
	D Thammi Raju	Review Meeting of NAHEP Project (Component 2)	KAB I, New Delhi.	Nov 27, 2019
	Alok Kumar	International workshop on Strengthening International Agribusiness Trade: Stakeholder Dialogue and Partnerships towards SDGs	TERI School of Advanced Studies, New Delhi	Nov 14-15, 2019

	Name	Programme	Venue	Period
November 2019	Alok Kumar	ISEE National Seminar on Socio-Digital Approaches for Transforming Indian Agriculture	CCSHAU, Hisar	Nov 20-22, 2019
	Alok Kumar	National level Brainstorming Workshop on NAHEP Results Framework Organized by National Agricultural Higher Education Project (NAHEP)	IARI, New Delhi	Nov 11, 2019
	Manju Gerard	National Conference on Integrative Plant Biochemistry and Biotechnology	IIRR, Hyderabad	Nov 8-9, 2019
	Manju Gerard	BioNest Review Meet	BIRAC, New Delhi	Nov 20, 2019
	Manju Gerard	Global BioIndia Road Show Meet	DBT-BIRAC, New Delhi	Nov 21-23, 2019
	Surya Rathore	International Workshop on Strengthening International Agribusiness Trade: Stakeholder Dialogue and Partnership towards SDGs	TERI School of Advanced Studies, New Delhi	Nov 14-15, 2019
	SK Soam	Consultative Group Meeting on Geographical Indications for four GI applications	Sampada Bhawan, New Delhi	Nov 22, 2019
	SK Soam	International Workshop on Strengthening International Agribusiness Trade: Stakeholder Dialogue and Partnership towards SDGs	TERI School of Advanced studies, New Delhi	Nov 14-15, 2019
	SK Soam	Global Bio-India Roadshow – 2019	ICAR-NAARM, Hyderabad	Nov 11, 2019
	SK Soam	Regional Workshop on Youth as Torchbearers of Business Oriented Agriculture in South India	PJTSAU, Hyderabad	Nov 21-22, 2019
	SK Soam	External Expert Committee Meeting conducted by MANAGE, Hyderabad for Grant of Advancement under MARs.	MANAGE, Hyderabad	Nov 02, 2019
	M Balakrishnan	Seminar on Artificial Intelligence and Block Chain for Efficient Public Service Delivery	The Taj Vivanta Hyderabad	Nov 19, 2019
	M Balakrishnan	International Workshop on Strengthening International Agribusiness Trade Stake Holder Dialogue and Partner towards SDGs	TERI School of Advanced Studies, New Delhi	Nov 14-15, 2019
	BS Yashavanth	International Conference on Agricultural Statistics	The Ashoka, New Delhi	Nov 18-21, 2019
	PD Sreekanth	International Conference on Agricultural Statistics	The Ashoka, New Delhi	Nov 18-21, 2019
	Ch Srinivasa Rao	Briefing Meeting for UNFCCC COP-25 in Chile under the Chairmanship of Joint Secretary (Climate Change), MoEFCC.	Indira Paryavaran Bhavan, New Delhi	Nov 01, 2019.
	Ch Srinivasa Rao	International Workshop on Strengthening Internal Agribusiness Trade: Stakeholder Dialogue and Partnerships towards SDGs under NAHEP.	TERI School of Advanced Studies, New Delhi	Nov 14-15, 2019
	Ch Srinivasa Rao	110 th Meeting of the Executive Council of National Academy of Agricultural Sciences	NAAS, New Delhi	Nov 27, 2019

	Name	Programme	Venue	Period
December 2019	S Senthil Vinayagam	Meeting on Digital Content Modules (Collaborative projects of NAARM & NDRI)	NDRI, Bengaluru	Dec 03, 2019
	S Senthil Vinayagam	Discussion meet cum Mentoring Workshop for AAS projects.	CSIR-CIMAP, Lucknow	Dec 05, 2019
	GRK Murthy	Sensitization Meeting with NDRI, Bengaluru Officials regarding Digital Content Modules	NDRI, Bengaluru	Dec 03, 2019
	GRK Murthy	4 th Consultative Workshop on Entrepreneurship Development through Farmer Led Innovations – A Case Study in Plantation Sector	IIPM, Bengaluru	Dec 02, 2019
	GRK Murthy	Sensitization Meeting with NDRI, Karnal Officials on Digital Content Modules (Collaborative Projects of NAARM & NDRI)	NDRI, Karnal	Dec 12, 2019
	Alok Kumar	89 th Annual Session of NASI and Symposium on Science & Technology based Entrepreneurship Development	ICAR-NAARM, Hyderabad	Dec 21-23, 2019
	Manju Gerard	IP Powering the 5 Trillion Economy, STEM	Hyderabad	Dec 06-07, 2019
	Manju Gerard	XIV Triennial National Conference of Indian Women Scientist Association	Hyderabad	Dec 11-13, 2019
	Manju Gerard	89 th Annual Session of National Academy of Science, India	ICAR-NAARM, Hyderabad	Dec 21-23, 2019
	BS Sontakki	International Conference on Extension for Strengthening Agricultural Research and Development	ICAR-JSS KVK, Suttur, Mysuru	Dec 14-15, 2019
	I Sekar	89 th Annual Session of NASI and Symposium on Science & Technology based Entrepreneurship Development	ICAR-NAARM, Hyderabad	Dec 21-23, 2019
	SK Soam	27 th Global Symposium dedicated to the World Project Management Forum (WPMF).	The Le Meridien, New Delhi	Dec 10-12, 2019
	N Srinivasa Rao	International Conference on Geo Smart India	HICC, Hyderabad	Dec 03-05, 2019
	S Ravichandran	Workshop on Fisheries R&D Infrastructure in Telangana	Department of Fisheries, Telangana	Dec 10, 2019
	S Ravichandran	89 th Annual Session of NASI & Symposium on Science and Technology based Entrepreneurship Development	ICAR-NAARM, Hyderabad	Dec 21-23, 2019
	BS Yashavanth	89 th Annual Session of NASI & Symposium on Science and Technology based Entrepreneurship Development	ICAR-NAARM, Hyderabad	Dec 21-23, 2019
	A Dhandapani	4 th KRISHI Nodal Officer's Workshop	NASC, New Delhi	Dec 10-11, 2019

6 Happenings at NAARM

6.1 Special Events

6.1.1 Expert Committee to Review Scientific Cadre Positions of NAARM

The National Committee to review the cadre strength of ICAR chaired by Secretary, DARE & Director General, ICAR, in its recommendations pertaining to the scientific cadre positions for ICAR-NAARM, envisaged that the Academy shall strive to evolve as a Think tank for ICAR in the field of agriculture, in addition to its role in building the capacity of individuals and institutions under the NARES. The Council constituted an expert committee to review and determine the cadre positions, disciplines and roles, as suitable for the Academy.

The Committee was chaired by Dr. B. Venkateswarlu, Former VC, VNMKV, Parbhani and the members were Dr. RRB. Singh, Director, NDRI, Karnal; Dr. RN. Chatterjee, Director, PDP, Hyderabad; Dr. Suresh Pal, Director, NIAP, Delhi; Dr. Himanshu Pathak, Director, NRRI, Cuttack; Dr. OP. Yadav, Director, CAZRI, Jodhpur. Dr. P. Krishnan, Principal Scientist, NAARM served as the Member Secretary.



Expert Committee Meeting to Review Scientific Cadre Positions at ICAR-NAARM

6.1.2 Interaction Meeting to Review Scientific Cadre Positions at NAARM

The Committee constituted for reviewing Scientific Cadre positions at NAARM conducted a detailed review of existing cadre positions of

the Academy and interacted with the scientists of the Academy and the Senior Officials at the Council / ASRB. The first meeting was held in New Delhi, following which the Committee met twice at NAARM, Hyderabad on 29 Jan 2019 and 18 Feb 2019. The draft report was presented to Dr. Trilochan Mohapatra, Secretary, DARE & Director General, ICAR on 3rd March 2019 in a meeting held in the Committee Room of DG, ICAR. The meeting was attended by Shri. Sushil Kumar, Additional Secretary (DARE) & Secretary, ICAR; Dr. B. Venkateswarlu, Former VC, VNMKV & Chairman of the Committee; Shri. Sujit K. Mitra, Director (Personnel), ICAR; Dr. P.K. Ghosh, National Coordinator, NAHEP Unit, Education Division, ICAR; Dr. A. Arunachalam, Principal Scientific Officer to Secretary (DARE) & DG (ICAR) and Dr. P. Krishnan, Member Secretary of the Committee. The Committee has finalized its report on the Scientific Cadre Positions for NAARM including a road map to aid in the transformation of the Academy into a Think Tank of ICAR.

6.1.3 The 89th Annual Session of NASI & Symposium

The “89th Annual Session of National Academy of Sciences, India (NASI)-cum-Symposium on Science and Technology based Entrepreneurship Development” was organized at Academy during Dec 21-23, 2019. The three-day event was jointly organized by the National Academy of Sciences, India and ICAR-NAARM, Hyderabad. The prestigious event witnessed gathering of many science stalwarts of the country namely Dr. Trilochan Mohapatra, Secretary, DARE and DG, ICAR; Prof. (MRs.) Manju Sharma, Former Secretary, Department of Biotechnology & Former President, NASI; Prof. G. Padmanabhan, President NASI and Former Director, IISc-Bengaluru; Dr.

Balram Bhargava, DG, ICMR and Secretary, DHR; Dr. Ram Gopal Rao, Director, IIT-Delhi and many others.

Dr. Mohapatra underlined the three revolutions namely - Green Revolution, White Revolution and Pulse Revolution that have been possible with the efforts of the scientists and farmers of the country. He also highlighted about the benefits reaped in terms of production gains, self-reliance, greater cut in imports and exports of surplus, etc. He emphasized on the need for linking the science, technology and innovation with the entrepreneurship for not only retaining youth in agriculture, but also to create employment by making the entire value chain in agriculture, a profitable proposition. While recounting the current incubators across the country being supported by the ICAR, DST, DBT and other agencies, the Director General urged for a paradigm shift towards the science and technology led entrepreneurship that is evidence based.



Dr. Trilochan Mohapatra, Secretary, DARE and DG, ICAR delivering the inaugural address at 89th Annual Session of NASI & Symposium, ICAR-NAARM

The Director General along with the other dignitaries inaugurated the exhibition showcasing the products and services of the Research Institutes, Start-Ups and other

companies. Prof. (Mrs.) Manju Sharma, Former Secretary, Department of Biotechnology & Former President, NASI highlighted about the activities being undertaken by NASI for the betterment of science in particular and for the benefit of society at large. Prof. G. Padmanabhan, President, NASI emphasized on the importance of research in basic sciences. Dr. A.V. Rama Rao, Chairman & MD, AVRA Labs, Hyderabad was awarded the P.C. Ray Memorial Award for his outstanding contributions in the field of science. Dr. Anil Kakodkar, Former Chairman of the Atomic Energy Commission of India and Former Secretary, Government of India chaired the panel discussions on Science, Technology and Entrepreneurship. He accentuated on the critical need of entrepreneurship for taking forward the science led technology developments. Earlier, Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM welcomed the dignitaries.

An event of mainstreaming the Biodiversity-UNDP was also organized during the occasion. The occasion also marked the release of a Souvenir and a Book on "Farm Innovations". The Young Scientists of biological and physical sciences were also felicitated with the NASI-Springer



Prof. (Mrs.) Manju Sharma, Former Secretary, Department of Biotechnology & Former President, NASI with other dignitaries at 89th Annual Session of NASI & Symposium, ICAR-NAARM

Young Scientist Awards and NASI-Swarna Jayanti Puraskar Awards during the occasion. The distinguished NASI Fellows, Scientists including of eminence personalities - Padma Vibhushan, Padma Bhushan, and Padma Shri Award Winners and students participated in the symposium.

6.1.4 The 248th Governing Body Meeting of ICAR

The 248th Governing Body Meeting of Indian Council of Agricultural Research was organized at the Academy on Nov 30, 2019. Dr. Trilochan Mohapatra, Secretary DARE and Director General ICAR chaired the Governing Body meeting. Dr. AK Singh, DDG (Agricultural Extension); Dr. RC Agrawal, DDG (Agricultural Education); Dr. AK Singh, DDG (Horticultural Science); Dr. Joykrushna Jena, DDG (Fisheries Science & Animal Science); Dr. K. Alagusundaram, DDG (Agricultural Engineering); Dr Ch Srinivasa Rao, Director, ICAR-NAARM along with senior dignitaries of ICAR marked their presence during the meeting.



Dr. Trilochan Mohapatra, Secretary DARE and Director General, ICAR with GB Members at ICAR-NAARM

6.1.5 Gallery Inauguration at Academy

Dr. Trilochan Mohapatra, Secretary (DARE) & Director General (ICAR) inaugurated the Gallery at the Academy on 31st August, 2019 in the presence of Dr. Ch Srinivasa Rao, Director, and Dr. S K Soam, Joint Director (I/c), NAARM. The Gallery was established keeping in view of visits of large number of students, faculty, scientist and officials of NARES across the country. The Secretary, DARE appreciated the efforts for showcasing

different activities of the Academy for the benefit of various stakeholders. Salient information displayed included genesis, history and transition of the Academy, organogram, focal themes of NAARM etc. Special focus was given on Centres of Excellence viz. a-IDEA, Technology Business Incubator and Centre for Open and Lifelong Learning in Agriculture (COLLAGe) to portray the activities of Academy's contribution to science and agricultural education. For the benefit of students, the information about academic programmes offered by the Academy is also provided. The visiting students are made to take a pledge for the welfare of the farming community; emphasizing promotion of good agricultural practices and water conservation, enhancing quality education, ensuring environmental safety, growing more trees, avoiding food wastage, reducing carbon footprint etc.



Dr. Trilochan Mohapatra, Secretary (DARE) & Director General (ICAR) Inaugurated NAARM Gallery

6.1.6 NAAS Hyderabad Regional Chapter Meeting

Meeting of Fellows of NAAS Hyderabad Regional Chapter was held on April 4, 2019 at NAARM. The Meeting was chaired by Dr Panjab Singh, President, NAAS and presided over by Dr Anil K Singh, Secretary, NAAS. In all 22 Fellows and Four Associates of NAAS belonging to Telangana and Andhra Pradesh participated in the meeting. Dr Ch. Srinivasa Rao, Convener made a brief

presentation on activities of NAAS Hyderabad chapter and calls for greater support from NAAS for strengthening its activities. Dr Panjab Singh felicitated Dr P. Appa Rao for being the recipient of K.C. Mehta Memorial Award and Dr Raman Meenakshi Sundaram for being elected as Fellow of NAAS. Padmashri Prof. E. A. Siddiq gave a talk on “Way Forward for Genomics in Agriculture: Policy Needs”. Dr A.K. Singh, Secretary, NAAS made a presentation on “Drought proofing policy in India.



Dr. Panjab Singh, Former Secretary, DARE & DG, ICAR & President, NAAS in NAAS Hyderabad Regional Chapter Meeting at ICAR-NAARM

6.1.7 NASI Hyderabad Chapter Meeting

The NASI Hyderabad Chapter meeting was held at ICAR-NAARM, Hyderabad on April 5, 2019 as a preparatory meeting for organizing 89th Annual Session of NASI and Symposium on Science & Technology based Entrepreneurship Development. It would be jointly organized by NASI and NAARM during 21-23 December 2019 at ICAR-NAARM, Hyderabad. The meeting was chaired by Dr. Ch Mohan Rao, Chairman, Hyderabad chapter of NASI. Dr Ch. Srinivasa Rao, Director, ICAR-NAARM made a presentation on the Academy’s activities and infrastructure



NASI Hyderabad Chapter Meeting at ICAR-NAARM

facilities available at the Academy for holding the NASI Annual Session and Symposium on Science & Technology based Entrepreneurship Development.

6.1.8 International Workshop ‘Strengthening International Agribusiness Trade’

TERI School of Advanced Studies in partnership with Western Sydney University, National Academy of Agricultural Research Management (NAARM) and National Agricultural Higher Education Project (NAHEP) organized a two-day international workshop on Strengthening International Agribusiness Trade in collaboration with Global Centre for Land-Based Innovation during Nov 14-15, 2019 in New Delhi. Prof. Ramesh Chand, member, NITI Aayog was the Chief Guest and Shri Ashok Kumar Dalwai (IAS), CEO of National Rainfed Area Authority was the Guest of Honour during inaugural session. There were 7 thematic sessions in workshop namely, increasing farm productivity through resource efficient techniques and practices; fostering academia-industry-government partnerships for excellence in higher agricultural education; community institutions in agri-value chains; climate change, food & nutritional security; building sustainable agri-food supply chains; doubling farmers income: technologies, business models and sustainable livelihoods; and innovations in agricultural marketing and social entrepreneurship. Around 300 participants participated in the event.



Participants at International Workshop on Strengthening International Agribusiness Trade in New Delhi

6.1.9 Sensitization Programme on e-Office File Management System

A sensitization programme on 'e-Office File Management System (e-file)' was organized at the Academy in collaboration with e-Office Support Team of ICAR-IASRI, New Delhi during Dec 5-7, 2019. The academy invited all master trainers, local administrators and nodal officers from all ICAR institutes situated at Hyderabad and IOPR, Pedavegi (Andhra Pradesh). A total of 83 master trainers and local admins from 10 institutes participated on 1st day of training. Dr. YS Murthy, Senior Technical Director, National Informatics Centre, Telangana State Centre, Hyderabad was the Chief Guest of inaugural event. On 2nd day, regular users of Administration and Finance of ICAR-NAARM were trained. On 3rd day all scientists and technical staff of ICAR-NAARM have been trained. The academy also organized a follow-up technical session on 9 Dec 9, 2019 for institutes who are facing some issues in creating the EMD (Employee Master Data). A total 126 participants have been trained during 3 days' period.

6.1.10 Quinquennial Review Team (QRT) Meeting

The Director-General, Indian Council of Agricultural Research (ICAR) constituted the Quinquennial Review Team (QRT) under the chairmanship of Dr. Mangla Rai, Former Secretary DARE & Director General, ICAR to review the work done by the National Academy of Agricultural Research Management (NAARM) for the period 2012-17. Several rounds of meetings were held during 2018 with different Stakeholders including retired ICAR Directors/ Scientists, VCs/Deans of SAUs, Scientists, and Officers of NAARM followed by two meetings in 2019. During the year 2019, the committee interacted with Senior Officials from the SMD, DDG (Edn.), ADG (Edn.), Director,

NAARM, Director, NIAP, Director, IASRI, Joint Director and Head, PME, NAARM on January 08, 2019, at NASC, New Delhi. The QRT report was presented to DDG (Education) on July 04, 2019, then it was also presented before the Director-General, Indian Council of Agricultural Research (ICAR) in the presence of other dignitaries of ICAR on October 01, 2019. The major recommendations of the QRT in reference of ICAR-NAARM are the Think Tank role of NAARM, strengthening the research focus, bringing uniqueness to Academic Programs, capacity building on MOOC, innovation in Capacity Building Programs, mandatory Mid-Career Trainings, functional autonomy to NAARM, linkages and collaboration and strengthening the infrastructure.



QRT Interface meeting at ICAR Headquarters on January 08, 2019

6.1.11 The 7th Research Advisory Committee (RAC) meeting

The third meeting of 7th Research Advisory Committee (RAC) of NAARM was conducted on Oct 28, 2019 under the Chairmanship of Dr. PK Joshi, Former Director for South Asia, International Food Policy Research Institute (IFPRI). Dr. Ch Srinivasa Rao, Director, NAARM welcomed and introduced the Chairman and members. He also presented an overview of the Academy. The RAC reviewed the research and capacity building progress, and suggestions were given for further improvements to attain global visibility. The RAC members viz. Dr. KC Bansal, Dr. C Shambu Prasad, Dr. Ishwar Murthy, Dr. RK Samanta, Shri V Venkateshwarlu and HoDs

as co-opted members and Incharges as special invitees from NAARM attended the meeting. Dr. SK Soam, Member Secretary presented action taken report.



7th Research Advisory Committee (RAC) Members at ICAR-NAARM

6.1.12 Institute Research Council Meeting

The 22nd IRC (Institute Research Council) meeting of the Academy was organized during May 28-29, 2019 under the Chairmanship of Dr. Ch. Srinivasa Rao, Director, NAARM. Dr. P. Krishnan, Head, PME and Member Secretary, IRC gave an overview of the projects under operation in the Academy. The PIs of, 36 ongoing and completed projects under 6 divisions, presented the summary of achievements made in their respective projects during September 2018 to March 2019, which included 4 projects from HRM division, 6 from ABM division, 7 from ICM division, 3 from XSM division, and 8 each from RSM and ESM divisions. Three new projects namely Economic Analysis of Eucalyptus and Subabul based Agroforestry Ecosystem in Telangana and Andhra Pradesh, Development of Market Intelligence System through Big Data Analytics and Agri Startups in India: Decoding the Determinants and Success Factors were also proposed in the meeting.



Institute Research Council Meeting at ICAR-NAARM

6.1.13 Academic Committee Meeting at Academy

Academic committee, a body of the faculty members meet at the Academy once in a month. In this forum, issues pertaining to the capacity building, research and other academic programmes are discussed and relevant decisions are arrived. This committee is chaired by the Director and the member Secretary, Dr.K.Kareemulla. During the year 2019, a total of twelve meetings were held. Some of the important agenda items taken were periodical review of publications (research papers & policy papers), status of impact studies, regular review of capacity building programmes with progress of previous month and planned programs for the ensuing month, status and progress of PGDMA, PGTMA, and PGETMA programmes. Similarly, the preparations for the special meetings like IRC, RAC and QRT were also discussed in the meeting. The plan for starting a new professional society by the faculty of NAARM i.e. AMARA was also discussed and its registration was completed. Elaborate preparations for the annual conference of National Academy of Science, India (NASI) which was conducted during December 2019 were also discussed in the Academic Committee meeting.



Academic Committee Meeting Chaired by Director, NAARM

6.1.14 Senior Officers' Committee Meeting at Academy

The senior officers' committee headed by Director, NAARM consists of Heads of Divisions, Joint Director, JD (Admin.), CFAO, and Section in-charges as members. The committee meets

once in a month and in 2019, there were twelve meetings held. Some of the agendas taken for discussion in the meeting include infrastructure development and maintenance, budget utilization under capital and revenue heads, resource generation, procurement of items, research and operation expenditure, conduct of academic programmes and special events, swatchhta activities. The Chairman of SOC meeting is Director, NAARM and member secretary is Joint Director (Administration).



Senior Officers Committee Meeting Chaired by Director, NAARM

6.1.15 BIRAC BIG's 15th Technology Evaluation Panel Round

Agri BioNEST of NAARM a-IDEA organized a Technical Evaluation Panel (TEP) round for BIRAC's BIG applicants during Dec 16-18, 2019. The TEP program was organized to select the innovators who applied for Biotechnology Ignition Grant Scheme of grants in aid up to ₹ 50 lakhs. The TEP is the second round of evaluation in the process to select the BIG innovators. NAARM TBI aIDEA has been inducted as a partner from 15th BIG call and there were total 31 applicants in this round, out of which 17 were selected for TEP round which is still continuing in other venues of BIG partners. In the TEP entrepreneurs and startups working on innovative Idea/POC/Prototype in Agri, Allied and Agribiotech sectors presented their case in front of the panel consisting of researchers, academicians, industry experts, investors, BIRAC team and representatives of other BIG partners.



BIRAC BIG's 15th Technology Evaluation Panel

6.1.16 Vigilance Awareness Week

The Academy observed 'Vigilance Awareness Week 2019' during Oct 28-Nov 2, 2019 on the theme "Integrity – A way of life" with great zeal and enthusiasm. During this week several activities viz. display of banners at strategic locations for creation of awareness for public, administration of integrity pledge, organization of different competitions (Debate, essay writing, drawing) for staff and students of the Academy were organized. Dr. T.K.K. Naidu, Professor and Head, Department of Forensic Medicine, Prathima Institute Medical Sciences and Vice Principal, Medico Legal Consultant, Karimnagar was invited as a Chief Guest for the valedictory function.

The ICAR- National Academy of Agricultural Research Management (NAARM) organized two days Training Workshop for Vigilance Officers of ICAR Institutes as a part of Vigilance Awareness Week – 2019. The training workshop has benefited 32 Vigilance Officers of various institutes of ICAR, conducted during 31st October, 2019 to 1st November, 2019. The participants were exposed to different vigilance procedures to be followed and preventive vigilance. The participants and the staff of the academy were sensitized on various aspects of vigilance i.e., duties and responsibilities of vigilance officers, disciplinary procedures, recruitment procedures, conduct rules, purchase and work procedures etc. Dr. B. Maria Kumar IPS (Retd.), Former DGP, Madhya Pradesh was the Chief Guest for the valedictory function.



Observance of Pledge during Vigilance Awareness Week

6.1.17 Pension Adalat and Meetings ICAR Pensioners at NAARM

As per Govt of India instructions, the Pension Authorization unit, ICAR-NAARM organized, several meetings along with Pensioners / Family Pensioners to resolve their grievances related to pension on January 22, January 28, April 27, and August 23, 2019. The meeting was held under the Chairmanship of Director, ICAR-NAARM. It was informed that the medical facilities may be extended to all the pensioners on par with the employees and well-furnished air-conditioned rooms and comfort to all the pensioners who are visiting to NAARM health centre. This year, Health Cards for Pensioners was also distributed to pensioners for their betterment.



Pensioners of ICAR at NAARM

6.1.18 Krishi Kala Utsav

A five-days Krishi Kala Utsav (KKU) was conducted Academy during Nov 4-8, 2019. Twenty-four students of the famous Sir JJ School of Arts, Mumbai were invited to create paintings and works of art on agriculture and allied sector. Forty-eight paintings including a stone craft

mural work were completed during the said days. ICAR-NAARM, Hyderabad has hosted this Krishi Kala Utsav for the third year in association NAAS-Hyderabad Chapter. Dr. T Mohapatra, Secretary DARE and Director General, ICAR also visited the KKU workshop hall as the artists were at work on Nov 8, 2019 and appreciated the initiative taken by NAARM and the talent of the artistes. He interacted with artists on their innovative presentation on various important aspects of agriculture and farmer's welfare. The artists performed their work to display modern technologies and virtues of traditional agriculture practices, environmental aspects, youth in agriculture, rural markets, soil and water conservation, importance of trees, integrated farming systems, human health from healthy foods, mechanized agriculture systems etc. The programme was inaugurated by Dr. Ch Srinivasa Rao, Director, NAARM and the valedictory function was graced by Dr. AK Pal, Joint Director (Retired) of ICAR-CIFE, Mumbai. Dr. Laxman Ahire and Shri P. Namdev, CTO's led by Dr. M. Krishnan, Head, ESM Division organised the event.



Canvas Painting by a Student of Sir JJ School of Arts, Mumbai

6.1.19 Entrepreneurship Development Programme for Student Entrepreneurs

The Academy organised an entrepreneurship development programme for the students of

agriculture and allied discipline during January 21-25, 2019. The programme was aimed at shaping the students' business ideas having a potential to scale-up in agribusiness sector. Besides, it also envisaged to scout for those students who have presented his/her ideas at different forums. With support of Caspian Impact Investment (CII), a-IDEA, took this initiative to look for college students at early stage who have some interest in entrepreneurship. Total 25 participants with different educational background from 7 states attended the programme. The 5-days programme had mix of pedagogy with sessions by domain experts, business game, activity, field visits to CCMB & T-Hub and interactions with successful startups.

6.1.20 Workshop on 'Lexis Nexis Total Patent One' and 'Lexis Nexis Patent Optimizer'

A workshop on 'LexisNexis Total Patent One-a Patent search tool' and 'LexisNexis Patent Optimizer-a patent drafting and analyzing tool' was organized on Feb 20, 2019 under ZTMU project for faculty and project staff of the Academy. Ms. G Tharraniiswari, resource person from LexisNexis India, described key features of these tools and provided basic training. Need for patent search and analysis for research purpose using Total Patent One tool was highlighted. Practical demonstration on patent drafting and analyzing using Patent Optimizer tool was carried out.

6.1.21 Sensitization workshop on Entrepreneurship Development for Students of BAU, Sabour, Bihar

Two days sensitization workshop on entrepreneurship for agriculture and allied students of Bihar Agricultural University was

conducted during May 15-16, 2019 at Dr Kalam College of Agriculture, BAU, Kishanganj. More than 150 students and faculty from all the five campuses (Sabour, Kishanganj, Noorsarai, Dumraon, Saharasa) under the university participated in the workshop. The workshop was inaugurated by Dr. AK Singh, Hon'ble Vice Chancellor of the University in the presence of Dr. RK Sohane, Director (Extension). These students were exposed to exciting opportunities emerging under startup ecosystem. Sessions included- scouting business ideas, business plan preparation, exploring funding sources, startup management, personal effectiveness through goal orientation, etc. Two young entrepreneurs from Bihar- Ms. Himani Mishra and Mr. Santosh Kumar shared their personal experiences and motivated the students to think entrepreneurship as an alternate career option. Eight student teams presented their business ideas, out of which two were selected for future hand-holding by a-IDEA.

6.2 Visit of Dignitaries at NAARM

6.2.1 Shri. Narendra Singh Tomar, Hon'ble Minister for Agriculture, Farmers Welfare and Rural Development, GoI Interacts with IFS Probationers of IGNFA, Dehradun

Shri. Narendra Singh Tomar, Hon'ble Minister for Agriculture, Farmers Welfare and Rural Development, GoI appealed to the IFS probationers to be compassionate with the rural, tribal and farming communities while upholding the core principles of environmental governance and sustainable utilization of the resources. Hon'ble Minister opined that agriculture and rural development have very close linkages with the environment, and complemented NAARM for designing this module for the IFS officers, which would have great significance in their field service.

The training program was coordinated by Dr. SK. Soam, Dr. Bharat S. Sontakki and Dr. P. Krishnan from NAARM. The participants overwhelmingly complimented the faculty of NAARM for their technical inputs, which would help substantially in their professional career.



Shri. Narendra Singh Tomar, Hon'ble Minister for Agriculture, Farmers Welfare and Rural Development, Gol, interacting with IFS Probationers (2018-20 Batch) from IGNS, Dehradun

6.2.2 Shri Gajendra Singh Shekhawat, then Hon'ble Union Minister of State for Agriculture and Farmers Welfare, Gol Interacts with Farmers, Scientists and other Stakeholders

Shri Gajendra Singh Shekhawat, Union Minister of State for Agriculture & Farmers' Welfare, had an interactive meeting at the Academy, with the farmers, farmer associations, scientists of all the ICAR Institutes located at Hyderabad on March 09, 2019. During the session, Shri Shekhawat accentuated that the government's priority is for



Shri Gajendra Singh Shekhawat, Hon'ble Union Minister of State for Agriculture and Farmers Welfare, Gol addressing Farmers, Scientists and other Stakeholders at NAARM

making the agriculture profitable by reducing the input cost and increasing the remunerative prices for enhancing the farmers' income. The Minister pointed out that, listening to the different stakeholders will help in enhancing the government's functioning and decision making process. The Minister also released a NAARM Hindi publication titled "AgRIM" during the occasion.

6.2.3 Shri Surya Pratap Shahi, Agriculture Minister, Uttar Pradesh visits ICAR-NAARM

Honorable Minister of Agriculture, Government of Uttar Pradesh Sri Surya Pratap Shahi visited ICAR-NAARM along with delegation members consisting Dr. S. Solomon, Vice Chancellor, Chandra Sekhar Azad University of Agriculture and Technology, Kanpur along with senior officials of Govt of UP visited NAARM on 29th Aug, 2019. Sri Surya Pratap Shahi in his interactions presented the overall objective of the visit is to come out with action plan for agriculture development in Uttar Pradesh in terms of research, quality teaching, training of stakeholders for enhanced capacity. He appreciated the critical role of NAARM in country agriculture development in terms of human resource management and research policies and suggested that NAARM support in developing training modules for senior agriculture officers/managers and Farmer Producer Organization of Uttar Pradesh. He also suggested to have national workshop inviting Agriculture Ministers of different states for developing collective road map for agriculture sector in India for achieving the Honorable Prime Ministers Mission of "Doubling the Farmers Income in India by 2022". Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM presented the activities of the academy in terms of research, education, training and research policies covering various sectors of agriculture in India. Dr. SK. Soam, JD In-Charge proposed vote

of thanks. Later the delegation visited various laboratory facilities including the TELAgE lab.



Sri Surya Pratap Shahi, Hon'ble Minister of Agriculture, Government of Uttar Pradesh interacting with Faculty at NAARM

6.2.4 Secretary (DARE) & Director General (ICAR) addresses the Scientist-Probationers (109th FOCARS)

Dr. Trilochan Mohapatra, Secretary (DARE) & Director General (ICAR) while addressing scientists- probationers, who were undergoing the 109th Foundation Course for Agricultural Research Service (FOCARS) at the Academy on Jan 18, 2019 said that 'Science Communication is important for agricultural scientists'. He called upon the young scientists to come up with innovative ideas to undertake research on prioritization for the benefit of the stakeholders. He also shared the importance of new frontier areas like artificial intelligence, block chain technology and IoT in agricultural research in the areas of precision agriculture and post-harvest management. Dr. Ch Srinivas Rao, Director emphasized the importance



Secretary (DARE) & Director General (ICAR) addressing the Scientist-Probationers (109th FOCARS)

of positive thinking and outlined the profile of the FOCARS 109th batch which comprises 32 trainees from 16 disciplines in the Agricultural Research Service (ARS) from various states in the country.

6.2.5 Secretary, DBT visits a-IDEA of ICAR-NAARM

Dr. Renu Swarup, Secretary, Department of Biotechnology, Ministry of Science & Technology, Gol and Chairperson, BIRAC visited a-IDEA on Feb 21, 2019. During her visit, she interacted with few start-ups who are being supported by a-IDEA in different theme areas of agri-value chain. She also visited a-IDEA partner, Agri Biotech Foundation (ABF) and inspected the web lab facilities and work of Start-ups. Director, NAARM who participated in the discussion, briefed that the Academy is a unique institute engaged in capacity building of NARES scientists, Research Policy (think-tank) of ICAR and Agri-business management education. Dr. K Srinivas, and CEO, a-IDEA briefed about the activities and progress of Incubator and Agri-BioNEST.



Visit of Dr. Renu Swarup, Secretary, DBT to a-IDEA at NAARM

6.3 Celebrations

6.3.1 NAARM Celebrates 44th Foundation Day

The 44th Foundation Day of ICAR-NAARM was celebrated on Sep 01, 2019. Dr. Trilochan Mohapatra, Secretary, DARE and Director

General, ICAR, Ministry of Agriculture and Farmers' Welfare, Government of India was the Chief Guest. He delivered the foundation day lecture and felicitated 8 innovative farmers from different states and retired faculty of the Academy. Dr. Yadlapalli Venkateswara Rao, Padma Shri Awardee-2019, and Editor, Rythunestham (popular Agricultural Telugu Monthly Magazine) was the Guest of Honor. On this occasion DG, ICAR, released a publication entitled "Inspire, Innovate and Change: Field Innovations of Farmer FIRST Project" and "Online Grievance Redressal System". Dr. T. Mohapatra presented awards to the winners of NAARM Awards for their outstanding contribution under different categories. Many dignitaries like the Former Directors of NAARM, serving and former Vice Chancellors of Agricultural Universities, Directors of ICAR

institutes in Hyderabad and those from ICAR-CTRI and ICAR-IIOPR, Heads and representatives of other agricultural institutions in Hyderabad graced the celebrations. The farmers who received awards were Sh. Deepak Mali of Madhya Pradesh, Sh. Shyam Dutt of Haryana, Sh. Chandi Ram of Haryana, Sh. Hakam Singh of Punjab, Smt. Savita Vaibhav Nalkar of Maharashtra, Smt. Shweta Maurya of Uttar Pradesh, Sh. Chhabbu Lal Paikara of Chhattisgarh, Sh. A. Jambulingam of Tamil Nadu and Sh. S Hari Babu of Telangana for their notable contributions in sustainable and innovative agricultural practices.



Dr. Trilochan Mohapatra, Secretary, DARE & Director General, ICAR Delivering Lecture on 44th Foundation Day of ICAR-NAARM



Dr. Trilochan Mohapatra, Secretary, DARE & Director General, ICAR Presenting Best Scientist Award-NAARM to Dr. P. Krishnan



Dr. Trilochan Mohapatra, Secretary, DARE & Director General, ICAR Presenting Best Faculty Award-NAARM to Dr. G R K Murthy



A Glimpse of the Audience view during NAARM Foundation Day Celebrations

6.3.2 ICAR-NAARM Organizes National Science Day

The Academy celebrated National Science Day on Feb 28, 2019 commemorating great contributions of noble laureate Sir CV Raman. The theme was:

'Science for the people and the people for the Science'. More than 100 students from Zilla Parishad High School, Shivarampally and Zilla Parishad High School, Budvel participated in the celebrations. Essay and elocution competitions were conducted for school students. Dr. Ch Srinivasa Rao, Director, ICAR-NAARM while addressing the students, said "Science has made significant strides through various inventions for ease of living. The young generation should use technology wisely for the benefit, and should not get addicted to modern electronic gadgets". Dr. SK Soam, Jt. Director (I/c) highlighted the scientific development in the field of agriculture.



Director, NAARM distributing prizes on National Science Day at NAARM

6.3.3 National Farmers' Day

The Academy celebrated National Farmers' Day (Kisan Diwas) on Dec 23, 2019, with the theme "Igniting Young Minds by Innovative Farmers". The day was celebrated by inviting two innovative farmer entrepreneurs – Shri Bachu Veera Reddy and Shri Hari Babu to motivate and inspire younger generations on the prospects of 'new-generation farming' as a career option among over 100 students of Bharatiya Vidya Bhavan's Vidhyashram, NIRD Campus, Rajendranagar, Hyderabad and P. Obul Reddy Public School, Jubilee Hills, Hyderabad. Both the farmers emphatically stressed on the need to infuse young blood in farming sector, well supported by their rich and satisfying experience as innovative

farmer and entrepreneur. These two innovative farmer entrepreneurs have won several awards at national, regional and state levels in recognition of their contributions to promoting eco-friendly farming for healthy living. Dr. Ch Srinivasa Rao, Director of the Academy felicitated the two innovative farmers and addressed the students on this occasion.



An innovative farmer felicitated by Director, ICAR-NAARM

6.3.4 NAARM Celebrates Independence Day

The Academy celebrated the 73rd Independence Day on 15th August, 2019. Dr. Ch. Srinivasa Rao, Director, NAARM hoisted the National Flag and addressed scientists and staff of the Academy. He highlighted the achievements made at the Academy during the year and lauded the efforts put forth by the staff of the Academy. He reiterated to focus on the mandated activities of the Academy in the face of challenges in agricultural sector on the whole and NARES in particular to take the Academy to greater heights in the coming years. Winners of various competitions received prizes from Director, NAARM during the event.



Flag hoisting ceremony on 73rd Independence day at NARRM

6.3.5 The 150th Birth Anniversary of Mahatma Gandhi and Swachhta Hi Seva

The 150th Birth Anniversary of Mahatma Gandhi along with *Swachhta Hi Seva* was celebrated from 11th Sep to 2nd October, 2019 at ICAR-NAARM. As part of these celebrations, many competitions and various activities of *Swachhta* were held for school children, staff and students of the academy. Some of the competitions include Debate Competition for PGDMA students on the theme 'Relevance of Gandhian Philosophy in 21st century'. Painting competition was held for the students of KRISHI High School. An Essay Competition in English on the topic, 'Relevance of Mahatma Gandhiji's Philosophy in the Present World' and an essay competition in Hindi on the topic "भारत में एकल उपयोग वाले प्लास्टिक से छुटकारा" (Get rid of single-use plastic in India) were held on 28th Sep and 1st Oct, 2019 respectively. A Quiz competition for faculty, administrative, technical staff, PGDMA students, SRF's and Young Professionals was held on 30th Sep, 2019. A film on Gandhi was shown to 80 students of 6th class to 9th class from Adarsh High School, Rajendranagar, Hyderabad. An awareness campaign was taken up in the campus on single use plastic ban by students and staff of the academy.



Dr. Trilochan Mohapatra, Secretary, DARE & Director General, ICAR Participating in Swachhta Hi Seva at NAARM



Glimpses of competitions conducted during celebration of 150th Birth Anniversary of Mahatma Gandhiji

6.3.6 Swachhta Pakhwada-19 Programme at NAARM

Swachta Pakhwada-19 programme was observed at ICAR-NAARM from 16th to 31st December 2019. During this period, swachta activities were taken up daily in the farm. Swachhata Pledge was taken up as a part of Swachta Pakhwada-19. A total of 100 students from Bhartiya Vidya Bhawan, Rajendranagar, 5.3.6 Swachhta Pakhwada-19 programme at NAARM Swachta Pakhwada-19 programme was observed at ICAR-NAARM from 16th to 31st December 2019. During this period, swachta activities were taken up daily in the farm. Swachhata Pledge was taken up as a part of Swachta Pakhwada-19. A total of 100 students from Bhartiya Vidya Bhawan, Rajendranagar, Hyderabad and P Obul Reddy Public School of Jubilee Hills Hyderabad participated on 23rd December'19. Faculty and staff also took the pledge. Various competitions were conducted as a part of the programme. Students, staff and faculty members were participated in competitions like elocution, singing and essay writing. The theme for Essay Writing Competition was 'My Contribution to Swachh Bharat Abhiyan' and for the elocution Competition, the theme was 'Plastic Waste Awareness and Management' and for the Singing Competition, the theme was on 'Any of the Mahatma Gandhi's Quotes

on 'Cleanliness'. Certificates were distributed to the winners on the day of the closing ceremony. During the programme, Swachhata boards were kept in all the buildings to create awareness among visitors.



Certificate distribution to the winners during Swachhata Programme



Swachhata pledge by Staff in the Academy

6.3.7 Performance of Padma Shri Dr. Nartaki Nataraj for SPICMACAY

Padma Shri Dr. Nartaki Nataraj gave a scintillating performance at the Academy for SPICMACAY on Feb 11, 2019. The programme was the first of its kind rendered by Dr. Nartaki after receiving the Padmasri Award from the Govt. of India on 26th Jan, 2019. She started the recital with *Sollukettu*, followed up with the piece, *Bo Shambo Shiva Shambo*, in Rag, Revathi. Her delectable movements, fleeting feet and her emotive eye expressions kept the audience rooted and spellbound. The programme was organised and conducted by the Sanskriti Club of PGDMA students of ICAR-NAARM along with

students from PJTSAU, PVNR TVU, MANAGE and from other neighbouring institutions located at Rajendranagar.



SPICMACAY Artists at Academy

6.3.8 International Women's Day Celebrations

International Women's Day was celebrated at the Academy on Mar 8, 2019. The whole day celebration revolved around the theme given by United Nations, 'Think equal, build smart, innovate for change'. Different activities were organized for women by the women cell of the Academy, which included visual quiz, games on women of eminence and expert lecture by Prof. Pari Plavi on 'Balance for Better'. She spoke on how women can balance their dual and triple roles & responsibilities with minimum stress. On this occasion, Dr. Ch. Srinivasa Rao, Director applauded the women's role as managers at home and work and released a book on 'Women in Agriculture (The Invisible Partners in Development)'. Around 75 women including scientific, technical and administrative staff participated in the programme. As a part of charity work to the underprivileged, members of women cell of the Academy distributed soap, hair oil and clothes to 120 inmates of "Aaramghar: A home for destitute".

6.4 Sports, Yoga and Meditation

6.4.1 The 5th International Yoga Day celebrated at ICAR-NAARM

The Academy celebrated the 5th 'International Day of Yoga' on June 21, 2019. About 250 participants including students from local high school, PGDMA students and trainees of Academy, employees of ICAR institutes in Hyderabad and

campus residents took part in the celebration. The yoga practice session was performed from 6:00 to 7:00 AM at the lake view lawn in front of the Auditorium as per revised Common Yoga Protocol (CYP) circulated by Govt of India. Dr. SK Soam, Joint Director (i/c) addressed the audience on the theme "Usage of Eco-friendly Products in the Practice of Yoga" being promoted for this year. Dr. A Debnath, Academy's Medical Consultant & Yoga Instructor conducted the session.



5th International Day of Yoga organized at ICAR-NAARM

6.4.2 ICAR-NAARM participated in Inter-Zonal and Inter Institutional Sports Tournaments

ICAR Inter Zonal Sports Tournament was held at IVRI, Izatnagar during February 25-28, 2019. Mr. Sham Bahadur and Mr. MK Samson participated in the event and Mr. Samson got prizes in Shotput and Discuss throw. NAARM Contingent participated in ICAR Inter Institutional Sports Tournament organized at CIFT, Kochi during 4-8 November 2019. The inauguration of the tournament started on 4th November 2019 at Maharaja Stadium and the chief guest was Sh. Suhas, IAS, Ernakulam District Collector. NAARM Contingent bagged many prizes. Shri MK Samson bagged first prize in Shotput and Discuss throw and runner-up in Carrom. Mrs. Rukmini got first prize in Shotput and Discuss throw and emerged as a winner in Carrom and runner-up in Table Tennis (Singles).



ICAR-NAARM Sports Contingent at ICAR-Inter-Institutional Sports Tournament

6.5 Campus Beautification of the Academy

Development and maintenance of the campus with further beautification was undertaken through several activities such as Lawns development, Renovation of existing Lawns through Re-grassing, Footpath areas development, Avenue Plantation and Pruning activities.

6.5.1 New Lawns Development

New lawns were developed in 2019 and nearly 7500 Sq.mt of area of new lawns were developed. New lawns were developed in new locations,



New Lawns Development at NAARM

which include, Children Park, near the campus lake and Buddha Statue, Tennis court and West gate (NIRD gate), in-front of main building Sports Complex, near Bank and Press buildings, back sides of Indoor Stadium building, International Guest House IGH 2 and also Shade Grass Lawns in-front of Faculty Centre and either side of the road towards D type quarters.

6.5.2 Lawns with Re-grassing

Lawns were renovated with re-grassing in an area of 15800 sq.mt at several locations including Faculty Building Lawns and the Lawns between IGH1 & 2 with Pop-up irrigation, IGH-1 Left side Lawns (Yoga Practice Lawn), Auditorium left and right side Lawns, Football ground lawns at Sports Complex and also Renovation of other Lawn at Sports Complex.

6.5.3 Development of Footpath areas with Flowering/Shade Plants

During the year, many footpath areas were developed in the campus with flowering/shade plants like Bougainvillea/ Pisonia Alba/ Tecoma Fireball/ Galfinia/ Hamelia/ Neerium/ Spider Lilly/ Almonda Creepers/ Eranthemum/ Dracena Mahatma etc., Under this, a total of 7000 Sqmt. of area was developed including the places like: both sides of road from Rose Garden to Security point of Guest Houses, From Bus stop to Agronomy Well and then from Agronomy Well to T Junction (up to Sports complex) and then to West Gate (NIRD Gate) and also from Security check post to road towards D type quarters.



Foot path areas

6.5.4 Cement Pots with Flowering Plants

As a part of beautification in the campus, around 300 Cement pots with flowering plants were kept at various locations including Auditorium, Conference Hall, Director's office, Administrative Building, Sports complex, Health Centre, Faculty Building, Faculty Centre, ABI building and also near all the Guest Houses.

6.5.5 Development of Round Brick Basins near Tree Trunks

Number of Round brick basins were developed near tree trunks and later filled with soil and flowering plants were grown. This was done at various locations including areas surrounding guest houses, Administrative Building, Faculty Building, Faculty Centre, Road from ABI building to Hostel and also near Health Centre, Sports Complex main building and Indoor stadium.

6.5.6 Hanging Pots with Shade Plants

Hanging pots with shade plants were kept near Lecture Halls, in-front of Conference Hall area, Auditorium vicinity and also near Walk-way from Conference Hall to Auditorium.

6.5.7 Avenue Plantation in the Campus

Avenue plantations were planted near guest houses, residential quarters, near office buildings, auditorum etc., including Coconut Plantations near Agronomy well, Tecoma, Neerium and Ixora plantation near guest houses, Temple Trees near Children park/water storage point, Custard Apple near E type Residential Quarters, Conocarpus plantations near T Junction and at Sports Complex/Auditorium etc. Planting of Ujoniya plants was taken up beside Faculty centre of the Academy. Planting of trees as a part of Tree Planting Programme was done at the Academy on 15th August 2019 by involving PGDMA Students. The theme of the programme was 'One Student One tree'.

6.5.8 Pruning of the Trees

Pruning was done in the campus in order to remove the dead branches, irregular and branches and to shape the plants. During the period many number of trees at several locations in the campus were pruned including Road side trees from ABI building to the areas surrounding Office Buildings, Guest Houses and Residential quarters and on both sides of roads from Bus stop to T Junction (Sports Complex) and from T Junction to West Gate and those located at sports complex.

6.5.9 Swachhta Activities

Number of activities were taken up under Swachh Bharat Abhiyan Programme like intensive cleaning at several locations including Cleaning of the road side areas of the office, office surroundings, in and surrounding areas of Sports Complex, Indoor stadium, Farm Office, Residential Quarters and also areas surrounding all Guest Houses. The orchard plot bunds and fence areas were cleaned. Weeding in the basins and cleaning the plots by tillage machines were taken up. Subabul and the weeds like Parthenium were removed from cultivated and non-cultivated plots of the Academy. Parthenium Awareness Week was observed in the campus during 16-22 August 2019.



Parthenium Cleaning drive programme during 16-22 August 2019

6.5.10 Recognition for Landscaping

In recognition of the several activities taken up for beautification of NAARM campus, Department of Horticulture, Government of Telangana awarded ICAR-NAARM with Best Landscaping Institute Award (under Central Government Institutions Category) and also Best Gardens Maintained Award under Educational Institute Category during Garden Festival-2019, Hyderabad.



Recognition for Landscaping

6.6 Glimpses of Beautification and Swachhta Activities at Academy



6.7 Visits of Distinguished Guest Faculty

Guest Faculty	Title of Lecture	Guest Faculty	Title of Lecture
Dr. Trilochan Mohapatra, Secretary (DARE) & Director General (ICAR) New Delhi	Dream to Destination	Dr. Suman P.S. Khanuja, Former Director, CIMAP (CSIR), Lucknow	Agri-Entrepreneurship towards Enhanced Farmers' Income
Dr. J.C. Katyal, Former DDG (Education), ICAR, New Delhi	Pursuing Scientific Research as a Career Option	Dr. B. Venkateswarlu, Former Vice Chancellor, VNMKV & Former Director, CRIDA, Hyderabad	Drought Proofing in India
Dr. Narendra Singh Rathore, Deputy Director General (Education), ICAR, New Delhi	Research Priorities and Challenges in Agricultural Higher Education	Dr. P.B. Kirti, Professor, Department of Plant Sciences, University of Hyderabad	Advances in Biotechnology in Agriculture Systems
Dr. A.K. Singh, Deputy Director General (Agricultural Extension) ICAR, New Delhi	Recent Developments in Agricultural Extension	Mr. Kanhaiya Chaudhary, Director (Admin), ICAR, Krishi Bhavan, New Delhi	ICAR Administration
Brahma Kumaris, Hyderabad	Academy for a Better World	Mr. S.K. Mitra, Director (Personal), ICAR, Krishi Bhavan, New Delhi	Reforms by Personnel Division of ICAR
Dr. T.K. Adhya, Director, South Asia Nitrogen Centre, New Delhi	Mitigation of Green House Gas Emissions in Rice based Eco-Systems in India	Prof. Panjab Singh, President, National Academy of Agricultural Sciences, New Delhi	Perspectives in Agricultural Research and Development
Dr. H. Pathak, Director, ICAR-National Rice Research Institute, Cuttack	Climate Change and Indian Agriculture	Dr. Shiv Prasad Kimothi, ADG (TC), Krishi Bhavan, New Delhi	National and International Collaboration of ICAR
Dr. Bandla Gangaiah, Head, Division of Natural Resource Management, CIARI, Port Blair	Research & Developmental Priorities for Island Ecosystem in India	Dr. A. Arunachalam, Principal Scientific Officer to Secretary (DARE), Krishi Bhavan, New Delhi	Agro forestry Policy in India: Status and Implementation
Dr. P.K. Joshi, Director South Asia, IFPRI, New Delhi	Indian Agriculture: Sustainable Development Goals	Mr. G.P. Sharma, Director, Finance, ICAR, KB, New Delhi	Budget Planning and Implementation
Dr. S. Senthil Kumar, Additional Professor, Indira Gandhi National Forest Academy, Dehradun	Bureaucracy and Science Interface and Conflict Management	Dr. P.V.K. Sasidhar Professor, IGNOU, New Delhi	Evaluation Tools and Techniques as relevant to Extension Research
Dr. S.K. Chaudhary, ADG (S&W), ICAR, Krishi Bhavan, New Delhi	Developments in Soil & Water Management Research	Mr. S. Suman Senior Manager, Dr Reddy's Foundation, Hyderabad	Innovative Extension Research Model for Large Scale Impact
Dr. A.K. Srivastava, Member, ASRB, Krishi Anusandhan Bhavan, New Delhi	Food and Nutritional Security Challenges and Opportunities in India	Dr. N. Reddy, Former Regional Director, CBWE, Ministry of Labour & Employment, New Delhi	Leadership lessons from Epics

6.8 Visits of Students/Trainees/Farmers from SAUs and Other Institutes

S.No	Details of College and University	Degree	Date of visit	No. of Students/ Farmers*
1	College of Forestry, Dr. Yashwanth Singh Permar University of Horticulture and Forestry, Nauni, Solan, HP	B.Sc. (Hons. Forestry) 3 rd year	January 1, 2019	39
2	College of Forestry, Dr. Yashwanth Singh Permar University of Horticulture and Forestry, Nauni, Solan, HP	B.Sc. (Hons) 3 rd year	January 3, 2019	41
3	BN College of Agriculture, Biswanath Chairali, Assam	B.Sc. (Ag) 3 rd year	January 9, 2019	36
4	College of Forestry, Dr. Yashwanth Singh Permar University of Horticulture and Forestry, Neri, Hamirpur, HP	B.Sc. Hons (Forestry) & (Biotechnology)	January 16, 2019	55
5	ATMA, Odisha	--	January 18, 2019	33*
6	Agricultural College, PJTSAU Aswaraopet, Khammam, Telangana	B.Sc. (Hons) 3 rd year	January 28, 2019	64
7	KRC College of Horticulture, Arabhavi, Karnataka	B.Sc. (Hort.) 4 th year	January 30, 2019	55
8	Department of Agricultural Extension, College of Agriculture, GKV, UAS, Bengaluru	M.Sc. (Ag) & Ph. D	February 7, 2019	19
9	Sri Siddeshwar FPO Company Ltd, Belagur, Karnataka	--	February 11, 2019	55*
10	College of Horticulture, University of Horticultural Sciences, Hosalingapur, Munirabad, Koppal, Karnataka	B.Sc. (Hort.) 4 th year	February 16, 2019	51
11	College of Horticulture, Bagalkot University of Horticultural Sciences	B.Sc. (Hort.) 4 th year	February 16, 2019	51
12	MGM College of Agricultural Bio-Technology, Gandheli, Aurangabad	B.Sc. (Agri. Biotec.) 4 th year	February 21, 2019	46
13	District Watershed Development Agency, Salem & Dindigul	--	February 21, 2019	85*
14	College of Home Science, ANGRAU, Guntur, AP	B.Sc. (Home Science)	March 11, 2019	90
15	SRM Institute of Science and Technology, Tamil Nadu	B.Sc. (Hons) 1 st year	October, 2019	240
16	National Institute of Rural Development and Panchayati Raj, Hyderabad	Stake holders/ Agro entrepreneurs	May 22, 2019	24
17	Late Ambadasrao Warpudkar College of Agriculture, Parbhani	Students	Feb 23, 2019	64
18	University of Agricultural Sciences, Bengaluru	B.Sc. (Ag) 3 rd year	June 11, 2019	83
19	Dr. Panjabrao Deshmukh Krishi Vidyapeeth, Akola	B.Sc. (Ag)	June 17, 2019	134

S.No	Details of College and University	Degree	Date of visit	No. of Students/ Farmers*
20	NS Agricultural College, Markapur	B.Sc. (Hons) 2 nd year	June 18, 2019	70
21	Vivekananda Agricultural College, Maharashtra	B.Sc. (Ag) 2 nd year	June 6, 2019	115
22	Bordikar College of Agriculture, Selu	B.Sc. (Ag) 4 th year	July 7, 2019	80
23	University of Horticulture and Forestry, Uttarakhand	B.Sc. (Ag)	July 13, 2019	43
24	Dr. Kalam Agricultural College, Bhagalpur	B.Sc. (Ag)	July 15, 2019	30
25	Bihar Agricultural University	B.Sc. (Ag)	July 1, 2019	33
26	Extension Education Institute	Scientists/KVKs	July 31, 2019	18
27	Bhola Paswan Shastri Agricultural College	BSc. (Ag.)	July 2, 2019	43
28	Abhiyamaan College of Agricultural and Research, Tamil Nadu	B.Sc. (Ag) 4 th year	September 3, 2019	107
29	S. Thangapazham Agricultural College, Vasudevanallur	B.Sc. (Ag) 4 th year	September 5, 2019	68
30	TNAU- Anbil Dharmalingam Agricultural College and Research Institute, Trichy	B.Sc. (Ag) 4 th year	September 5, 2019	99
31	The Indian Agricultural College, Radhapuram	B.Sc. (Ag) 4 th year	September 5, 2019	35
32	Don Bosco College Of Agriculture, Tamil Nadu	B.Sc. (Ag) 4 th year	September 6, 2019	38
33	TNAU- Agricultural College & Research Institute, Killikulam	B.Sc. (Ag) 4 th year	September 10, 2019	111
34	RVS Agricultural College, Thanjavur	B.Sc. (Ag) 4 th year	September 11, 2019	99
35	Sethu Bhaskara Agricultural College and Research Foundation, Karaikuda	B.Sc. (Ag) 4 th year	September 12, 2019	71
36	Krishna College of Agriculture & Technology, Tamil Nadu	B.Sc. (Ag) 4 th year	September 16, 2019	41
37	J.K.K. Munirajah College of Agricultural Science, Tamil Nadu	B.Sc. (Ag) 4 th year	September 16, 2019	86
38	PRIST , Thanjavur	B.Sc. (Ag) 4 th year	September 16, 2019	65
39	JSA College of Agriculture and Technology	B.Sc. (Ag) 4 th year	September 16, 2019	52
40	Tamil Nadu Agricultural University, Vazhavachanur	B.Sc. (Ag) 4 th year	September 17, 2019	65
41	RVS Padmavathy College of Horticulture, Tamil Nadu	B.Sc. (Hort.) 4 th year	September 18, 2019	40
42	Horticultural College and Research Institute, Coimbatore	B. Tech (Hort)	September 18, 2019	39
43	PGP College of Agricultural Sciences, Namakkal	B.Sc. (Ag) 4 th year	September 19, 2019	106

S.No	Details of College and University	Degree	Date of visit	No. of Students/ Farmers*
44	Tamil Nadu Agricultural University, Coimbatore	B.S (ABM) 4 th year	October 10, 2019	36
45	Pandit Jawaharlal Nehru College of Agriculture and Research Institute, Karaikal	B. Sc (Hons) Agriculture 4 th year & B.Sc. (Hons.) Hort 4 th year	October 10, 2019	154
46	University of Agricultural Sciences, Dharwad	B.Sc. (Ag) 4 th year & B.Sc. (Ag. Maco.) 4 th year	November 1, 2019	53
47	Vanavarayar Institute of Agriculture, Tamil Nadu	B.Sc. (Ag) 4 th year	November 2, 2019	113
48	University of Agricultural Sciences, Dharwad	B.Sc. (Ag) 4 th year & B.Sc. (Ag. Maco.) 4 th year	November 4, 8, 11, 15, 18, 2019 (6 batches)	366
49	College of Sericulture, Chintamani	B.Sc. (Ag) 3 rd year	November 4, 5, 2019 (2 batches)	76
50	Thanthai Roever Institute of Agriculture and Rural Development, Perambalur	B.Sc. (Ag) 4 th year	November 5, 2019	52
51	Uttar Banga Krishi Viswavidyalaya, West Bengal	B.Sc. (Hons.) Agriculture	November 7, 2019	28
52	Nammazhavar College of Agriculture and Technology	B.Sc. (Ag) 4 th year	November 7, 2019	42
53	Horticultural College and Research Institute, TNAU, Peryakulam, Tamil Nadu	B.Sc. (Hort.) 4 th year	November 8, 2019	71
54	SRS Institute of Agriculture and Technology, Tamil Nadu	B.Sc. (Ag) 4 th year	November 8, 2019	20
55	Horticulture College & Research Institute, TNAU, Tamil Nadu	B.Sc. (Hort.) 4 th year	November 8, 2019	28
56	SRM Institute of Science & Technology, Tamil Nadu	B.Sc. (Hons.) Agriculture 1 st year	November 11-15, 2019	230
57	Agricultural College and Research Institute, Pudukkottai, Tamil Nadu	B.Sc. (Ag) 4 th year	November 12, 2019	70
58	Adhiparasakthi Agricultural College, Kalavai-TNAU	B.Sc. (Ag) 4 th year	November 18, 2019	81
59	Nalanda College of Agriculture, Tamil Nadu	B.Sc. (Ag) 4 th year	November 21, 2019	40
60	University of Agricultural Sciences, Bengaluru	B.Sc. (Ag) 4 th year	November 21, 2019	55
61	SRM Institute of Science & Technology, Tamil Nadu	B.Sc. (Hons.) Agriculture	November 25, 2019	25
62	Agricultural College and Research Institute, Coimbatore	B.Sc. (Ag) 4 th year	December 5, 2019	115

S.No	Details of College and University	Degree	Date of visit	No. of Students/ Farmers*
63	Kumaraguru Institute of Agriculture, Tamil Nadu	B.Sc. (Ag) 4 th year	December 5, 2019	56
64	Tamil Nadu Veterinary and Animal Sciences, Thanjavur	B.V.Sc. & AH., 4 th year	December 6, 2019	33
65	College of Horticulture, Kerala Agricultural University, Thrissur	B.Sc. Hons. (Ag.) 4 th year	December 7, 2019	62
66	Gandhigram Rural Institute, Tamil Nadu	B.Sc. (Ag) 4 th year	December 9, 2019	28
67	Agricultural College and Research institute, Thanjavur	B.Sc. (Ag) 4 th year	December 10, 2019	63
68	Forest College and Research Institute, Mettupalayam	B.Sc. (Seri.) 4 th year	December 11, 2019	28
69	Tamil Nadu Agricultural University, Coimbatore	B. Tech. (AIT)	December 23, 2019	21
70	Veterinary College and Research institute, Namakkal, Tamil Nadu	B.V.Sc. & AH., 4 th year	December 24-25, 2019	55
71	College of Agriculture Engineering and Technology, PAU, Ludhiana	B.Tech (Ag.)	December 29, 2019	29
72	Kerala Agricultural university, Thiruvananthapuram	B.Sc. (Hons.) Agriculture	December 30, 2019	102
73	Dr. Y.S. Parmar University of Horticulture & Forestry, Himachal Pradesh	B.Sc. (Hons.) Hort 3 rd year	December 31, 2019 & January 1 st 2020.	92
Total Number of Students / Farmers Visited the Academy during the Year 2019				5043

The Academy is established in a green serene and sprawling campus of about 50 hectares. It has various facilities like Faculty Center, Academic Block, Administrative Block, Library, Medical and Health Centre, Sports Complex, Children Park, Guest houses and Trainees Hostel, Housing and Residential Quarters.

7.1 Library

The NAARM Library has about 31440 books, 71 International Journals, 33 International Online Journals, 26 free access Online Journals, and 72 Indian Journals. The library is also strengthened with 12 online databases / softwares such as EBSCO Business Source Premier, Indiastat.com, Commodities, Economic Outlook, Districts of India, iThenticate, Lexis Nexis.com, Total Patent, Patent Optimizer, Orbit.com, SciVal and Web of Sciences. The Library has many of its publications in digital formats like CD, VCD and its own digital repository (EPrints@NAARM). The Academy is also a member of CeRA, and has access to journals and databases through CSIRO, Springer and Open J-Gate etc. Library management has been digitized through the open source software "KOHA". The Academy also has a rich repository of online patent search providers, GIS software and other statistical software.



Library at ICAR-NAARM

7.2 Health Centre

NAARM Health center provides preventive and curative treatment for the employees and their dependents, trainees, students and pensioners and their dependents. On an average 40 patients are seen per day and medicines are provided to the patients. Emergency treatments are provided to the staff members and their dependents of other ICAR institutes. More than 700 ICAR pensioners have registered at NAARM Health center for regular OPD treatment and obtain referral services. Periodical health camps are organized at health center with investigations and specialist doctor's consultation at health center. Seven camps were conducted in 2019 for the benefit of NAARM staff, their family members, trainees and pensioners. Health talks by the reputed specialist doctors of recognized corporate hospitals are conducted at the academy from time to time. A part time lady medical officer has been appointed at Health center. Periodical Blood donation camp are organized by health center with the help of Red Cross blood bank. A Cardiologist visits health center once a month for two hours to take care of cardiac patients with free consultation.



Health Center at Academy

7.3 Sports Complex

The sports complex at NAARM has several facilities which includes a stadium for outdoor events like cricket, football, lawn tennis and various athletic events. Indoor complex comprises of badminton court, table tennis, billiard/snooker and gymnasium. This facility is used by the NAARM staff and trainees and with the use of this facility, NAARM sports contingent have won prizes in inter zonal sports meet and inter institutional sports events.



Sports Complex at NAARM

7.4 Children Park

Children park at NAARM was established for the benefit of children of staff who reside in the campus, who spend their leisure time in the open and fresh air of the park. It was inaugurated by Dr. Trilochan Mohapatra, Secretary, DARE & Director, General, ICAR. The children's park has several playthings for them to play with and enjoy. The facilities in the park include swings, slides, rods for exercise and many other such items that provide for children the required exercise and enjoyment benefitting both physical and mental health at tender age.



Dr. Trilochan Mohapatra, Secretary, DARE & Director General, ICAR Inaugurated Children Park at NAARM

7.5 NAARM Gallery

The Gallery was established keeping in view of visits of large number of college and school students, faculty, scientists and officials of NARES across the country on day to day basis. Some of the salient information displayed in the gallery include genesis, history and transition of the Academy, focal themes of NAARM, academic programmes offered by the Academy, challenges in agriculture sector etc. Significant emphasis on Centres of Excellence viz. a-IDEA, Technology Business Incubator and Centre for Open and Lifelong Learning in Agriculture (COLLAGe) was laid to portray the activities of Academy's contribution to science and agricultural education. Taking a pledge for the welfare of the farming community; emphasizing promotion of good agricultural practices and water conservation, enhancing quality education, ensuring environmental safety, growing more trees, avoiding food wastage, reducing carbon footprint etc. is given importance to the students who visit the gallery. The Secretary, DARE & Director General, ICAR who inaugurated the gallery appreciated the efforts for showcasing activities and accomplishments of the Academy for the benefit of various stakeholders in agriculture.



NAARM Gallery Inaugurated by Dr. Trilochan Mohapatra, Secretary, DARE and Director General, ICAR

7.6 Orchard Development

The Orchard crops, Floriculture, Polyhouse, and Nursery are well established at NAARM campus. The orchard crops such as mango (2.0 ha), sweet lime (1.5 ha), sapota (0.25 ha), amla (0.25 ha), custard apple (1.0 ha), phalsa (0.5 ha), and few other fruit crops were grown in total area of 6.5 hectares. New plantations were taken up in the orchard block of the campus. The Guava varieties viz. Allahbad Safeda, Lucknow-49, and Thai Guava were grown in an area of 0.5 hectares. Karonda planting was taken up all along the border of the plot. Nutri-garden was established in the orchard plot (H-7) with nutritionally important crops like Drumstick (PKM-1), Papaya, Amla, Rose apple,

Dragon fruit, Carrombola etc. The Pomegranate (*var. Bhagwa*) and Fig (*var. Pune*) planted in H-6 orchard plot started bearing fruits. Gap filling with improved varieties of mango and custard apple was done in the existing orchards and the plants are now well established.



Sweet lime in fruiting stage



Amla in fruiting stage



Sapota in fruiting stage

7.6.1 Intercropping of Groundnut and Vegetable crops in Orchard plots

In the mango orchard, three high yielding groundnut varieties *ICGV-03043*, *ICGV-171024*, *ICGV-86105* with high oil content released by ICRISAT were grown as intercrops. Vegetables such as tomato, bhendi, bottle gourd, ridge gourd, bitter gourd, dolichos bean, french bean, cluster bean and leafy vegetables were grown in an area of 0.25 ha as intercrops.



Groundnut (Var: ICGV-03043) grown as intercrop in mango orchard

7.7 Floriculture & Nursery

Rose, Jasmine, gladiolus, seasonal flower beds were cultivated in an area of 0.5 ha in the Floriculture block of the campus, which meets

the periodical requirement of floral arrangement and floral bouquets at campus. A rose garden extension block was established with around 200 bushes of 45 new rose varieties (45 varieties). The Academy bagged 14 different prizes in various categories, which includes the First prizes (6 No.), Second prizes (5 No.) and Third prizes (3 No.) in the 37th All India Rose Convention & Annual Rose Show at the Institute of Engineers, Visvesvaraya Bhavan, Hyderabad during 19-21 January 2019.



Improved Rose varieties grown at NAARM



Prizes won by the Academy in 37th All India Rose Convention in Hyderabad during 19-21 January 2019

Multiplication of ornamental plants like *hibiscus*, *golden durantha*, *bougainvillea*, *almanda* etc. was taken up at the nursery. Besides nearly 1000 indoor plant species of potted plants available are *Ferns*, *Dieffenbachias*, *Dracaena*, *Pedilanthus*, *Palms*, *Aralias*, *Eranthemums*, *Sanceveria*, *Kalaonche*.



Polyhouse at ICAR-NAARM



Plants grown in nursery at NAARM

7.8 Agro-Technology Park

An initiative for setting up a Technology Demonstration Park was made in the Academy pursuant to the recommendations of the 8th Research Advisory Committee and the main objective of this initiative is to showcase the agri-allied technologies of ICAR to the trainees and visitors of the Academy and to create an awareness and instill interest among them. The Academy has provisioned for adopting many diverse agriculture and other green technologies in the serene campus of about 50 ha. With an extensive site survey and consultations, the Academy has developed a detailed plan for documenting and cataloguing the technologies, introducing new technologies with an emphasis on green and eco-friendly technologies, and developing publicity material for showcasing the activities to trainees and other visitors. Currently, 25 technologies have been documented for display and showcasing.

ICAR-National Academy of Agricultural Research Management Rajendranagar, Hyderabad-500030, Telangana, India https://naarm.org.in			
AGRO-TECHNOLOGY PARK			
S. No.	Technologies	Details	Area / plot
1	Integrated Nutrient Management	FYM, Vermicompost & fertilizers for nutrient use efficiency and improvement of soil and farm produce quality	Orchard plots
2	Integrated Pest Management	Insect traps, pheromones, Bio-agents	Sweet lime, mango, rose gardens
3	Integrated Weed Management	Power weeder / mulching	Rose garden, papaya, drumstick
4	Integrated Farming Systems	Fish-duck based IFS	Water pond
5	Crop-diversification	Agro-forestry & Agro-horticulture systems	Orchards & plantations
6	Conservation of bio-diversity	Agri-biodiversity park	Near rear gate
7	Rain water harvesting	Rain water harvesting	Lake & farm pond
8	Lining of farm pond	Rain water conservation	Lake & farm pond
9	Micro-irrigation	Drip and sprinkler irrigation systems for water use efficiency	Orchard crops & Landscape gardens
10	Nutrient use efficiency	Placement of fertilizers/root zone application	Orchard plots
	Pit management	Salinity reclamation	Rose garden/ sweet lime plot
12	Organic Mulching	<i>Glyricidia/ Pongamia</i> Green leaf manuring	Orchard plots
13	Land treatments	Ridge and furrow system for water conservation	Row crops/ vegetable crops
14	Roof water harvesting	Roof water harvesting pits/recharging ground water	Near office buildings
15	Solid waste management	Nutrient recycling	Vermi-composting units
16	Soil Health Cards	Site specific nutrient application	For orchards & floriculture crops
17	Medicinal & Aromatic Park	Medicinal & aromatic plants	Herbal Garden
18	High value crops	Floriculture- Rose garden (named varieties) and Coconut plantation	Floriculture block/ A-1 plot
19	Nutri-garden	<i>Drumstick, Papaya, Amla</i>	H-7 plot
20	Protected cultivation	Ornamental plants, flower seedlings/ vegetable nursery	Naturally ventilated poly house
21	Farm Mechanization	Bush cutter, Telescopic tree pruner, power operated lawn mower, hedge cutter, power chain saw, rotavator, post hole digger, battery operated sprayer	Used in orchards & landscape gardens
22	Crop cultivars	High yielding/ disease tolerant varieties	Fruit crops & Rose varieties
23	Plant propagation	Ornamental plant propagation & development of potted plants	Nursery
24	Organic garden	Vegetable crops	Kitchen garden in H.O.R
25	Landscaping with Different lawn-grass	<i>Korean carpet grass, Fescue grass, Paspalam shade grass, Saint Augustine shade grass</i>	Landscape gardens of offices/guest houses/ Sports complex

7.9 Lake and Rainwater Harvesting

To harness maximum quantity of rain water in the NAARM lake, rain water harvesting was undertaken. Keeping this in view, clearing the catchment area, diverting the water harvested from roof tops, rain water drains etc. were implemented so as to attain the lake to its full potential by the end of monsoon. This also helps in recharging the wells nearby during dry season and help in enhancing the ground water table. This enables irrigating the gardens and lawns around the lake. Nearly, 40,000 fish seedlings of different species including Rohu, Katla, Grass Carp and Common Carp were re-

leased into the lake waters in October 2019. The lake water and the surrounding lush green lawns add further to the aesthetic beauty and becoming an eco-friendly campus.



An aesthetic view of NAARM Lake

8.1 Recognition

8.1.1 Visits Abroad

Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM participated in 49th Session of Intergovernmental Panel on Climate Change (IPCC) as a part of UNDP project under MoEFCC. He has participated as a member of the Government of India Delegation to participate in IPCC held at Kyoto, Japan from 8th -12th May 2019. Dr.Ch. Srinivasa Rao participated as a member, Government of India Delegation in UNFCCC Global Climate Change Negotiations held at Bonn, Germany from June 17-June 27, 2019. He also participated as a Government of India Delegation Member in United Nations Climate Change Conference UNFCCC COP-25/ CMP-15/CMA-2-2019 sponsored by ICAR held at Madrid, Spain from December 2-13, 2019.

Drs. R.V. S Rao, D. Thammi Raju and N. Srinivasa Rao, Principal Scientists of NAARM visited four European countries from October 12 - 21, 2019 as a part of Executive Development Programme on Developing Effective Organizational Leadership for Senior Officers of ICAR. The scientists along with senior officers of ICAR visited Netherlands, Germany, Belgium and Switzerland. The interactions and training sessions were organized

at Wageningen University and Research, The Netherlands; Carus - Research facility of the Department of Animal Sciences, Wageningen University; Royal Flora Holland, Aalsmeer; Dutch Ministry of Agriculture, Nature and Food Quality, Hague, The Netherlands; German Development Institute (DIE) and Development Research Institute (ZEF), University of Bonn, Germany; Ghent University, Ghent; Joint Research Centre, European Commission, Brussels, Belgium; European Commission & European Innovation Partnerships-Agri Network, Brussels, Belgium; Research Institute of Organic Agriculture, Frick, Switzerland and World Trade Organization, Geneva, Switzerland. Dr D Thammi Raju and Dr N Srinivasa Rao, as a part of NAHEP participated in the programme for i) Fostering/ further strengthening ICAR collaboration with Higher Education Institutions (HEIs) ii) Understanding and adopting International best higher education practices in India and iii) Creating/facilitating opportunities for Indian scientists for scientific/ academic exchange and to make use of channels including EU competitive grants & to understand trade practices in agriculture & allied areas & to channelize research efforts to facilitate higher agricultural exports.



**Dr. Ch. Srinivasa Rao with delegates
in the 49th Session of IPCC
in Kyoto, Japan**



**Dr. Ch. Srinivasa Rao at UNFCCC
Global Climate Change Negotiations
in Bonn, Germany**



**Dr. Ch. Srinivasa Rao
representing India at UNFCCC
in Madrid, Spain**

8.1.2 Awards Received by the Faculty of NAARM

Name of Staff	Name of award	Date	Purpose	Awarding body
Ch Srinivasa Rao	ICAR Award of 'Dr Rajendra Prasad Puraskar'- 2019	16 th July 2019	For Publishing Best Technical Book in Hindi in Natural Resource Management and Agricultural Engineering.	Indian Council of Agricultural Research, New Delhi
Ch Srinivasa Rao	Life Time Achievement Award	26 th Dec 2019	For Outstanding Contribution in the Field of Science and Technology	Andhra Pradesh Akademi of Sciences (APAS)
Ch Srinivasa Rao	Ch. Dimag Singh Award	22 nd Oct 2019	For Outstanding Contribution in Soil Science	Astha Foundation, Meerut, U.P.
SK Soam	Rythunestham Award- 2019 (Farmer Friend Award-2019)	22 nd Sep 2019	For Distinguished Contribution in Agricultural Research Management.	Rythunestham Foundation, Hyderabad
Alok Kumar	ISEE Fellow Award	22 nd Nov 2019	For Outstanding work in the Field of Extension Research and Education	Indian Society of Extension Education, New Delhi
VV Sumanth Kumar	Outstanding Scientist Award	20 th Oct 2019	For Research Initiatives in Sustainable Agriculture & Allied Sciences	Astha Foundation, Meerut
MB Dastagiri	National Award of Excellence- 2019	26 th Feb 2019	For Distinction of being one of the Outstanding Scientists.	Global Management Council, Ahmedabad

8.1.3 Other Recognitions Received by the Faculty of NAARM

- Dr. Ch Srinivasa Rao, Director, ICAR-NAARM is elected as Fellow of National Academy of Sciences, India (FNASI).
- Dr. Ch Srinivasa Rao, Director, ICAR-NAARM is elected as Honorary Fellow of Andhra Pradesh Academy of Sciences (APAS), Amaravati, Andhra Pradesh.
- Dr. Ch Srinivasa Rao, Director, ICAR-NAARM is nominated by President, ICAR Society as Member, Selection-cum-Standing Committee for Emeritus Scientist Scheme of ICAR for a period of three years with effect from December 2019.
- Dr. Ch Srinivasa Rao, Director, ICAR-NAARM is nominated as Member, ICAR Selection Committee for Netaji Subhas – ICAR International Fellowships – 2019-20.
- Dr. Ch Srinivasa Rao, Director, ICAR-NAARM is nominated as Chairman, Committee for screening of proposals for new Agribusiness Incubation Centres in ICAR Institutes under Component-II (Incubation Fund) of NAIF Scheme.
- Dr. Ch Srinivasa Rao, Director, ICAR-NAARM is selected as Member, Executive Board of International Dryland Development Commission (IDDC), Cairo, Egypt – 2019-20.
- Dr Ch. Srinivasa Rao, Director, ICAR-NAARM is nominated as Member, Senior Officers Committee, ICAR.
- Dr. Ch Srinivasa Rao, Director, ICAR-NAARM is nominated as Member, NAAS Sectional Committee for NRM.
- Dr. Ch Srinivasa Rao, Director, ICAR-NAARM is selected as President, Association for Management of Agricultural Research and Agripreneurship (AMARA), Hyderabad

- Dr. Sumanta Kundu, PhD student of Dr. Ch. Srinivasa Rao received 'The Mosaic Company Foundation Award for Best Doctoral Thesis'.
- Dr. Ch Srinivasa Rao, Director, ICAR-NAARM participated in the 49th Session of Intergovernmental Panel on Climate Change (IPCC) as Indian Delegation Member at Kyoto, Japan during May 8-12, 2019.
- Dr. Ch Srinivasa Rao, Director, ICAR-NAARM participated as a Government of India Delegation Member in UNFCCC Global Climate Change Negotiations at Bonn, Germany during June 17-27, 2019.
- Dr Ch. Srinivasa Rao, Director, ICAR-NAARM participated as a Government of India Delegation Member in United Nations

Climate Change Conference UNFCCC COP-25/CMP-15/CMA-2 – 2019 at Madrid, Spain during December 2-13, 2019.

- Dr Ch. Srinivasa Rao, Director, ICAR-NAARM bagged ICAR Award of 'Dr Rajendra Prasad Puraskar' for the year 2019 for publishing Best Technical Book in Hindi in Agriculture and allied Sciences in the area of Natural Resource Management and Agricultural Engineering.
- Dr Ch. Srinivasa Rao, Director, ICAR-NAARM is nominated by DG, ICAR as Member of the Advisory Committee of the Initiative for Climate Action Transparency Project on Adaptation (ICTA-A) of The Energy and Resources Institute (TERI), New Delhi.



Dr Ch. Srinivasa Rao, Director, ICAR-NAARM receiving NASI fellowship from Prof. G. Padmanabhan, President NASI



Dr Ch. Srinivasa Rao, Director, ICAR-NAARM receiving 'Ch. Dimag Singh Award' from Astha Foundation, Meerut, U.P.



Dr. Ch Srinivasa Rao, Director, ICAR-NAARM receiving Life Time Achievement award from APAS, Amaravati, Andhra Pradesh



Dr Ch. Srinivasa Rao, Director, ICAR-NAARM Receiving ICAR Award of 'Dr Rajendra Prasad Puraskar'- 2019

- Dr Ch. Srinivasa Rao, Director, ICAR-NAARM is nominated as Member, Regional Advisory Group (RAG) of NABARD for the year 2019-20.
- Dr Ch. Srinivasa Rao, Director, ICAR-NAARM is nominated as Member of Board of Management, Acharya N.G. Ranga Agricultural University (ANGRAU), Guntur as ICAR nominee.
- Dr Ch. Srinivasa Rao, Director, ICAR-NAARM is nominated as Member of Board of Management, Professor Jayashankar Telangana State Agricultural University (PJTSAU), Hyderabad as ICAR nominee.
- Dr. I. Sivaraman, PhD Student of Dr. M. Krishnan received 'The Jawaharlal Nehru Award for Outstanding Research in Social Sciences' for the year 2018-19.
- Dr. MB Dastagiri has been recognised by IASR as one of 500 Most Influential Experts in Agriculture for the Year 2019.
- Dr. MB Dastagiri received Certificate of Honour award at University of Hyderabad for paper presentation at 4th International Conference on Advances in Agriculture & Animal Sciences towards Global Food Security on 28th Aug 2019.
- Dr. G R K Murthy received Best Faculty Award for contribution in Teaching on 1st September, 2019 at ICAR-NAARM, Hyderabad.
- Dr. P Krishnan received Best Scientist Award for contribution in Research on 1st September, 2019 at ICAR-NAARM, Hyderabad.



Dr. S.K. Soam receiving Rythunestham award 2019 from Shri Venkaiah Naidu, Hon'ble Vice President of India



Smt. K.K. Rukmani Ammal receiving the Best Administrative Staff - ICAR Award Scheme-2018 during 91st ICAR Foundation Day.



Smt. V. Saroja receiving the Best Skilled Supporting Staff - ICAR Award Scheme-2018 during 91st ICAR Foundation Day.

8.1.4 Honours and Awards Received by the PGDMA Students and Others

Name of Student	Name of Award/ Fellowship/ Event	Date	Position	Awarding body
Aayush Aswathi Utkarsh Kokate C H Ranjit	Paramarsh 2019- Case study Competition	11 th Nov 2019	1 st	IIM Raipur
Gayatri Gopan	13 th Annual RMAI Awards for Best Summer Projects in Rural Marketing	27 th Sep 2019	3 rd	Rural Marketing Association of India
Naveen Yadav Ramya Nargund	Krishi Chanakya- Dattansh, the Data Analytics	17 th Aug 2019	2 nd	National Institute of Agricultural Extension Management
K Sai Prajeeth Reddy Siddharth P	Dhwani 2019	18 th Jan 2019	1 st	Shri Dharmasthala Manjunatheshwara Institute for Management Development

PGDMA Students Bagged 2nd Prize at ICAR-IARI, New Delhi

Ashish Pandey, Divyanshu Yadav, Venkatesh Karlapudi, Shivanand Masali and Sushant Chandra of PGDMA 2019-21 bagged 2nd prize in “Impactful Innovations for Indian Agriculture” at EIMA AGRIMACH 2019 jointly organized by FICCI, FEDERUNACOMA ITALY, Ministry of Agriculture & Farmers Welfare and Indian Council of Agricultural Research held at IARI Pusa, New Delhi during Dec 5-7, 2019.



PGDMA Students with Prizes at EIMA AGRIMACH 2019 in New Delhi

PGDMA Students Bagged 2nd Prize at IIM, Ahmedabad

Aayush Awasthi and Utkarsh Kokate of PGDMA 2019-21 bagged 2nd prize in business plan

(B-Plan) competition in PITCHERS 2.0 2019 organized at IIM Ahmedabad.



PGDMA Students received second prize at IIM, Ahmedabad

PhD Student of Dr. M. Krishnan Received the Jawaharlal Nehru Award for Outstanding Research in Social Sciences

Dr. I. Sivaraman, Scientist (Fisheries Extension), ICAR- Central Institute of Freshwater Aquaculture, Bhubaneswar, and PhD student of Dr. M. Krishnan, ICAR- NAARM received the Jawaharlal Nehru award for outstanding research in social sciences for the year 2018 -19, from Dr. T. Mohapatra, Secretary DARE and Director General ICAR and Dr. Panjab Singh, Former Secretary DARE and Director General ICAR, during the ICAR's 91st Foundation Day celebrations held at the National

Agricultural Science Centre Complex, Pusa, New Delhi on 16th July, 2019. Dr. Sivaraman worked for his Ph.D. thesis entitled 'A Critical Analysis of Better Management Practices in Shrimp Farming in Andhra Pradesh' at the Division of Fisheries Economics, Extension and Statistics of ICAR - Central Institute of Fisheries Education, Mumbai, under the guidance of Dr. M. Krishnan, Professor & Head, Education Systems Management Division, ICAR- NAARM, Hyderabad.



**Dr. I. Sivaraman, PhD Student of Dr. M. Krishnan
Receiving the Jawaharlal Nehru Award**

8.1.5 ICAR-NAARM in Media

1. News coverage about the curtain raiser on 89th Annual Session of the National Academy of Sciences, India (NASI) and Symposium on Science and Technology based Entrepreneurship Development' scheduled during December 21-23, 2019 at ICAR-NAARM, Hyderabad. Telangana Today, The Hindu; Hindi Milap, Vaartha, Velugu, Sakshi, Andhra Jyothi, Praja Paksha, Andhra Bhoomi dated December 21, 2019.
2. News item about the visit of Sri Narendra Singh Tomar, Hon'ble Union Home Minister for Agriculture and Farmers' Welfare, to ICAR-NAARM and interaction with the IFS probationers of the Training Programme on 'PRA, communication strategies and watershed management' on September

27, 2019 in Andhra Jyothi, Velugu, Sakshi, Namasthe Telangana, Indian Express, Praja Paksha, Eenadu daily newspapers, dated September 29, 2019.

3. News coverage about the Agri-Udaan 3.0 organized by ICAR-NAARM and celebrations of 44th Foundation Day of ICAR-NAARM on September 1, 2019 in the dailies of Hindi Milap, Hindu, Eenadu, Andhra Jyothi, Velugu, Sakshi, Namasthe Telangana, Mana Telangana, Andhra Praha, dated September 2, 2019
4. Special news coverage about the Component 2A project of World Bank-funded NAHEP Project on 'Investments in Indian Council of Agricultural Research Leadership in Agricultural Higher Education' given to ICAR-NAARM, Hyderabad in The Hindu BusinessLine and The Hans India, dated May 20, 2019 and Eenadu telugu daily newspaper dated June 10, 2019.
5. News item about the National Workshop on 'Academic Excellence through Building Partnerships and Resource Generation' under component 2A of the World Bank funded ICAR-NAHEP project on Investments in Indian Council of Agricultural Research Leadership in Agricultural Higher Education' held at ICAR-NAARM on April 30 and May 1, 2019. The Hans India, English daily newspaper; Eenadu, Vaartha, Andhra Jyothi, Andhra Prabha, Telugu daily newspapers; Hindi Milap and Swatantra Vaartha Hindi daily newspapers covered it on May 2, 2019.
6. News coverage about the visit of Shri Gajendra Singh Shekhawat, Hon'ble Union Minister of State for Agriculture and Farmers' Welfare to ICAR-NAARM on March 8, 2019. This news appeared in New Indian

Express English daily newspaper, Eenadu, Vaartha, Andhra Jyothi, Andhra Prabha, Andhra Bhoomi and Praja Paksha, Telugu daily newspapers; and Hindi Milap Hindi daily newspapers dated March 9, 2019.

7. News item about the celebrations of National Science Day organized by ICAR-NAARM on February 28, 2019 in Mana Telangana, Andhra Jyothi, Vijaya Kranthi, Eenadu, Saakshi, Vaartha, Prajapaksham, Andhra Bhoomi telugu daily newspapers dated March 1, 2019 and Swatantra Vaartha, Hindi Milap Hindi daily newspaper and the New Indian Express English daily newspaper dated March 1, 2019.
8. News Item on PGDETM admission Notification in Sakshi- Telugu newspaper, Hindi Milap- Hindi daily newspaper,

The Hindu, Sunday Times- daily English Newspapers dated October 19, 2019.

The press releases in connection with many important training programmes / events / meetings / research were issued to reach out to the masses at a larger scale. News coverage came in print media of national and local dailies of different languages of English, Hindi and Telugu besides electronic channels. The special news coverage appeared in main editions of Andhra Jyothi, Mana Telangana, Eenadu, Indian Express and Hindi Milap. The news on several training programmes organized by ICAR-NAARM appeared in ICAR website. The following is the list of NAARM programmes / meetings / events which found place in both print and electronic media.

Press Coverage of Programmes / Events of ICAR-NAARM	Date of Media Coverage
Training Programme on 'Entrepreneurial Skill Development among Agri-Graduates of ANGRAU' Press coverage in Dailies of both Telangana and Andhra Pradesh states	Feb 15, 2019 & Mar 6, 2019
Special News Coverage on National Science Day celebrations (The Hindu, Indian Express, Hindi Milap and Swatantra Vaartha)	Feb 28, 2019 & Mar 1, 2019
International Women's Day celebrations at NAARM	Mar 9, 2019
Interaction Session of Shri Gajendra Singh Shekhawat, Hon'ble Minister of State for Agriculture and Farmers Welfare, Government of India, with Farmers, Scientists, and other Stake Holders at NAARM	Mar 9, 2019
National Workshop on 'Investments in Indian Council of Agricultural Research Leadership in Agricultural Higher Education' under NAHEP	May 1 & 9, 2019
Special Coverage on NAHEP (Business line and Hans India)	May 16, 2019
Agri Udaan 3.0 Accelerator Programme in Hyderabad	Sep 2, 2019
NAARM Foundation Day Celebrations at NAARM, Hyderabad	Sep 2-4, 2019
Special coverage on NAARM (Mana Telangana & Andhra Jyothi)	Sep 2, 2019
Training Programme on 'Participatory Rural Appraisal (PRA), Communication Strategies and Watershed Management' for IFS Probationers	Sep 27, 2019
Field Training on 'Participatory Rural Appraisal and Watershed Management' for 62 IFS Probationers of 2018 batch in Collaboration with Panchayat Officials and Villagers of Gadda Mallaiah Guda village in Yacharam Mandal of Ranga Reddy district	Sep 29, 2019
150 th Birth Anniversary Celebrations of Mahatma Gandhi at ICAR-NAARM, Hyderabad	Oct 3, 2019
NASI Annual Session and Symposium at ICAR- NAARM, Hyderabad	Dec 21-23, 2019

Glimpses of Media Releases



అంతర్జాతీయ సాంకేతికతల అభివృద్ధి కోసం ప్రపంచ శాస్త్రాల అభివృద్ధి

రాజ్‌గఢ్‌లో, డిసెంబర్ 28 ప్రభుత్వం ప్రస్తుత ముందు జీవంలో ఎవరైనా దేశం ఎంతో అభివృద్ధి చెందుతుంది అని భావిస్తే, వ్యవసాయం మరియు పరిశ్రమల ద్వారా సమగ్ర ప్రగతిని సాధించే ప్రయత్నం చేయాలి. వ్యవసాయం మరియు పరిశ్రమల ద్వారా సమగ్ర ప్రగతిని సాధించే ప్రయత్నం చేయాలి. వ్యవసాయం మరియు పరిశ్రమల ద్వారా సమగ్ర ప్రగతిని సాధించే ప్రయత్నం చేయాలి.



వైద్యశాస్త్ర ప్రదర్శనలు స్వచ్ఛాత్యక్తతను వెలికితీస్తాయి

వైద్యశాస్త్ర ప్రదర్శనలు స్వచ్ఛాత్యక్తతను వెలికితీస్తాయి. వైద్యశాస్త్ర ప్రదర్శనలు స్వచ్ఛాత్యక్తతను వెలికితీస్తాయి. వైద్యశాస్త్ర ప్రదర్శనలు స్వచ్ఛాత్యక్తతను వెలికితీస్తాయి.



కృషి కో పోషకతా కే రూప మే దేఖే : గజేంద్ర

కృషి కో పోషకతా కే రూప మే దేఖే : గజేంద్ర. కృషి కో పోషకతా కే రూప మే దేఖే : గజేంద్ర. కృషి కో పోషకతా కే రూప మే దేఖే : గజేంద్ర.



ICAR meet to discuss Fall armyworm menace

ICAR meet to discuss Fall armyworm menace. ICAR meet to discuss Fall armyworm menace. ICAR meet to discuss Fall armyworm menace.



నార్కల్లో ముగిసిన నాసీ వార్షిక సదస్సు
రాజేంద్రవర, డిసెంబర్ 23 (ఆంధ్రజ్యోతి): రాజేంద్రవరలోని నేషనల్ అకాడమీ ఆఫ్ సైన్సెస్ రీసెర్చ్ సెంటర్‌లో (నార్కల్) మూడు రోజులపాటు జరిగిన నేషనల్ అకాడమీ ఆఫ్ సైన్సెస్ ఇండియా (నాసీ) 89వ వార్షిక సదస్సు సోమవారం ముగిసింది. ఈ సందర్భంగా విద్యార్థులకు పైన్ట్ చే నిర్వహించి వ్యవసాయ విద్యపై అభిగాహన నిర్వహించారు. కార్యక్రమంలో నాసీ అధ్యక్షుడు దాక్షన్ జీపద్మ నాథ్, మాజీ అధ్యక్షుడు చంబాకర్, కార్యదర్శి సర్వధియో, నార్కల్ డైరెక్టర్ కార్తీక్ సీహెచ్ శ్రీనివాసరావు, చలువురు శార్వత్ త్తలు పాల్గొన్నారు. కాగా, నార్కల్ ఏర్పాటు చేసిన బుచ్చిడి విగ్రహం ప్రత్యేక ఆకర్షణగా నిలిచింది.



నార్కల్లో
ఏర్పాటు
చేసిన
బుచ్చిడి
విగ్రహం

8.1.6 TV/Radio Talk Delivered

S.No	Name of the Person	Topic of the Talk	Date	Name of the Channel and Program
1.	Ch. Srinivasa Rao	About 89 th Annual Session of the National Academy of Sciences, India (NASI) and Symposium on Science and Technology based Entrepreneurship Development during December 21-23, 2019 at NAARM, Hyderabad	21 st Dec 2019	ETV News Channel (Telugu)
2	Ch. Srinivasa Rao	About the 89 th Annual Session of the National Academy of Sciences, India (NASI) and Symposium on Science and Technology based Entrepreneurship Development during December 21-23, 2019 at NAARM, Hyderabad	23 rd Dec 2019	DD Yadgiri (Telugu)
3	Ch. Srinivasa Rao	About Nela Aarogyam – II - Soil Health and Soil Conservation at MANAGE, Hyderabad.	2019	https://www.youtube.co/watch?v=yFbDN2IoEtl

8.1.7 Invited Lectures Delivered by the Faculty and Staff

Name of Staff	Lecture	Programme/ Organized by
A Dhandapani	SPSS and Crop Stats Packages for Data Analysis	Training Programme on Digital Applications by EEI, Hyderabad on May 31, 2019
	Data Analysis in AICRP Trials	Orientation Programme to the AIRCP (Oilseeds) breeders by ICAR-IIOR, Hyderabad on Aug 23, 2019
B Ganesh Kumar	Fish and Meat Value Chains	Training Programme on 'Value Chain Management for Agricultural Produce' by MANAGE, Hyderabad on Jan 08, 2019
	Linking Farmers to the Market for Sustainable Livelihood through Animal Husbandry	Training Programme on 'Extension Management Approaches for Sustainable Livelihood through Animal Husbandry' by MANAGE, Hyderabad on Jan 22, 2019
	Marketing Strategies for Livestock Commodities	Training Programme on 'Recent Advances in Livestock Sector' for the Extension Functionaries of State Department of Animal Husbandry, Tamil Nadu by MANAGE, Hyderabad on Feb 01, 2019 & Jun 12, 2019
	Linking Livestock Farmers to Market Issues and Challenges	Training Programme on Extension Management Approaches for Livestock Development MANAGE, Hyderabad during Aug 21, 2019- Nov 15, 2019
	Linking Farmers to Market	Training Programme on New Dimensions in Extension Management for the Officers of State Department of Animal Husbandry by MANAGE, Hyderabad on Oct 17, 2019
	Agribusiness Opportunities in Livestock Sector	Training Programme on Agriculture as Business for Rural Youth by MANAGE, Hyderabad on Aug 23, 2019
	Supply Chain Management in Fish and Meat	Training Programme on Extension Skills for Supply Chain Management in Agri and Allied Sectors by MANAGE, Hyderabad on Oct 30, 2019
	Livestock Based Livelihoods in Rural Areas	International Training Programme on Natural Resource Management for Sustainable Rural Livelihoods by NIRD&PR, Hyderabad on Sep 12, 2019
	Livestock, Dairy & Small Ruminant Farming for Improved Income and Better Climate Resilience	Certificate Course on Sustainable Livelihoods and Adaptation to Climate Change (SLACC) by NIRD&PR, Hyderabad during Jun 20- Sep 03, 2019
	Value Chain Dynamics in Dairy Sector	Training Programme on Strategic Leadership and Change Management' by ASCI, Hyderabad on Aug 22, 2019
BS Sontakki	Student Centered Learning	Training Programme for Newly Recruited Assistant Professors in SKLTSHU, Hyderabad on Mar 18, 2019
	Basics of Scientific Writing	Training Programme for Newly Recruited Assistant Professors in SKLTSHU, Hyderabad on Mar 18, 2019
	Management of Trainees in Cross-cultural Context: Implications for Practice	Capacity Enhancement for Organizing Training Programs on Agricultural Extension for International Participants by MANAGE, Hyderabad on Jul 30, 2019
	Designing and Standardization of Methodology for Developing Vulnerability Index and Model for Enhancing Income of Women in Agriculture	Brainstorming Workshop on AICRP on Women in Agriculture at PJTSAU, Hyderabad during 29-30 April 2019

Name of Staff	Lecture	Programme/ Organized by
Ch Srinivasa Rao	Synergy of Research-Technology - Policy Implementation for Resilient Rainfed-Dryland Production Systems	13 th International Conference on Development of Drylands - Converting Dryland Areas from Grey into Green by IDDC and Arid Zone Research Association of India, Jodhpur during February 11-14, 2019
	Novel Approaches for Doubling Farmers Income through Sustainable Agricultural Production Systems	Annual National Conference for PG Students (NCPGS-2019) by ANGRAU, Tirupati on March 07, 2019
	Developing Appropriate Eco-System for Establishing Rural Enterprises.	NABARD Foundation Day Seminar on Engaging Rural Youth Gainfully by NABARD, New Delhi on July 12, 2019
	Farmer Producer Organizations for Sustainable Technology Dissemination	3 rd Annual AGRIBIOTECH INDIA SUMMIT 2019 by Inventicon Business Intelligence, Hyderabad on July 25, 2019
	Climate Resilient Tobacco Based Production System in India: Adaptation and Mitigation Strategies for Farm Income Stability	National Symposium on Approaches and Strategies for Augmenting Tobacco Farmers Income' by CTRI, Rajahmundry during July 19-20, 2019
	Integrated Farming: Key for Enhanced Farmers Income in India	4 th International Conference on Agriculture and Animal Husbandry by University of Hyderabad, Hyderabad on Aug 28, 2019
	Making (Indian) Drylands a Good Niche for Carbon Sequestration through Soil and Crop Management Practices: Potential Vs. Realism	Workshop on Carbon Management In Soils Through Resource Conservations Technologies (Rct): Issues and Strategies by Centre of Advanced Agricultural Science and Technology (Caast) on Conservation Agriculture, BCKV, Mohanpur, West Bengal on Nov 02, 2019
D Thammi Raju	Application of Ict & Social Media for Effective Advisory Services	Extension Management Approaches For Sustainable Livelihood Through Animal Husbandry by MANAGE, Hyderabad on Jan 22, 2019
	Application of Ict for Effective Livestock Extension Services	Recent Advances in Livestock Sector by MANAGE, Hyderabad on Feb 01, 2019
	Basics of Teaching and Learning and Innovative Teaching Methods	Training on Digital Teaching Techniques by University of Agricultural Sciences, Bangalore on Feb 26, 2019
	Curriculum Design and Lesson Planning	Training on Digital Teaching Techniques by University of Agricultural Sciences, Bangalore on Feb 26, 2019
	Micro Teaching	Training on Digital Teaching Techniques by University of Agricultural Sciences, Bangalore on Feb 26, 2019
	Application of ICT for Effective Livestock Extension Services	Recent Advances in Livestock Sector by MANAGE, Hyderabad on Mar 1, 2019
	Innovative Teaching Methods	Training Programme for newly recruited Assistant Professors by College of Horticulture, Sri Konda Lakshman Telangana State Horticultural University, Hyderabad on Mar 15, 2019

Name of Staff	Lecture	Programme/ Organized by
D Thammi Raju	Curriculum Design and Lesson Planning	Training Programme for newly recruited Assistant Professors by College of Horticulture, Sri Konda Lakshman Telangana State Horticultural University, Hyderabad on Mar 19, 2019
	Application of ICT for Effective Livestock Extension Services	Recent Advances In Livestock Sector by MANAGE, Hyderabad on Apr 09, 2019
	Online Education for Enhancing Efficiency of Faculty	MOOCs, e-Content Development and Open Educational Resources UGC-Human Resource Development Centre by Maulana Azad National Urdu University, Hyderabad on Jun 19, 2019
	ICT and Social Media for Effective Advisory Services in Livestock Sector	Extension Management Approaches for Livestock Development by MANAGE, Hyderabad on Aug 21, 2019
	ICT and Social Media for Effective Advisory Services in Livestock Sector	Extension Management Approaches for Livestock Development by MANAGE, Hyderabad on Nov 14, 2019
GRK Murthy	Developing MOOCs in Indian Languages	Refresher Course in Modern Indian Languages by MANUU, Hyderabad on Dec 13, 2019
	E-learning in Higher Education	42 nd Orientation Programme for the Lecturers/Assistant Professors of different Degree Colleges and Universities by MANUU, Hyderabad on Nov 18, 2019
	MOOCs: Use as Digital Learning Media	Training-cum-Workshop on Improving e-Governance in Agriculture by MANAGE, Hyderabad on Dec 5, 2019
	Artificial Intelligence in Spraying	Training Programme on Good Agricultural Practices in Spraying by NIPHM, Hyderabad on Sep 30, 2019
	e-Learning	Workshop on MOOCs, e-Content Development and Open Educational Resources by MANUU, Hyderabad on Jun 13, 2019
	e-Learning: Tools and Techniques	National Level Capacity Building Workshop for Librarians of SAUs by PJTSAU, Hyderabad on Feb 08, 2019
	Online Learning and MOOCs	Innovative Methods for Effective Extension Delivery by EEI, Hyderabad on Jul 31, 19
KH Rao	Role Play – A training Methodology	International Training Programme on Training Methodology for Development Professionals by NIRD & PR, Hyderabad on Jan 31, 2019
	Human Resources Management for Development Officials of Telangana State Government	Orientation Training for Mandal Development Officers by Telangana State Institute for Rural Development, Hyderabad on Dec 15, 2019
	Entrepreneurship Development in Production. Processing and Marketing of Milk and Milk products	Re-Settlement Program for Defence, Navy, Airforce Personnel by Institute of Cooperative Management, Hyderabad on Dec 22, 2019
K Kareemulla	Approaches to Research Methodology	41 st Orientation programme for the Lecturers/ Assistant Professors of different Degree Colleges and Universities by MANUU, Hyderabad on Oct 3, 2019

Name of Staff	Lecture	Programme/ Organized by
MB Dastagiri	International Trade Economic Analysis for Indian Agriculture during WTO Regime: Effects on Production, Exports, Imports and Poverty	4 th International Conference on 'Advances in Agriculture & Animal Sciences towards Global Food Security' Ending Conferences, USA in Association with Life Science Society of Hyderabad by University of Hyderabad, Telangana on Aug 28, 2019
M Balakrishnan	Designing of Guide RNAs and Selection of CRISPR Vectors	Workshop on "Gene Editing for Enhancing Plant Productivity and Stress Tolerance by ICAR-IIRR, Hyderabad during Nov 10- 12, 2019
P Ramesh	Psychology of Teaching and Learning	UGC-HRD-Orientation Program for Assistant Professors of different Universities by Maulana Azad National Urdu University, Hyderabad on Nov 27, 2019
	Psychological Principles of Teaching and Learning	UGC-HRDC-Orientation Program for the Lecturers/ Assistant Professors of different Degree Colleges and Universities by Moulana Azaad National Urdu University (MANUU), Hyderabad on Jan 22, 2019
PD Sreekanth	Geospatial Technologies in Agriculture	Google Earth Engine user Meet by Google, Bangalore during Mar 15-16, 2019
	Geospatial Technologies in Sustainable Agriculture and Food Security	4 th International conference on Agriculture and Animal Husbandry by University of Hyderabad, Hyderabad during Aug 28-30, 2019
	Drought and Surface Water Monitoring: Remote Sensing Technologies for Decision Making	International Conference on Agricultural Statistics by ICAR-IASRI New Delhi, during Nov 18-21, 2019
	Geospatial Technologies in NRM	Remote Sensing & Geospatial Information System in Plant Health Management by NAPHM Hyderabad during Dec 3-7, 2019
	Seaweed Culture & Processing - Policies, Laws & Opportunities	Training Workshop on 'Seaweed Cultivation and Processing Technology' by CSIR-Central Salt and Marine Chemicals Research Institute (CSMCRI) Bhavnagar on Oct 10, 2019
P Venkatesan	Farmer Producer Organizations (FPOs) and their Success	Market Led Extension for Agriculture and Allied Sectors by EEI, Hyderabad
	Training Need Analysis	Training of Trainers by Central Academy of State Forest Services (CASFOS), GOI
	Communication Skills	Communication Chapter by AC & RI (TNAU), Madurai
	Effective Presentation	Human Resource Management for the In-service SFS Officers of State Forest Department by Central Academy of State Forest Services (CASFOS), GOI
Ranjit Kumar	Conversion of Research Output to Business Outcome	AICRP Workshop on 'Women in Agriculture' by PJTSAU, Hyderabad on Apr 29, 2019
	Adoption of Natural Farming and its Effect on Crop Yield and Farmers' Livelihood in India	Brainstorming Workshop on 'Zero Budget Natural Farming- Myth or Reality' by NAAS, New Delhi on Aug 21, 2019

Name of Staff	Lecture	Programme/ Organized by
Sanjiv Kumar	Data Analysis	Workshop on Research Methodology and data processing in Social Sciences by MANUU, Hyderabad on Jan 30, 2019
	Data Analysis using MS-Excel	Workshop on MOOCs, e-Content Development and Open Educational Resources by MANUU, Hyderabad on Mar 14, 2019
	Statistical Data Analysis using Excel	ICAT for Agriculture Development by MANAGE, Hyderabad on Apr 23, 2019
SK Soam	Project Logical Framework	Project Management Skills for Officers of Agriculture and Allied Sector by MANAGE, Hyderabad on June 26, 2019
S Senthil Vinayagam	New Dimension in Agricultural Extension Management: Innovations	Training Program on Computer Applications in Agriculture by MANAGE, Hyderabad on Feb 20, 2019.
	Sustainability Issues in Agriculture & Food Security	International Training Programme on Participatory Planning for Poverty Reduction and Sustainable Development by NIRD, Hyderabad on Mar 14, 2019
	Open Educational Resources	Workshop on MOOCs, e-Content Development and Open Educational Resources by MANUU, Hyderabad on Mar 15, 2019
	Technology Blended Learning & OER	Workshop on MOOCs, e-Content Development and Open Educational Resources by MANUU, Hyderabad on Jun 18, 2019
	Agri-Preneurship Development through Education & Extension	Skill Development Training Program on Agro Based Entrepreneurship by NIRD, Hyderabad on May 16, 2019
	Transformation of Innovation in Agriculture to Agribusiness	Skill Development Training Program on Agro Based Entrepreneurship by NIRD, Hyderabad on May 16, 2019
	Approaches for Doubling the Farmer's Income	Training Program on Agricultural Extension Approaches for Agricultural Extension Officials' by MANAGE, Hyderabad on Jun 17, 2019
	Leveraging Agricultural Extension Mechanism for promotion of Agribusiness among Rural Youth	Training Program on Agriculture as Business for Rural Youth by MANAGE, Hyderabad on Aug 20, 2019
	IT for Agriculture and Rural development	International Training Programme on 'Good Governance for Management of Rural Development Programmes' by NIRD, Hyderabad on Sep 05, 2019
	Extension Approaches for DFI & Transfer of Dryland Farm Technologies	Model Training Course on Good Agricultural Practices for Biotic Stress Management and Productivity Improvement in Millets As Nutricereals by IIMR, Hyderabad on Sep 25, 2019
	Extension Strategies for Enhancing Market Access	Extension Strategies for Linking Farmers to Market by MANAGE Hyderabad on Oct 18, 2019
	Market Led Extension Strategies in Doubling the Farmers' Income	Consultancy Training Program on 'New Dimensions of Agricultural Marketing' by MANAGE, Hyderabad on Nov 26, 2019

Name of Staff	Lecture	Programme/ Organized by
Surya Rathore	Skill Oriented Trainings at ICAR - NAARM	State Level Workshop on Skill Development in Agriculture & Allied Sectors by State Agricultural Management Extension Training Institute, Telangana, Hotel Marriott, Hyderabad on Jan 29, 2019
	Monitoring, Evaluation & Impact Assessment of Extension Programmes	International Conference on Advances in Agriculture & Allied Sciences Research by Samagra Vikas Welfare Society, Lucknow from Rama University, Kanpur (U.P.) during Feb 23- 24, 2019
	Unleashing the potential of ICT for Graduate Learning	Training programme on Communication Management for Entrepreneurship Development by College of Community & Applied Sciences, MPUA&T, Udaipur (Rajasthan) on Apr 22, 2019
	Strengthening Academia-Industry Linkages in Agricultural and Community Sciences	National Workshop on Prospect of Enrolment and Evocation in Home Science and Agricultural Education, Rajasthan University, Jaipur (Rajasthan) during Nov 18-19, 2019
	Nurturing Entrepreneurial Ecosystem in Agricultural Universities	ISEE National Seminar on Socio-Digital Approaches for transforming Indian Agriculture by Indian Society of Extension Education, Banda University of Agriculture & Technology & CCSHAU, Hisar, (Haryana) during Nov 20- 22, 2019
S Ravichandran	Experimental Data Analysis Using SAS	Refresher Course by Madras Veterinary College, Chennai on Feb 27, 2019
	Nonlinear Statistical Modelling for Pests and Diseases Forecasting	World Bank Funded Project 'Modelling and ICT Applications in Forecasting Pest and Diseases' by GKVK, Bangalore during Feb 12-13, 2019
	Big Data Analytics in Agriculture in the Current Status and Emerging Needs	World Bank Funded Project 'Modelling and ICT Applications in Forecasting Pest and Diseases' by GKVK, Bangalore during Feb 12-13, 2019

8.2 Publications

8.2.1 Research Papers in Refereed Journals

- Abhilash, K.R., Sankar, R., Purvaja, R., Deepak, S.V., Sreeraj, C.R., Krishnan, P., Sekar, V., Biswas, A.K., Kumarapandiyam, G., Ramesh, R. (2019). Impact of long-term seaweed farming on water quality: a case study from Palk Bay, India. *Journal of Coastal Conservation*. 23(2): 485–499.
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- 8.2.2 Books and Monographs**
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- Sumanthkumar V.V. and Ram Kiran Dhulipalla (2019). *GreenPHABLET™ powered by the Green SIM - An Appliance for knowledge revolution in farming community* at International Conference on Global Research Initiatives for Sustainable Agriculture & Allied Sciences during 20-22 October 2019 by Astha Foundation at NAARM, Hyderabad.
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8.2.7 Research Project Report and Working Papers

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8.2.8. Policy Briefs

Murthy GRK, Senthil Vinayagam S, Thammi Raju D, Krishnan M, Pandey PS, Srinivasarao Ch. (2019). *Digital Learning for Vocational and Informal Education in Agriculture*. ICAR-National Academy of Agricultural Research Management, Hyderabad.

Srinivas K., Ranjit Kumar, Manju Gerard and Srinivasa Rao, Ch. (2019). *Fostering Entrepreneurship in Agriculture through Incubation Centres in National Agricultural Research & Education System (NARES)*. Policy brief No. 5, August 2019. ICAR-NAARM, Hyderabad.

8.2.9 Copyrights

Title	Registration Number & Date	Authors
Agro-biodiversity Knowledge Management Portal	SW-12734/2019 19 August 2019	M Balakrishnan, SK Soam and PD Sreekanth
AHP Analyser	SW-12766/2019 26 Aug 2019	SK Soam, Ch Sridhar Babu, T Raj Kumar and B Raghupathi
Project Logframe Writer	SW-12751/2019 22 Aug 2019	SK Soam, B Raghupathi and T Raj Kumar
Research Concept Writer	SW-12735/2019 19 Aug 2019	SK Soam, B Raghupathi and T Raj Kumar

8.3 Other Roles

Faculty	Other Roles
Alok Kumar	Joint Convenor in organizing ISEE National Seminar-2019 on Socio-Digital Approaches for Transforming Indian Agriculture held at CCSHAU, Hisar, Haryana during November 20-22, 2019.
	Co-Chairman, Technical Session on Policies and Economics of Profitable Agriculture of the International Conference on Global Research Initiatives for Sustainable Agriculture & Allied Sciences during October 20-22, 2019.
	Member, Workshop Organizing Committee, International Workshop on Strengthening International Agribusiness Trade: Stakeholder Dialogue and Partnerships towards SDGs, during November 14-15, 2019 at TERI School of Advanced Studies, New Delhi
	Rapporteur, Technical Session on Clean Energy and Bio-economy of 89 th Annual Session of NASI and Symposium on Science & Technology based Entrepreneurship Development during December 21-23, 2019 organized at ICAR-NAARM, Hyderabad
	Rapporteur, Plenary Session of ISEE National Seminar-2019 on Socio-Digital Approaches for Transforming Indian Agriculture held at CCSHAU, Hisar, Haryana during November 20-22, 2019
D Thammi Raju	Member of Organizing Committee constituted for 248 th ICAR Governing Body
	Member of a Committee on Utilization of budget under NAHEP component 1C – Innovation Grants, under NAHEP, ICAR
	Bid Evaluation Committee of NRC Meat, Hyderabad
	Expert Member for 30 th Extension Education Advisory Council of TANUVAS, Chennai
	Thesis Evaluator in SVVU, AAU
	Reviewer for International Journal of Livestock Research
	Reviewer for International Transactions in Operational Research
GRK Murthy	Doctoral Thesis Evaluator in the Farm Machinery and Power specialisation for TNAU, Coimbatore
	Nominated by ICAR – NAARM to a-Idea as Member.
I Sekar	Coordinator, Session on Science and Technology based Entrepreneurship Development in NASI Symposium during December 21-23, 2019 organized jointly by The National Academy of Sciences, India (NASI) and ICAR-NAARM, Hyderabad.
	Expert member of Evaluation Committee of Career Advancement Scheme for Scientist at CIBA, Chennai on March 3, 2019.
	Editor, Journal of Association for Management of Agricultural Research and Agripreneurship (AMARA), Hyderabad.
	Thesis Evaluator in Agricultural Economics for Tamil Nadu Agricultural University, Coimbatore, Tamil Nadu.
	Member, a-Idea, nominated by ICAR- National Academy of Agricultural Research Management, Hyderabad.

Faculty	Other Roles
KH Rao	Expert member in Dairy Technology and Dairy Business Management from November 1-3, 2019 organized by National Testing Agency, Ministry of HRD, Govt. of India
	Coordinated Exhibition of NAARM Career Development and Academic Programs in Kisan mela and exhibition from November 14-16, 2019 at CMFRI, Cochin
	Evaluator, External examiner for Viva-voce examination in Program on Industrial Co-operative Management ICM, Hyderabad on October 9, 2019
M Balakrishnan	Reviewer, NAAS Proceedings, India Section B: Biological Sciences
	External examiner for Bharthisadan University and Bharathiyar University, Tamil Nadu and Ph.D. Thesis Evaluation for Anna University, Tamil Nadu
	Chairman, Technical Session II Animal Husbandry, International Conference on Agriculture and Animal Husbandry on Advances in Agriculture and Animal Sciences towards Global Food Security from August 28-30, 2019 at University of Hyderabad
P Krishnan	Expert member in the Brainstorming Meeting on Fisheries Informatics held at National Fisheries Development Board (NFDB), Hyderabad on 11 th Jan 2019.
	Expert Member in the Committee to select Scientists for National Centre for Sustainable Coastal Management (NCSCM), Chennai during 04-05 Mar 2019
	Participated in the First Meeting of the Project Monitoring Committee (PMC) held at New Delhi on 07 Mar 2019 by DBT
	Participated in the Fourteenth Core Project Advisory Committee Meeting held on April 15, 2019 at TERI, New Delhi
	Participated in the Expert Consultation Convened by UNDP, New Delhi at National Biodiversity Authority (NBA), Chennai on 16 July 2019
	Participated in the meeting held in the Chamber of DDG (Edn), ICAR for discussing the EFC 2020-25 of ICAR-NAARM on 09 Aug 2019
	Convened an Expert Consultation on Strengthening Natural Resource Management (SNRM) - A Framework for Mainstreaming Biodiversity Conservation in the Development Schemes of Agricultural Sector in India
	Convened an Expert Consultation on Developing a Framework for Enhancing Efficiency of Agricultural Research Management in ICAR on 12 Sep 2019
	Expert member in the Assessment Committee for the Discipline of Fisheries for the Promotion of Teachers under Career Assessment Scheme at Dr. Rajendra Prasad Central Agricultural University, Pusa, Bihar on 18 Sep 2019
	Session Chair during the International Conference on Asia Pacific Aquaculture (APA-2019) organized by World Aquaculture Society at Chennai Trade Centre, Chennai during 19-21 June 2019
	Co-Convener in the Stakeholders' Meeting on Entrepreneurship Development held at National Research Centre on Orchids, Pakyong, Sikkim on 07 June 2019
	Expert Member in the Working Group formed by the National Institute for Disaster Management (NIDM), New Delhi on Preparation of National Agriculture Disaster Management Plan for MoAGFW on 12 July 2019.
	Member Secretary in the Committee constituted by ICAR for reviewing the Scientific Cadre of ICAR-NAARM.
	Member in the Expert Committee constituted by National Board for Fisheries Development (NFDB) for drafting the National Mariculture Policy

Faculty	Other Roles
Sanjiv Kumar	Subject Editor of Agri business and Rural Management in Pantnagar Journal of Research
S Senthil Vinayagam	Member of Organizing Committee constituted for 248 th ICAR Governing Body
	Member of Ranking Framework and Proforma developed by NAAS
	Reviewer of Journal of Rural Development, NIRD & Soybean Research, Indore
VV Sumanth Kumar	Treasurer, a-Idea.
	Treasurer, AMARA Society
	Nodal Officer for Ek Bharat Shrestha Bharat.

9 FINANCIALS

9.1 Budget Allocation and Expenditure during Jan-Dec 2019 (Rs. in lakh)

S.No.	Head of Account	Budget	Expenditure
A. Grant in Aid - Capital			
1	Works	210.13	210.13
2	Equipments	76.37	76.37
3	Information Technology	21.76	21.76
4	Library Books & Journals	100.88	100.88
5	Vehicles & Vessels	30.40	30.40
6	Furniture & Fixtures	21.94	21.84
Total Capital		461.37	461.37
B. Grants in Aid - Salaries (REVENUE)			
1	Establishment Expenses		
2	i.Establishment Charges	2277.87	2277.87
3	ii.Wages	174.14	174.14
4	iii.OTA	0.16	0.16
Total Salaries		2452.16	2452.16
C. Grant in Aid - General (REVENUE)			
1	Pension & Other Retirement Benefits	3717.18	3714.18
2	Domestic TA / Transfer TA	77.19	77.19
3	Research & Operational Expenses	196.59	196.59
4	Administrative Expenses	1651.65	1651.65
5	HRD	23.35	23.35
6	Miscellaneous Expenses	148.46	148.46
Total Grants in Aid - General		5811.42	5811.42
Grand Total (Capital+Salaries+Revenue)		8724.95	8724.95

9.2 Resource Generation

9.2.1 Budget of Sponsored Research Projects

(Rs. In Lakhs)

S.No	Name of the Project	Budget outlay	Fund available/ received	Expenditure
1	RKVY - Telangana State Govt. Developing SAIDP Project	24.72	4.66	4.66
2	Developing Guidelines & Methodologies for IP Policies for PJTSAU	5.00	3.26	3.26
3	PJTSAU Food Consumption Project	10.17	4.64	0.00
4	Environmental Impact Assessment of Tobacco curing	20.87	7.05	7.05
5	R&D Strategies and Policies under SERB, DST	12.42	4.42	4.42
6	Agricultural Research and Development Infrastructure in Select Indian States under NSTMIS Scheme of DST	30.00	12.59	8.34
7	RGMVP, Raibareli	2.12	2.12	2.12
8	IFPRI Sponsored Project - ASTI in Indian NARES- ASTI Project	28.35	14.14	2.18
9	Consultancy Project on Adoption of Insect Resistance Management (Mhyco, Monsanto)	30.68	20.03	14.30
10	NIPHM Project on Plant Biosecurity through MOOC Platform	7.20	6.88	4.24
11	MANAGE Project on Agri-Warehousing Management	9.90	8.37	0.00
12	PJTSAU Digital Content for MOOCS	5.31	5.31	0.81
13	ITC Economic Analysis of Eucalyptus and Subabul based Agroforestry Ecosystems in Telangana & A.P	11.28	10.95	4.66
14	Profiling the Resources Devoted in Agricultural Research and Development of National framework for Evidence-based S&T Governance in India Scheme of DST	24.88	21.00	13.88
15	Skill Development in Bioinformatics Infrastructure Facility on Agril. Bio Technology under DBT	19.42	19.42	12.89
16	Adoption of Natural Farming and its effect on Crop Yield and Farmers Livelihood in India NITI Ayog Project	35.00	12.60	12.60
17	Developing MOOC Courses on Commercial Dairy farming & Milk processing & Value Edition	23.36	13.00	6.17
18	NIPHM Project on Developing online course on Rodents & Household Pest Management	7.43	3.40	0.00
19	IFPRI Sponsored Project on Developing a National Dashboard for Evidence-Based Governance of Agricultural Research in India	20.67	13.78	0.00
20	Strengthening Natural Resource Management (SNRM)	29.50	10.80	2.38
21	ICAR-NAARM-BCIL, Strengthening Natural Resource Management (SNRM)	29.50	26.10	0.00
Total		387.78	224.52	103.96

9.2.2 Off - Campus and Sponsored Programmes

(Rs. In Lakhs)

Sl.No	Name of the Programme	Total Budget Received
1	Emotional Intelligence for Personal & Work Excellence for Faculty of CIFE, Mumbai from 11-13 Feb, 2019	1.63
2	FDP at UAHS, Shivamogga, Karnataka from 4-8 Feb, 2019	2.23
3	Digital Teaching Techniques for UAS, Bangalore from 19-23 Feb, 2019	2.94
4	DWRP in Agril.Res. for Faculty members of RVSKVV, Gwalior from 7-9 March, 2019	2.60
5	Entrepreneurial Skill Development Among Agri-Graduates of ANGRAU-2 programmes from 14.02.2019 to 06.03.2019 & 20.03.2019 to 09.04.2019	22.50
6	ICT & its Application in Livestock Production and Health under NAHEP for students of SVVU, Tirupathi 16-23 May, 2019	4.28
7	Training Programme on Scientific Communication on DWRP for Faculty members of Bihar Animal Sciences University, Patna 31 -05 -2019 to 02-06 2019 & 16-18 Aug, 2019.	6.00
8	Training Programme on DWRP in Agriculture / Agricultural engineering Research for PJTSAU faculty of College of Agriculture, Awararaopeta, Saifabad, Jagityal, Sangareddy & (2 programs) 5-7 April, 2019 & 26-28 April, 2019.	12.31
9	Training Programme on Science Communication and Scientific presentation of PJTSAU faculty of College of Agriculture, Awararaopeta, Saifabad, Jagityal, Sangareddy (2 programs) 16-18 May, 2019 & 11-13 July, 2019.	
10	Training Programme under NAHEP for National Knowledge Management Centre for Agriculture Education & Research University Librery, PJTSAU, Hyderabad 16-20, July, 2019	5.12
11	Off Campus Programme of Competency Enhancement of Admin & Fin staff of Agril. University, Jodhpur from 29 th June, 2019 to 05 th July, 2019	3.53
12	M/s Coromandal International Ltd, CIL- NAARM collabroating Training Programme for Young Agri-business Professionals-3 Programs	10.27
13	M/s Coromandal International Ltd, CIL- NAARM collabroating Training Programme for Retail sales Managers (RSMs) SEED program-3 Programs	
14	Induction course for newly Recruited Assistant professor of Telangana State Forest Academy held from 8-17 May, 2019.	3.07
15	Training Programme for Officers of NABARD on Agriculture & Agri Market Scenario in India from 26-28 June, 2019	2.25
16	Training on Entrepreneurial Skill Development of Graduating Students of SVVU, Tirupathi from 21-26 October, 2019	4.45
17	M/s Coromandal International Ltd, CIL- NAARM collabroating Training Programme for New entrants Assimilation Training Phase-I from 22-26 July, 2019	1.76
18	PRA & Communication Strategies including Integrated Watershed Management in the 2018-20 batch of IFS Probationary officers from 26-29 September, 2019 IGNFA, Dehradun	7.09
19	Off Campus Training Programme on 'Market Intelligence' at HUC, Chennai from 13-18 Dec, 2019	3.00
20	Off Campus Training Programme on DWRP for Faculty members of MPUAT, Udaipur, Rajastan from 03-05 January, 2020	2.87
21	Off Campus Training Programme on DWRP for Faculty members of Kamadhenu University, Amereli, Gujarat from 29-31 January, 2020	2.87
Total		100.78

9.2.3 Resource Generation from Educational Programmes

S.No	Educational Programmes	Gross Receipts (Rs. In Lakhs)
1	PGDMA 2017-19	28.18
2	PGDMA 2018-20	117.03
3	PGDMA 2019-21	187.55
4	PGDTMA 2019	8.69
5	PGDETM-I	1.71
	Total	343.16

9.2.4 Resource Generation from Other Activities

S.No	Receipts from Other Activities	(Rs. In Lakhs)
1	Farm Produce	3.26
2	Guest House and Quarters	37.41
3	Institute Training Programmes	63.78
4	Services rendered	6.42
5	Miscellaneous Receipts	200.04
	Total	310.91

9.2.5 Net Receipts from Sponsored Research Projects

S.No	Sponsored Research Projects	(Rs. In Lakhs)
1	RKVY (TS) Project SAIP	4.31
2	RGMVP, Raibareli	0.69
3	Developing Guidelines & Methodologies for IP Policies for PJTSAU	0.65
4	Environmental Impact Assessment of Tobacco Curing	3.74
5	Net Receipts from Educational Programme PGDMA 2017-19	117.96
6	Net Receipts from IRGS Programmes	12.80
7	Adoption of Insect Resistance Management in BT cotton Cultivation in India	0.90
8	Profiling the Resources Dev. to Agril. R&D -S&T Governance in India DST Scheme	1.56
	Total	142.61

9.3 Budget Allocation and Expenditure in Plan Schemes

(Rs. In Lakhs)

Sl. No.	Name of the Scheme	Budget Allocation	Funds Available / Received including Opening balance for 2019-20 (Upto Dec'19)	Expenditure (Upto Dec'19)
1	National Agril. Innovation Fund (Component-I)-ZTMC	12.75	12.44	8.36
2	National Agril. Innovation Fund (Component-II)-ABI	32.00	27.50	5.44
3	Technology Enhanced Learning in Agril. Edn - TELAgE Project	9.19	6.89	4.66
4	XII Plan EFC Main Scheme KRISHI Project	26.21	24.33	12.37

Sl. No.	Name of the Scheme	Budget Allocation	Funds Available / Received including Opening balance for 2019-20 (Upto Dec'19)	Expenditure (Upto Dec'19)
5	Management & Impact Assessment of Farmer First Project	11.79	10.73	6.43
6	ICAR Extra Mural Research Project Competency Analysis among Faculty of Agricultural University in Enhanced Learning	4.34	4.34	1.49
7	NASF, New Delhi - Entrepreneurship Development Through Farmer Led Innovations	7.00	4.72	2.82
8	Summer / Winter Short courses	9.17	8.85	2.39
9	NAHEP	554.21	473.86	119.35

10 Personnel

10.1 Human Resources at the Academy (as on 31.12.2019)

Category	Sanctioned Strength	In position	Vacant
Scientific - RMP	2	1	1
Scientific - Faculty	60	33	27
Technical -	**34	30	4
Administrative	49	32	17
Skilled Support Staff	33	29	4
Total	178	125	53

** Excludes 13 positions fallen under dying cadre

Research Management Positions

1. Dr Ch. Srinivasa Rao (Director)
2. Dr S K Soam (Incharge Joint Director)

Scientific Staff

Agribusiness Management Division 1. Dr. Ranjit Kumar, Head 2. Dr. B. Ganesh Kumar, Principal Scientist 3. Dr. Tava Srinivas, Principal Scientist 4. Dr. N. Sivaramane, Principal Scientist 5. Dr. P.C. Meena, Principal Scientist 6. Dr. Sanjiv Kumar, Scientist	Extension Systems Management Division 1. Dr. Bharat S. Sontakki, Head 2. Dr. (Smt.) N. Sandhya Shenoy, Principal Scientist 3. Dr. Surya Rathore, Principal Scientist 4. Dr. P. Venkatesan, Principal Scientist	3. Dr. S. Ravichandran, Principal Scientist 4. Dr. M. Balakrishnan, Principal Scientist 5. Dr. N. Srinivasa Rao, Principal Scientist 6. Dr. P.D. Sreekanth, Principal Scientist 7. Dr. B.S. Yashavanth, Scientist
Education Systems Management Division 1. Dr. S. Senthil Vinayagam, Head (I/c) 2. Dr. G.R. Ramakrishna Murthy, Principal Scientist 3. Dr. D. Thammi Raju, Principal Scientist 4. Dr. V.V. Sumanth Kumar, Senior Scientist	Human Resources Management Division 1. Dr. R.V. S Rao, Head (I/c) 2. Dr. P. Ramesh, Principal Scientist 3. Dr. K.H. Rao, Principal Scientist 4. Dr. Alok Kumar, Principal Scientist	Research Systems Management Division 1. Dr. I. Sekar, Head 2. Dr. K. Kareemulla, Principal Scientist 3. Dr. K. Srinivas, Principal Scientist 4. Dr. M.B. Dastagiri, Principal Scientist 5. Dr. P. Krishnan, Principal Scientist 6. Dr. Manju Gerard, Principal Scientist 7. Dr. Umesh Hudedamani, Scientist 8. Dr. Ramesh Naik, Scientist
	Information and Communication Management Division 1. Dr. SK Soam, Head 2. Dr. A. Dhandapani, Principal Scientist	

Administrative Staff

Joint Director (Admn.) & Registrar

Shri. C. Muralidharan

Chief Finance & Accounts Officer

Sh. S. George (till July 31, 2019)
Sh. Z.H. Khilji (from August 28, 2019)

Senior Administrative Officer

Sh. J.N.L. Das

Asst. Admn. Officer

1. Sh. P.G. Kohad
2. Sh. M. Dinesh
3. Mrs. K.K. Rukmani Ammal

Private Secretary

1. Sh. N. Raghunath
2. Mrs. A. Mercy
3. Mrs. T. Vanisri

Junior Accounts Officer

1. Mrs. N. Vijayalakshmi

Assistants

1. Sh. T. Srinivas
2. Sh. G. Raj Reddy
3. Sh. C. Phani Raj
4. Mrs. B. Padma Saroja
5. Sh. P. Srinivasu
6. Sh. K.R. Ghana shyam
7. Sh. R. Chandra Babu
8. Sh. C. Julius Samuel
9. Sh. M. Sridhar

Personal Assistants

1. Sh. T. V. Ramadas
2. Mrs. Y. Anuradha
3. Mrs. S. Sesha Sai
4. Mrs. V. Shailaja
5. Mrs. S. Shanthi

Upper Division Clerks

1. Ms. Y. Gayathri
2. Sh. M.K. Samson
3. Mrs. Rajashri Bokde
4. Sh. K. Suryanarayana

Lower Division Clerk

1. Sh. P. Swamy
2. Sh. M. Narsing Rao
3. Sh. M. Ashok
4. Sh. C. Bikshapathi

Technical Staff

Grade T-9 (Category-III)

1. Dr. M.A. Basith, Chief Technical Officer (Jr. Farm Superintendent)
2. Dr. P. Vijender Reddy, Chief Technical Officer (Research Assistant)
3. Sh. P. Namdev, Chief Technical Officer (Arts)
4. Sh. K.V. Kumar, Chief Technical Officer (Statistical Assistant)
5. Sh. Sohail Ahmed Khan, Chief Technical Officer (Junior Engineer-Civil)

Grade T (7-8) (Category-III)

1. Mrs. G. Aneja, Assistant Chief Technical Officer (Assistant Editor)
2. Sh. M. Shekhar Reddy, Assistant Chief Tech. Officer (Dark Room Assistant)
3. Dr. Ahire Laxman Mahar, Assistant Chief Technical Officer (Horticultural Technical Assistant)
4. Sh. M. Ravi, Sr. Technical Officer (Photographer-cum-Artist)
5. Sh. Sham Bahadur, Asst. Chief Technical Officer (Catering incharge)
6. Sh. P. Mohan Singh, Asst. Chief Technical Officer (Computer Assistant)
7. Ms. Savithri Murali, Asst. Chief Technical Officer (Catering in-charge)

Grade T-6 (Category-III)

1. Sh. Ravi Viswanathan, Sr. Technical Officer (Editor cum-Information Officer)

Grade T-5 (Category-II)

1. Sh. N. Naresh Kumar, Technical Officer (VCR-cum-Tele cine Operator)
2. Sh. K. Obulapathi, Technical Officer (Technician Electrical)
3. Sh. K. Shivaiah, Technical Officer (Technician)
4. Sh. D. Rajagopal Rao, Technical Officer (Media Operator)
5. Sh. S. Rajukumar, Technical Officer (Electronic Computer Operator)
6. Sh. T. Laxman, Technical Officer (Driver)
7. Sh. A. C.P. Rama Nageswara Rao, Sr. Technical Assistant (Lineman)

Grade T-4 (Category-II)

1. Sh. Pitla Srinivas, Sr. Technical Assistant (Proof Reader)
2. Sh. G. Muthyalu, Sr. Technical Assistant (Driver)
3. Sh. N. Ashok, Sr. Technical Assistant (Driver)
4. Sh. N. Prabhakar Rao, Sr. Technical Assistant (Plumber)
5. Sh. M. Srinivasa Rao, Sr. Technical Assistant (Pump Driver)
6. Sh. K. Veera Narasaiah, Sr. Technical Assistant (Carpenter cum-Painter)

Grade T-3 (Category-II)

1. Sh. B. K. Venkatram, Technical Assistant (Pump Driver)
2. Sh. R. Siva Prasad, Technical Assistant (Driver)
3. Sh. Dhanam Murahari Technical Assistant (A.V. Operator)

Skilled Support Staff

1. Sirigiri Venkatesham
2. Mrs. B. Santhamma
3. Mrs. S. Shakuntala
4. Sh. Kumba Satyanarayana
5. Sh. K. Pentaiah
6. Sh. P. Yadaiah
7. Sh. J. Chandraiah
8. Sh. M. Krishnaiah
9. Sh. R. Sattaiah
10. Ms. Indirapati Bharatamma
11. Sh. C. Chandramouli
12. Sh. G. Dasaratha
13. Ms. Nazia Parveen
14. Sh. L. Satyanarayana
15. Ms. C. Kausalya
16. Sh. M. Ganesh Kumar
17. Sh. Chilumula Venkatesham
18. Sh. Sirigiri Narsimha
19. Sh. Kyasam Satyanarayana
20. Sh. B. Premdas
21. Mrs. A.Saroja
22. Sh. S. Nayab Rasool
23. Sh. B. Ashok
24. Sh. K. Daniel
25. Smt. K. Mariyamma
26. Sh M. Satyanarayana
27. Sh D. Srisailam
28. Sh B. Ratnaiah
29. Sh D. Babaiah

Appointments / New Joinings

1. Shri D. Babaiah, CLTS has been regularized against the vacant post of Skilled Support Staff w.e.f. 22.02.2019 (FN)
2. Shri B. Ratnaiah, CLTS has been regularized against the vacant post of Skilled Support Staff w.e.f. 22.02.2019 (FN)
3. Shri C. Muralidharan, joined the Academy as Joint Director (Admn.) & Registrar w.e.f. 13.05.2019 (FN)

4. Shri Z. H.Khilji, joined the Academy as Chief Finance & Accounts Officer w.e.f. 28.08.2019 (FN)
5. Dr. Umesh Hudedamani, joined the Academy as Scientist w.e.f. 25.11.2019
6. Dr. Tavva Srinivas, joined the Academy as Principal Scientist w.e.f. 02.12.2019
7. Dr. M. Ramesh Naik, joined the Academy as Scientist w.e.f. 03.12.2019

Promotions

1. Smt. S. Shanthi, Stenographer Grade III has been promoted as Personal Assistant w.e.f. 28.02.2019 (FN)
2. Smt. V.S hailaja, Stenographer Grade III has been promoted as Personal Assistant w.e.f. 28.02.2019 (FN)
3. Dr .N. Sivaramane, Sr. Scientist (Agribusiness & Marketing Management) has been promoted under CAS to the grade of Principal Scientist – RGP10000 w.e.f. 28.03.2018
4. Dr. P.C. Meena, Sr. Scientist (Financial Management) has been promoted under CAS to the grade of Principal Scientist – RGP10000 w.e.f.24.04.2017
5. Dr. P. Venkatesan, Sr.Scientist (Rural Sociology) has been promoted under CAS to the grade of Principal Scientist – RGP10000 w.e.f. 05.05.2017
6. Shri P. Mohan Singh, T-6 STO (Computer Assistant) has been Promoted to the Grade of T (7-8) ACTO w.e.f. 01.01.2016

7. Shri P. Namdev, T (7-8) ACTO (Artist) has been promoted to the Grade of T-9 CTO w.e.f. 27.07.2017
8. Shri B.K. Venkatram, T-3 Technical Assistant (Pump Driver) has been promoted to the grade of T-4 Senior Technical Assistant w.e.f. 29.06.2019
9. Smt. K.K. Rukmani Ammal, PA has been promoted as Assistant Administrative Officer w.e.f. 05.09.2019.
10. Shri A.C.P. Rama Nageswara Rao, T-4 Senior Technical Assistant (Lineman) has been promoted to the grade of T-5 Technical Officer w.e.f. 01.01.2013
11. Shri V.V. Sumanth Kumar, Scientist has been promoted under CAS to the grade of Sr. Scientist (RGP8000) w.e.f. 14.10.2014 and Sr. Scientist (RGP9000) w.e.f. w.e.f. 16.04.2017.
12. Shri Sohail Ahmad Khan,T (7-8) Assistant Chief Technical Officer [Jr. Engineer (Civil) has been promoted to the Grade of T-9 Chief Technical Officer w.e.f. 01.01.2019
13. Smt. Savithri Murali, T-6 Senior Technical Officer (Catering In charge) has been promoted to the Grade of T (7-8) Assistant Chief Technical Officer, w.e.f.01.01.2017

Financial Up gradations (under MACP Scheme for Admn. and Skilled Support Staff)

1. Shri M. Narsing Rao, LDC, 3rd MACP, 2400 (as per 6 CPC). Level: 4 (as per 7 CPC) from 26.08.17
2. Shri C. Bickshapathi, LDC, 3rd MACP, 2400 (as per 6 CPC) Level: 4 (as per 7 CPC), from 26.08.17
3. Shri M. Ashok, LDC, 3rd MACP, 2400 (as per 6 CPC), Level: 4 (as per 7 CPC) from 26.08.17
4. Mrs. B. Padma Saroja, Assistant, 3rd MACP, 4600 (as per 6 CPC), Level: 7 (as per 7 CPC) from 26.08.17
5. Shri P. Srinivasu, Assistant, 3rd MACP, 4600 (as per 6 CPC), Level: 7 (as per 7 CPC) from 11.01.18
6. Mrs. A. Mercy, Private Secretary, 3rd MACP, 4800 (as per 6 CPC), Level: 8 (as per 7 CPC) from 01.09.18
7. Mrs. Vanisri T, Private Secretary, 3rd MACP, 4800 (as per 6 CPC), Level: 8 (as per 7 CPC) from 01.09.18
8. Shri T. V. Ramdas, Personal Assistant, 3rd MACP, 4800 (as per 6 CPC) from 01.09.18
9. Mrs. Y. Anuradha, Personal Assistant, 3rd MACP, 4800 (as per 6 CPC), Level: 8 (As per 7 CPC) from 01.09.18
10. Mrs. Gayathri Y, UDC, 3rd MACP, 4200 (as per 6 CPC) Level: 6 (as per 7 CPC) from 08.01.19
11. Shri M. K. Samson, UDC, 3rd MACP, 4200 (as per 6 CPC), Level: 6 (as per 7th CPC) from 25.01.19
12. Mrs. B. Bharatamma, Skilled Support Staff (Retd.), 3rd MACP, Level: 4 (as per 7 CPC) from 26.08.17

13. Shri S. Jangaiah, Skilled Support Staff (Retd.), 3rd MACP, Level: 4 (as per 7 CPC) from 26.08.17
14. Shri G. Pentaiah, Skilled Support Staff, 3rd MACP Level: 4 (as per 7 CPC) from 26.08.17
15. Shri G. Dasaratha, Skilled Support Staff, 3rd MACP Level : 4 (as per 7 CPC) from 26.08.17
16. Shri J. Chandraiah, Skilled Support Staff, 3rd MACP Level: 4 (as per 7 CPC) from 27.08.17
17. Shri C. Chandramouli, Skilled Support Staff, 3rd MACP Level: 4 (as per 7 CPC) from 26.08.17
18. Late Krishnaiah M. Ex-Skilled Support Staff, 3rd MACP Level : 4 (as per 7 CPC) from 26.08.17
19. Shri R. Sattaiah, Skilled Support Staff, 3rd MACP Level: 4 (as per 7 CPC) from 26.08.17
20. Shri L. Satyanarayana, Skilled Support Staff 3rd MACP Level: 4 (as per 7 CPC) from 25.06.18
21. Mrs. C. Kausalya, Skilled Support Staff 2nd MACP Level: 3 (as per 7 CPC) from 04.07.18
22. Mrs. I. Bharatamma, Skilled Support Staff 3rd MACP Level: 4 (as per 7 CPC) from 06.03.19

Grant of Advance Increment under Technical Services

1. Dr. V. Murali, Ex- T-9, Chief Technical Officer (Farm Superintendent), One Increment (@ Rs.468/-) in the grade of T-9 from 01.09.2006
2. Dr. M. A. Basith, T-9, Chief Technical Officer (Jr. Farm Superintendent), One Increment (@ Rs.1, 053/-) in the grade of T-9 from 01.07.2017

Retirements

1. Shri Phool Kumar, SSS, retired on superannuation w.e.f. 31.01.2019 (AN)
2. Shri Ashish Roy, Joint Director (Admn.) & Registrar retired on superannuation w.e.f. 28.02.2019
3. Smt. C. Narsu Bai, Skilled Support Staff (SSS), retired on superannuation w.e.f. 30.06.2019 (AN)
4. Dr. V. Murali, T-9 (holding the post of T-6), retired on superannuation w.e.f. 30.06.2019 (AN)
5. Smt. Jessie Ecclicia, Assistant, retired voluntarily w.e.f. 03.07.2019 (FN)
6. Shri S. George, Chief Finance & Accounts Officer retired on superannuation w.e.f. 31.07.2019 (AN)
7. Shri B.S.N. Murthy Technical Officer retired on superannuation w.e.f. 31.08.2019 (AN)
8. Shri S. Jangaiah, Skilled Support Staff retired on superannuation w.e.f. 30.09.2019 (AN)
9. Smt. B. Bharathamma Skilled Support Staff retired on superannuation w.e.f. 30.11.2019 (AN)
10. Dr. M. Krishnan, Head of the Division, Education Systems Management retired on superannuation w.e.f. 30.11.2019 (AN)

Transfers

1. Shri W. Sreenivasa Bhat, Administrative Officer was relieved w.e.f. 20.07.2019 (AN) on his promotion as Senior Administrative Officer and posting at ICAR-CIFT, Kochi
2. Dr. V.K.J. Rao, Principal Scientist was relieved w.e.f. 30.11.2019 (AN) on his own request transfer to IIHR, Bangalore

Clearance of Probationary Period (PP) and Confirmation

1. Dr. (Mrs.) Manju Gerard Probation Period cleared w.e.f. 13.10.2018 and confirmed w.e.f. 14.10.2018 in the post of Principal Scientist

Others – Best Performer Awards for 2018-19

Scientist / Faculty

1. Dr. G.R.R. Murthy
Best Faculty (PGDMA)
2. Dr. P. Krishnan
Best Scientist

Admn.& Finance

1. P.G. Kohad, Officer
2. G. Raj Reddy, Non-Officer

Technical

1. D. Rajagopal Rao
T-1 to T-5 Grade
2. M. Shekhar Reddy
T-6 and above

SSS

1. Nazia Parveen, Female
2. S.N. Rasool, Male

CLTS

1. K. Lakshmi, Female
2. M.S. Ravi, Male

Award for the decade

1. M. Dinesh, Admn.inc. Finance
2. B.S.N. Murthy, Technical
3. C. Chandra-mouli, SSS
4. Md. Sadat, CLTS

Special Category Awards

(Covering all the categories of employees from Faculty to CLTS)

Special recognitions

1. J.N.L. Das for Swachch Bharat Abhiyan
2. Dr. L.M. Ahire for Infrastructure
3. K. Obulapathi for Energy Conservation
4. S. George –Outstanding award for Institution building

ICAR Cash Award Scheme 2018 from Administrative & Skill Support Staff

(Cash Award & Certificates presented on 16.07.2019)

1. Mrs. Rukmani Ammal under Administrative category
2. Mrs. A. Saroja under Skilled Support Category

10.2 List of Committees

Vigilance

1. Dr. S. Senthil Vinayagam, Vigilance Officer (w.e.f. 19-03-2019)

Higher Education Committee

1. Dr. Ch. Srinivasa Rao, Director - Chairman
2. Dr. S. K. Soam, Head, ICM Division & Joint Director I/C - Member
3. Dr. Bharat Sontakki, Head, XSM Division - Member
4. Dr. M. Krishnan, Head, ESM Division - Member
5. Dr. I. Sekar, Head, RSM Division - Member
6. Dr. Ranjit Kumar, Head, ABM Division - Member
7. Dr. R.V.S. Rao, Head, HRM Division - Member
8. Dr. P. Krishnan, Principal Scientist & Head, PME Cell - Member

9. Dr. P. Ramesh, Principal Scientist & Controller of Examinations (CoE) - Member
10. Shri. C. Muralidharan, Joint Director (A)& Registrar - Member
11. Dr. P.C. Meena, OIC, Training & PGS Unit - Member - Secretary
12. Coordinators of PGDMA, PGD-TMA, PGD-ETM - Coopted Members

Board of Studies for PGDMA, PGDTMA, and PGDETM

1. Dr Ch. Srinivasa Rao, Director - Chairman
2. Dr. S. K. Soam, Head, ICM Division & Joint Director I/C - Member
3. Dr. Bharat Sontakki, Head, XSM Division - Member
4. Dr. Ranjit Kumar, Head, ABM Division - Member
5. Dr. K H. Rao, Principal Scientist & In-charge, Placement Cell - Member

6. Dr. A. Dhandapani, Principal Scientist, ICM Division - Member
7. Dr. D. Thammi Raju, Principal Scientist, ESM Division - Member
8. Shri C. Muralidaran, Joint Director (A) & Registrar - Member
9. Shri Z.H. Khilji, CFAO - Member
10. Shri. K. R. Ghanshyam, AAO, Training & PGS Unit - Member
11. Dr. P.C. Meena, Principal Scientist, ABM - Member secretary.
12. Coordinators of PGDMA, PGD-TMA, PGD-ETM - Coopted Members

Purchase Advisory Committee

1. Dr. I. Sekar, Head, RSM Division - Chairman
2. Dr. A. Dhandapani, Principal Scientist ICM Division - Member
3. Joint Director (Admin) and Registrar - Member
4. CFAO - Member
5. Storage and Purchase Officer - Member Secretary

Library Advisory Committee

1. Dr. S.K. Soam, Head -ICM Division and JD I/C - Chairman
2. All HoDs - Member
3. Dr. P. Krishnan, Pr. Scientist, RSM Division - Member
4. Dr. V.V. Sumanth Kumar, Senior Scientist, ESM Division - Member
5. Joint Director (A) & Registrar - Member
6. Chief Finance & Accounts Office - Member
7. Dr. V. Murali, CTO - Member Secretary
8. Dr. N. Srinivasa Rao, Principal Scientist - Member Secretary

Annual Report Committee

1. Dr. I. Sekar, Head, RSM Division - Chairman
2. Dr. Bharat Sontakki, Head, XSM Division - Member
3. Dr. Ranjit Kumar, Head, ABM Division - Member
4. Dr. S. Senthil Vinayagam, Head In-charge, ESN Division - Member
5. Dr. P. Ramesh, Principal Scientist, HRM Division - Member.
6. Dr. K. Srinivas, Principal Scientist, RSM Division - Member
7. Dr. P. D. Sreekanth, Principal Scientist, ICM Division - Member
8. Dr. Alok Kumar, Principal Scientist, HRM Division - Member Secretary

Anti-Ragging Committee

1. Joint Director/ Dean - Chairman
2. Dr. I. Sekar, Head, RSM Division - Member
3. Joint Director (Admn.) & Registrar - Member
4. Academic Coordinators, PGDM (A) Programme - Members
5. Office In-charge, Training & P.G. Studies - Member Secretary

Anti-Ragging Squad

1. Dr. (Mrs.) Surya Rathore, Pr. Scientist - Chairperson
2. Dr. B. Ganesh Kumar, Pr. Scientist - Member
3. Chief Finance & Accounts Officer - Member
4. Officer In-charge, Hostel - Member Secretary

Ombudsman / Grievance Committee of the Academy

1. Director - Chairman
2. Joint Director and Dean - Member
3. Joint Director (Admn.) & Registrar - Member
4. Academic Coordinators of PGDM (A) Programme - Member



5. Senior Admn. Officer - Member
6. Shri S. Rajukumar, Technical Officer - Member
7. Dr. L.M. Ahire, ACTO & In-charge, Training Unit - Member Secretary

Committee for Regulatory Affairs

1. Dr. N. Sivaramane, Pr. Scientist - Chairman
2. Senior Admn. Officer - Member
3. Shri S. Rajukumar, Technical Officer - Member
4. Dr. L.M. Ahire, Asst. Chief Technical Officer - Member
5. Shri K.R. Ghanshyam, AAO Member Secretary

Swachh Bharat Abhiyan Implementation Committee

1. Dr. R.V.S. Rao, In-charge Head, HRM Division - Chairman, Farm Advisory Committee
2. Dr. M.A. Basith, CTO & OIC, Farm - Member
3. Shri J.N.L. Das, Senior Admn. Officer - Member
4. Dr. L.M. Ahire, ACTO, Academic Cell - Member
5. Shri M. Shekhar Reddy, ACTO & OIC Outsourcing Service - Member
6. Dr. P. Vijender Reddy, CTO and In-charge, Academic Cell - Member Secretary

Farm Advisory Committee / Landscape and Beautification Committee

1. Dr. R.V.S. Rao, I/C Head-HRM Division - Chairman
2. Dr. P. Ramesh, Pr. Scientist, HRM Division - Member
3. Dr. G.R.K. Murthy, Pr. Scientist, ESM Division - Member
4. Dr. Surya Rathore, Pr. Scientist, XSM Division - Member
5. Dr. B.S. Yashavanth, Scientist, ICM Division - Member
6. Dr. V. Murali, CTO Member

7. Dr. Ahire Laxman Mahar, ACTO - Member
8. Shri P. Mohan Singh, STO - Member
9. Shri M. Shekhar Reddy, AC - Member
10. Shri T. Laxman - Member
11. Dr. M.A. Basith, AC - Member Secretary

Newsletter Committee

1. Dr. Ranjit Kumar, Head-ABM Division - Chairman
2. Dr. Sanjiv Kumar, Scientist, ABM Division - Member
3. Dr. B.S. Yashavanth, Scientist, ICM Division - Member

Campus Residents Welfare Committee

1. Dr. S.K. Soam, Head -ICM Division and JD I/C - Chairman
2. Dr. M. Balakrishnan, Principal Scientist - Member
3. Dr. Manju Gerard, Principal Scientist, RSM Division - Member
4. Dr. B.S. Yashavanth, Scientist, ICM Division - Member
5. Shri. Shitanshu Kumar, SAO, ICAR-IIOR, Hyderabad - Member
6. Shri. Ch. Suresh Kumar, Assistant, ICAR-IIMR, Hyderabad - Member

Examination Unit

1. Dr. P. Ramesh, Principal Scientist, HRM Division - CoE
2. Dr. Manju Gerard, Principal Scientist, RSM Division, In-charge - CoE
3. Shri J.N.L. Das, Senior Admn. Officer
4. Shri Pitla Srinivas, Senior Technical Assistant
5. Mrs. Padma Saroja, Assistant
6. Shri B. Ashok, SSS, Training Unit

Hostel Management Committee

1. Dr. S.K. Soam, Joint Director I/c. - Chairman
2. Dr. Surya Rathore, Principal Scientist - Member
3. Dr. M. Balakrishnan, Principal Scientist - Member
4. Joint Director(A) & Registrar - Member
5. Chief Finance & Account Officer - Member
6. Dr. Yashavanth B.S., Scientist - Member
7. Smt. Savithri Murali, Sr.Tech. Officer - Member
8. Sh. Sham Bahadur, Asst. Chief Tech. Officer - Member Secretary

Institute Technology Management Committee

1. Dr. Ch. Srinivasa Rao, Director, NAARM - Chairman
2. Dr. S. K. Soam, Jt. Director, I/c - Member
3. Dr. G.R.K. Murthy, Pr. Scientist, ESM Division - Member
4. Dr. K. Srinivas, Pr. Scientist, RSM - Member
5. Joint Director(Admin) & Registrar - Member
6. Sh. S. George, CFAO - Member
7. Dr. R.M. Sundaram, Pr. Scientist, IIRR Member
8. Dr. Manju Gerard, Pr. Scientist, RSM Division - Member Secretary

Sports Complex Maintenance Committee

1. Dr. P Vijender Reddy, ACTO - Chairman
2. Dr. M. A. Basith, CTO - Member
3. Sh. Sohail Ahmd Khan, ACTO - Member
4. Dr. Laxman M Ahire, ACTO - Member

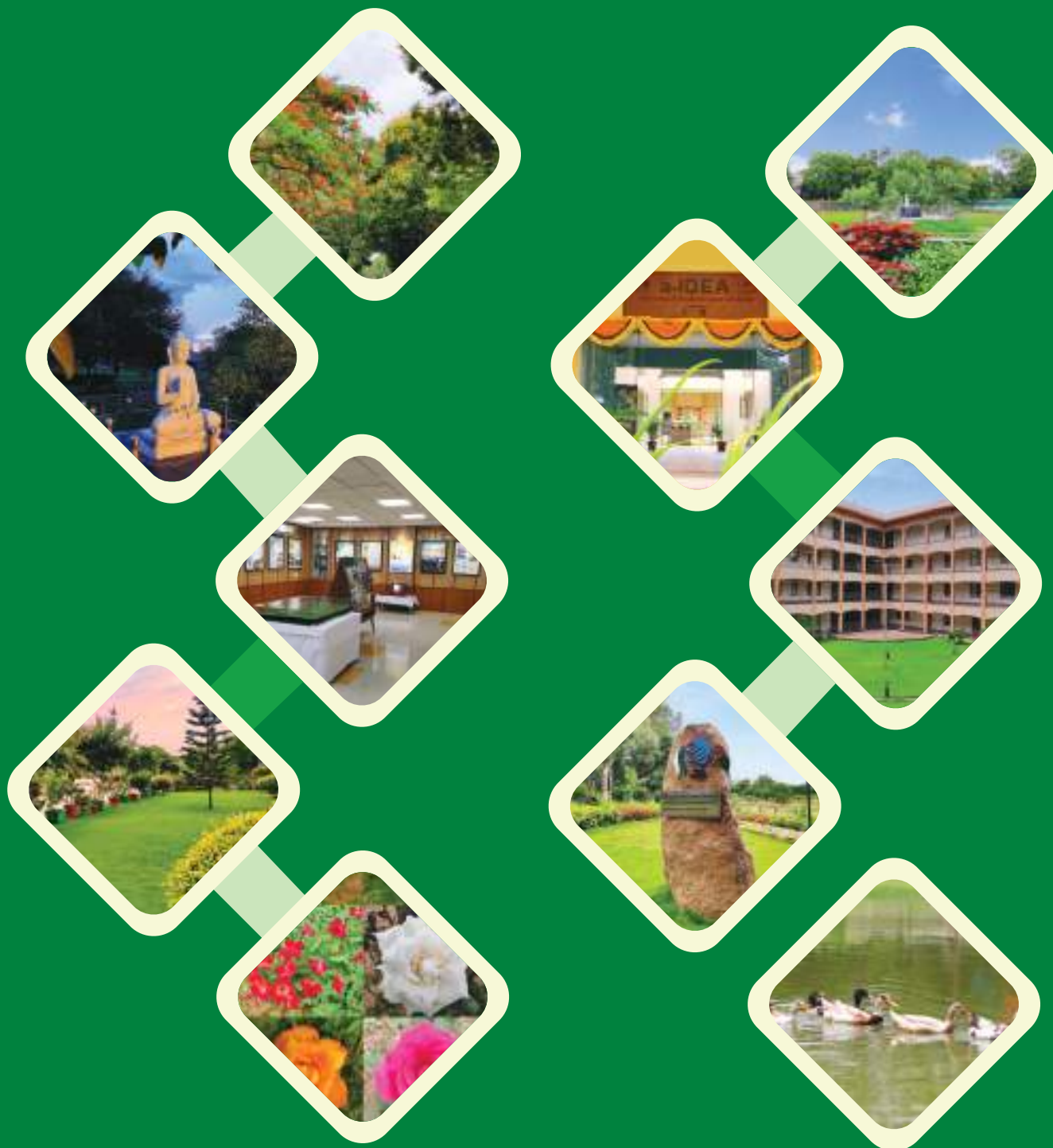
5. Sh. M Shekhar Reddy, ACTO - Member
6. Sh. P. Mohan Singh, STO - Member
7. Sh. Sham Bahadur, ASTO - Member
8. Sh. M.K. Samson, UDC - Member
9. Smt. K.K. Rukmani Ammal, PA- Member Secretary

Campus Canteen Committee

1. Dr. Manju Gerard, Pr. Scientist, RSM Division - Chairperson
2. Dr. P.C. Meena, Principle Scientist, ABM - Member
3. Sh. P.G. Kohad, AAO - Member
4. Smt. Y. Gayarthri, UDC - Member
5. Representative from FOCARS - Member
6. Two Representatives each from PGDMA I & II - Member
7. Secretary, IJSC - Member
8. Sh. K. Veeranarsaah, Tech. Assistant, Member Secretary

Internal Complaints Committee

1. Dr. Surya Rathore, Principal Scientist, XSM - Chairperson
2. Mrs. Vjayalaxmi, Jr. Account Officer - Member
3. Mrs. G. Aneja, Asst Chief Technical Officer - Member
4. Mrs. P. Padmavathi, Katurba Gandhi National Memorial Trust - Member
5. Mrs. KK Rukmani Ammal, AAO, S&P Officer - Member Secretary



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